Praxia Update

Daily orthopaedic literature digest · Sunday, 11 October 2026 · 521 new articles across 10 journals

American Journal of Sports Medicine

Volume 54, Issue 12 · October 2026 · 31 new articles

More Than a Decade of Data: The Maturation of Hip Preservation Through the Sports Medicine Lens.

Spiker AM

PubMed · DOI

Outcomes and Return-to-Sport Rates of Competitive Athletes Treated for Femoroacetabular Impingement Syndrome With Contemporary Hip Arthroscopy Techniques: A Matched Comparison to Early Techniques.

Schab AR, Rana K, Wallace I, Quesada-Jimenez R, Jimenez AE, Kuhns BD, Domb BG

BACKGROUND: Hip arthroscopy for the treatment of femoroacetabular impingement syndrome (FAIS) and labral tears has evolved over time with varying methods of labral and capsular treatment.

PURPOSE: To report patient-reported outcomes (PROs) and return-to-sport (RTS) rates of athletes undergoing primary hip arthroscopy with contemporary techniques and compare their outcomes to those of athletes treated with early techniques.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective analysis of the records of patients who underwent primary hip arthroscopy for the treatment of FAIS and labral tears was performed. All patients reported preoperative participation in sports within 1 year of surgery. The contemporary arthroscopy (CA) cohort underwent capsular repair, and labral treatment was either repair or reconstruction with knotless anchors. The early technique (ET) cohort received labral treatment with either a knotted repair or debridement and did not have capsular repair. Patients had a 2-year minimum follow-up for PROs or an endpoint. The cohorts underwent 1:1 propensity matching for age, sex, body mass index, and level of competition. Clinically important thresholds, PROs, secondary surgeries, and RTS rates were compared.

RESULTS: A total of 98 hips (92 patients) were included in the study. Athletes in both cohorts demonstrated significant improvement in P […]

PubMed · DOI

Psychiatric Comorbidity Is Associated With Increased Failure Rates and Inferior Clinical Outcomes After Hip Arthroscopy for Femoroacetabular Impingement Syndrome at a Minimum 10-Year Follow-up.

Cameron RK, Jazrawi I, Montgomery SR, Hollis R, Budis E, Samsonov A, Stein S, Youm T

BACKGROUND: Psychiatric comorbidity has been linked to inferior orthopaedic outcomes, but studies evaluating its impact on survivorship after hip arthroscopy for femoroacetabular impingement syndrome (FAIS) in the long term remain limited.

PURPOSE: To evaluate the association between preoperative psychiatric comorbidity, particularly among those who are using psychotropic medications, and failure, patient-reported outcome measure (PROM) scores, and failure-free survivorship after hip arthroscopy for FAIS at a minimum 10-year follow-up.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective review was performed on 243 adult patients who underwent primary hip arthroscopy for FAIS between 2010 and 2014. A preoperative psychiatric comorbidity diagnosis and psychotropic medication

use were identified through a medical record review. Failure was defined as revision hip arthroscopy or conversion to total hip arthroplasty. Kaplan-Meier analysis was used to assess failure-free survivorship. Logistic regression analyses were performed to evaluate associations between psychiatric comorbidity and failure. PROMs, including the modified Harris Hip Score (mHHS) and Non-Arthritic Hip Score (NAHS), were administered preoperatively and at 10-year follow-up. A post hoc power analysis was performed using a 2-proportion model. Significance was considered at P < .05.

RESULTS: […]

PubMed · DOI

Labral Repair With Concomitant Endoscopic Gluteus Medius and/or Minimus Repair: Outcomes at a Minimum 5-Year Follow-up Relative to a Propensity Score-Matched Control Group.

Sparks CA, Brinkman JC, Monty TL, Nho SJ

BACKGROUND: Gluteal tendon and labral tears commonly co-occur and can be addressed in a single surgical procedure. The concomitant management of gluteal tendon and labral tears has shown favorable outcomes at short-term, mid-term, and long-term follow-up, with comparable outcomes to isolated labral repair at 2 years. Yet, there remains a limited understanding of the outcomes of concomitant gluteal tendon and labral repair compared to isolated labral repair at mid-term follow-up.

PURPOSE: To compare patient-reported outcomes (PROs), achievement of clinically significant outcomes, and reoperation-free survival between concomitant gluteal tendon and labral repair compared to isolated labral repair.

STUDY DESIGN: Retrospective cohort study; Level of evidence, 3.

METHODS: Patients who underwent concomitant endoscopic gluteal repair and arthroscopic labral repair between January 2012 and November 2019 with ≥5-year follow-up were included. Patients who underwent concomitant gluteal and labral repair were propensity score matched to patients who underwent isolated labral repair by age, sex, and body mass index. PROs were assessed preoperatively and at ≥5 years postoperatively, including those for the Hip Outcome Score-Activities of Daily Living (HOS-ADL), Hip Outcome Score-Sports Subscale (HOS-SS), modified Harris Hip Score (mHHS), 12-item International Hip Outcome Tool (iHOT-12), a […]

PubMed · DOI

Periacetabular Osteotomy and Combined Open Femoral Head-Neck Junction Osteochondroplasty: A Concise Follow-up of a Previous Report at a Mean 10-Year Follow-up.

Shah VP, O'Connor KP, Harris M, Nepple JJ, Pascual-Garrido C, Clohisy JC

BACKGROUND: Hip dysplasia is a well-established cause of hip dysfunction and secondary osteoarthritis in young adults. Periacetabular osteotomy (PAO) is a common surgical treatment for symptomatic acetabular dysplasia. Recent awareness of concomitant impingement in dysplastic hips has led to inclusion of osteochondroplasty (OCP) for femoral head-neck abnormalities in select cases to improve survivorship and clinical outcomes.

PURPOSE: To compare outcomes, reoperation rates, radiography, survivorship, and complications at a mean of 10 years after PAO with and without femoral head-neck OCP.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: The authors conducted a retrospective review of the data of 38 patients (38 hips) who underwent PAO with OCP compared to a matched control group of 42 patients (42 hips) who underwent PAO only between 2000 and 2007. Outcomes included the modified Harris Hip Score (mHHS) and Western Ontario and McMaster Universities Osteoarthritis Index score. Failure was defined as clinical failure (failure to meet mHHS minimal clinically important difference [MCID] or Patient Acceptable Symptom State [PASS]), conversion to total hip arthroplasty (THA), reoperation, and composite failure (either clinical failure or reoperation).

RESULTS: The mean follow-up was 10.6 years (range, 6.9-16.3 years) for the PAO+OCP cohort and 11.8 years (range, 7.2-17.2 […]

PubMed · DOI

Absolute Risk of Anterior Cruciate Ligament Graft Reruptures From the Posterior Tibial Slope Falls With Increasing Age: A Risk Stratification Analysis.

Pineda T, Mazy D, Ramos-Rojas J, Cance N, Dan MJ, Demey G, Dejour DH

BACKGROUND: Posterior tibial slope (PTS) and young age are well-established independent predictors of an anterior cruciate ligament (ACL) graft rerupture. Although both factors have been studied individually, whether age changes the relative effect of PTS, its absolute clinical impact, or both has not been formally examined across the full age range of patients undergoing ACL reconstruction (ACLR).

PURPOSE: To determine whether age modifies the relative or absolute rerupture risk associated with PTS after ACLR, and to identify age thresholds at which clinically relevant absolute risk levels are reached for given slope values.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective cohort of 585 patients who underwent primary ACLR with hamstring tendon autografts was analyzed, comprising a low-risk group (isolated ACLR) and a high-risk group (ACLR with lateral extra-articular tenodesis), with a minimum follow-up of 6 years. The Firth penalized logistic regression was used to model the joint effects of PTS and age on graft reruptures in pooled and group-specific analyses. A PTS×age interaction term was evaluated via the likelihood ratio test to assess whether the relative effect of PTS differed across age strata. Predicted absolute rerupture probabilities were derived from additive models, and age thresholds below which the risk fell under predefined clinica […]

PubMed · DOI

Increased Posterior Tibial Slope and Lateral Extra-articular Tenodesis Are Independently Associated With Revision ACL Reconstruction in a Young High-Risk Population.

van der Wal WA, de Groot TM, Tamana L, Tsoulgkounidis N, Veraart JT, Hoogeslag RAG, van der Hoeven H, Engelsma Y et al.

BACKGROUND: Increased posterior tibial slope (PTS) has been associated with higher revision rates after anterior cruciate ligament reconstruction (ACLR). Whether the addition of lateral extra-articular tenodesis (LET) mitigates the risk associated with increased PTS is unclear.

PURPOSE: To determine the prevalence of increased PTS (≥12°) in a young high-risk primary ACLR population and evaluate whether LET is associated with lower revision rates in patients with increased PTS.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This study included patients aged ≤25 years undergoing primary ACLR across 3 high-volume centers with ≥2 years between surgery and data collection. PTS was measured on lateral radiographs for medial and lateral plateaus. Primary analyses consisted of 4 prespecified multivariable logistic regression models with different PTS operationalizations as continuous predictors per 1° increase (medial PTS, lateral PTS, the maximum [PTSmax], and the lateral-medial difference), adjusted for patient and surgical covariates including graft type, with surgical center as a fixed effect. Secondary analyses used PTS dichotomized at 12°; revision rates were compared between isolated ACLR and ACLR with LET, with the treatment effect quantified as absolute risk reduction and number needed to treat.

RESULTS: A total of 780 patients were included; 84 (10.8%) underwe […]

PubMed · DOI

Association of Slope-Reducing Osteotomy With Return to Preinjury Level of Sport After Failed Anterior Cruciate Ligament Reconstruction in Elite Athletes.

Martorell-de Fortuny L, Perelli S, Cance N, Mazy D, Monllau JC, Dejour DH

BACKGROUND: Increased posterior tibial slope (PTS) has been identified as a major risk factor for failure after anterior cruciate ligament reconstruction (ACLR). In athletes with high PTS undergoing revision ACLR, slope-reducing osteotomy (SRO) has been proposed to reduce anterior cruciate ligament graft forces and improve outcomes, but data regarding return to sport in elite athletes remain limited.

PURPOSE: To evaluate whether combining SRO with revision ACLR would allow elite athletes with increased PTS to return to their preinjury level of sport and, further, to assess clinical, functional, and radiographic outcomes.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: This retrospective study included elite athletes with failed ACLR and PTS ≥10° who underwent revision ACLR combined with supratuberosity SRO between 2011 and 2023 at 2 centers. Return-to-sport (RTS) rate and time, clinical and functional outcomes (International Knee Documentation Committee [IKDC], Lysholm, Tegner, Anterior Cruciate Ligament-Return to Sport After Injury scale [ACL-RSI]), and radiographic parameters (PTS, hip-knee-ankle angle, and Caton Deschamps index) were evaluated.

RESULTS: Of the 22 ACLRs with SRO performed, 19 patients met the inclusion criteria. All athletes returned to competition at their preinjury level after a mean of 12.1 ± 2.3 months (range, 10-20 months), with a mean ACL- […]

PubMed · DOI

Quadriceps Tendon Autograft Harvest Technique Affects Both Graft Strength and Cross-sectional Area.

Bi AS, Augustin EJ, Smelley CC, Angotti ML, Acheampong KK, Sachdev D, Shewman E, Yanke AB

BACKGROUND: Bone-patellar tendon-bone (BTB) autografts have been considered the gold standard for anterior cruciate ligament reconstruction, although the popularity of quadriceps tendon (QT) autografts is increasing.

HYPOTHESIS/PURPOSE: The purpose of this study was to compare the ultrasound-derived cross-sectional area (CSA) and biomechanical strength of BTB and QT grafts harvested with different techniques. The authors hypothesized that ultrasound CSA measurements would be reliable and that QT grafts would be larger and stronger than BTB grafts, without differences between QT harvest methods.

STUDY DESIGN: Controlled laboratory study.

METHODS: Thirty grafts from 15 cadaveric knees included 15 BTB grafts harvested with a 9-mm double-blade scalpel, 7 full-thickness QT grafts harvested with the same 9-mm double-blade scalpel (DB-QT), and 8 QT grafts harvested with a proprietary 9-mm cylindrical QT harvester (CH-QT). CSA was measured at 3 regions with ultrasound by 2 raters, followed by cyclic loading and pull-to-failure testing. Between-group differences were analyzed, and associations between CSA and biomechanical outcomes were assessed with linear regression.

RESULTS: Ultrasound CSA demonstrated good to excellent reliability (ICC2,1 = 0.70-0.87; ICC2,k = 0.82-0.93). CSA differed between graft types (mean ± SD; BTB, 0.44 ± 0.08 cm2; CH-QT, 0.64 ± 0.10; DB-QT, 0.97 ± 0.22; a […]

PubMed · DOI

Biomechanical Effect of Medial Collateral Ligament Repair With and Without Suture Augmentation on Knee Laxity and Anterior Cruciate Ligament Force.

von Rehlingen-Prinz F, Shamritsky DZ, Rilk S, Kunze KN, Mueller MM, Demopoulos B, Imhauser CW, DiFelice GS

BACKGROUND: The medial collateral ligament (MCL) is the most frequently injured knee ligament. MCL repair is a surgical technique intended to acutely reduce medial side laxity that can be performed in isolation or combined with suture augmentation (SA).

PURPOSE: To assess the effect of MCL repair with and without SA on (1) valgus laxity and anteromedial rotational laxity and (2) force carried by the anterior cruciate ligament (ACL) and (3) to quantify load-sharing between the MCL repair and SA components.

STUDY DESIGN: Controlled laboratory study.

METHODS: Seven cadaveric knees were tested using a 6 degrees of freedom robotic manipulator in 4 states: (1) intact, (2) MCL cut, (3) isolated MCL repair, and (4) MCL repair combined with SA. Valgus laxity and anteromedial rotational laxity were measured at 0°, 15°, 30°, and 90° of flexion. In situ ACL force and forces carried by the MCL repair components were assessed at peak valgus moment using principle of superposition.

RESULTS: Valgus laxity after isolated MCL repair was greater than the intact state at all flexion angles >0° (P≤ .01). The addition of SA to MCL repair significantly reduced valgus laxity at 15° and 30° of flexion only (P≤ .02). Anteromedial rotational laxity after both MCL repairs was greater than the intact state at 90° of flexion (P < .01),

whereas no differences were detected at 0° and 30°. For both MCL rep […]

PubMed · DOI

Effect of Multiple Centralization Sutures on Meniscal Extrusion and Contact Pressures in a Cadaveric Model of Medial Meniscus Root Repair.

Andersen M, Tollefson LV, Ngobi J, Carlson MR, LaPrade CM, Fladberg B, Frihagen F, Inderhaug E et al.

BACKGROUND: Adding a centralization suture (CS) during repair of a posterior medial meniscus root (PMMR) tear may decrease meniscal extrusion and osteoarthritis progression. However, the number and optimal placement of CSs remain unclear.

HYPOTHESIS: Increasing numbers and changing localization of CSs would not influence the degree of postoperative extrusion or joint contact pressures.

STUDY DESIGN: Controlled laboratory study.

METHODS: Meniscal extrusion and tibiofemoral contact pressures were assessed using a 3-dimensional motion capture system and pressure sensors in 9 human cadaveric knees. A nonanatomic root repair was created to simulate persistent meniscal extrusion after a PMMR repair. Each knee was tested under 10 conditions: (1) intact, (2) PMMR tear, (3) root repair, (4) CS at the posterior border of the medial collateral ligament (MCL) (CS1), (5) CS midway between CS 1 and CS 3 (CS2), (6) CS at the midpoint between the MCL and posterior root attachment (CS3), and (7-10) combinations of these CSs (all CSs were in addition to the root repair). Analysis of variance testing was used for analyses of pressure data and extrusion measurements.

RESULTS: There was no significant difference in medial contact pressure between the intact state and the states with root repair in combination with any CS configuration ($P > .05$). Contact pressures were reduced compared to the ro […]

PubMed · DOI

The Association of Advanced Patellofemoral Chondrosis With Outcomes After Posterior Medial Meniscus Root Repair.

Shen M, Chen L, Padon B, Ruff G, Kaplan DJ, Strauss E, Jazrawi L, Alaia M

BACKGROUND: Meniscus root repair has been shown to improve symptoms and protect against progressive knee arthritis by restoring more normal weightbearing force distribution. While advanced tibiofemoral chondrosis is known to negatively affect root repair outcomes, there is a paucity of literature describing the impact of concomitant patellofemoral chondrosis (PFC).

PURPOSE: To evaluate outcomes of meniscus root repair in patients with and without advanced PFC.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This single-center retrospective study included patients who underwent isolated posterior medial meniscus root repair (PMMRR) between July 2013 and February 2023, with a minimum 2-year follow-up. Based on intraoperative arthroscopic findings, patients were stratified into 2 groups: those with PFC, defined as Outerbridge grade ≥3, and those without PFC (Outerbridge grade < 3). Clinical outcomes, including meniscal retear, conversion to unicompartmental knee arthroplasty (UKA) or total knee arthroplasty (TKA), and revision meniscal surgery, were obtained through chart review. Patient-reported outcomes (PROs) were collected from patients without retear or revision meniscal surgery and who did not convert to UKA or TKA and included the Knee injury and Osteoarthritis Outcome Score-12 (KOOS-12), International Knee Documentation Committee (IKDC) subjective score, and […]

PubMed · DOI

More Horizontal ACL Posterolateral Bundle Position in Osteochondritis Dissecans of the Lateral Femoral Condyle.

Lehoczky G, Khoo B, Lin Z, Shea KG

BACKGROUND: The pathogenesis of osteochondritis dissecans (OCD) is multifactorial. An association of the horizontal anterior cruciate ligament (ACL) position with the anatomic proximity of its posterolateral (PL) bundle to the OCD of the lateral femoral condyle (LFC) has not been previously found.

HYPOTHESIS: In OCD of the LFC, a more horizontal ACL and PL bundle position can be found, compared to that of control patients.

STUDY DESIGN: Case-control study; Level of evidence, 3.

METHODS: The authors retrospectively analyzed magnetic resonance imaging scans of 90 knees in 90 patients, creating 3 age- and sex-matched cohorts in a 1:1:1 ratio. Group 1 had OCD of the LFC. As controls, group 2 had OCD of the medial femoral condyle and group 3 had no structural knee pathology. Patients <18 years of age with no previous surgeries and intact ACLs were included, and patients with discoid meniscus were excluded. On the coronal plane, the height of the PL bundle insertion was measured relative to physeal height and LFC height. Femorotibial angle was also calculated to verify consistency of the measurement plane.

RESULTS: The mean age was 12.5 years (range, 8-16 years; SD, 2.24 years), with 67% male patients. There was a statistically significant difference between the height of the PL bundle insertion in patients with OCD of the LFC relative to the physis height ($P < .001$) and the LFC […]

PubMed · DOI

Patella Alta and Trochlear Dysplasia as a Determinant of Patellar Tendon Lateralization and Inclination: A Cross-sectional MRI Study in Adolescents With Patellofemoral Instability.

Crippa M, Beltrame G, Beber SA, Matarangas AI, Lu S, Lijesen E, Jaramillo D, Green DW

BACKGROUND: Patellofemoral instability (PFI) is traditionally assessed using tibial tubercle-trochlear groove (TT-TG) distance, patellar height indices, and trochlear morphology. However, soft tissue-based measures of patellar tendon orientation may provide complementary insight into patellar containment and extensor mechanism malalignment. Patellar tendon-lateral trochlear ridge distance (PT-LTR) has been proposed as a tendon-based marker of lateralization. The authors introduce the patellar tendon inclination angle (PTIA): a magnetic resonance imaging (MRI)-based parameter designed to quantify the obliquity of the patellar tendon relative to the femoral condyles.

PURPOSE: To define and validate PTIA and to compare its performance with PT-LTR. Secondary objectives were to assess correlation of PTIA and PT-LTR with established morphologic risk factors for PFI and to evaluate their diagnostic accuracy for identifying PFI.

STUDY DESIGN: Case-control study; Level of evidence, 3.

METHODS: MRI studies of 284 knees (PFI, n = 142; controls, n = 142) were included in this analysis. PTIA was developed and measured by orthopaedic surgeons, and their agreement was calculated. PT-LTR, sulcus angle, patellar tilt, lateral patellofemoral angle, TT-TG distance, and Caton-Deschamps index (CDI) were measured for all knees as well. Correlations between PT-LTR/PTIA and other classic measuremen […]

PubMed · DOI

IL-1β-Induced Inflammatory Transcriptome and TGF-β1 Attenuation of Chemokine Expression in ACL Remnant Fibroblasts.

Cai L, Brophy RH, Tycksen ED, Rai MF

BACKGROUND: The high incidence of posttraumatic osteoarthritis after anterior cruciate ligament (ACL) injury and reconstruction suggests that biological mechanisms beyond joint instability contribute to disease progression. After injury, ACL-resident cells are exposed to inflammatory mediators within the joint and may influence the local inflammatory microenvironment; however, their global transcriptomic responses to inflammatory stimuli remain underexplored.

HYPOTHESIS: Interleukin-1 beta (IL-1β) induces inflammatory transcriptomic responses in ACL remnant-derived (ACLr) fibroblasts, and transforming growth factor beta 1 (TGF-β1) attenuates these responses.

STUDY DESIGN: Descriptive laboratory study.

METHODS: Primary ACLr fibroblasts were isolated from ACL remnants (n = 10) collected at the time of knee surgery. Near-confluent cultures were treated with IL-1β (10 ng/mL) for 24 hours to model an inflammatory joint environment. Genome-wide transcriptional changes were assessed by RNA sequencing using an Illumina NovaSeq-6000, with selected targets confirmed by microfluidic digital polymerase chain reaction (PCR). To evaluate the effects of TGF-β1, ACLr fibroblasts pretreated with IL-1β were exposed to TGF-β1 (10 ng/mL) for 48

hours, and chemokine transcript levels were assessed using digital PCR.

RESULTS: IL-1β induced broad inflammatory transcriptional responses characteriz […]

PubMed · DOI

Suprascapular Nerve Release Improves the Histological Healing Morphology and Biomechanical Strength of the Rotator Cuff in a Rat Chronic Massive Rotator Cuff Tear Model.

Zhang W, Guo Y, Wang H, Huang R, Wang Y, Sun Y

BACKGROUND: While suprascapular nerve (SSN) entrapment is widely recognized as a complication of massive rotator cuff tears (MRCTs), its role as an independent factor affecting tendon healing remains underreported. Additionally, the therapeutic benefit of SSN release in MRCT repair remains controversial.

PURPOSE: To investigate the incidence of SSN entrapment via histological stain and whether SSN release enhances rotator cuff healing in a rat model of delayed MRCT repair.

STUDY DESIGN: Controlled laboratory study.

METHODS: A chronic MRCT model was established in rats via bilateral transection of the supraspinatus and infraspinatus tendons. After 4 weeks, 5 rats (10 shoulders) were euthanized to assess SSN entrapment incidence via immunofluorescence staining. The remaining 28 rats were randomized into 2 groups: tendon repair (TR) group (n = 14; bilateral tendon reattachment) and tendon repair + nerve release (TR+NR) group (n = 14; same repair with concurrent SSN release). Postoperative evaluations at 4 and 8 weeks included biomechanical testing (maximum load and stiffness) and histological analysis (healing quality, nerve regeneration).

RESULTS: SSN entrapments were observed in 7 of 10 shoulders using Neurofilament-200 staining. The TR+NR group demonstrated significantly higher maximum load values at 8 weeks postrepair ($P < .05$). Additionally, the TR+NR group achieved signi […]

PubMed · DOI

The Lark-Loop Technique for Onlay Biceps Tenodesis: A Biomechanical, Histological, and Radiological Study in a Rabbit Model.

Zhou M, Li X, Zhang J, Deng X, Wang Z, Hou J, Yan Y, Long Y et al.

BACKGROUND: Arthroscopic cortical surface tenodesis has been established as an effective technique for managing lesions of the long head of the biceps tendon (LHBT). However, traditional suture techniques frequently provide insufficient initial mechanical strength, which may contribute to tenodesis failure. To address this limitation, the Lark-Loop suture technique was developed.

PURPOSE: To compare the biomechanical, histological, and radiological outcomes of Lasso-Loop, Lark-Loop, and Krackow suture techniques for cortical surface tenodesis of the LHBT in a rabbit model.

STUDY DESIGN: Controlled laboratory study.

METHODS: A total of 75 New Zealand White rabbits underwent bilateral proximal biceps tenodesis using a cortical surface fixation model. Animals were randomly assigned to 1 of 3 groups: (1) Lasso-Loop suture (control group; n = 25), (2) Lark-Loop suture (experimental group; n = 25), and (3) Krackow suture (positive control group; n = 25). At 4, 8, and 12 weeks postoperatively, 21 rabbits from each group (n = 63 total) were sacrificed for evaluation: 3 rabbits per group were used for magnetic resonance imaging and histological analysis, and 4 rabbits per group for micro-computed tomography and biomechanical testing. An additional 12 rabbits (4 per group) were used for initial biomechanical testing at week 0. One-way analysis of variance and Bonferroni post hoc corre […]

PubMed · DOI

Minimum 5-Year Follow-up After Arthroscopic Latarjet: Outcomes and Complications.

Berge T, Garratt AM, Moatshe G, Øines AG, Blich I, Støen R, Ludvigsen TC, Vuleti■ F et al.

BACKGROUND: The Latarjet procedure is widely used for recurrent anterior shoulder instability with glenoid bone loss. Arthroscopic techniques are increasing; however, there is limited evidence for mid- to long-term outcomes.

PURPOSE: To evaluate clinical outcomes, recurrence rates, and quality of life at a minimum of 5 years' follow-up after undergoing the arthroscopic Latarjet procedure.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Patients who underwent the arthroscopic Latarjet procedure using double-screw fixation at Oslo University Hospital between 2014 and 2019 were prospectively registered. Descriptive data and Western Ontario Shoulder Instability Index (WOSI) scores were recorded preoperatively and at 1 and 5 years postoperatively. At 5 years, complications and health-related quality of life (HRQoL; EQ-5D-5L) were assessed.

RESULTS: Of 107 patients who underwent the arthroscopic Latarjet procedure, 93 cases met the inclusion criteria. After loss to follow-up, 85 cases were included. The median follow-up time was 5.2 years (range, 5.1-9.4 years). The median age at surgery was 27.8 years (range, 15.6-49.8 years), and 78 of 85 (92%) cases were male. Of the 85 cases, 56 (66%) had >10 dislocations before surgery, and 36 (42%) were revision procedures. At the 5-year follow-up, 2 of 85 patients experienced recurrent dislocations, and 8 of 85 reported subluxat […]

PubMed · DOI

Arthroscopy for Femoroacetabular Impingement in Tonnis Grade 2 Arthritic Hips Results in Similar Clinical and Radiographic Outcomes but Lower Survivorship Than Tonnis Grades 0 and 1: A Systematic Review and Meta-analysis.

Krivicich L, Driscoll A, Hayes-Lattin M, Guenthner G, Yen YM, Salzler M

BACKGROUND: An expanding body of literature is available on arthroscopy for femoroacetabular impingement syndrome (FAIS). Studies have attempted to relate Tonnis grade to postoperative outcomes but do not often provide a complete picture of the clinical, radiographic, and failure outcomes for a given Tonnis cohort.

PURPOSE: To systematically review and meta-analyze the impact of preoperative arthritic changes, as measured by Tonnis grade, on clinical, radiographic, and survivorship outcomes in patients undergoing hip arthroscopy for FAIS.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: PubMed, EMBASE, and Web of Science Core Collection databases were queried for investigations associating preoperative Tonnis grade with postoperative outcomes after hip arthroscopy for FAIS. Eligible articles compared patient-reported outcomes (PROs), radiographic arthritis progression, or rate of secondary surgical interventions between patients with various Tonnis arthritis grades. Continuous random-effects models were used to compare PROs between arthritic grades as represented by standardized mean difference (SMD), whereas restricted maximum likelihood random-effects models compared event rates of Tonnis grade progression and secondary surgical interventions, reported as odds ratio (OR).

RESULTS: A total 29 investigations were included in this systematic review, of which 1 […]

PubMed · DOI

Low Fragility Index Undermines Confidence in ACL Graft Superiority: A Systematic Review and Fragility Index Analysis.

Mahatme RJ, Moore SA, Gangavaram A, Anderson W, Campa W, Grawe BM

BACKGROUND: Graft selection is a key factor in anterior cruciate ligament reconstruction (ACLR), yet the statistical robustness of comparative outcomes remains unclear. Reliance on P values may overstate confidence in clinical differences.

PURPOSE: To assess the statistical fragility of dichotomous outcomes in randomized controlled trials (RCTs) comparing hamstring tendon (HT), bone-patellar tendon-bone (BPTB), quadriceps tendon (QT), and allograft ACL grafts.

STUDY DESIGN: Systematic review and fragility analysis; Level of evidence, 1.

METHODS: A systematic search of PubMed, Embase, and Medline (June 2025) identified RCTs comparing 2 ACL graft types with >1 dichotomous outcome. Outcomes were analyzed for fragility index (FI), reverse FI (rFI), fragility quotient (FQ), and reverse FQ (rFQ); those with zero events in both groups were excluded. Median FI and rFI values were compared across graft types and outcome categories, and significant outcomes were stratified by graft

type.

RESULTS: A total of 29 RCTs (200 outcomes) were included; 179 outcomes (89.5%) were analyzed, and 21 (10.5%) showed significant results. Overall median FI and rFI were both 2.0 (interquartile range: FI 1.0-5.0, rFI 1.0-4.0). In 26 studies (89.7%), loss to follow-up exceeded or equaled the FI or rFI. HT versus BPTB was the most common graft pair comparison (72.1%). All graft comparisons had high fragi […]

PubMed · DOI

Are There Sex Disparities in Return to Sport After Anterior Cruciate Ligament Reconstruction? A Systematic Review and Exploratory Meta-analysis.

Elliott DA, Gill SS, Gupte CM

BACKGROUND: Sex disparities in return-to-sport (RTS) outcomes after anterior cruciate ligament reconstruction (ACLR) remain poorly reported. Female athletes may demonstrate higher rates of anterior cruciate ligament (ACL) injuries than male athletes yet exhibit lower RTS rates.

PURPOSE: To compare postoperative RTS outcomes between male and female patients after primary ACLR.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: A systematic search of the PubMed, MEDLINE, Embase, Scopus, and Web of Science databases was conducted from database inception through September 2025, following PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Studies directly comparing RTS outcomes between males and females after primary ACLR with ≥12 months of follow-up were included. Data were synthesized using an exploratory random-effects meta-analysis. Odds ratios with 95% confidence intervals were calculated for dichotomous outcomes, and mean differences for continuous variables.

RESULTS: In total, 25 studies including 7183 athletes (2892 females, 4291 males) met the inclusion criteria. Female athletes demonstrated significantly lower odds of RTS at 12 months (OR, 0.76; 95% CI, 0.62-0.92; P = .005) and up to and including 24 months (OR, 0.72; 95% CI, 0.61-0.85; P < .001). No significant sex-based differences were observed for RTS to competition […]

PubMed · DOI

Fibrin-Based Scaffold Augmentation During Meniscal Repair May Enhance Healing and Functional Outcomes: A Systematic Review.

Skaik K, Kahlon H, Bouchard MD, Vivekanantha P, Belzile EL, Matache BA, Abouali J

BACKGROUND: Biologic augmentation techniques, such as fibrin-based scaffolds, have been proposed to enhance healing of meniscal tears, particularly in avascular zones. However, clinical outcomes remain variably reported.

STUDY DESIGN: Systematic review; Level of evidence, 4.

PURPOSE: To evaluate the healing rate, complication rate, and patient-reported outcome scores after meniscal repair augmented with fibrin-based scaffolds.

METHODS: A systematic review was conducted according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, searching PubMed, Embase, and Ovid MEDLINE through October 15, 2025. Studies evaluating fibrin-based augmentation during meniscal repair were included. Primary outcomes were healing and complication rates. Secondary outcomes included International Knee Documentation Committee (IKDC), Lysholm, and Tegner scores.

RESULTS: Of 757 studies, 15 met inclusion criteria, comprising 438 patients with meniscal tears in mostly avascular zones. Of these, 345 underwent fibrin-based augmentation; the rest did not. There were 4 fibrin-based methods identified: fibrin clot from peripheral venous blood (FC-PVB; n = 252), fibrin clot from bone marrow aspirate concentrate (FC-BMAC; n = 30), platelet-rich fibrin (PRF; n = 15), and PRF combined with platelet-rich plasma (PRF+PRP; n = 48). Among patients who received FC-PVB, 77.5% a […]

PubMed · DOI

Arthroscopic vs Open Bankart Repair for Anterior Shoulder Instability: A Systematic Review and Meta-analysis.

Boutros M, Awad G, Saad JP, Khatib R, Smadi Z, Elhassan B

BACKGROUND: Bankart repair is commonly performed for traumatic anterior shoulder instability. Modern arthroscopic techniques have advanced considerably, raising the question of whether their clinical performance now matches that of traditional open repair.

PURPOSE: To compare contemporary arthroscopic and open Bankart repair across instability-related, functional, perioperative, and complication outcomes.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: A systematic search of the PubMed, Scopus, Cochrane Library, and Google Scholar databases was conducted through December 2025. A total of 32 comparative studies met inclusion criteria. Extracted outcomes included postoperative dislocation, subluxation, overall instability, apprehension, patient-reported outcome measures (Rowe, American Shoulder and Elbow Surgeons [ASES], University of California Los Angeles [UCLA], Western Ontario Shoulder Instability Index [WOSI], and visual analog scale [VAS] scores), postoperative range of motion (ROM), loss of motion relative to baseline, operative time, infection, and reoperation rates. A subgroup analysis stratified studies by publication period (before 2010 vs 2010 onward) to explore the influence of evolving surgical techniques over time.

RESULTS: No differences were found between arthroscopic and open Bankart repair in postoperative dislocation (P = .05), subluxation […]

PubMed · DOI

Microfragmented Adipose Tissue as an Alternative to Platelet-Rich Plasma for Intra-articular Injection in Knee Osteoarthritis: Letter to the Editor.

Baria MR, Barker T

PubMed · DOI

Microfragmented Adipose Tissue as an Alternative to Platelet-Rich Plasma for Intra-articular Injection in Knee Osteoarthritis: Response.

Park YB, Kim JH

PubMed · DOI

Efficacy of Inferior Extensor Retinaculum Reinforcement in Arthroscopic Ankle Lateral Ligament Repair: Letter to the Editor.

Maffulli N, Spiezia F, Oliva F, Feng SM

PubMed · DOI

Rethinking Obesity-Driven Urgency in Rotator Cuff Repair: Response.

Wu Y, Liu Q, Zhang J, Yin Z, Bai L

PubMed · DOI

Efficacy of Inferior Extensor Retinaculum Reinforcement in Arthroscopic Ankle Lateral Ligament Repair: Response.

Lee SH, Lim GH, Kang HT, Kim YE

PubMed · DOI

Further Considerations Regarding Return to Sports After Robotic-Assisted Patellofemoral Arthroplasty in Young Patients: Response.

Noyes FR, Fleckenstein CM, Nolan J

PubMed · DOI

Further Considerations Regarding Return to Sports After Robotic-Assisted Patellofemoral Arthroplasty in Young Patients: Letter to the Editor.

Liu Y, Bian J

PubMed · DOI

Rethinking Obesity-Driven Urgency in Rotator Cuff Repair: Letter to the Editor.

Qu X, Yang D, Tang X, Chen B

PubMed · DOI

BJJ

The Bone & Joint Journal

Volume 108-B, Issue 10 · October 2026 · 14 new articles

Periprosthetic joint infection : the evidence we still lack.

Mancino F, Haddad FS

PubMed · DOI

Diaphyseal zonal fixation in revision total knee arthroplasty.

Sculco PK, El-Hassan M, Shichman I, Lee E, Morgan-Jones R, Haddad FS

The concept of zonal fixation for the distal femur and proximal tibia has revolutionized our understanding of how to best achieve durable fixation in revision total knee arthroplasty. However, the diaphyseal zone covers most of the femur, and achieving durable diaphyseal fixation is essential for optimizing outcomes for patients undergoing multiple revisions and distal femoral replacements (DFRs), since this may be the only anatomical area available for fixation. Therefore, we detail a standardized methodology that divides the diaphysis into three diaphyseal zones: the distal diaphysis, the isthmus, and the proximal diaphysis. This new classification, which expands on the principles of zonal fixation, and further defines the regions of the diaphysis, could offer a more comprehensive approach to preoperatively plan and analyze diaphyseal fixation, and evaluate how it relates to complex revision and DFR implant survivorship.

PubMed · DOI

New guidelines for the prevention of periprosthetic joint infection : a ten-year update.

Ferrini A, He M, Papalia R, He W, Gehrke T, Parvizi J

Periprosthetic joint infection (PJI) remains one of the most serious complications of arthroplasty. Since the publication of major guidelines, several aspects of the prevention of PJI have been refined, although high-quality evidence remains limited in many areas. The aim of this annotation was to provide an updated overview of current preventive strategies, and to summarize the latest recommendations in perioperative practice.

PubMed · DOI

In vivo 3D hip kinematics during racket swinging after total hip arthroplasty : evaluation of dynamic range of motion and liner-to-neck contact.

Nakao Y, Hamai S, Murakami K, Harada S, Kawahara S, Ikebe S, Higaki H, Nakashima Y

AIMS: Interest in sports participation after total hip arthroplasty (THA) has grown, and some patients return to playing racket sports after THA. However, dynamic hip kinematics and the risk of liner-to-neck contact during racket swinging remain unclear. This study aimed to evaluate in vivo, 3D dynamic hip kinematics during racket swinging after THA, and to identify factors associated with reduced liner-to-neck clearance, which may indicate an increased risk of liner-to-neck contact.

METHODS: A total of 19 patients (24 hips) who were playing racket sports after primary THA were evaluated. Dynamic radiological images of forehand and backhand swings were acquired using a flat panel detector. Hip joint kinematics were analyzed using a model-image registration technique. Liner-to-neck contact and clearance were assessed using postoperative CT data and kinematic analysis. Multivariable regression analyses identified factors associated with reduced posterior clearance.

RESULTS: The mean total axial hip rotation during racket swinging was 32°, with the hip maintained in slight flexion and abduction throughout the motion. Liner-to-neck contact occurred in two of 24 hips (8%), all at the elevated rim during maximum external rotation (ER). No bone-to-bone contact was observed. Posterior clearance below 10°, indicating an increased risk of posterior contact, was observed in nine hips (2 […]

PubMed · DOI

Indications for revision following Birmingham Hip Resurfacing : insights from the largest 25-year series from a design centre.

Nasser AAHH, Samra I, Treacy RBC, Hussain S

AIMS: The literature reports excellent medium- to long-term results of Birmingham Hip Resurfacing (BHR). However, revision risks after hip resurfacing vary across implant types, particularly as a result of variations in adverse reactions to metal debris (ARMD). We aimed to characterize the indications for BHR revision and to identify factors associated with specific failure modes.

METHODS: We identified all patients who underwent a BHR at our institution between July 1997 and July 2022. Patients were categorized by sex, age, and BHR head size. Indications for revision were grouped into consensus clinical categories, and Kaplan-Meier survival analysis was performed.

RESULTS: Out of a total of 4,513 BHR cases over 25 years, 392 (8.7%) were revised. Overall implant survival in males was 95.3%. A total of 4.7% (n = 148) of males were revised compared with 17.8% (n = 244) of females ($p < 0.001$). The most common indication for revision was aseptic loosening (n = 98; 2.2%). A total of 92 patients (2.0%) underwent a revision for adverse reaction to metal debris (ARMD). Female sex was significantly associated with revision for ARMD ($p < 0.001$). Revision for periprosthetic fractures occurred early while revision for ARMD occurred late (median 0.6 years (IQR 0.1 to 5.6) vs 10.5 (IQR 8.1 to 14.0), respectively; $p < 0.001$). Revision rates in femoral heads < 50 mm were significantly higher […]

PubMed · DOI

Higher femoral periprosthetic fracture rate with dual-mobility compared with single-mobility total hip arthroplasty : a retrospective propensity-matched study of 780 uncemented hips.

Komat M, Sabot L, Simion F, Lechien D, Bhogal H, Gan El E, Diaby S, Hernigou J

AIMS: Dual-mobility implants have reduced dislocation rates in total hip arthroplasty (THA), particularly in high-risk populations. However, under conditions of high stress, such as a fall or forced range of motion, the energy that would otherwise cause dislocation is instead transmitted distally along the femoral component and into the surrounding bone. Therefore, concerns remain regarding their potential impact on the risk of periprosthetic fracture (PPF).

METHODS: In a single-centre study, we retrospectively analyzed 1,415 patients who underwent uncemented THA with either a dual-mobility (DM-THA; 425 hips) or ceramic-on-ceramic (CoC-THA; 990 hips) bearing between January 2017 and December 2024. Propensity score matching (PSM) was performed for age, sex, BMI, and surgical indication, yielding a matched cohort of 260 DM-THA and 520 CoC-THA patients. Incidences of PPF, dislocation, and mortality were compared.

RESULTS: PPF occurred significantly more frequently in the DM-THA group than in the CoC-THA group (4.0% (17/425) vs 0.8% (8/990); odds ratio (OR) 4.8 (95% CI 1.86 to 12.4); $p = 0.001$), a finding confirmed after PSM (OR 3.5 (95% CI 1.3 to 8.6); $p = 0.007$). Femoral neck fracture (FNF) as the surgical indication was independently associated with an increased risk of PPF in both the overall and matched cohorts. In contrast, DM-THA was associated with a significantly lower d […]

PubMed · DOI

The impact of changing Coronal Plane Alignment of the Knee classification on early functional outcomes after primary robotic-assisted total knee arthroplasty.

Zaidi F, Goplen CM, Akbarshahi M, MacDessi SJ, Petterwood J, McEwen P, Munro JT, Monk AP

AIMS: We aimed to determine the impact of changing a patient's Coronal Plane Alignment of the Knee (CPAK) phenotype on early functional outcomes after robotic-assisted total knee arthroplasty (RA-TKA).

METHODS: We conducted a retrospective multicentre cohort study of 297 patients undergoing primary RA-TKA. Standardized pre- and postoperative long-leg radiographs were obtained. Radiological measures included the lateral distal femoral angle (LDFA) and medial proximal tibial angle (MPTA). Pre- and postoperative CPAK classification was derived by determining the arithmetic hip-knee-ankle (HKA) angle and joint line obliquity (JLO). Oxford Knee Scores (OKSs) were assessed at preoperative baseline and at three months, six months, and one year postoperatively. We also investigated the isolated impact of changes in LDFA and MPTA on OKS.

RESULTS: Preoperatively, the most common classification was CPAK type II (30.0%; n = 89), followed by type I (27.6%; n = 82). Postoperatively, the majority of patients were CPAK type V (30.0%; n = 89), followed by type IV and type II. Overall, 30.6% (n = 91) of patients remained in the same preoperative CPAK classification, while 62.3% (n = 185) shifted by one CPAK classification and 6.4% (n = 19) moved two phenotypes following RA-TKA. Patients who moved across two CPAK phenotypes after RA-TKA demonstrated significantly smaller improvements in OKS at […]

PubMed · DOI

Ten-year insert wear and the migration of uncemented tibial and femoral components with either conventional polyethylene or sequentially annealed highly crosslinked polyethylene: a blinded randomized controlled trial using radiostereometric analysis.

de Ridder R, Kaptein BL, Pijls BG, Nelissen RGHH, Pasma JH, Oomen R, Kaptijn HH

AIMS: The aims of this randomized controlled trial (RCT) was to compare the long-term insert wear and migration of the tibial and femoral components between highly crosslinked polyethylene (HXLPE) inserts and conventional polyethylene (PE) inserts in an uncemented Triathlon fixed insert cruciate-retaining total knee arthroplasty (TKA).

METHODS: A double-blinded RCT was conducted including 96 TKAs. The wear of the insert, and migration of both the tibial and femoral components, was measured with model-based radiostereometric analysis up to ten years postoperatively. Patient-reported outcome measures (PROMs) were collected preoperatively and at follow-up.

RESULTS: There was no progression of (medial) wear (from 0.15 mm at five years to 0.13 mm at ten years) for the HXLPE group after five years, whereas the wear rate increased for the PE group (from 0.22 mm at five years to 0.46 mm at ten years). There was no clinically relevant difference in terms of migration of the tibial and femoral component or PROMs between the HXLPE and PE groups at ten years. Migration of the tibial component (maximal total point motion; MTPM) after the ten-year follow-up for the HXLPE group was 0.99 mm (95% CI 0.83 to 1.18), and for the PE group, was 1.05 mm (95% CI 0.87 to 1.24). Migration of the femoral component (MTPM) after the ten-year follow-up for the HXLPE group was 1.16 mm (95% CI 0.99 to 1.35) […]

PubMed · DOI

Validation of the European Bone & Joint Infection Society recommendations for debridement, antibiotics, and implant retention in hip and knee periprosthetic joint infection : a study from specialist and non-specialist hospitals in the UK.

Hotchen AJ, Richards J, Thumbadoo RP, Tan SJH, Foster P, Canbilen SW, Patel B, Palmer A et al.

AIMS: The aim of this study was to externally validate the European Bone & Joint Infection Society (EBJIS) recommendations for the use of debridement, antibiotics, and implant retention (DAIR) in the treatment of periprosthetic joint infection (PJI), examining the independent associations of the selection of patients and the operative technique with failure after DAIR.

METHODS: This was a multicentre retrospective cohort study involving seven UK hospitals and including adults with hip or knee PJI treated with DAIR. Patients were classified by EBJIS recommendations (DAIR-indicated, risk factors, or not recommended) and operative technique (exchange of components vs no exchange). The primary endpoint was the failure of treatment as defined by the need for further surgery for infection, long-term suppressive

antibiotics, or death within 90 days, amputation, or arthrodesis. Sensitivity analyses were stratified by joint and repeated, excluding long-term suppressive antibiotics from the composite endpoint.

RESULTS: A total of 411 patients (165 hips; 246 knees) were analyzed: 82 (20.0%) DAIR-indicated, 103 (25.1%) with risk factors, and 226 (55.0%) not recommended. Modular components were exchanged in 213 (51.8%). Over a median 2.1 years of follow-up (IQR 0.58 to 5.5), failure occurred in 193 patients (47.0%). Compared with the DAIR-indicated group, the DAIR not recommended group ha […]

PubMed · DOI

Preoperative clinical and imaging factors associated with arthroscopically confirmed occult syndesmotic instability in patients undergoing surgery for osteochondral lesions of the talus.

Jia S, Fu K, Li Z, Hao C, Fang Z

AIMS: Osteochondral lesions of the talus (OLTs) are common ankle disorders. Recent clinical observations suggest that a subset of patients with OLTs also exhibit occult syndesmotic instability (OSI) intraoperatively, implying a potential association between the two conditions. This study aimed to investigate the potential clinical association between OLTs and OSI, and to identify preoperative factors associated with OSI in patients with OLTs.

METHODS: A retrospective review was performed on patients diagnosed with OLTs who underwent ankle arthroscopy between January 2021 and June 2023. Based on arthroscopic probe assessment of distal tibiofibular stability, patients were divided into an OLT with OSI group and an OLT without OSI group. Demographic characteristics, arthroscopic findings, imaging results, physical examination outcomes, and preoperative American Orthopaedic Foot and Ankle Society ankle-hindfoot and visual analogue scale scores were compared between groups. Univariate and multivariate logistic regression analyses were performed to identify independent predictors, and receiver operating characteristic analysis determined diagnostic cut-off values.

RESULTS: Among 175 patients with OLTs, 36 (20.6%) demonstrated OSI. Univariate analysis showed that BMI, symptom duration, height of syndesmotic fluid, lesion size, presence of osteophytes, and Bassett's ligament hypertro […]

PubMed · DOI

Evaluation of the clicky hip as a screening tool in the diagnosis of developmental dysplasia of the hip.

Poacher AT, Bilal M, Smith A, Scourfield L, Hawes E, Ross M, Yasin T, Carpenter EC

AIMS: The early diagnosis of developmental dysplasia of the hip (DDH) is important because treatment in early infancy is usually nonoperative. A consensus statement from the British Society for Children's Orthopaedic Surgery identified ongoing uncertainty regarding the predictive value of a documented clicky hip. The aim of this study was to evaluate the association between a clicky hip at the newborn or six- to eight-week examination and pathological DDH requiring treatment in a selective screening programme in the UK.

METHODS: This was a retrospective analysis of prospectively collected registry data for all live births in the catchment area of a single tertiary centre between January 2016 and December 2023. Infants underwent examination at birth and again at six to eight weeks, with ultrasound screening offered for those with primary risk factors or abnormal findings on examination, including a clicky hip. The primary outcome was pathological DDH requiring treatment. Multivariable logistic regression was used to estimate the adjusted odds ratios (ORs) for DDH. Positive predictive value (PPV), risk difference (RD), and the number needed to screen (NNS) were calculated as programme-level measures.

RESULTS: Of 42,338 infants who were analyzed, 7,564 (17.9%) underwent ultrasound screening. Pathological DDH requiring treatment was identified in 245 (5.8 per 1,000 live births). […]

PubMed · DOI

From bowing to fracture : radiolucent lines as a predictor of progression in congenital pseudarthrosis of the tibia.

Zhang S, Wang T, Zeng L, Xiao B, Mei H, Mao X, Tan Q, Yang G

AIMS: Congenital anterolateral bowing of the tibia represents the prefracture stage of congenital pseudarthrosis of the tibia. However, specific radiological factors which are predictive of fracture remain poorly defined. The aim of this study, therefore, was to investigate whether the presence of radiolucent lines could serve as a predictive factor for the development of a fracture in patients with Crawford types I to III congenital pseudarthrosis of the tibia.

METHODS: We retrospectively reviewed 251 children with congenital pseudarthrosis of the tibia who were treated between 2015 and 2020. Demographic, clinical, and radiological data were collected, and the presence and extent of radiolucent lines were systematically assessed. Inter- and intraobserver agreement were evaluated using κ coefficients. Group comparisons used the independent-samples t-test, chi-squared test, or Fisher's exact test, as appropriate. Independent predictive factors of fracture were identified with multivariable logistic regression. Fracture-free survival was evaluated using Kaplan-Meier analysis.

RESULTS: A total of 142 children with Crawford types I to III congenital pseudarthrosis of the tibia met the inclusion criteria. The assessment of radiolucent lines showed high inter- and intraobserver agreement. Radiolucent lines were strongly associated with the occurrence of a fracture. The risk of frac […]

PubMed · DOI

Corrigendum.

Xiong S, Li Y, Liu Z, Zhao F, Jiang D, Hu Y, Jiao C, Xie X et al.

PubMed · DOI

Corrigendum.

Busby C, Nightingale J, Deacon C, Deakin D, Valdes A, Ollivere BJ, Moran C

PubMed · DOI

Clinical Orthopaedics & Related Research

Volume 484, Issue 10 · October 2026 · 31 new articles

Editorial: Readers Have a Role in Ensuring Our Patients Can Trust the Claims We Make.

Gebhardt MC, Manner PA, Porcher R, Rimnac CM, Wongworawat MD, Leopold SS

PubMed · DOI

Editor's Spotlight/Take 5: Synovial Fluid Leukocyte Count and Differential Are Poor Standalone Rule-in Tests for Periprosthetic Joint Infection: Results of a Methodological Audit and Reanalysis of the 2025 Meta-analysis Underpinning the Unified Periprosthetic Joint Infection Criteria.

Leopold SS

PubMed · DOI

Synovial Fluid Leukocyte Count and Differential Are Poor Standalone Rule-in Tests for Periprosthetic Joint Infection: Results of a Methodological Audit and Reanalysis of the 2025 Meta-analysis Underpinning the Unified Periprosthetic Joint Infection Criteria.

Parthasarathi A, Setoguchi S, DeConti DK, Kupchak P, Donnelly P, Kardos K, McLaren AC, Fillingham YA et al.

BACKGROUND: The Unified Periprosthetic Joint Infection (PJI) Criteria were developed and endorsed by the European Society of Clinical Microbiology and Infectious Diseases, the European Bone and Joint Infection Society, the Infectious Diseases Society of America, and the Musculoskeletal Infection Society and formally presented at the 2025 International Consensus Meeting. These criteria recommended that synovial fluid white blood cells (WBC) > 3000 cells/μL or polymorphonuclear cells (PMN) > 75% each should be considered individually sufficient to confirm a diagnosis of PJI as standalone rule-in tests, stating specificity of > 95%. These thresholds and performance align with a 2025 meta-analysis published on behalf of the Unified PJI Definition Task Force,

reporting specificities of 98.5% and 97.8%, respectively. Because such near-perfect specificities conflict with prior meta-analyses and known causes of false-positive leukocyte testing, independent validation of the supporting evidence is warranted.

QUESTIONS/PURPOSES: (1) Is the 2025 meta-analysis published on behalf of the Unified PJI Definition Task Force methodologically rigorous and reproducible under independent appraisal and audit? (2) Do WBC count and PMN percentage demonstrate sufficient and generalizable rule-in performance for PJI?

METHODS: An independent consortium with expertise in PJI diagnostics, epidemiology, […]

PubMed · DOI

Not the Last Word: The Epistemology, Metaphysics, and Practical Philosophy of Resident Selection.

Bernstein J

PubMed · DOI

The Forward Movement: Amplifying Black Voices on Race and Orthopaedics-The Hidden Organizational Tax of Bias.

Owusu-Akyaw K

PubMed · DOI

Medicolegal Sidebar: The Risks of Testifying as an Expert Witness.

Bal BS

PubMed · DOI

Beyond the Bone: Shared Decision-making in the Super-aged Era.

Kim TK

PubMed · DOI

Cochrane in CORR ®: Cryotherapy Following Total Knee Replacement.

Abdel Khalik H, Khan M

PubMed · DOI

What Is the Relationship Between Clinician-reported and Patient-reported Outcomes in Orthopaedic Surgery?

Aepala MR, Kiani S, Yeramosu T, Sabharwal S

BACKGROUND: Patient-reported outcome measures (PROMs) and clinician-reported outcome measures (CROMs) are designed to capture different dimensions of treatment response: the patient's experience and the clinician's assessment, respectively. Because of that, some degree of discordance between them is expected. However, the nature and magnitude of PROM-CROM correlations across orthopaedic subspecialties have not been systematically characterized. Understanding where they converge and where they diverge may help clinicians and researchers select instrument pairings that capture complementary dimensions of treatment response for different musculoskeletal conditions.

QUESTIONS/PURPOSES: (1) Do CROMs correlate differently with condition-specific PROMs compared with general health PROMs? (2) Do PROMs show distinct correlations with hard CROMs (objective, performance-based measures without patient-reported outcomes) versus soft CROMs (objective, performance-based measures including patient-reported domains)? (3) How do PROMs and CROMs compare in their ability to detect change over time as measured by responsiveness indices such as effect size, standardized response mean, and other related metrics?

METHODS: We systematically searched Medline via PubMed, Embase, and Web of Science Core Collection in July 2025. Two reviewers independently performed title, abstract, and full-text screeni […]

PubMed · DOI

CORR Insights®: What Is the Relationship Between Clinician-reported and Patient-reported Outcomes in Orthopaedic Surgery?

Poeran J

PubMed · DOI

What Is the Association Between Adding Genicular Nerve Block to Adductor Canal Block and Infiltration Between the Popliteal Artery and Capsule of the Knee and Perioperative Outcomes After Primary TKA?

Giannakis P, Yu S, Illescas A, Rowe JE, Liu J, Marx RG, Della Valle AG, Memtsoudis SG et al.

BACKGROUND: Limited evidence exists on the functional and analgesic efficacy of adding a genicular nerve block to an adductor canal block (ACB) plus infiltration between the popliteal artery and posterior knee capsule (IPACK) for TKA. Filling this knowledge gap would be important because if this approach were to be effective, patients could benefit from an additional opioid-free, motor-sparing analgesic modality.

QUESTIONS/PURPOSES: (1) Is the addition of a genicular nerve block to an ACB/IPACK associated with better pain management during hospitalization? (2) Is the addition of a genicular nerve block to an ACB/IPACK associated with better pain management in the postanesthesia care unit (PACU)? (3) Is the addition of a genicular nerve block to an ACB/IPACK associated with earlier time to physical therapy (PT) clearance? (4) Is the addition of a genicular nerve block to an ACB/IPACK associated with lower incidence of opioid refills within 90 days postoperatively?

METHODS: This was a retrospective, propensity score-matched cohort study including patients undergoing TKA from January 2021 to December 2024 at a high-volume academic institution specializing in inpatient and outpatient arthroplasty. All adults undergoing primary elective unilateral TKA for primary osteoarthritis with administration of an ACB/IPACK, with or without a genicular nerve block, were assessed for eligibil […]

PubMed · DOI

CORR Insights®: What Is the Association Between Adding Genicular Nerve Block to Adductor Canal Block and Infiltration Between the Popliteal Artery and Capsule of the Knee and Perioperative Outcomes After Primary TKA?

Webb ML

PubMed · DOI

What Are the Substantial Clinical Benefit Thresholds for KOOS Pain, Physical Function, and Joint Replacement Scores After Primary TKA?

Elmenawi KA, Pasqualini I, Khan ST, Hadad MJ, Piuzzi NS, Cleveland Clinic Adult Reconstruction Research Group (CCARR)

BACKGROUND: The Centers for Medicare & Medicaid Services (CMS) have mandated the collection of patient-reported outcome measures (PROMs) for inpatient TKA with a submission deadline of September 2025. Not achieving a 20-point improvement in substantial clinical benefit (SCB) on the Knee Injury and Osteoarthritis Outcome Score (KOOS) for joint replacement (JR) will negatively impact hospital reimbursement. However, SCB thresholds for KOOS subscales and factors associated with not achieving SCB remain poorly defined.

QUESTIONS/PURPOSES: We retrospectively analyzed a large TKA database at a multisite tertiary healthcare system and asked: (1) What are the SCB thresholds for the KOOS pain, physical function short form (PS), and JR subscales at 1 year after primary TKA? (2) What patient factors are associated with failure to achieve the SCB on these KOOS subscales?

METHODS: This was a retrospective analysis of a longitudinally maintained institutional registry, which evaluated the results of 15,638 patients who underwent primary TKA from 2016 to 2022. Of these, 89% (13,850) of patients completed baseline PROMs, and 79% (10,988 of 13,850) of patients completed 1-year follow-up; data from those patients were used in this analysis. Demographics, comorbidities, as well as baseline and 1-year KOOS scores were recorded. The median (IQR) age of these patients was 67 years (61 to 73), 60% […]

PubMed · DOI

CORR Insights®: What Are the Substantial Clinical Benefit Thresholds for KOOS Pain, Physical Function, and Joint Replacement Scores After Primary TKA?

Yau WP

PubMed · DOI

Reduced Cerebellar Activation With Eyes Closed Is Associated With Delayed Peroneal Reaction Time in Patients With Chronic Ankle Instability.

Hou Z, Zeng X, Huang W, Jiang D

BACKGROUND: Chronic ankle instability (CAI) is a common consequence of lateral ankle sprain and is characterized by recurrent episodes of giving way and impaired neuromuscular control. Neuromuscular control impairments (such as decreased postural stability and delayed muscle responses) are dominant during reduced vision. However, the neural mechanisms underlying impaired sensorimotor integration under reduced vision remain unclear. As a key structure involved in sensorimotor processing, the role of the cerebellum during vision-related neuromuscular control impairments has not been well defined.

QUESTIONS/PURPOSES: (1) Are there any differences in cerebellar activation across different visual conditions in patients with CAI compared with a control group? (2) Are there any associations between vision-related cortical activation changes and static and dynamic neuromuscular control deficits in patients with CAI?

METHODS: This cross-sectional study enrolled physically active adults with unilateral CAI and control participants matched for age, sex, and activity level. Between May and September 2025, a total of 55 individuals were screened. Of these, 64% (35 of 55) were considered eligible based on the inclusion criteria, consisting of 57% (20 of 35) in the CAI group (nine females, mean ± SD age 28 ± 7 years) and 43% (15 of 35) in the control group (eight females, mean age 25 ± 4 ye […]

PubMed · DOI

CORR Insights®: Reduced Cerebellar Activation With Eyes Closed Is Associated With Delayed Peroneal Reaction Time in Patients With Chronic Ankle Instability.

Guyton GP

PubMed · DOI

Do Patients Who Undergo Conversion of Failed Osteoarticular Allografts to Megaprostheses Have Worse Outcomes Than Patients Treated With Primary Megaprostheses at Initial Resection?

Aksoy T, Mosher HA, Torralbas Fitz SJ, Louka JG, Geiger EJ, Hornicek FJ, Crawford BM, Temple HT

BACKGROUND: Osteoarticular allografts are used for knee reconstruction after tumor resection, but the number of these types of reconstruction has declined because of frequent complications and perceived better results with tumor endoprostheses. Failed osteoarticular allografts can be salvaged with megaprosthesis conversion, but it remains unclear whether the outcomes of the patients who undergo conversion differ from those of patients who have primary megaprosthesis reconstructions at the time of tumor resection. Also, there is lack of evidence on the comparison of implant survival. This study aimed to fill the gaps in our current knowledge about conversion megaprosthesis to inform patients and guide treatment accordingly.

QUESTIONS/PURPOSES: (1) Did the survivorship free from revision or amputation differ between patients who initially received an osteoarticular allograft and underwent a conversion to a megaprosthesis and patients who underwent primary megaprosthesis reconstruction? (2) Did the groups differ in terms of Musculoskeletal Tumor Society-93 (MSTS-93) scores, knee flexion angle, and extensor lag?

METHODS: Between January 1997 and December 2023, we treated 276 patients with megaprosthesis for bone sarcoma of the distal femur and proximal tibia, including patients who had a failed osteoarticular allograft in the distal femur or proximal tibia and were subsequently c […]

PubMed · DOI

CORR Insights®: Do Patients Who Undergo Conversion of Failed Osteoarticular Allografts to Megaprostheses Have Worse Outcomes Than Patients Treated With Primary Megaprostheses at Initial Resection?

Moran K

PubMed · DOI

Development and Internal Evaluation of EAST-BMS (East Asian Survival Tool for Bone Metastasis Surgery): A Multinational Machine-learning Survival Prediction Model for Patients Undergoing Surgery for Nonspinal Bone Metastasis in East Asia.

Yang E, Lin HC, Shimomura S, Lee J, Kim HS, Yen HK, Iwata S, Lin WH et al.

BACKGROUND: Patients undergoing surgery for bone metastases typically have advanced disease, and postoperative survival varies substantially. Accurate survival estimation is important for surgical decision-making and patient counseling. Several prognostic models have been externally validated in East Asian populations, but these tools were originally developed in Western cohorts and do not incorporate region-specific epidemiology or treatment patterns.

QUESTIONS/PURPOSES: (1) To develop, internally evaluate, and select a machine learning-based survival prediction model for patients undergoing surgery for nonspinal bone metastases using a multinational East Asian cohort. (2) To compare the performance of the selected model with that of an established Western prognostic tool developed by the Skeletal Oncology Research Group (SORG). (3) To identify which clinical features carried the greatest importance in the new model that we developed.

METHODS: All patients who underwent surgery for nonspinal bone metastases at three tertiary referral centers in the Republic of Korea, Taiwan, and Japan between January 2009 and December 2022 were included. In total, 1045 patients met the inclusion criteria. The median (range) age at surgery was 64 years (19 to 96), 46% (478 of 1045) of patients were female, and the femur was the most common metastatic site (66% [690]). Data for 3-month, 6-mont […]

PubMed · DOI

CORR Insights®: Development and Internal Evaluation of EAST-BMS (East Asian Survival Tool for Bone Metastasis Surgery): A Multinational Machine-learning Survival Prediction Model for Patients Undergoing Surgery for Nonspinal Bone Metastasis in East Asia.

Groot OQ

PubMed · DOI

Are Levels of Blood Metal Nanoparticles Elevated After Spinal Fusion and Implantation in Patients With Adolescent Idiopathic Scoliosis?

Wang Y, Li T, Xiang G, He S, Deng A, Guo C, Yin Y, Zeng Z et al.

BACKGROUND: Implantation of titanium alloys (Ti-6Al-4V) or cobalt-chromium-molybdenum alloys (Co-Cr-Mo) for long-segment spinal fixation is the standard surgical treatment for adolescent idiopathic scoliosis (AIS), but long-term presence of the implants in the body may lead to the release of metal nanoparticles. Unlike orthopaedic internal fixation surgeries, spinal implants generally remain in the human body for a lifetime. The release of metal nanoparticles remains a potential clinical issue, but currently there is a lack of strong data on the topic. This study aimed to fill this crucial knowledge gap by providing longitudinal evidence on the dynamics of metal ion release. This is the first step in determining safety characteristics and clinical thresholds.

QUESTIONS/PURPOSES: (1) Are serum Ti, Co, and Cr concentrations in patients with AIS after implantation elevated compared with in patients without implantation over a 4-year period? (2) Does surgical removal of implants lead to a reduction in systemic metal nanoparticle load? (3) Is there a correlation between metal ion levels and implant burden? (4) Does implant type (Ti-6Al-4V versus Co-Cr-Mo) influence serum metal concentrations?

METHODS: This was a single-center retrospective cohort study performed at an urban tertiary referral hospital from January 2018 to December 2020. A total of 336 patients with AIS were include […]

PubMed · DOI

CORR Insights®: Are Levels of Blood Metal Nanoparticles Elevated After Spinal Fusion and Implantation in Patients With Adolescent Idiopathic Scoliosis?

Thompson GH

PubMed · DOI

Isolated Proximal Femoral Valgus Osteotomy for Hinge Abduction Improves Femoroacetabular Relationship in Patients With Perthes Disease.

Rhee I, Park BK, Shah M, Lim CK, Park S, Park KB, Park H, Kim HW

BACKGROUND: Proximal femoral valgus osteotomy is a recognized salvage procedure for managing hinge abduction in Perthes disease. Although proximal femoral valgus osteotomy aims to improve joint congruency, previous studies have been limited by heterogeneous methodologies and concomitant procedures, making it difficult to determine its isolated effect. Furthermore, no study of which we are aware has specifically examined how proximal femoral valgus osteotomy influences acetabular development or lower limb alignment over time.

QUESTIONS/PURPOSES: (1) Is isolated proximal femoral valgus osteotomy associated with improvements in femoral sphericity and proximal femur development at skeletal maturity? (2) Is it associated with promotion of acetabular development? (3) Is it associated with changes in the lower limb mechanical axis?

METHODS: Between November 2000 and December 2020, we treated 33 patients using proximal femoral valgus osteotomy for painful hinge abduction deformity associated with Perthes disease, which was initially suspected through serial clinical examinations and confirmed during examination under anesthesia combined with arthrographic findings. All 33 patients who met diagnostic criteria during the study period were treated with the same operation (except as specified below), and all were accounted for at a minimum of 5 years of follow-up and considered potential […]

PubMed · DOI

CORR Insights®: Isolated Proximal Femoral Valgus Osteotomy for Hinge Abduction Improves Femoroacetabular Relationship in Patients With Perthes Disease.

Ellsworth BK

PubMed · DOI

Medial Meniscus Posterior Root Shows More Severe Histologic Degeneration in Female Patients With Advanced Osteoarthritis and Varus Alignment: A Histopathologic Analysis of Human Menisci.

Dzidzishvili L, Terzi MÁ, Nájar EE, Suárez Franck ME, García-Oltra E, Aliaga F, López Pérez V, Fernández-Ruiz PL et al.

BACKGROUND: Medial meniscus posterior root (MMPR) tears are more commonly observed in female patients with preexisting osteoarthritis (OA); however, the histopathologic basis for medial root predilection and female predominance remains unclear.

QUESTIONS/PURPOSES: (1) Is histopathologic degeneration associated with knee OA that selectively affects the MMPR? (2) Is greater histologic degeneration of the meniscus root insertion observed in female patients compared with male patients? (3) Do patients with varus alignment and severe OA exhibit greater tissue degeneration at the meniscus root insertion?

METHODS: This histologic study analyzed menisci harvested from patients with OA undergoing TKA between August 2024 and July 2025. Patients with a history of prior meniscal surgery, rheumatologic or gouty arthritis, or previous cruciate or collateral ligament injury and those who declined participation were excluded. Of 122 eligible patients, 1% (1) declined, the meniscal root could not be retrieved in 18% (22), and tissue quality was insufficient in 11% (13). The final cohort included 86 patients, yielding 138 posterior meniscal roots: 69 MMPR and 69 lateral (LMPR). Of those, 63% (54) of patients were female. Mean ± SD age did not differ between the MMPR and LMPR groups (72 ± 7 versus 73 ± 7 years; p = 0.28). The severity of OA, assessed by Kellgren-Lawrence (KL) grade, was similar […]

PubMed · DOI

Letter to the Editor: Guest Editorial: Recalling a Recall.

Doyle TR, Hennessy O, Keogh CJ, Hughes AJ, Kenny PJ, Sheridan GA, Murphy CG

PubMed · DOI

Reply to the Letter to the Editor: Guest Editorial: Recalling a Recall.

Cornell C

PubMed · DOI

Letter to the Editor: Editorial: AI-assisted Letters to the Editor-Scope of a Growing Ethical and Practical Concern, and CORR 's Approach to Managing It.

Cahill PJ, Coleman TA, Gogel S

PubMed · DOI

Letter to the Editor: Only Nine Percent of Orthopaedic Clinical Trials Report and One Percent Analyze a Social Determinant of Health: A Systematic Review.

Lee TC

PubMed · DOI

Reply to the Letter to the Editor: Only Nine Percent of Orthopaedic Clinical Trials Report and One Percent Analyze a Social Determinant of Health: A Systematic Review.

Brauer S, Paul RW, Berberian W, Alberta FG

PubMed · DOI

Letter to the Editor: Cemented Fixation in Arthroplasty for Hip Fractures Does Not Increase Cardiopulmonary Complications: A Secondary Analysis of the HIP ATTACK Trial.

Qin K, Chen D, Sun P, Liang W

PubMed · DOI

JBJS (American)

Volume 108, Issue 19 · October 2026 · 16 new articles

Erratum: Comparison of Large Language Models with Rules-Based Natural Language Processing Algorithms for Extracting Data from Operative Notes.

Yang L, Mulford KL, Girod-Hoffman MM, Khela M, Khosravi A, Crossman DM, Kanabar A, Saniei S et al.

PubMed · DOI

Shaping the Future of Orthopaedics: A Call to Contribute to JBJS Special Issues.

Bhandari M

PubMed · DOI

Good News: The Easier, More Efficient Technique for Pain Management Is Equivalent: Commentary on an article by Jinlei Li, MD, PhD, et al.: " Glucocorticoid-Enhanced Fascial Plane and Peripheral Nerve Blocks Versus Periarticular and Local Infiltration Analgesia in Total Hip Arthroplasty. A Prospective Randomized Controlled Trial ".

Marsh JL

PubMed · DOI

Bridging the Gap: Technology-Assisted THA Reduces Dislocation Risk for Low-Volume Surgeons: Commentary on an article by Avinash Iyer, BS, et al.: "Technology Assistance Mitigates the Volume-Dependent Risk of Hip Dislocation Following Total Hip Arthroplasty".

Ricciardi BF

PubMed · DOI

Early Diagnosis of Pediatric Septic Hip Infection Using MRI: Commentary on an article by Ahmer Ahmad Khan, et al.: " The Prevalence of Pediatric Septic Arthritis of the Hip with Concomitant Osteomyelitis. A Retrospective Study of 58 Consecutive Cases Investigated Using MRI ".

Kamegaya M

PubMed · DOI

A Reflection of Trends Across Modern Academia: Commentary on an article by Yiqiao Zhang, MD, et al.: "High-Grade 3-Column Osteotomies Are Not Interchangeable. Schwab-Grade-III PSO Versus Schwab-Grade-V VCR in Treating Severe Kyphoscoliosis with an Average 9-Year Follow-up".

Cahill PJ

PubMed · DOI

Informed Consent for Robotic-Assisted Reverse Shoulder Arthroplasty: Framing the Conversation and Closing the Expectation Gap.

Menendez ME, Sanchez-Sotelo J

PubMed · DOI

Limb Rehabilitation and Functional Recovery Following Cast Immobilization.

Kamal H

PubMed · DOI

A System in Motion: The Evolution of Orthopaedic Care in Romania.

Dragosloveanu S, Nedelea DG, Scheau C

➢ Romanian orthopaedic practice has demonstrated a growing uptake of digital infrastructure, including the implementation of robotic-assisted surgical workflows and emerging applications of artificial intelligence. ➢ Romania ranks among the European countries with the highest hospital bed capacity per capita, supporting the provision of sustained care for elderly patients, including those with fragility fractures and oncologic orthopaedic conditions and those undergoing complex revision surgeries. ➢ The Romanian Arthroplasty Register was modernized and relaunched in 2025 as an initiative to align with other long-standing national European counterparts.

PubMed · DOI

Principles of Level Selection for Instrumentation in Adult Cervical Spinal Deformity Surgery: Guidelines for Treatment.

Sardar ZM, Coury JR, Bautista AG, Dong K, Riew KD, Reyes JL

➢ For isolated cervical deformities in patients with good bone health, the upper instrumented vertebra can be C3 or C4. ➢ For patients with global cervical deformity, poor bone health, planned 3-column osteotomies, or long fusion constructs, C2 may be an appropriate upper instrumented vertebra. ➢ For patients with focal kyphosis, optimal sagittal alignment, and preserved bone stock, a lower instrumented vertebra cranial to the cervicothoracic junction is appropriate. ➢ For patients with poor cervical sagittal alignment, a lower instrumented vertebra in the proximal thoracic spine, ranging from T1 to T3, may be appropriate. ➢ For combined cervical and thoracolumbar spinal deformities, a lower instrumented vertebra in the lower thoracic or upper lumbar spine is appropriate. The

sagittal stable vertebra or first lordotic vertebra concepts can be considered.

PubMed · DOI

Harnessing the Biologic Power of Bacteriophages to Improve the Treatment of Periprosthetic Joint Infection: Barriers and Future Directions.

Abdelbary H, Malko S, Taha M, Crawley AM, Powell JN, Abedi AA, Suh GA, Citak M

➢ Periprosthetic joint infection (PJI) remains the most common cause of failure after total joint arthroplasty, with current standard therapies associated with failure rates between 10% and 30%. ➢ The 2025 International Consensus Meeting in Istanbul, Türkiye, highlighted the potential benefits of bacteriophage (phage) therapy in managing PJI. ➢ One of the barriers to integrating phage therapy into standard clinical PJI care is the knowledge gap between preclinical research and clinical practice. ➢ This Expert Review summarizes the key therapeutic advantages of using phages as an adjuvant therapeutic platform for PJI. It also highlights the current challenges that are continuing to limit the integration of phages into standard PJI care.

PubMed · DOI

Glucocorticoid-Enhanced Fascial Plane and Peripheral Nerve Blocks Versus Periarticular and Local Infiltration Analgesia in Total Hip Arthroplasty: A Prospective Randomized Controlled Trial.

Li J, Rubin LE, Krishnan R, Blessing M, Townsend D, Tung WS, Zhao X, Treggiari M et al.

BACKGROUND: The purpose of this study was to compare an anterior quadratus lumborum block (aQLB) plus a lateral femoral cutaneous nerve block (LFCNB) with periarticular and local infiltration analgesia (PALIA) in total hip arthroplasty (THA), with both modalities using dual glucocorticoids: hydrophilic dexamethasone sodium phosphate (DEX) and lipophilic methylprednisolone acetate (MPA).

METHODS: A total of 192 patients were randomized to either PALIA or aQLB+LFCNB and received 60 mL of 0.2% ropivacaine, 10 mg of DEX, and 80 mg of MPA. The mean age of the 188 included patients was 61 years, 46% were male, 96% were non-Hispanic, and 82% were White. The primary outcome was opioid consumption, measured as oral morphine milligram equivalents (oMME), on postoperative day (POD) 1. Secondary outcomes included opioid consumption on POD 2, fasting serum glucose, white blood-cell count, Brief Pain Inventory (BPI) pain severity and interference, and functional recovery measures, including Activity Measure for Post-Acute Care (AMPAC) and Patient-Reported Outcomes Measurement Information System (PROMIS) scores, from POD 0 to 1 year.

RESULTS: Ninety-three patients in the aQLB+LFCNB group and 95 patients in the PALIA group were included in the final analysis. There was no significant difference in the primary outcome, oMME on POD 1, between the aQLB+LFCNB group (median, 29.84 [interquartile […]

PubMed · DOI

Predictors of Loss to Long-Term Follow-up After Shoulder Surgery.

Soule S, Smith A, Staten T, Kirkham M, Zhang C, Presson A, Joyce CD, Tashjian RZ et al.

BACKGROUND: Loss to follow-up is among the most important sources of bias in clinical research. Thus, the purpose of this study was to identify factors associated with a failure to complete a 2-year follow-up after shoulder surgery. We hypothesized that older patients would demonstrate higher follow-up rates and that patients who underwent instability surgery would have lower follow-up rates.

METHODS: A retrospective chart review was conducted on 1,028 consecutive patients who underwent shoulder surgery performed by a single surgeon between 2017 and 2022. All patients were contacted by the surgeon via telephone, text message, and email at 2 years postoperatively. Variables analyzed included demographic characteristics, insurance status, surgical procedure type, Social Deprivation Index (SDI), and distance from the surgical site to home. Multivariable logistic regression analysis was performed to determine which factors were associated with 2-year follow-up.

RESULTS: Of the 1,028 patients, 507 (49%) completed a 2-year follow-up. The mean patient age was 51 years, 585 patients (57%) were male, and 443 patients (43%) were female. Ninety-four percent of patients were White. Patients who completed follow-up were older (mean age, 53 compared with 49 years; $p < 0.001$), more likely to

have undergone an arthroplasty, and more often insured by Medicare. Loss to follow-up was associated […]

PubMed · DOI

Technology Assistance Mitigates the Volume-Dependent Risk of Hip Dislocation Following Total Hip Arthroplasty.

Iyer A, Telang SS, Culler MC, Yun AG, Oakes DA, Lieberman JR, Heckmann ND

BACKGROUND: Lower surgeon case-volume has been associated with a greater risk of postoperative complications such as dislocation following total hip arthroplasty (THA). However, robotic assistance and computer navigation may mitigate the volume-dependent risk of instability. This study sought to compare dislocation rates between lower-volume surgeons performing technology-assisted (TA) THAs and higher-volume surgeons utilizing conventional instrumentation (CI).

METHODS: The Premier Healthcare Database was queried to identify adult patients who underwent primary elective THA from 2016 to 2023. Surgeons with <10% technology use formed the CI group, and surgeons with ≥90% technology use formed the TA group. These groups were further subdivided into higher-volume (HV) and lower-volume (LV) on the basis of surgeon annual case-volume, using a previously validated threshold of 109 cases/year. Mixed-effects modeling was used to compare the 90-day risk of dislocation between patients treated by low-volume surgeons using TA (LV-TA group) and high-volume surgeons using CI (HV-CI group).

RESULTS: A total of 669,098 patients undergoing THA were identified. Of these, 5,447 patients were treated by LV-TA surgeons and 190,550, by HV-CI surgeons. Notably, LV-TA surgeons achieved a similar rate of dislocation compared with HV-CI surgeons (0.48% versus 0.42%, p = 0.510). After controlling for c […]

PubMed · DOI

The Prevalence of Pediatric Septic Arthritis of the Hip with Concomitant Osteomyelitis: A Retrospective Study of 58 Consecutive Cases Investigated Using MRI.

Khan AA, Paris E, Sbampato V, De Marco G, Tabard-Fougère A, Vazquez O, Steiger C, Dayer R et al.

BACKGROUND: This observational study systematically used magnetic resonance imaging (MRI) to determine the prevalence of concomitant osteomyelitis and its influence on clinical outcomes in cases of pediatric septic arthritis (SA) of the hip.

METHODS: We retrospectively analyzed the demographic, clinical, microbiological, and radiographic data of 58 children treated for SA of the hip who underwent systematic MRI between 2000 and 2025. Patients were categorized into 2 groups: isolated septic arthritis and septic arthritis with concomitant osteomyelitis. The clinical and laboratory parameters, causative pathogens, and treatments were compared between the groups.

RESULTS: Concomitant osteomyelitis was identified with MRI in 43% (25) of the 58 patients, while radiographs detected it in only 16%. Demographic, clinical, and inflammatory parameters were statistically similar between the groups. Kingella kingae was the most commonly identified pathogen (37.9%), and Staphylococcus aureus and Streptococcus spp. were more frequently associated with repeat surgery. No significant differences in complication rates, treatment duration, or outcomes were found between the groups.

CONCLUSIONS: The systematic use of MRI revealed concomitant osteomyelitis in >40% of cases of pediatric SA. However, the presence of osteomyelitis was not associated with worse outcomes, suggesting that factors rela […]

PubMed · DOI

High-Grade 3-Column Osteotomies Are Not Interchangeable: Schwab-Grade-III PSO Versus Schwab-Grade-V VCR in Treating Severe Kyphoscoliosis with an Average 9-Year Follow-up.

Zhang Y, Wang S, Bai Z, Zhang Y, Niu X, Liu S, Han Y, Zhuang Q et al.

BACKGROUND: Pedicle subtraction osteotomy (PSO) and vertebral column resection (VCR) are powerful techniques for correcting severe spinal deformities. Although PSO has been proposed as a viable alternative to VCR, their comparative efficacies and safety profiles require further elucidation.

METHODS: This single-center retrospective study analyzed 169 patients (mean age, 22 years; 84 male; 169 ethnic Chinese) with severe kyphoscoliosis who underwent primary corrective surgery via PSO (n = 85) or VCR (n = 84). Radiographic parameters, surgical data, intraoperative neuromonitoring (IOM) changes, Scoliosis Research

Society (SRS)-22 scores, and complications were compared between groups.

RESULTS: Both techniques significantly improved all radiographic parameters and SRS-22 scores ($p < 0.001$). The VCR group demonstrated superior correction of the major curve (65.5% versus 56.9%, $p = 0.003$), segmental kyphosis (68.1% versus 61.5%, $p = 0.03$), and apical vertebral rotation (48.5% versus 34.4%, $p = 0.001$). At the critical osteotomy stage, 105 (62.1%) of 169 patients experienced IOM signal decline. The neurological complications rate was significantly higher in the VCR group (13 of 84 versus 7 of 85, $p = 0.038$), as was the overall complication rate (43 of 84 versus 29 of 85, $p = 0.008$). Each 1° increase in correction achieved with VCR was associated with a 1.6% higher risk of complicati […]

PubMed · DOI

JA

The Journal of Arthroplasty

Volume 41, Issue 10S1 · October 2026 · 68 new articles

Proceedings of the 2025 American Association of Hip and Knee Surgeons Annual Meeting.

Buller LT

PubMed · DOI

The James A. Rand Young Investigator's Award: "Nickel-Free" Oxinium Hypoallergenic Versus Standard Cobalt-Chrome-Containing Total Knee Arthroplasty: Is There a Difference in Synovial Metal Ions at Minimum of Two-Year Follow-Up?

Neufeld ME, Sheridan GA, Sidhu A, Howard LC, Greidanus NV, Masri BA, Garbuz DS

BACKGROUND: Allergy and adverse local tissue reaction (ALTR) to metal debris may both be modes of failure in total knee arthroplasty (TKA). Yet, the intra-articular synovial fluid metal ion levels in well-fixed implants beyond the first postoperative day remain unknown. The aim of this study was to compare intraarticular synovial fluid levels of metal ions in patients who had a hypoallergenic TKA implant versus a matched cohort with a standard cobalt-chromium (Co-Cr)-containing TKA implant at a minimum 2-year follow-up.

METHODS: A retrospective, matched, case-control study was performed using prospectively collected data from a single institution. A total of 22 cases (metal allergy/sensitivity with Oxinium hypoallergenic TKA implants) and 18 controls (standard Co-Cr TKA implants) were enrolled, who had a median follow-up of 4.9 years (range, 2.2 to 6.1). Cases and controls were matched for age, sex, body mass index, American Society of Anesthesiologists classification, indication for primary TKA, femoral component design, patellar resurfacing, and time from TKA surgery ($P > 0.05$). Synovial fluid from aseptic aspirations of the study knees was analyzed for metal ion levels of cobalt (Co), chromium (Cr), nickel (Ni), and titanium (Ti). Well-fixed implants were confirmed radiographically, and functional outcomes were analyzed.

RESULTS: The median Ni ion level in the synovial flu […]

PubMed · DOI

Practice Management Strategies Among the Current Members of the American Association of Hip and Knee Surgeons.

Lieberman JR, Buller LT, Meneghini RM

BACKGROUND: At the 2025 Annual Meeting of the American Association of Hip and Knee Surgeons (AAHKS), a survey was conducted to determine practice management strategies among the AAHKS members.

METHODS: An app was used by the AAHKS members to answer both multiple-choice and "yes" or "no" questions related to practice management issues.

RESULTS: Overall, private practice (41%) is still the most common form of practice by AAHKS members, but hospital employment (31%) has increased. Relative value units are now more common (35%) than fee for service (32%) with respect to compensation. Outpatient total joint arthroplasty is now the norm.

CONCLUSIONS: The AAHKS member survey provides critical information regarding the practice patterns of AAHKS members, compensation, and the shift to same-day surgery.

PubMed · DOI

Navigating the Ambulatory Surgery Center Setting: How to Go From Operating to Thriving.

Meneghini RM, Lombardi AV, DeCook CA, Ast MP, Dundon JM

This symposium described how hip and knee surgeons can thrive in the ambulatory surgery center (ASC) setting and focused on patient selection, patient care, efficiency, financial profitability, and enabling technology. Faculty highlighted how to safely increase the volume of surgical candidates through proper screening and protocols; described how to optimize types of patients not previously considered ASC candidates and transition them into the ASC setting; and outlined methods to optimize perioperative and clinical throughput and surgical efficiency. Furthermore, the symposium addressed contracts, negotiation, implants, and strategies to maximize profits and minimize costs without disrupting care, including approaches to financial management and partnership development. Also, the symposium addressed robotics and enabling technology in the ASC and how to utilize technology in a cost-efficient manner to limit inventory and create efficiency in the operative suite.

PubMed · DOI

Dogma Under the Knife: Challenging the Status Quo in Primary Total Knee Arthroplasty.

Padgett BM, Terhune EB, Chalmers BP, Pagnano MW, Hannon CP

BACKGROUND: Primary total knee arthroplasty (TKA) is among the most successful procedures in orthopaedic surgery, reliably providing pain relief, functional improvement, and durable implant survivorship. Despite these achievements, approximately 10 to 20% of patients report persistent dissatisfaction following primary TKA, even in the absence of technical failure. This disconnect between implant longevity and patient-perceived success has prompted renewed examination of the foundational principles that have historically guided TKA practice. Advances in implant design, biomaterials, surgical technology, and a changing patient population have challenged long-standing dogma related to alignment philosophy, fixation strategy, bearing surface selection, and patellar resurfacing.

METHODS: This symposium manuscript reviews contemporary evidence addressing the evolving paradigms in primary TKA, with a focus on individualized alignment strategies, the growing role of cementless fixation, expanded bearing design options, and selective approaches to patellar resurfacing.

RESULTS: In each of these areas, the goal is not to promote a single new standard, but rather to provide an evidence-based framework to support thoughtful, patient-specific decision-making.

CONCLUSIONS: As the field continues to evolve, moving beyond rigid dogma toward personalized reconstruction strategies may represe […]

PubMed · DOI

Solving Intraoperative Challenges During Primary Total Hip Arthroplasty: Prevention, Recognition, and Management.

Ledford CK, Taunton MJ, Cohen EM, Dattilo JR, Hamilton WG

BACKGROUND: Intraoperative challenges during total hip arthroplasty can importantly impact surgical outcomes and patient satisfaction. Effective management of these complications is essential to avoid adverse events and optimize recovery.

METHODS: This symposium reviews four key areas where intraoperative difficulties commonly arise: surgical exposure, acetabular preparation, femoral management, and restoration of leg length, offset, and stability.

RESULTS: Techniques for improving exposure, avoiding acetabular fixation errors and fractures, managing femoral complications such as perforation and poor bone quality, and restoring anatomical alignment are discussed. Emphasis is placed on prevention, recognition, and intraoperative problem-solving.

CONCLUSIONS: A structured approach to intraoperative challenges in total hip arthroplasty can enhance surgical efficiency, reduce complications, and improve patient outcomes. This symposium review provides actionable insights and algorithms to guide surgeons through complex scenarios safely and effectively.

PubMed · DOI

The Multiply Revised Knee: Techniques for What to Do Next.

Rozell JC, Bedard NA, Wolfstadt JI, Sculco PK, Gililland JM

The multiply revised total knee arthroplasty (TKA) poses unique surgical challenges. Patients who present to the office with a failed TKA or revision TKA expect durable, long-term fixation and improved function, yet bone loss and extensor mechanism insufficiency can easily complicate the reconstruction. The aim of this American Association of Hip and Knee Surgeons symposium was to provide up-to-date guidance for rerevision TKA with a specific focus on (1) safe exposure; (2) improved fixation with impaction grafting; (3) improved fixation using porous metal structural augmentation; and (4) managing the patella. After reviewing this article, the reader should feel comfortable with standard and extensile exposures of a revision knee arthroplasty, understand the technique and outcomes for impaction bone grafting and the use of porous metal augmentation, and develop an algorithm for treating and optimizing patello-femoral kinematics through augmentation and reconstruction techniques.

PubMed · DOI

A Concise Update on Decision-Making and Optimal Techniques in Revision Total Hip Arthroplasty.

Cohen-Rosenblum AR, Heckmann ND, Manktelow AR, Haddad FS

This symposium, presented at the 2025 American Association of Hip and Knee Surgeons Annual Meeting, provides an overview of concepts and techniques used during revision total hip arthroplasty.

PubMed · DOI

A Proposed Alternative Sampling Strategy for Centers for Medicare & Medicaid Services Arthroplasty Patient-Reported Outcome Performance Measure Compliance.

Lyman S, Chin AS, Alschuler DM, Brill CO, Fontana MA, MacLean CH, McLawhorn AS

BACKGROUND: The Centers for Medicare & Medicaid Services (CMS) mandates reporting of Hip Disability and Osteoarthritis Outcome Score for Joint Replacement and Knee Injury and Osteoarthritis Outcome Score for Joint Replacement to support a hospital-level patient-reported outcome performance measure. The current CMS requires matched preoperative and postoperative patient-reported outcome measures for ≥ 50.0% of all eligible patients, regardless of hospital volume. This may yield unreliable performance estimates for low-volume centers and potentially biased and unnecessarily large estimates for large hospitals. The purpose of this study was to determine the smallest sample required to produce statistically valid patient-reported outcome performance measure estimates.

METHODS: We calculated the minimum number of completed patient-reported outcome measures needed in order to achieve a 95% confidence interval width 10.0% (± 5.0%) for hospitals with five, 50, 500, and 5,000 eligible total hip arthroplasty (THA) and total knee arthroplasty (TKA) cases annually, assuming a conservative substantial clinical benefit achievement of 50.0%. These values were compared with current CMS requirements. Sensitivity analyses explored the effects of confidence interval width, hospital volume, and frequency of substantial clinical benefit attainment on required sample sizes.

RESULTS: The CMS's volu […]

PubMed · DOI

Hip and Knee Disability and Osteoarthritis Outcome Scores Minimal Clinically Important Differences Vary by Payer Type: Implications for Mandatory Patient-Reported Outcome Reporting.

Pfeil AN, Hryc CF, Patel AR, LeBrun DG

BACKGROUND: Recent mandates from the Centers for Medicare and Medicaid Services have tied patient-reported outcome measure (PROM) metrics to hospital reimbursement for total hip arthroplasty (THA) and total knee arthroplasty (TKA). Concurrently, the proportion of patients meeting certain PROM metrics at each hospital will be publicly available. Threshold values such as the minimal clinically important difference (MCID) may vary across patient subpopulations. The Centers for Medicare and Medicaid Services proposal to apply blanket

threshold values to all hospitals may skew results depending on the population being served. We hypothesized that socioeconomic parameters, such as insurance type, may influence thresholds, such as MCID for the Hip Disability and Osteoarthritis Outcome Score, Joint Replacement (HOOS-JR), and Knee Disability and Osteoarthritis Outcome Score, Joint Replacement (KOOS-JR).

METHODS: There were 852 THA and 1,439 TKA patients who had HOOS-JR and KOOS-JR scores at preoperative and 1-year time points, respectively, and were categorized by payer type: commercial, Medicare (including Medicare Replacement), and Medicaid. The MCID thresholds for each payer were calculated using a distribution-based method (0.5 × SD of H/KOOS-JR) via a bootstrapping procedure with 1,000 iterations per group and random sampling. The results were then compared with both analyses of v […]

PubMed · DOI

How Long is Orthopaedic Team Work Time During the Surgical Encounter for Outpatient Total Joint Arthroplasty?

Kraus KR, Deckard ER, Buller LT, Meneghini RM

BACKGROUND: Outpatient total joint arthroplasty (TJA) volume has increased substantially, while cuts to reimbursement for primary TJA have been continually reduced by the Centers for Medicare and Medicaid Services (CMS). This study quantified the total time an orthopaedic team dedicated to patient care on the day of surgery for outpatient primary TJA.

METHODS: A total of 14,564 primary total hip (41.6%) and knee (58.4%) arthroplasties (THAs and TKAs) performed at 15 statewide facilities were retrospectively reviewed. A total of 45.6% of patients were discharged on the same day of surgery. The total work time was operationally defined as the time in minutes the orthopaedic team spent providing care from patient arrival to discharge, including preoperative tasks, intraoperative services, and discharge preparation.

RESULTS: The mean total work time spent per arthroplasty was 226.5 minutes (THA 232.2, TKA 222.4), with a mean operating room time of 128.5 minutes (THA 134.3, TKA 124.3), a mean setup time of 34.7 minutes (THA 36.9, TKA 33.0), and a procedure time 86.4 minutes (THA 89.8, TKA 83.9). The mean total work time for THAs was significantly longer compared to TKAs ($P < 0.001$). The mean total work time was 211.4 minutes for patients discharged the same day and 239.0 minutes for those discharged after an overnight stay ($P < 0.001$).

CONCLUSIONS: The average total work time for […]

PubMed · DOI

Certificate-of-Need Legislation That Targets Construction of Ambulatory Surgery Centers Is Associated With Increased Patient Migration Out-of-State for Primary Hip and Knee Arthroplasty.

Carlock KD, Potluri AS, Ahmad F, Yadav AS, DeBenedetti A, Della Valle CJ, Gerlinger TL, Sporer SM

BACKGROUND: With the migration of arthroplasty to the ambulatory surgery center (ASC) setting, it is critical to understand the impact of certificate-of-need (CON) laws limiting ASC construction on the manner patients receive care. This study examined location of care within a practice that straddles the border of a CON state (Illinois) and non-CON state (Indiana) to determine the proportion of patients traveling away from their home state for surgery. The cost of care for cases that migrated out-of-state was investigated.

METHODS: Patients who had a home address in Illinois or Indiana and underwent primary hip or knee arthroplasty at ASCs affiliated with our institution from 2016 to 2024 were retrospectively reviewed. Patients who underwent surgery outside of Illinois or Indiana, or at a non-ASC facility, were excluded. Location of surgery, facility fee reimbursement, and driving distance between patients' home addresses and ASC locations were compared. Patients were treated at eight ASCs, four in Illinois and four in Indiana. Of the 7,503 patients included, 6,505 (86.7%) were from Illinois and 998 (13.3%) from Indiana.

RESULTS: Patients from Illinois (CON state) were 3.7 times more likely to have surgery out-of-state than those from Indiana (non-CON state) (28.4 versus 7.7%, $P < 0.001$). Of the patients who had surgery out-of-state, 67.3% from Illinois and 100% from Indiana […]

PubMed · DOI

Trends in Reimbursement for One- versus Two-Stage Exchange Arthroplasty for Periprosthetic Joint Infection.

Forlenza EM, Serino J, Hietpas K, Fehring KA, Della Valle CJ, Fehring TK

BACKGROUND: Emerging evidence suggests similar efficacy of one- and two-stage revision for periprosthetic joint infection (PJI). However, there is concern that extended surgical time and lower reimbursement for one-stage exchange may disincentivize surgeons from performing this procedure.

METHODS: Reimbursement data from 200 patients enrolled in a randomized controlled trial evaluating one-versus two-stage exchange for PJI between 2016 and 2022 was collected. The final cohort consisted of 98 patients undergoing one-stage exchange and 101 patients undergoing two-stage exchange. There was one patient excluded from the final analysis due to incomplete reimbursement data. Reimbursements were converted to 2022 inflation-adjusted dollars. The mean reimbursements for one- and two-stage exchanges were compared, and subgroup analyses based on procedure type (total hip arthroplasty or total knee arthroplasty), insurance status (Medicare or private), and Current Procedural Terminology codes were performed. A separate opportunity cost analysis was performed to determine the theoretical cumulative reimbursement surgeons could expect after treating 10 cases of PJI via either one- or two-stage exchange.

RESULTS: The mean combined reimbursement for a two-stage exchange was significantly greater than for a one-stage exchange ($4,342.40 versus $3,382.40; P < 0.001). The Current Procedural Term […]

PubMed · DOI

Evaluating the Utility and Acceptability of an Artificial Intelligence-Powered Care Aid for Perioperative Patient Inquiries in Joint Arthroplasty Surgery.

Zepeda KE, Burgio C, Yared T, Vachhani S, Bloom E, Steinberg J, Ast MP, Mayman DJ et al.

BACKGROUND: Conventional educational materials for total joint arthroplasty frequently exhibit deficiencies in readability and personalization. This investigation assessed a custom-tailored artificial intelligence (CTAI) platform to evaluate its efficacy in enhancing preoperative patient education within arthroplasty surgery.

METHODS: This prospective, observational pilot study employed a two-phase sequential design: a 10-patient a priori cohort followed by implementation with 30 additional participants undergoing primary total hip or knee arthroplasty. Participants accessed the platform during the preoperative period and early postoperative recovery. There were 32 patients (50.0% knee, 50.0% hip) who completed the preoperative survey. Outcomes were captured using structured surveys administered at the presurgical screening visit and again at the 6-week follow-up using a phase-specific postoperative instrument that retained core items and incorporated recovery-focused domains. Measures included feasibility (e.g., platform access and engagement) and patient-reported experience, including usefulness, understandability of responses, perceived completeness of answers, and perceived preparedness/readiness.

RESULTS: The majority (84.8%) successfully engaged with the CTAI; 59.4% submitted one to four queries, while 25.1% submitted five or more. Surgeon-provided educational materials […]

PubMed · DOI

Standardizing Safe Cefazolin Use in Total Joint Arthroplasty: A Multidisciplinary Triage Protocol.

Guido JM, Hudock NL, Frye BM, Parrish KL, Lastinger AM, Howard CA, Matyus VL, Leadman KM et al.

BACKGROUND: Cefazolin is the first-line perioperative antimicrobial for total joint arthroplasty (TJA) due to its efficacy against typical skin pathogens, favorable safety profile, and cost-effectiveness. Its use reduces the risk of periprosthetic joint infection (PJI) by over 30% compared to second-line antimicrobials. Approximately 15% of patients report a penicillin allergy, although true allergy is confirmed in only 1 to 10% of these cases. Cefazolin's unique molecular structure of the R1 side chain prevents immediate immunoglobulin E (IgE)-mediated cross-reactivity with penicillin or other cephalosporins. Severe cutaneous adverse reactions or delayed hypersensitivity reactions to beta-lactam antimicrobials are rare, less understood, and involve T-cell-mediated pathways. Despite evidence suggesting no immediate IgE-mediated cross-reactivity risk, it is still common practice to use second-line antimicrobials or preoperative allergy testing in patients with a reported penicillin allergy or cephalosporin allergy.

METHODS: In January 2025, a multidisciplinary team implemented a novel algorithmic triage protocol for patients with penicillin allergy and cephalosporin allergy undergoing TJA. Patients who had a history of cefazolin allergy, anaphylaxis, or angioedema to any cephalosporin or severe cutaneous adverse reactions/delayed hypersensitivity reactions to any beta-lactam we […]

PubMed · DOI

A Randomized Controlled Trial to Compare a Mobile-Bearing Cementless and Cemented Unicompartmental Knee: Results of an Investigational Device Exemption Study in the United States.

Hamilton WG, Redfern RE, Klaassen MA, Berend KR, Lackey WG, Berend ME, Anseth SD, Emerson RH et al.

BACKGROUND: Cementless implants have seen a resurgence in knee arthroplasty to improve long-term bone fixation. The purpose of this study was to report the results of the US Investigational Device Exemption study comparing the mobile bearing cementless and cemented unicompartmental knee arthroplasty (UKA).

METHODS: A single-blind, multicenter, randomized controlled trial was performed to compare a medial cementless and cemented UKA implant. There were 378 patients allocated 2:1 to receive the cementless (n = 241) or cemented (n = 137) device between November 2013 and November 2018. The primary end points were identified to demonstrate noninferiority of the cementless device in terms of survivorship, radiographic results, and Knee Society Clinical Assessment and Functional Scores.

RESULTS: There was no statistical difference in Kaplan-Meier 2-year survivorship between the cementless (94.0%, 95% confidence interval [CI] 89.9 to 96.8) and cemented cohorts (97.5%, 95% CI: 93.0 to 99.5, P = 0.19); similar survival persisted at five years (91.7%, 95% CI: 86.4 to 95.0 versus 95.6%, 95% CI: 88.4 to 98.4, P = 0.24). Examination of radiographic success revealed noninferior performance of the cementless prosthesis (93.8 versus 97.4%, P = 0.19). Knee Society function (89.9 ± 13.0 versus 89.8 ± 14.0, P = 0.95) and clinical assessment (95.5 ± 8.5 versus 95.6 ± 6.9, P = 0.92) scores at two […]

PubMed · DOI

Excellent Implant Survivorship and Knee Function of a Fixed Lateral Unicompartmental Knee Arthroplasty: A Consecutive Single-Center Series of 1,015 Knees at a Mean Follow-Up of Five Years.

Waldstein W, Rühle V, Schweizer C, Pulatkan A, Abdulai S, Krug T, Herre J, Aldinger PR et al.

BACKGROUND: Isolated lateral-compartment knee osteoarthritis (OA) remains a condition challenging to treat. Lateral unicompartmental knee arthroplasty (UKA) may offer therapeutic advantages over total knee arthroplasty (TKA); however, it remains infrequently performed. This study determined implant survivorship and reoperation-free survival in a consecutive single-center series of primary lateral UKA. Furthermore, patient-reported outcome measures (PROMs) and radiological outcomes were assessed.

METHODS: In this retrospective analysis, 1,015 knees in 927 patients (75% women) who had a mean follow-up of 5.1 years (range, 2.0 to 8.9) were included (mean age 70 years, range, 37 to 94; mean body mass index 27.5, range, 15 to 48). All patients were contacted via telephone to record implant survivorship, any reoperation, and the functional outcome (follow-up rate: 92%). Preoperative and postoperative radiographs were obtained following a standardized protocol. The Oxford Knee Score and the University of California, Los Angeles Activity Score were documented. Patients who had a previous ipsilateral medial UKA or patellofemoral arthroplasty and patients who declined consent were retrospectively excluded. Cumulative survival was calculated using Kaplan-Meier survival analysis.

RESULTS: The cumulative 9-year survival rate was 97.6% (95% confidence interval: 96.6 to 98.6) with implant r […]

PubMed · DOI

Robotic Unicompartmental Knee Arthroplasty Demonstrates Improved Recovery During the First 90 Postoperative Days Compared With Robotic Total Knee Arthroplasty: A Prospective Remote Monitoring Study.

Schneider AM, Bendich I, Hannon CP, King J, Riegler V, Barrack RL, Nunley RM

BACKGROUND: Unicompartmental knee arthroplasty (UKA) and total knee arthroplasty have been extensively compared; however, prior studies have largely relied on manual surgical techniques and patient-reported outcomes collected at discrete time points, limiting characterization of recovery during the initial postoperative period. With increasing adoption of robotic assistance and wearable technology, higher-resolution assessment of recovery is possible. This study compared recovery following robotic-assisted UKA (rUKA) and robotic-assisted TKA (rTKA) using continuous remote patient monitoring during the first 90 postoperative days.

METHODS: We prospectively enrolled 80 patients undergoing rUKA and 94 patients undergoing rTKA for isolated medial compartment osteoarthritis. Patients utilized wearable activity monitors and a smartphone app for two weeks preoperatively and 90 days postoperatively. Outcomes included daily visual analog scale (VAS) pain scores, daily opioid consumption, step counts, patient-reported outcome measures, and time to walking-aid cessation. Pain trajectories were analyzed using linear mixed-effects models adjusting for baseline pain, demographic variables, comorbidity burden, and surgeon.

RESULTS: Preoperative VAS pain was higher in the rUKA cohort. Adjusted pain trajectories differed between groups, with lower pain in the rUKA cohort during early recovery […]

PubMed · DOI

Does Patellar Resurfacing Impact Patient-Perceived Outcomes Following Primary Total Knee Arthroplasty? A Propensity-Matched Analysis.

Freeman IA, Shimizu MR, Ilyas MH, Sampson WT, Mannina CM, Kwon YM

BACKGROUND: Patellar resurfacing is performed in up to 87% of total knee arthroplasties (TKAs) in the United States. While reducing anterior knee pain is a theoretical benefit, there are no established best practices for resurfacing in primary TKA. Previous studies comparing resurfacing-related patient-reported outcome measures (PROMs) were limited by sample size and a lack of propensity matching. This study aimed to compare the impact of patellar resurfacing on complication rates and clinically meaningful PROMs in propensity-matched primary TKA patients using an institutional database.

METHODS: A total of 5,197 consecutive patients who underwent TKA were categorized based on patellar resurfacing. A 1:2 match by age, body mass index, sex, race, comorbidities, and implants was conducted. Commonly used PROMs were extracted. The mean PROM scores were recorded for each cohort. Differences between the mean preoperative and postoperative scores (delta scores) were calculated. The distribution-based minimal clinically important difference values were computed. Postoperative complication rates and proportions of clinical improvement, no change, and worsening were recorded. Of the eligible patients, 1,176 resurfaced patients were matched to 588 non-resurfaced patients. There were no demographic differences between matched groups ($P > 0.05$).

RESULTS: Preoperative Patient-Reported Outco […]

PubMed · DOI

Patella Resurfacing Is Associated With Improving Six-Month Outcomes in Kinematically Aligned Total Knee Arthroplasty: A Propensity Score-Matched Analysis.

Ng Y, Chen J, Jared Chua HY, Jin Tay DK, Pang HN, Yeo SJ, Lincoln Liow MH

BACKGROUND: Kinematic alignment (KA) seeks to restore each patient's native knee anatomy. Limited research compares functional outcomes between KA robotic-assisted total knee arthroplasty (rTKA) and conventional KA-TKA (cTKA). This study examined differences in early and 6-month functional outcomes between imageless, table-mounted rTKA and cTKA.

METHODS: We conducted a retrospective, propensity score-matched cohort study including 291 primary KA-TKAs from January 2018 to October 2024. In cTKA, patella resurfacing was routinely performed, but not in rTKA. There were 98 KA-rTKA patients matched with 98 KA-cTKA patients. Early postoperative (i.e., static/dynamic pain score and ambulation distance) and 6-month functional outcomes (e.g., Knee Society Function and Knee score, Oxford Knee Score [OKS], and Short Form-36) were analyzed. Group differences in functional scores, proportion attaining minimal clinically important difference, and coronal plane alignment of the knee (CPAK) restoration were analyzed at a 95% confidence level, with significance at $P < 0.05$.

RESULTS: At 6 months, cTKA showed significantly improved Knee Society Function score (P = 0.049), Knee Society Knee score (P < 0.001), and OKS (P = 0.015). The cTKA trended toward higher Short Form-36 outcome measures, with significant differences in physical function (P = 0.002), role physical (P = 0.034), and physical com […]

PubMed · DOI

Establishing Minimal Clinically Important Difference and Substantial Clinical Benefit Thresholds for Objective Gait Metrics After Total Knee Arthroplasty.

Redfern RE, Khan ST, Archibeck MJ, Peters CL, Anderson MB, Piuzzi NS

BACKGROUND: Assessment of objective gait recovery in patients who are undergoing total knee arthroplasty (TKA) is becoming increasingly prevalent. However, few studies determined minimal clinically important difference (MCID) or substantial clinical benefit (SCB) thresholds in arthroplasty populations for clinical interpretation. This analysis aimed to calculate these values for various gait metrics following TKA.

METHODS: A secondary analysis of a longitudinal observational cohort of patients who underwent primary TKA was performed. Objective gait metrics were collected via wearables and smartphones from preoperatively to one year postoperatively. The EuroQol 5-Dimension 5-Level was completed by 2,146 study patients at preoperative and one year stages and used as the anchor to calculate MCID and SCB of each metric utilizing the receiver operating curve analysis (ROC-AUC) with Youden's Index. Baseline characteristics and outcomes were compared as a function of meeting the gait speed MCID.

RESULTS: The MCID thresholds were step count (1,227 steps), gait speed (0.067 meters per second), step length (0.02 meter), double support time (0.5%), and walking asymmetry (1%). The corresponding thresholds for SCB were step count (1,630 steps), gait speed (0.067 meters per second), step length (0.02 m), double support time (4%), and walking asymmetry (3%). At one year, 79% of patients met […]

PubMed · DOI

The Impact of Spinal Anesthetic Dose on Lengths of Stay in Primary Total Knee Arthroplasty.

Kuznetsov M, Playter KP, Travers HI, Ly SM, Hollenbeck BL, Erdman JS, Smith EL

BACKGROUND: The implementation of neuraxial anesthesia in total knee arthroplasty (TKA) has been shown to shorten hospital lengths of stay (LOSs) compared to general anesthesia. Despite these advances, LOS for TKA in a hospital outpatient department (HOPD) setting remains high. We examined the association between the dose of neuraxial anesthesia and LOS in patients undergoing primary TKA.

METHODS: A retrospective review of internal medical records for patients who underwent primary TKA at a single institution from June 1, 2024, to December 31, 2024, was performed. Demographic data, comorbidities, surgeon, anesthesiologist details, neuraxial anesthetic details, and LOS were collected. For uniformity, only patients who received bupivacaine were included. Univariate and multivariable analyses were conducted. A total of 1,486 primary TKA patients were included in the study.

RESULTS: The largest increase in LOS occurred at a dosage greater than 1.6 mL. Patients who received 1.6 mL or less of neuraxial bupivacaine had a 6-hour shorter LOS than those who received more than 1.6 mL (P < 0.001). Multivariate regression analyses revealed that the dose of neuraxial anesthetic (β = 8.2, P = 0.03) and age (β = 0.31, P = 0.004) were predictors of increased LOS. Being a man was predictive of lower LOS (β = -6.56, P = 0.003).

CONCLUSIONS: In this study of primary HOPD TKA, dose of bupivacain […]

PubMed · DOI

Finding the Sweet Spot: A Mixed-Effects Analysis of Four Tourniquet Strategies and Blood Conservation in 15,394 Primary Total Knee Arthroplasties.

Mallett KE, Weinblatt AI, Ondek NT, Öettl FC, Lan R, Lyman S, Chalmers BP, Gonzalez Della Valle A

BACKGROUND: Tourniquet use in total knee arthroplasty (TKA) remains variable, and its effect on perioperative blood loss in the era of tranexamic acid is debated. This study compared four contemporary tourniquet strategies

and their associations with hemoglobin drop, calculated blood loss, and transfusion risk.

METHODS: We retrospectively reviewed 15,394 primary TKAs performed from 2019 to 2023 at a high-volume academic center. Tourniquet use was categorized as follows: no-tourniquet, selective use during cementation only, incision-to-cementation (reference), or incision-to-closure. Outcomes included perioperative hemoglobin (Hgb) drop, calculated blood loss using Nadler and Mercuriali's equations, and postoperative transfusion. Mixed-effects linear and logistic regression models quantified associations after adjustment for age, sex, body mass index, surgical time, Charlson Comorbidity Index, fixation method, preoperative hemoglobin, American Society of Anesthesiologists classification, and postoperative venous thromboembolism prophylaxis, with surgeon as a random intercept.

RESULTS: Compared with incision-to-cementation, the no-tourniquet group experienced a greater Hgb drop (+0.46 g/dL; 95% confidence interval (CI) 0.38 to 0.54; P < 0.001) and the highest median calculated blood loss (785 mL), with an adjusted increase of +211.3 mL (95% CI 169.8 to 252.7; P < 0.001). Select […]

PubMed · DOI

Standardized Sagittal Alignment Targets Fail to Restore Native Medial and Lateral Posterior Tibial Slope in Primary Total Knee Arthroplasty.

Hwang R, Quevedo-Gonzalez FJ, Burgio C, Zepeda KE, Debbi EM, Mayman DJ

BACKGROUND: Prior studies report high variability of the posterior tibial slope (PTS) in patients undergoing total knee arthroplasty (TKA). However, these studies are often performed using radiographs and do not differentiate between the medial and lateral PTS. The purpose of this study was to report the medial and lateral PTS using computed tomography (CT) on patients undergoing primary TKA.

METHODS: Retrospective analysis was performed on 591 consecutive patients undergoing a robotic-assisted TKA from May 2023 to November 2024. Preoperative CT scans were obtained as part of the standard CT-based planning workflow and used to measure the medial and lateral PTS relative to the tibial mechanical axis.

RESULTS: The mean medial PTS was 6.9° (range, -5.9 to 25.8), while the mean lateral PTS was 9.3° (range, -3.4 to 24.6; P < 0.01). The medial PTS was greater for women (mean 7.5°, range, -3.9 to 25.8) than for men (mean 6.4°, range, -5.9 to 21.1; P < 0.01). The lateral PTS was 9.1° (range, -3.4 to 24.6) for women and 9.6° (range, -0.4 to 23.7) for men (P = 0.09). There were 45% of patients who had a medial PTS of greater than 3° from the mean, while there were 40% of patients who had a lateral PTS of greater than 3° from the mean. Using a standardized sagittal alignment target of 3° of PTS would result in greater than 4° deviation from the native medial and lateral PTS in 51 and 7 […]

PubMed · DOI

Cryoneurolysis Reduces Postoperative Opioid Use in Patients Undergoing Total Knee Arthroplasty.

Toavs TL, Sudheendra VH, Barrett T, Lentine B

BACKGROUND: Cryoanalgesia is increasingly used in total knee arthroplasty (TKA) as a nonopioid pain control modality. This study evaluated the effect of cryoanalgesia on postoperative opioid use and recovery of range of motion in patients who underwent TKA for end-stage knee osteoarthritis.

METHODS: This retrospective case-controlled study included patients who received cryoanalgesia prior to primary, unilateral TKA by one of two fellowship-trained arthroplasty surgeons at a single institution. Exclusion criteria included unicompartmental knee arthroplasty, bilateral TKA, revision knee arthroplasty, revision or reoperation after index arthroplasty, and history of preoperative opioid use. The cryoanalgesia group received ultrasound-guided cryoneurolysis of the anterior femoral cutaneous nerve (FCN), intermediate FCN, medial FCN, and infrapatellar branch of the saphenous nerve prior to TKA. A total of 60 cryoanalgesia cases were matched with 60 controls using age ± five years, sex, and body mass index ± 1.5. The mean age was 69 years (range, 49.5 to 88.1), and the mean body mass index was 32.2 (range, 22.9 to 44.3). There were 84 patients (70%) who were women. Independent sample t-tests were used to compare cumulative postoperative opioid morphine milligram equivalents (MMEs) and recovery of knee flexion between groups.

RESULTS: Patients who received cryoanalgesia required sign […]

PubMed · DOI

The Impact of Photographic Evidence on Postoperative Range of Motion Following Total Knee Arthroplasty: A Randomized Control Trial.

Farooq H, Howard J, Johnson B, Winter J, Dickens B, Brown NM

BACKGROUND: Improved function is a key goal following total knee arthroplasty. Providing patients who have postoperative photographic evidence of their knee range of motion (ROM) may encourage engagement in rehabilitation and improve outcomes.

METHODS: A randomized controlled trial was conducted in which 27 patients received photographs of their knee in maximum flexion and extension immediately after surgery, while 28 patients did not. The primary outcome was knee flexion at six weeks postoperatively. The secondary outcomes included knee extension, physical therapy metrics, length of stay, and patient-reported outcomes via the Knee injury and Osteoarthritis Outcome Score (KOOS) survey. Baseline demographic characteristics and preoperative knee ROM were similar between groups, with no significant differences observed.

RESULTS: At six weeks, patients who received photographs demonstrated significantly greater median knee extension (0 versus 3.0■, P = 0.01. Although several outcomes, such as average knee flexion (118 versus 115■, P = 0.12), distance walked on postoperative day one (30 versus 26 feet, P = 0.6), duration of physical therapy (10 versus 11 weeks, P = 0.7), number of sessions (16 versus 20 sessions, P = 0.5), time to achieve 110° flexion (4.5 versus five weeks, P = 0.2), and KOOS primary score (58.2 versus 52.1, P = 0.5), trended favorably for the photographic eviden […]

PubMed · DOI

Postoperative Swelling Did Not Return to Preoperative Levels in Most Patients Two Years After Total Knee Arthroplasty.

Barton KI, Carson NM, Jennings JM, Dennis DA, Forster JE, Hogan CA, Stevens-Lapsley JE, Bade MJ

BACKGROUND: Swelling after a total knee arthroplasty (TKA) can result in pain, decreased range of motion, muscular inhibition, limitations in function, and negative perception of recovery. The objectives of this study were to determine how swelling changes after TKA and if swelling is associated with patient-reported outcomes, functional outcomes, or impairment measures.

METHODS: This is a subanalysis of a two-arm, parallel, randomized controlled trial. Participants (n = 135) were recruited (mean age 64 years, 60% women, mean body mass index 28.7). The TKA participants were recruited and assessed pre-TKA and at 10 weeks, six months, and two years post-TKA. Demographic information was collected at the preoperative visit. Patient-reported outcomes (Western Ontario and McMaster Universities Arthritis Index (WOMAC) and patient satisfaction), functional outcomes (6-minute walk test [6MWT] and 30-second sit-to-stand test [30STS]), and impairment measures (quadriceps muscle strength, pain during functional measures, and knee range of motion) were collected at each clinic visit. Swelling was assessed using bioelectrical impedance analysis (BIA) and quantified post-TKA. Significance for all statistical tests was accepted at $P < 0.05$.

RESULTS: Mean swelling pre-TKA was 2.06 ± 7.8% and then was 15.3 ± 10.6% at 10 weeks, 9.6 ± 7.3% at six months, and 5.4 ± 7.1% at two years after TKA. Af […]

PubMed · DOI

One Patient, Two Knees, No Difference: Magnesium Sulfate Fails to Improve Total Knee Arthroplasty Outcomes.

Chen PA, Hsueh YC, Lin YC, Chen SY, Chang CH, Hsieh PH, Yang KH, Peng SH

BACKGROUND: Whether the addition of magnesium sulfate (MgSO4) to intra-articular analgesic cocktails improves early pain after total knee arthroplasty (TKA) remains uncertain.

METHODS: This prospective, double-blind, within-patient randomized controlled trial enrolled 72 adults undergoing simultaneous bilateral primary TKA. For each patient, one knee received an intra-articular cocktail with MgSO4, and the contralateral knee received the identical cocktail without MgSO4. The first-operated knee (left or right) was independently randomized to balance order effects. Baseline pain and function were assessed preoperatively in both knees. Primary outcomes were numeric rating scale (0 to 10) scores for pain at rest (six, 12, 24, 48, and 72 hours) and during motion (12, 24, 48, and 72 hours). The secondary outcomes were range of motion and active

straight-leg raise (24 to 72 hours). Paired analyses of between-knee differences are reported as mean paired differences with 95% confidence intervals (CIs).

RESULTS: At 24 hours, the mean paired difference in numeric rating scale was +0.35 at rest and +0.21 during motion (95% CI: -0.10 to 0.80 and -0.13 to 0.55, respectively). From six to 72 h (rest) and 12 to 72 hours (motion), absolute differences were small (less than 0.35), and all 95% CI included zero. Range of motion and active straight-leg raise did not differ between knees at 24, 4 […]

PubMed · DOI

United States Trends in Utilization and One-Year Revision Risk of Cementless Total Knee Arthroplasty: An Analysis of American Joint Replacement Registry Data.

Hilow HJ, Tanenbaum JE, Zaniletti I, Versteeg GH, Manning DW, Edelstein AI

BACKGROUND: Cementless total knee arthroplasty (TKA) has emerged as an alternative to cemented fixation. We used the American Joint Replacement Registry (AJRR) to assess trends in cementless TKA utilization and associated one-year revision rates.

METHODS: We analyzed primary TKAs recorded in the AJRR from 2012 to 2024 with available fixation data. Annual proportions of cementless fixation were calculated, and trend analyses were performed using Kendall Tau-b tests. The one-year revision risk was evaluated in cases linkable to Centers for Medicare and Medicaid Services (CMS) data among patients aged 65 years and older who had at least two years of potential follow-up. A multivariable generalized linear model was used to estimate one-year revision risk, adjusting for patient and hospital characteristics and clustering by institution. A subanalysis examined specific reasons for revision.

RESULTS: Of 1,178,783 primary TKAs from 2012 to 2024, 127,781 (10.8%) were cementless. Use increased from 1.6 to 19.0% across all regions ($P < 0.002$). In the Centers for Medicare and Medicaid Services-linked cohort (n = 685,274), 52,742 (7.7%) were cementless. The one-year revision rates were 1.61% for cementless and 1.53% for cemented TKAs ($P = 0.162$). After adjustment, cementless fixation was associated with increased revision risk (adjusted odds ratio (aOR) = 1.12; 95% confidence interval: 1. […]

PubMed · DOI

Cementless Total Knee Arthroplasty in the Morbidly Obese Patient Using a Highly Porous Tibial Baseplate: Minimum 10-Year Follow-Up.

Nadar AC, Gul EH, Smith LS, Henningsen JD, Lustig S, Yakkanti MR, Malkani AL

BACKGROUND: Morbidly obese patients undergoing cemented primary total knee arthroplasty (TKA) have demonstrated higher revision incidence. Cementless TKA implants have demonstrated good five-year results in this patient population. The purpose of this study was to review 10-year results of primary TKA in morbidly obese (body mass index ≥ 39.5) patients using a highly porous, cementless tibial baseplate.

METHODS: This is a retrospective study of 114 TKAs in morbidly obese patients who underwent primary cementless TKA with a minimum 10-year follow-up. There were 25 TKAs lost to follow-up, leaving 89 TKAs in 74 patients who had a mean follow-up of 124.7 months (range, 116.9 to 141.9). The average age was 57 years (range, 36 to 73), and the average body mass index was 46.1 (range, 39.5 to 63.9). Clinical results, patient-reported outcome measures, radiographs, and complications were reviewed.

RESULTS: There were nine revisions, including two for tibial component aseptic loosening (2.3%), two for instability (2.3%), two for deep infection (2.3%), two for extensor mechanism rupture (2.3%), and one for patellar dislocation (1.1%). The two aseptic loosening cases occurred within eight months of the index TKA. The Knee Society Score improved from a mean of 36.2 to 88.3, 86.5, and 90.4 at 2-, 5-, and 10-year follow-up, respectively. The Knee Society Score function score improved from a […]

PubMed · DOI

Preoperative Contralateral Joint And/Or Back Pain Impact on Achieving the Substantial Clinical Benefit for Hip Disability and Osteoarthritis Outcome Score-Joint Replacement.

Elmenawi KA, Jevnikar BE, Oyem PC, Jin Y, Krebs VE, Molloy RM, Deren ME, Cleveland Clinic Adult Reconstruction Research (CCARR) et al.

BACKGROUND: The Centers for Medicare & Medicaid Services recently mandated reporting of patient-reported outcome measures following primary total hip arthroplasty (THA), with emphasis on the substantial clinical benefit (SCB) threshold for the Hip Disability and Osteoarthritis Outcome Score-Joint Replacement (HOOS-JR). The impact of preoperative back and/or contralateral hip pain on achieving SCB and other validated patient-reported outcome measure thresholds remains unclear.

METHODS: We retrospectively analyzed a prospective cohort of 4,412 Medicare primary elective unilateral THAs (2016 to 2023). Patients were stratified by preoperative back and/or contralateral hip pain and severity. Outcomes included achievement of SCB (22-point improvement in HOOS-JR), minimal clinically important difference, patient-acceptable symptom state, and satisfaction at one year. Multivariable logistic regression was adjusted for demographic, clinical, and socioeconomic factors.

RESULTS: Patients who had both back and contralateral hip pain had the highest odds of failing to achieve SCB (odds ratio (OR) 3.1, 95% confidence interval (CI) 2.4 to 4.1; P < 0.001), followed by isolated contralateral hip pain (OR 1.7, 95% CI 1.1 to 2.6; P = 0.01) and isolated back pain (OR 1.6, 95% CI 1.2 to 2.0; P = 0.002). Risk increased proportionally with pain severity (mild/moderate back pain: OR 1.7; severe back […]

PubMed · DOI

Obesity and Chronic Opioids Were Not Negatively Associated With Achieving Centers for Medicare and Medicaid Services Mandated Thresholds for Hip Disability and Osteoarthritis Outcome Score for Joint Replacement Scores After Primary Total Hip Arthroplasty.

Buller LT, Luo TD, Deckard ER, Meding JB, Meneghini RM

BACKGROUND: The Centers for Medicare and Medicaid Services (CMS) will begin mandating Hip Disability and Osteoarthritis Outcome Score, Joint Replacement (HOOS JR) improvement thresholds for total hip arthroplasty (THA) starting in 2026 and have established a 22-point improvement target. Surgeon and institutional data will be publicly reported, and some may avoid patients who have risk factors for poor outcomes such as chronic opioid use. The purpose of this study was to identify predictors of HOOS JR improvement based on the CMS-mandated HOOS JR improvement threshold.

METHODS: A total of 1,682 consecutive primary THAs were retrospectively reviewed. Patient data, including the presence of fibromyalgia, autoimmune disease, osteonecrosis, depression, chronic narcotic usage, and lumbar spine disease were manually extracted from a statewide health care system. Predictors of improvement in HOOS JR total scores were evaluated accounting for the minimal clinically important difference and substantial clinical benefit thresholds established by CMS. Multivariable statistical analyses were performed with P < 0.05 indicating statistical significance.

RESULTS: At a mean clinical follow-up of 22 months, mean improvement in HOOS JR score from preoperative baseline was 36.0 points (range, -44 to 100). A total of 76.8% of patients met the HOOS JR substantial clinical benefit of 22 suggested b […]

PubMed · DOI

Efficacy of Intraosseous Tranexamic Acid in Primary Total Hip Arthroplasty: A Prospective Randomized Controlled Trial.

Chen Y, Cao L, Ji B, Li G, Zhang X

BACKGROUND: Tranexamic acid (TXA) reduces blood loss and transfusion requirements in patients who underwent primary total hip arthroplasty. However, the optimal route of TXA administration remains unclear. This randomized controlled trial evaluated whether intraosseous (IO) TXA administration was noninferior to intravenous (IV) or topical administration.

METHODS: In this equal-proportion, noninferior randomized controlled trial, 126 patients undergoing primary total hip arthroplasty from October 2024 to April 2025 were randomized to IO (20 mg/kg TXA applied in the cancellous bone of the femur and ilium), IV (20 mg/kg TXA applied five minutes before incision), or topical (20 mg/kg TXA applied before suture) group. The primary outcome was hemoglobin reduction on the day of surgery (DOS) and postoperative days one to three (POD one to three). The secondary outcomes included blood loss, transfusion rate, and adverse events. Demographics were similar among the groups.

RESULTS: The mean hemoglobin reduction was comparable on the DOS, POD one, POD two, and POD three (P = 0.431, 0.532, 0.479, and 0.443, respectively). Exploratory analyses showed lower mean calculated total blood loss in the IO group on the DOS and POD one than in the other groups (DOS and POD one: P = 0.005 and 0.016, respectively), although the clinical relevance of this finding remains uncertain. Transfusion rates […]

PubMed · DOI

Up to 10-Year Follow-Up of Thin Polyethylene Liners in Total Hip Arthroplasty.

Leipman JH, Salehi N, Restrepo C, Smith EB, Purtill JJ, Fillingham YA

BACKGROUND: Using thin polyethylene liners with large-diameter femoral heads in total hip arthroplasty (THA) may decrease the risk of instability. This study investigated the 1-to-10-year survivorship of thin highly cross-linked polyethylene (HXLPE) liners.

METHODS: This was a retrospective cohort study of patients who underwent primary THA for osteoarthritis at a single institution. Patients who received an HXLPE liner with a thickness of 3.9, 5.9, or 7.9 mm and a variety of femoral head and cup sizes were included. Surgical information, patient-reported outcome measures, and 1- to 10-year outcomes were collected. Of the 4,951 cases included, 92 utilized a 3.9-mm liner, 2,690 utilized a 5.9-mm liner, and 2,169 utilized a 7.9-mm liner, with respective mean lengths of follow-up of 4.5 years (range, 1.2 to 12.2), 6.1 years (range, 1.0 to 12.1), and 6.1 years (range, 1.0 to 12.4).

RESULTS: There was no significant difference in revisions or reoperations among the cohorts. Among those revised, 7.9 mm liners were more likely to have instability as the revision indication when compared to 5.9 mm liners (0.1 versus 0.04%). The 3.9 mm group had no revisions for instability or polyethylene wear. At six months postoperatively, the Hip Disability and Osteoarthritis Outcome Score Joint Replacement (HOOS-JR) was significantly greater in patients who had a 3.9 mm liner (P = 0.004). At two, […]

PubMed · DOI

Triple-Taper Stems in Obese Total Hip Arthroplasty Patients: Complication and Revision Rates From a United States Registry.

Sambare TD, Royse KE, Son SJ, Fasig BH, Paxton EW, Eisemon EO, Kelly MP

BAACKGROUND: Obesity is a known risk factor for complications and revision following total hip arthroplasty (THA). A triple-taper femoral stem design may improve metaphyseal engagement and rotational stability. This study investigated outcomes in obese patients undergoing cementless primary THA by stem design.

METHODS: We conducted a retrospective cohort study of 25,112 obese patients undergoing unilateral, cementless THA for osteoarthritis between 2018 and 2024. Use of triple-taper increased over the study period (25.4% in 2018 versus 58.3% in 2024). Patients were categorized by stem type per the Khanuja-Mont 2011 classification: type 1 (single taper), type 2 (dual taper), and type 3C (triple-taper) and stratified by obesity class (class I: body mass index 30.0 to 34.9; classes II to IV: body mass index ≥ 35.0). Multivariable Cox proportional hazard models assessed cause-specific revision risk, adjusting for confounders and surgeon clustering.

RESULTS: At 5 years, cumulative aseptic revision probability was lowest for type 3C stems. Among class I obesity patients, type 3C stems were associated with lower aseptic revision risk than type 1 (hazard ratio = 0.47, 95% confidence interval = 0.31 to 0.71, P = 0.0004) and type 2 stems (hazard ratio = 0.39, 95% confidence interval = 0.17 to 0.91, P = 0.029). Similar findings were observed for class II to IV obesity. Type 3C stems wer […]

PubMed · DOI

Cemented Femoral Stem Design Is Not Associated With Differences in All-Cause Revision or Specific Failure Modes After Total Hip Arthroplasty for Femoral Neck Fracture.

Gusho CA, Hoskins W, Riley C, Kelly M, Hegde V, Kagan R, Ghanem ES

BACKGROUND: Cemented stem fixation in total hip arthroplasty (THA) for femoral neck fracture (FNF) has increased in utilization. Cemented stems reduce the periprosthetic fracture risk compared to cementless fixation, but it is unclear whether revision varies by stem design in this cohort. We compared taper-slip versus composite beam cemented stems among FNF patients undergoing THA using a modern classification system.

METHODS: Patients ≥ 65 years of age undergoing primary THA for FNF between 2012 and 2021 with a minimum 2-year follow-up were screened within the American Joint Replacement Registry. A total of 3,479 cases (mean [standard deviation] with follow-ups of 78.8 months (28.3)) met inclusion. The primary analysis compared all-cause and 90-day revision between taper slip (n = 599; 17.2%) and composite beam (n = 2,880; 82.8%) stems. The secondary analyses compared revision for aseptic loosening, fracture, infection, dislocation, and mortality. Cox proportional regressions with competing risks of death evaluated the association of revision with stem design when adjusted for confounders.

RESULTS: Taper-slip stems had an all-cause revision rate of 1.8% (n = 11 of 599) versus composite beams (3.2%; n = 92 of 2,880; P = 0.07). There were no statistically significant differences observed in all-cause revision (hazard ratio [HR] 1.81; 95% confidence interval (CI) 0.96 to 3.40; […]

PubMed · DOI

Revision Total Joint Arthroplasty at the Ambulatory Surgery Center: A Single Institutional Experience.

Acuña AJ, Potluri AS, Yadav AS, Alfonsi SA, Forlenza EM, Della Valle CJ

BACKGROUND: The safety and efficacy of outpatient primary total joint arthroplasty (TJA) in appropriately selected patients have been established. As rates of revision TJA (rTJA) continue to increase, there is growing interest regarding the complications and revision incidence among procedures performed in the outpatient setting. The purpose of this study was to report outcomes of revision TJA procedures performed at an ambulatory surgery center (ASC).

METHODS: A retrospective case series review of 181 patients undergoing aseptic rTJA at a free-standing ASC between January 1, 2017, and July 1, 2024, at a single large academic referral center was performed. Patients who did not have a minimum 1-year follow-up were excluded (n = 28). Our primary outcome was the incidence of complications within 90 days of rTJA, with our secondary outcome as the incidence of rerevision at final follow-up. A majority of patients underwent modular component exchanges (75.7%), with instability being the most common revision etiology (71.8%). The mean follow-up was 3.4 years (range, 1.1 to 8.6). A Kaplan-Meier survival analysis was performed to identify survivorship free of all-cause rerevision at two and five years postoperatively.

RESULTS: There was one patient (0.6%) who developed a deep vein thrombosis within 90 days postoperatively. There were no other medical complications seen among included […]

PubMed · DOI

Single- versus Dual-Component Revision Following Total Knee Arthroplasty Failure for Isolated Tibial or Femoral Loosening.

Nwankwo TN, Olson NR, Ray JR, Jaicks CM, Sershon RA

BACKGROUND: Aseptic loosening is a leading cause of failure in total knee arthroplasty (TKA). When loosening is limited to either the femoral or tibial component, the choice between single- and dual-component revision remains a debated topic. This investigation examines outcomes between single- versus dual-component revision for isolated component loosening in TKA.

METHODS: A retrospective review was conducted of 288 first-time revision TKAs performed for isolated femoral or tibial aseptic component loosening between February 2005 and December 2024 by 9 high-volume arthroplasty surgeons. The mean follow-up was 4.5 years (range, zero to 17.0). Patients were stratified by revision strategy: single- or dual-component. The primary outcome was survivorship and rerevision rate. The secondary outcomes included complications, postoperative range of motion (ROM), and patient-reported outcome measures (PROMs). Multivariable Cox regressions evaluated factors associated with rerevision risk.

RESULTS: Survivorship analysis showed comparable revision rates between both groups (P = 0.615). There were no differences found between the overall rerevision rate for single-component revision (11.2%, 16 of 134) and dual-component revision (9.7%, 15 of 154). The 90-day complications were similarly low in both groups (P > 0.05). Cox regressions identified no independent predictors of rerevision, inc […]

PubMed · DOI

The "Distal Femoral Replacement Downgrade": Technique of Using Bicondylar Femoral Cones With a Hinge Total Knee Arthroplasty Revision in Patients Who Have Massive Distal Femoral Bone Loss as an Alternative.

Chalmers BP, Afzal S, Bhatti P, Anatone AJ, Blevins JL, Debbi EM, Sculco PK, Gausden EB

BACKGROUND: Massive femoral bone loss with collateral insufficiency in revision total knee arthroplasty is often managed with distal femoral replacement (DFR), but this procedure sacrifices bone stock and carries notable risks. We described outcomes of a bone-preserving alternative (the "DFR downgrade") that pairs bicondylar, metadiaphyseal-engaging femoral cones with a hinge construct.

METHODS: We performed a retrospective review of consecutive patients who underwent revision total knee arthroplasty using a "DFR downgrade" technique for severe femoral bone loss between February 2016 and July 2024. There were 20 patients (mean age, 69 years (range, 58 to 86); 65% men) included, with a mean follow-up of 14.5 months (range, 0.9 to 43.7). Indications were aseptic loosening in 75%, infection in 10%, ligamentous instability in 10%, and arthrofibrosis in 5%. Demographics, Anderson Orthopaedic Research Institute classification, operative details, postoperative complications, and clinical and radiographic outcomes were collected. The Kaplan-Meier method was used to estimate reoperation-free and rerevision-free survivorship. Femoral bone loss was classified as Anderson Orthopaedic Research Institute (IIA in 10%, IIB in 30%, and III in 60% of cases.

RESULTS: Survivorship free from revision for aseptic loosening was 100% at one and two years. Survivorship free from any rerevision was 85 […]

PubMed · DOI

Subsidence Starts Distally: Scaffolding Versus Reconstitution Closure of the Extended Trochanteric Osteotomy in Revision Total Hip Arthroplasty.

Jolissaint JE, Rodriguez S, Sanchez LA, Weiner TR, Gausden EB, Chalmers BP, Rodriguez JA, Sculco PK

BACKGROUND: Extended trochanteric osteotomy (ETO) facilitates safe removal of well-fixed femoral stems during revision total hip arthroplasty. The optimal ETO closure method, scaffolding (open ETO during canal preparation) or reconstitution (ETO reduced prior to preparation), remains unclear, especially regarding subsidence of tapered splined titanium stems (TSTS). We evaluated the impact of the ETO closure technique on postoperative TSTS subsidence, hypothesizing no significant difference.

METHODS: We retrospectively reviewed 66 revision total hip arthroplasties (2016 to 2020) performed with ETO and TSTS implantation. There were 35 cases that used reconstitution and 31 that used scaffolding. Subsidence was measured radiographically from immediate postoperative to final follow-up. Subsidence greater than five mm was deemed clinically relevant. Statistical analyses included Mann-Whitney U tests, Chi-square tests, and multivariable logistic regressions.

RESULTS: Average subsidence was higher in the reconstitution group (6.0 ± 3.6 mm) than in the scaffolding group (4.3 ± 2.9 mm, P = 0.018). Rates of significant subsidence (greater than five mm) were similar (31.4 versus 32.3%, P = 0.958). Scaffolding achieved greater distal bicortical contact (50 ± 5.4 versus 28 ± 4.8 mm; P = 0.002), which inversely correlated with subsidence (r = -0.26, P = 0.037). Contact length within the ETO […]

PubMed · DOI

First Revision Total Hip Arthroplasty for Instability: 10-Year Outcomes and Risk Factors Associated With Rerevision.

Yapp LZ, Howard LC, Greidanus NV, Garbuz DS, Masri BA, Neufeld ME

BACKGROUND: Recurrent instability remains a leading cause of failure after revision total hip arthroplasty (rTHA). This study assessed the 10-year survivorship of the first rTHA when undertaken for instability and identified factors associated with failure.

METHODS: We retrospectively identified all consecutive rTHA performed for instability at our institution between 2000 and 2022. Of the 678 rTHA identified, 284 (41.9%) were the first rTHA and included in the study. Rerevisions and oncology cases were excluded. The mean age at revision was 65 years, and 63.2% were women. The mean follow-up was 11.4 years (range, two to 22). A total of 84 (29.6%) hips underwent at least one rerevision. Survival free from instability-related and all-cause rerevision at 10 years was calculated using Kaplan-Meier analysis.

Multivariable Cox proportional hazard regression modeling was used to assess factors associated with 10-year survival.

RESULTS: The 10-year survivorship was 82.2% for instability and 73.6% for all-cause rerevision. Dual mobility (DM) components had higher 10-year instability-free survival than both standard bearings and constrained liners (P = 0.04). However, when considering "large" (≥ 36 mm) femoral heads, there was no difference in the revision rate when compared to DM components (P = 0.59). Constrained liners were associated with a higher risk of revision for instability […]

PubMed · DOI

Does an Isolated Elevated Erythrocyte Sedimentation Rate Warrant Further Work-Up for Periprosthetic Joint Infection after Total Joint Arthroplasty?

Antonioli SS, Khury F, Duke AJ, Haider MA, Aggarwal VK, Schwarzkopf R, Hepinstall M

BACKGROUND: Erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) are widely used as screening tools for periprosthetic joint infection (PJI) following total knee and hip arthroplasty (TKA and THA, respectively). However, the significance of an isolated elevated ESR with a normal CRP, especially in cases with low clinical suspicion, remains unclear.

METHODS: We retrospectively reviewed 19,872 primary TKAs and 12,143 primary THAs performed between March 2012 and March 2023 at a high-volume academic orthopaedic hospital. Of these, 656 TKA and 253 THA patients ≥ 18 years old who underwent elective total joint arthroplasty for osteoarthritis had an elevated ESR (higher than 30 mm/hour) and a normal CRP (less than or equal to 10 mg/L) at least 90 days postoperatively. These cutoffs were chosen in concordance with the International Consensus Meeting 2018 diagnostic criteria for PJI. Patients who were already undergoing PJI treatment were excluded, as it was inferred the serum biomarkers were not drawn for screening purposes. The final cohorts consisted of 641 TKA and 252 THA patients. Data collected through manual chart review of clinical progress notes included presenting symptoms, inflammatory markers, and PJI diagnoses. We calculated the incidence of PJI within the cohort of patients who had an isolated, elevated ESR.

RESULTS: Of the 641 TKA patients who had an isola […]

PubMed · DOI

Operative Cultures Rarely Necessitate Change to Periprosthetic Joint Infection Antibiotic Plan Based on Monomicrobial Preoperative Aspiration.

Ducharme MT, Elmenawi KA, Poilvache H, Hannon CP, Abdel MP, Tande AJ, Bedard NA

BACKGROUND: Periprosthetic joint infections (PJI) are complex to manage and frequently result in prolonged hospitalization while awaiting outpatient antibiotic therapy coordination. Moving this work to the preoperative setting for nonemergent PJI cases could potentially provide more efficient and cost-effective care. This study sought to evaluate the appropriateness of antibiotic treatment plans based upon monomicrobial preoperative cultures.

METHODS: We identified 925 PJIs (312 hips of 613 knees) in 831 patients treated at a single institution between 2013 and 2022. Preoperative aspirations were culture negative in 171 joints (18%) and had polymicrobial growth in 107 joints (12%), leaving 641 joints (70%) with a single-organism PJI based on preoperative culture from which an antibiotic treatment plan could be made. Operative cultures were evaluated for concordance with preoperative cultures. An infectious disease specialist reviewed discordant results for clinical significance and whether a change in antibiotic therapy was necessary.

RESULTS: Among the 641 PJIs with a single organism identified preoperatively, an antibiotic plan based on preoperative culture was appropriate for 97% of PJIs. Only 17 PJIs (3%) had discordant operative cultures requiring a change in antibiotic plan. Compared with preoperative culture results, operative cultures were concordant in 484 PJIs (76%) […]

PubMed · DOI

Kidney Injury and Mortality Are Minimal Following Two-Stage Treatment of Periprosthetic Joint Infection.

Epley RL, Stoops TK, Deckard ER, Buller LT, Meneghini RM

BACKGROUND: The current gold standard treatment for chronic periprosthetic joint infection (PJI) is a two-stage surgery with antibiotics between implant resection and reimplantation. Recently, claims have been reported of nephrotoxic antibiotic exposure and acute kidney injury (AKI) during the two-stage PJI treatment. This study evaluated kidney function and mortality before, during, and after two-stage revisions for PJI using a standardized protocol.

METHODS: Clinical data from 138 hips and knees undergoing two-stage treatment for PJI were retrospectively reviewed. Standardized perioperative clinical protocols were used for all patients, including hypotensive anesthesia and tranexamic acid to minimize operative blood loss, aggressive fluid resuscitation, medical optimization, and six weeks of intravenous antibiotics prior to reimplantation. The incidence of AKI, serum creatinine, estimated glomerular filtration rate, blood urea nitrogen, and overall mortality were collected from a statewide healthcare system. P-values ≤ 0.05 were considered statistically significant.

RESULTS: Overall, the incidence of AKI was 9.4% (13 of 138), which was 4.3% in patients who did not have preexisting kidney disease. There were no significant differences observed in mean serum creatinine (1.10, 1.12, 1.13 mg/dL), estimated glomerular filtration rate (78.6, 77.7, 74.8 mL/minute/1.73 m2), or bloo […]

PubMed · DOI

The One-Year Infection Rates After Vancomycin Powder and Dilute Povidone-Iodine Lavage in High-Risk Primary Total Joint Arthroplasty: A Multicenter Randomized Controlled Trial.

Saba BV, VPIP Study Group, Long WJ, Higuera CA, Dundon J, Cooper HJ, Dennis DA, Chen AF et al.

BACKGROUND: Given the severe morbidity, mortality, and substantial cost of periprosthetic joint infection (PJI), substantial research has been conducted to compare perioperative and postoperative infection-prevention strategies. To our knowledge, there are no studies to date that have evaluated the 1-year efficacy of intraoperative vancomycin powder and/or dilute povidone-iodine lavage versus saline lavage in total joint arthroplasty. We previously reported no significant group differences at 3 months in a large multicenter randomized controlled trial. The present study reports 1-year outcomes of the same cohort.

METHODS: In this prospective, multicenter trial, 2,053 high-risk patients undergoing primary, unilateral total hip arthroplasty (THA) or total knee arthroplasty (TKA) were randomized to one of four intraoperative protocols: vancomycin powder, dilute povidone-iodine, vancomycin povidone-iodine protocol (VPIP, combination), or saline lavage control. The primary outcome was 1-year PJI resulting in septic revision surgery. Analyses were conducted on a per-protocol basis using Chi-square tests stratified by procedure. At 1 year, complete follow-up was available for 798 THA and 1,032 TKA patients after accounting for withdrawals, loss to follow-up, and nine unrelated deaths.

RESULTS: In the THA cohort, PJI occurred in 0.5% of vancomycin patients, 1.7% of iodine patients, 1 […]

PubMed · DOI

Classification and Outcomes of 214 Periprosthetic Patellar Fractures in Primary Total Knee Arthroplasty.

Ducharme MT, Selemon NA, Bedard NA, Spangehl MJ, Springer BD, Hidden KA, Yuan BJ, Berry DJ et al.

BACKGROUND: Long-term data following periprosthetic patellar fractures around primary total knee arthroplasty (TKA) are lacking. The purpose of this study was to describe the classification, management, and outcomes of periprosthetic patellar fractures in, to our knowledge, the largest series to date.

METHODS: We identified 214 (213 cemented and one uncemented) periprosthetic patella fractures (205 patients) around a primary TKA sustained between 1989 and 2020. There were eight intraoperative fractures and 206 postoperative fractures. Postoperative fractures were classified according to the Ortiguera and Berry classification. There were 76% Type I fractures, 12% Type II fractures, 6% Type IIIA fractures, and 6% Type IIIB fractures. The mean patient age was 66 years (range 22 to 90), 56% were men, and their mean body mass index was 32 (range, 21 to 54). The mean patient follow-up was seven years (range, two to 19).

RESULTS: The incidence of intraoperative patellar fracture was 0.02%, occurring most frequently during exposure, implant preparation, or trialing (75%). The 10-year survivorships free of any revision and any reoperation after intraoperative fractures were both 88%. Among the 206 postoperative fractures, the majority (66%) occurred atraumatically, while 31% occurred after ground level falls. Among postoperative fractures, the 10-year

survivorship free of any revision […]

PubMed · DOI

"Two Strikes, You're Out": Contemporary Revision Patterns Following Instability After Posterior and Anterior Approach Total Hip Arthroplasty.

Driscoll DA, Burgio C, Viggiano M, Yared T, Trenchfield D, Sokrab R, El-Hassan M, Vigdorchik JM et al.

BACKGROUND: Dislocation remains one of the most challenging complications after total hip arthroplasty (THA). Previous studies suggested that three or more dislocations were a threshold for revision; however, these data predate contemporary implants and modern surgical approaches. We used a contemporary cohort of primary THA patients to evaluate the relationship between dislocation recurrence, timing of instability, and subsequent revision for instability.

METHODS: There were 40,315 patients who underwent primary THA at our institution from 2016 to 2024. Among these patients, those who sustained a dislocation event were included. Keyword and diagnosis code search were performed to identify patients, and charts were reviewed to confirm dislocation. A total of 285 patients (0.7%) were included at a mean 5.7-year follow-up, of whom 242 (84.9%) ultimately underwent revision for instability. Instability was categorized by recurrence (one, two, or ≥ three dislocations) and timing ([early ≤ 90] versus late [greater than 90 days)]). Multivariable logistic regressions were performed to identify factors independently associated with risk of revision.

RESULTS: Revision risk increased sharply with recurrence, rising from 69.1% after one dislocation to 93.9% after two and 98.5% after three or more. Patients who had two dislocations had significantly greater odds of revision than those who […]

PubMed · DOI

Concordance of Dislocation Direction and Surgical Approach Is an Approach-Dependent Risk for Recurrent Instability Following Total Hip Arthroplasty.

Ruderman LV, Jing C, Deckey DG, Bolognesi MP, Ryan SP, Schwartz AM

BACKGROUND: Hip instability after total hip arthroplasty (THA), while uncommon, is a leading cause for revision. The purpose of the study was to quantify the risk of recurrent hip dislocation in relation to concordance of dislocation directionality and surgical approach.

METHODS: Included patients underwent anterior-based or posterior-based elective primary THA and sustained at least one dislocation treated at a single institution. Surgical approach, timing, direction, and management of dislocations, including closed reduction and reoperation, were obtained. Concordance was defined as dislocation in the same direction as the surgical approach. There were 112 patients (114 hips) who experienced dislocation after primary THA between January 1, 2013, and November 1, 2025. Chi-square analyses were performed to assess differences in dislocation recurrence, depending upon approach and dislocation directionality.

RESULTS: The posterior approach was most common (62.3%), followed by anterior (27.2%) and direct lateral (10.5%). The median time to first dislocation was 41 days [interquartile range 18, 103.5], and most were posteriorly dislocated (64.9%). There were five patients (4.4%) who required an open reduction and revision after index dislocation. There were 61 patients who experienced at least two dislocations, accounting for 56.0% of patients who were not revised at the initial […]

PubMed · DOI

What Factors Increase the Risk of Dislocation After Primary Total Hip Arthroplasty Using the Direct Anterior Approach?

Murphy MP, Forti CR, Ho H, Hopper RH, Hamilton WG

BACKGROUND: While the direct anterior approach (DAA) for total hip arthroplasty (THA) is associated with a lower risk of dislocation, this complication has not been eliminated and can be acutely disabling when it occurs. This study aimed to identify patient, surgical, and radiographic factors that contribute to dislocation risk for the DAA based on primary THAs performed over a 15-year period.

METHODS: Using an institutional database, we identified 8,658 DAA THAs performed by four surgeons from March 2009 through May 2024. A Cox proportional hazards regression was performed to determine how patient demographics and implant characteristics were associated with dislocation. Patients who dislocated were also compared to a matched control cohort without dislocation to evaluate the influence of radiographic parameters, including acetabular cup orientation, offset, limb length changes, and preoperative acetabular coverage.

RESULTS: The dislocation rate among all primary THAs was 0.5% (45 of 8,656). Women had a higher dislocation rate than men (0.7 versus 0.2%), and Cox regression identified an increased risk among women with 28 mm (hazard ratio 7.5; 95% confidence interval 3.2 to 17.6) and 32 mm heads (hazard ratio 3.8; 95% confidence interval 1.9 to 7.5). Radiographic analysis showed no significant differences in total offset (P = 0.13) or limb-length change (P = 0.41), but contro […]

PubMed · DOI

Divergent Dynamics: Unicompartmental Knee Arthroplasty Enables Greater Propulsive Joint Moments Than Total Knee Arthroplasty.

Kuntze G, Korley R, Abelseth G, Ronsky J, Johnston K

BACKGROUND: Current evidence indicates better functional performance following unicompartmental (UKA) compared to total knee arthroplasty (TKA). This randomized controlled trial addressed the differential effects of UKA and TKA on functional joint mechanics and patient-reported outcome measures (PROMs).

METHODS: A total of 38 patients were recruited, and joint moments data were available for 28 patients allocated to either UKA or TKA study arms. Instrumented gait analysis was performed before and one year after surgery. Dependent variables were patient-specific changes in peak sagittal ankle, knee, and hip moments during push-off and maximum coronal knee and hip moments. Patients completed the Oxford Knee Score and Western Ontario and McMaster University Arthritis Index before and six weeks, three months, and one year after surgery. Statistical analyses were conducted using linear mixed effects models for biomechanics and PROMs data.

RESULTS: Unicompartmental knee arthroplasty patients displayed significantly larger increases in sagittal hip (β = 0.11 Nm/kg·height; 95% confidence interval (CI), 0.04 to 0.18; P = 0.003), knee (β = 0.05; 95% CI, 0.01 to 0.09; P = 0.010), and ankle (β = -0.13; 95% CI, -0.25 to -0.01; P = 0.042) moments postoperatively than TKA patients. Further, UKA patients displayed significantly smaller changes in coronal knee moments (β = -0.12; 95% CI -0.19 […]

PubMed · DOI

Effect of Diluent on Molecular Iodine Release From On-Site Diluted Povidone-Iodine.

Meehan JP, Lum ZC, Moses C, Kessler J, Bayer IS

BACKGROUND: The global expansion of antibiotic-resistant organisms has led to a renewed interest in antiseptic principles of infection prevention. On-site dilutions of povidone-iodine (PI) with normal saline have increased substantially, but little information exists regarding the concentration of the singular active biocidal agent, i.e., molecular (free) iodine, in these formulations. The purpose of this study was to analyze the differences in the amount of molecular iodine released from on-site dilutions of five different commercially available 10% PI products and three different dilution ratios in normal saline and in sterile water.

METHODS: Potentiometric analysis of commercial solutions used an iodine-selective electrode with an Ag/AgCl reference and platinum working electrodes. Calibration was conducted with 0.345 g/L aqueous iodine solutions. Samples were diluted with relevant solvents. On electrode equilibration, electrochemical potentials were recorded. There were six different measurements taken from each dilution rate, and free iodine levels were measured. We used two-way analyses of variance to determine the significance of dilution rate and solution type.

RESULTS: Increased dilutions of 0.35, 0.22, and 0.10% PI in normal saline resulted in significant increases in the values of molecular iodine (P < 0.001). There was also a significant effect from type of diluent […]

PubMed · DOI

Should Inter-Prosthetic Screws Be Placed When Prophylactically Plating a Femur?

Beck AP, Feller HJ, Trotter CA, Roth JD, Doro CJ

BACKGROUND: Apposing intramedullary femoral implants can increase the risk of an interprosthetic femoral fracture. The aim of this study was to determine optimal screw configuration when prophylactically plating a femur with an interprosthetic segment of bone.

METHODS: Intramedullary implants were placed antegrade and retrograde to create two- and 10-cm inter-prosthetic distance (IPD) at the middiaphysis of 21 composite femora. Broad, curved, 18-hole 4.5 mm plates were applied to testing groups. Either one or zero screws (IP-1 or IP-0, respectively) were placed in the inter-prosthetic segment of bone. Femora were loaded to failure under lateral, four-point bending. We used one-factor analyses of variance with post hoc Tukey tests to compare the force-at-failure (FAF) and energy-to-failure (ETF). Digital image correlation was used to determine the surface strain distributions.

RESULTS: The FAF was greatest for 2- and 10-cm IP-0 (4,831 ± 213 and 4,437 ± 234 N, respectively) compared to all other conditions ($P < 0.001$). The ETF of 2- and 10-cm IP-0 (19.97 ± 1.15 and 23.57 ± 4.18 J, respectively) was significantly greater than the ETF of 10- and 2-cm IP-1 (5.89 ± 0.89 and 7.76 ± 2.59 J, respectively; $P < 0.0004$). The FAF and ETF were similar between nonplated native femora and those with 10-cm IPD ($P > 0.06$). Strain curves and heat mapping illustrated the concentration of strain […]

PubMed · DOI

Biomechanical Assessment of Pin-Site Fractures in Intra- Versus Extraincisional Pinning Technique for Robotic Total Knee Arthroplasty: A Cadaver Study.

Liau ZQG, Shao-Rong Pang A, Kee Ern ET, Peng Ng MS, Meng CS, Kumarsing RA, Chan CK, Vachalam D et al.

BACKGROUND: Robotic-assisted total knee arthroplasty offers enhanced precision, but introduces new challenges, including pin-site fracture risk from tracker fixation, with incidence rates of 0.06 to 4.8%. This study aimed to evaluate the effects of intra- versus extra-incisional pin placement on torsional rigidity to fracture failure in cadaver specimens.

METHODS: Twenty matched cadaver femora and tibiae were randomized to undergo either intraincisional/metaphyseal or extraincisionalincisional/diaphyseal pinning. Specimens were cemented into jigs and subjected to torsional load-to-failure in two stages, first at 100 Nm, then 300 Nm, with two separate machines, with continuous monitoring of torque, rotation, and time. Fracture documentation and trendline analysis were performed to calculate torsional rigidity and coefficient of determination (R2). Fracture characteristics, including pin-site involvement, and fracture patterns were also recorded.

RESULTS: Mean torsional rigidity was significantly higher for intraincisional femora at both 100 Nm (14.03 versus 10.91 Nm/degree, $P = 0.048$) and 300 Nm (14.46 versus 10.91 Nm/degree, $P = 0.048$). 80% of extraincisional femora fractured when subjected to 100 Nm of torsion, than 20% of intraincisional femora ($P = 0.058$). Mean torsional rigidity was 8.24 Nm/degree for intraincisional compared to 6.35 Nm/degree for extraincisional pinning […]

PubMed · DOI

Is Fretting Corrosion at the Modular Junction of Revision Total Knee Arthroplasty Actually a Problem?

Hall DJ, Wong J, Wright JL, Hochstatter JP, Burnett RA, Miller A, Levine BR, Pourzal R et al.

BACKGROUND: Corrosion at modular junctions in total knee arthroplasty (TKA) implants is recognized, but its clinical relevance remains unclear. This study evaluated corrosion at modular stem-bore junctions and examined whether severity correlated with radiographic evidence of adverse local tissue reactions or osteolysis.

METHODS: We evaluated 196 retrieved modular stem and bore components from 98 revision TKA patients, including taper (n = 73) and threaded (n = 25) designs. The median time in situ was 24 months for taper and 15 months for threaded junctions ($P = 0.052$). Periprosthetic joint infection and aseptic loosening were the most common indications for component explantation. Fretting corrosion at the modular junction was graded using a modified Goldberg Score (mGS) and quantified using an optical coordinate measuring machine. Radiographs were evaluated. Statistical analyses included Mann-Whitney U-, Chi-Square, and Spearman's correlation tests.

RESULTS: Taper junctions demonstrated greater corrosion than threaded designs ($P < 0.001$). The median mGS for modular tapers was 4 for femoral components, 3 for femoral stems, and two for both tibial components and stems. Threaded junctions showed minimal damage across all interfaces with a median score of 1. Damage

patterns included fretting wear, corrosion features, and dark deposits. Among cobalt-chromium (CoCr) femoral bore […]

PubMed · DOI

Fully Randomized Predictor-Outcome Pairings Using a National Database Yield Frequent Statistical Significance Without Clinical Meaning.

Grayson W, Chakraborty A, Brown NM

BACKGROUND: Large national databases have enabled extensive outcomes research in arthroplasty. Their vast sample sizes, however, raise concerns regarding the identification of statistically significant, but clinically meaningless associations. We hypothesized that completely random pairings between database variables, without any clinical rationale, would still frequently yield statistically significant associations, which were driven purely by sample size rather than any meaningful relationship. This study examines the extent of false discovery risk in data-driven analysis of large clinical datasets.

METHODS: A retrospective cross-sectional analysis was performed utilizing a large national database. Patients who underwent total knee arthroplasty (TKA) or total hip arthroplasty (THA) were identified based on the Current Procedural Terminology (CPT) codes. There were 20 predictor-outcome variable pairs randomly selected using a seeded random generator applied across all variables in the dataset. Appropriate statistical tests were applied based on the variable types: Pearson correlations for continuous-continuous comparisons, Welch two-sample t-tests for continuous-binary comparisons, and Chi-square tests for binary-binary comparisons. Among 955,092 total cases (total hip arthroplasty: 373,528 [39.1%], total knee arthroplasty: 581,564 [60.9%]), mean age of 66 years (range, 17 to […]

PubMed · DOI

Predicting Patient-Reported Outcome Measures, Satisfaction, Health-care Utilization, Mortality, and Return to Work After Total Knee Arthroplasty Using Machine Learning: A 14,900-Patient Study.

Khan ST, Sharma A, Patel SV, Jevnikar BE, Rotroff D, Cleveland Clinic Center of Excellence in Osteoarthritis Group, Piuzzi NS

BACKGROUND: Patient-reported outcome measures and health-care utilization metrics are increasingly used to evaluate the success of total knee arthroplasty (TKA). Predictive models may improve preoperative planning, patient counseling, and resource allocation. This study developed and validated machine learning models to predict postoperative patient-reported outcomes, satisfaction, health-care utilization, mortality, and return to work after primary TKA.

METHODS: A prospective cohort of 14,900 patients who underwent primary unilateral TKA at a large tertiary academic center from 2016 to 2022 was analyzed. Patients undergoing bilateral TKA or missing baseline patient-reported outcome measures were excluded. Random Forest and XGBoost models were developed using baseline demographic, clinical, socioeconomic, and surgical variables. Outcomes included postoperative Knee injury and Osteoarthritis Outcome Score (KOOS) Pain, Physical Function, Joint Replacement, and Quality of Life subscales; patient acceptable symptom state; length of stay; discharge disposition; 90-day readmission; 1-year mortality; and return to work. Model performance was evaluated using root mean square error for continuous outcomes and accuracy for categorical outcomes.

RESULTS: Predictive performance was moderate to strong across outcomes. For KOOS outcomes, root mean square error ranged from 15.13 to 23.49. A […]

PubMed · DOI

The Effect of Dual Mobility Articulations on Re-Revision After Revision for Dislocation.

Zamzam ME, Hodson NM, Zhu Q, Zheng H, Hallstrom BR, Charters MA

BACKGROUND: Recurrent instability remains a challenge after revision total hip arthroplasty (THA). This study evaluated whether dual mobility (DM) acetabular constructs used during revision for instability are associated with lower rerevision rates than fixed-bearing constructs.

METHODS: A retrospective analysis of the Michigan Arthroplasty Registry Collaborative Quality Initiative identified 796 patients who underwent revision of elective primary THA for dislocation; resurfacing, conversion, and urgent cases were excluded. Cumulative percent revision (CPR) curves were compared using log-rank testing. A Cox proportional hazards frailty model analyzed time to rerevision across DM, non-DM, and constrained liner groups, adjusting for revision year/era and site-level clustering. Multivariable logistic regression served as a sensitivity analysis for odds of rerevision within two years.

RESULTS: The two-year and five-year CPRs were 12.6% (95% CI [confidence interval], 10.1 to 15.0) and 17.1% (95% CI, 14.0 to 20.0), respectively. Dual mobility constructs had significantly lower CPR than non-DM (P = 0.022). At one year, CPR was 5.1% (95% CI, 2.1 to 8.0) versus 11.3% (95% CI, 7.7 to 14.8); at two years, 7.3% (95% CI, 3.6 to 10.8) versus 14.2% (95% CI, 10.2 to 18.0). By five years, confidence intervals overlapped (12.7 versus 20.2%). On adjusted analysis, non-DM constructs had signific […]

PubMed · DOI

Kneeling After Total Knee Arthroplasty: Does Resurfacing the Patella Affect the Ability to Kneel?

Hohmann AL, Bash HD, DeSimone CA, Fraval A, Lonner JH

BACKGROUND: Difficulty and pain with kneeling is a common postoperative complaint after total knee arthroplasty (TKA), but the cause of this issue has not been identified. The purpose of this study was to investigate the effect of patella resurfacing on a patient's kneeling ability after TKA with consideration of the preoperative presence of patellar osteoarthritis (P-OA).

METHODS: This was a single-center study of patients who underwent TKA at least one year prior to survey administration. Patellar resurfacing was routinely performed in the presence of patellar arthritis; whether to resurface a patella without arthritis was based on surgeon judgment. Patients were surveyed on preoperative and postoperative ability to kneel and satisfaction with ability to kneel, and Knee Injury and Osteoarthritis Outcome Score for Joint Replacement was collected preoperatively and with survey distribution. There were 132 knees in the no P-OA, unresurfaced group; 110 in the P-OA, resurfaced group; and 121 in the no P-OA, resurfaced group.

RESULTS: Preoperative and postoperative Knee Injury and Osteoarthritis Outcome Score-Joint Replacement was equivalent across groups. All cohorts reported worsening of their kneeling ability after TKA with no differences in reported difficulty in kneeling across cohorts. Patients who did not have resurfaced patellae reported nonsignificantly lower mean pain s […]

PubMed · DOI

Cement Viscosity Is Associated With Aseptic Loosening After Primary Total Knee Arthroplasty: An Analysis of the American Joint Replacement Registry.

Harris AB, Donnelly PC, Jennings JM, Martin JR, Khanuja HS, Hegde V

BACKGROUND: The mechanical properties of polymethylmethacrylate cement in total knee arthroplasty (TKA) can be influenced by both viscosity and the addition of antibiotics. Although smaller studies have reported high survivorship with commercially available formulations of polymethylmethacrylate in primary TKA, the independent and combined effects of viscosity and antibiotic additives on revision risk remain subject to debate.

METHODS: A retrospective analysis was performed using data from the American Joint Replacement Registry for all patients undergoing primary cemented TKA between 2012 and 2022 in patients aged ≥ 65 years. Patients were categorized by cement type (antibiotic versus nonantibiotic) and viscosity (low-, medium-, or high-viscosity cement [LVC/MVC/HVC]). Cox proportional hazards models were used to evaluate the risk of revision surgery, adjusting for multiple factors, including cement viscosity and antibiotic use.

RESULTS: A total of 233,841 TKAs were included. Both HVC and MVC had significantly higher cumulative percent revision for aseptic loosening compared with LVC, while MVC had the highest rate of aseptic loosening (hazard ratio 2.32, 95% confidence interval 1.89 to 2.85). Antibiotic cement was used in 81,462 cases (35%). Compared with nonantibiotic cement, antibiotic cement was not associated with a difference in revision for aseptic loosening or peripr […]

PubMed · DOI

The AAHKS Clinical Research Award: Maximizing Bearing Size Markedly Reduces Dislocations in Primary Total Hip Arthroplasty.

Wang E, McCormick K, Di Gangi C, Di Pauli von Treuheim T, Meftah M, Schwarzkopf R, Hepinstall MS

BACKGROUND: Modern polyethylene allows larger bearings in fixed-bearing total hip arthroplasty (THA), but any stability benefits of fully maximizing bearing diameter (e.g., 36-millimeter (mm) in 48/50-mm cups) are not well-established. We hypothesized that maximizing bearing diameter reduces odds of dislocation in primary fixed-bearing THA.

METHODS: We retrospectively reviewed all patients who underwent fixed-bearing THA at a large, urban, academic institution between 2016 and 2022. We noted cases receiving the largest bearing available from any manufacturer for the acetabular diameter: 28 mm in 40/42 mm, 32 mm in 44/46-, 36- in 48/50-, or 40- in 52/54/56 mm. Larger cups were excluded because proportionately larger bearings were unavailable. Multivariate analyses using least-absolute-shrinkage-and-selection-operator logistic regression were performed to explore the association between maximized bearing diameter and dislocation risk while controlling for confounders.

RESULTS: Bearing diameter was maximized in 835 (9.8%) of 8,607 patients, whereas 7,309 (84.9%) received the second-largest bearing available. There were 79 dislocations (0.9% overall); none occurred with maximized bearing diameters (P = 0.003). On univariate analyses, dislocation risk also varied with intraoperative technology use, surgical approach, and liner geometry (P = 0.017, P = 0.008, P = 0.007, respectivel […]

PubMed · DOI

A Contemporary Comparison of Modern Cementless to Cemented Femoral Stems for the Treatment of Femoral Neck Fractures.

Chen XT, Springer BD, Borkar S, Porter SB, Spaulding A, Trousdale RT, Goulding KA, Wilke BK

BACKGROUND: Cemented femoral fixation is considered the gold standard in hip arthroplasty for femoral neck fractures (FNFs), given lower periprosthetic fracture risk than cementless fixation. However, one major concern with cementation is bone-cement implantation syndrome (BCIS), a phenomenon involving cardiopulmonary distress attributed to cement pressurization. This study compared outcomes based on fixation method and stem design and evaluated the utilization of cemented versus cementless fixation following FNF.

METHODS: A retrospective review identified 2,829 patients who underwent total hip arthroplasty or hemiarthroplasty for FNF between December 2017 and January 2024 within a multi-institutional hospital enterprise. Cementless fixation was performed in 598 patients (21%), who were younger (75 versus 83 years; P < 0.001) and more likely men (66 versus 56%; P < 0.001) than those who received cemented fixation. Patients were propensity-matched for age, sex, body mass index, and comorbidities to compare fixation type and stem design. Following propensity matching, 346 matched pairs were analyzed. Outcomes including BCIS, periprosthetic fracture, and mortality were compared using Fisher's exact tests and a conditional logistic regression model.

RESULTS: Cemented fixation was associated with fewer periprosthetic fractures (4 versus 7%; P < 0.04) but more Grade 1 BCIS (43 vers […]

PubMed · DOI

Jumbo Femoral Heads in Total Hip Arthroplasty: Improved Early Stability Without Added Complications.

Kim BI, Cheng R, Grabov EH, Debbi EM, Vigdorchik JM, Mayman DJ, Shen TS

BACKGROUND: Larger femoral heads have been proposed to reduce dislocation risk following total hip arthroplasty (THA). While the benefit of increasing head size from 28 or 32 mm to ≥ 36 mm is well established, it remains unclear whether jumbo (40 or 44 mm) femoral heads confer additional stability over traditional large (36 mm) heads. This study aimed to determine whether jumbo heads improve stability or alter early complication rates.

METHODS: We identified 6,413 primary THAs with large-diameter femoral heads (36-, 40-, or 44-mm) implanted at a single institution between February 1, 2016 and December 31, 2022. Cases were stratified by femoral head diameter into the traditional large-head (36 mm, n = 5,897) and jumbo-head (40 or 44 mm, n = 516) groups. The median follow-up was 2.4 years (range, 0.1 to 9.8; interquartile range, 1.3 to 4.1). The primary outcome was prosthetic hip dislocation at the last clinical follow-up. Time-to-event analyses were performed using multivariable

Cox models with surgeon-level random effects, adjusting for patient-specific factors, year of surgery, surgical approach, and robotic assistance.

RESULTS: In the full follow-up period, jumbo heads were not associated with a lower adjusted dislocation risk (hazard ratio: 0.55 [95% confidence interval: 0.13 to 2.34], P = 0.139) when compared with 36 mm heads. When analysis was truncated to the first post […]

PubMed · DOI

High and Sustained Release of the Novel Echinocandin Rezafungin From Bone Cement: An In Vitro Comparison With Liposomal Amphotericin B and Micafungin.

Prats-Peinado L, Peral Y, Díaz-Navarro M, Márquez-Gómez M, Vaquero-Martín J, Muñoz P, Guembe M, Sanz-Ruíz P

BACKGROUND: Fungal periprosthetic joint infections are difficult-to-treat infections due to the complexity of fungal biofilm and to the limited number of antifungal agents that both elute effectively and remain active when incorporated into polymethylmethacrylate (PMMA) bone cement. Rezafungin (REZA), a next-generation, long-acting echinocandin with antibiofilm activity, may overcome these limitations. This study evaluated the in vitro elution kinetics of REZA from PMMA, assessed whether antibiofilm activity was maintained after elution, and compared its performance with micafungin (MICA) and gold standard liposomal amphotericin B (AmB).

METHODS: A commercial gentamicin-loaded bone cement was mixed with REZA or MICA at 0.5 and 1% and with AmB at 0.5%. Standardized samples (n = 3 per group) were incubated in phosphate-buffered saline under agitation. Eluates were collected at 10 time points from one hour to six weeks and quantified by liquid chromatography-mass spectrometry (AmB) or high-performance liquid chromatography (MICA and REZA). Cumulative drug release (area under the curve [AUC]) was calculated for each group. The antifungal activity of first-hour eluates of each drug was tested against 12 strains of Candida spp., using a 24-hour biofilm inhibition assay.

RESULTS: The REZA achieved the highest and most sustained elution, with 24-hour concentrations of 1,789 µg/mL (1% […]

PubMed · DOI

No Improvement in Infection or Complication Rate With Extended Oral Antibiotic Prophylaxis After Primary Total Joint Arthroplasty.

Raju A, Jahagirdar OB, Pour AE, Golden M, Molloy I, Grauer JN, Wiznia DH

BACKGROUND: Periprosthetic joint infection (PJI) leads to increased complications and cost after total hip and knee arthroplasty (THA and TKA). Extended oral antibiotic prophylaxis (EOA) has attracted interest as a measure to prevent PJI among high-risk patients, but efficacy and risks of additional complications are uncertain. The goal of this retrospective cohort study was to evaluate these aspects of EOA in THA and TKA.

METHODS: Adult patients who underwent primary THA or TKA between 2015 and 2025 were abstracted from an institutional database. The EOA protocol was a 7-day course of oral antibiotics directly following surgery given based on the discretion of the treating surgeon. Patients were extracted into two cohorts corresponding to whether they did or did not receive EOA. The cohorts were matched 1:4 on age, sex, and comorbidity. The incidence rate of PJI, along with various other complications, was assessed via univariate and multivariable analysis with Bonferroni correction of $P < 0.004$ applied. A total of 2,935 (1,205 THA; 1,730 TKA) patients were included following the inclusion criteria and matching. There were 587 (229 THA; 358 TKA) of these patients were in the EOA cohort, while 2,348 (976 THA; 1,372 TKA) were not.

RESULTS: After EOA, PJI rates did not significantly differ from non-EOA patients at 90 days ($P = 0.366$) and 1 year ($P = 0.254$), even among morbidly […]

PubMed · DOI

The AAHKS Surgical Techniques and Technologies Award: What Is the Ideal Technique and Implant in Aseptic Revision Total Knee Arthroplasty? An Analysis of Second Revision and Component Loosening Rates From the Australian Orthopaedic Association National Joint Replacement Registry.

Hoskins W, Gusho C, McAuliffe M, Wall C, Xu Q, Vince KG

BACKGROUND: Failed total knee arthroplasties (TKAs) can be revised via a range of techniques and fixation strategies. This study assessed the cumulative percent second revision (CP2R) rates based on what components were revised and the stem extensions and porous metal augments (cones/sleeves) used.

METHODS: Data from the Australian Orthopaedic Association National Joint Replacement Registry were analyzed for patients undergoing revision TKA (RTKA) for aseptic failures, following an initial diagnosis of osteoarthritis, from September 1999 to December 2024. The components revised (femur, tibia, and/or liner) and use of stems and cones or sleeves and their location were recorded. Complete revision where femoral and tibial stems and either cones or sleeves were used became the referent that all other revision constructs were compared to. The two primary outcomes were (1) CP2R for all-cause revision and (2) CP2R for aseptic loosening. Age, sex, body mass index, American Society of Anesthesiologists score, bearing surface, bearing type, and surgeon volume were included as covariates in Cox proportional hazard models. A total of 22,238 aseptic RTKAs met inclusion criteria, with 14 different revision construct permutations and 2,565 as the referent (mean age 70 years [median 71]; 59.1% women, mean follow-up from RTKA 4.5 years (range 0 to 22.3). The referent was compared with the othe […]

PubMed · DOI

Does the Magnitude of Change in Coronal Plane Alignment of the Knee Affect Clinical Outcomes in Functional and Mechanical Alignment Total Knee Arthroplasty?

Kawaguchi K, Tay ML, van Rooyen R, Walker ML, Farrington WJ, Bayan A, Young SW

BACKGROUND: Changes to native coronal plane alignment of the knee (CPAK) phenotypes following total knee arthroplasty may influence outcomes, particularly with mechanical alignment (MA). This relationship has not been evaluated with functional alignment (FA). This study compared the magnitude of coronal alignment change between MA and FA and whether this influenced patient-reported outcomes.

METHODS: This substudy analyzed data from a prospective, single-center, single-blinded randomized controlled trial involving 236 patients (119 FA and 117 MA). Long-leg radiographs were used to calculate the arithmetic hip-knee-ankle angle (aHKA), joint line obliquity (JLO), and CPAK phenotype preoperatively and postoperatively. Alignment change was assessed using absolute differences (change in aHKA, change in JLO), and a dose-response analysis was conducted. Clinical outcomes included the Forgotten Joint Score (FJS) and Oxford Knee Score at 1- and 2-yeas follow-up.

RESULTS: The FA resulted in smaller changes in alignment compared to MA (Δ aHKA P = 0.01; change in JLO P = 0.01) and more frequent restoration of the preoperative CPAK phenotype (42.9 versus 10.3%, P = 0.01). In the MA group, restoration of the CPAK phenotype was associated with significantly higher FJS, whereas no such relationship was observed in the FA group. A dose-response relationship was identified in the MA group: gre […]

PubMed · DOI

An Unsustainable Reimbursement Model: A 12-Year Analysis of Compensation for Revision Total Hip Arthroplasty.

Bernabe KD, Lizcano JD, Villa JM, Piuzzi NS, Barsoum WK, Higuera-Rueda CA

BACKGROUND: Revision total hip arthroplasty (rTHA) is a complex and resource-intensive procedure. However, physician and hospital compensations have not evolved to reflect such complexity. Therefore, we sought to describe trends in (1) physician, (2) 90-day episode-of-care, and (3) procedure type reimbursements during a 12-year period.

METHODS: Using a national claims database, we identified patients who underwent aseptic or septic rTHA between 2010 and 2022 using current-procedural terminology (CPT) and international-classification-of-diseases 9/10 codes. Debridement, antibiotics, and implant retention procedures were excluded. A total of 57,821 aseptic and 17,379 septic procedures were identified. The mean dollar reimbursement was calculated each year and adjusted for inflation. Patients were grouped by insurance type: (1) Commercial, (2) Medicare-Advantage (MCR-A), and (3) Medicare (MCR). A subgroup analysis compared septic revisions based on procedure types.

RESULTS: Commercially insured septic procedures showed the highest increase in physician (mean difference change of $1,525.3, P < 0.001) and 90-day episode-of-care (mean difference change of $529.9, P < 0.001) reimbursements during the 12-year period. Although compared to 2010, physician reimbursement in 2022 had a

slight increase in the MCR and MCR-A groups, 90-day episode-of-care reimbursement was significantly decr […]

PubMed · DOI

Methylprednisolone Taper After Total Knee Arthroplasty: A Prospective Single-Blinded Randomized Trial.

Premkumar A, Arellano E, Heo KY, Manz WJ, Hrudka B, Nour G, Fuqua A, Rieger E et al.

BACKGROUND: The use of short-term postoperative corticosteroid regimens has been reported to improve pain control and decrease opioid consumption following total knee arthroplasty (TKA). The purpose of this study was to evaluate the effect of a methylprednisolone taper (MPT) on reducing patients' subjective pain levels and opioid consumption in the early postoperative period following primary TKA.

METHODS: This was a prospective single-blinded randomized controlled trial of patients undergoing primary TKA for osteoarthritis. Patients were randomized to receive a standardized multimodal pain control regimen alone or the same regimen with a 6-day oral MPT (i.e., Medrol Dosepak). Patient pain and nausea Visual Analog Scale scores, as well as opioid consumption data, were collected for 7 days, while knee range of motion was collected preoperatively and at the 3-month follow-up visit. An a priori power analysis was conducted, and 69 patients were enrolled from October 2023 to March 2025, with 34 randomized into the control group and 35 into the MPT group.

RESULTS: Patients in the MPT group reported significantly decreased nausea levels on postoperative day 1 ($P < 0.05$). Daily and cumulative oral morphine equivalents were decreased in the MPT group over the first postoperative week, with a mean cumulative oral morphine equivalent of 95.5 versus 116.9, though this did not reach stat […]

PubMed · DOI

Journal of Orthopaedics & Traumatology

Volume 27, Issue 1 · June 2026 · 54 new articles

Fractures of the coronoid process: state of the art.

Marinelli A, Riva M, Coliva F, Minerba M, Carbone G, Guerra E

Coronoid fractures are rarely isolated and are much more frequently associated with other osseous or ligamentous structures injuries. On the basis of the coronoid fracture patterns, described by the O'Driscoll classification, it is possible to recognize three main patterns of injury that differ on traumatic mechanism and on associated lesions: posterolateral rotatory instability, posteromedial rotatory instability, and axial load injuries. The management of coronoid fractures is challenging and varies according to characteristics of the fracture, associated lesions, and amount of elbow instability. In general, operative treatment is indicated in every case the fracture is at least 50% of the whole coronoid, whether the sublime tubercle is involved, and whether the ulno-humeral joint is not perfectly reduced. In conclusion, the correct management of the coronoid, especially in the setting of complex elbow instability, represents a predictive factor for patient outcomes and functional results. The stability of the elbow, rather than the size of the coronoid fragment, is the main parameter for surgical indication, aimed to fix the coronoid and/or repair the associated lesions.

PubMed · DOI

Total hip arthroplasty in Italy: an observational, population-based study on surgical volume growth from 2001 to 2023 and forecasts until 2050 with six different statistical models.

Ciminello E, Cuccu A, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L et al.

BACKGROUND: The number of total hip arthroplasty (THA) procedures has been steadily increasing worldwide, driven by aging population, improvements in surgical techniques and implant design. This study aimed to analyze the temporal trends of elective THA in Italy since 2001-2023 and forecast THA volumes up to 2050 to provide insights for healthcare planning.

MATERIALS AND METHODS: International Classification of Diseases, 9th Revision, Clinical Modification (ICD9-CM) coding system was used to extract records of interest (elective THA) from the Italian National Hospital

Discharge Record database. Six statistical models were applied to forecast future THA volumes: logistic regression; Poisson regression; logarithmic regression; inverse/power regression; Poisson log-normal regression; and hierarchical Poisson regression with temporal effects (HPTE). Model performances were assessed by using error metrics and internal validation on the basis of a rolling-origin approach. An out-of-sample validation was conducted to ensure a robust assessment of forecasting reliability. THA volume forecasts were provided with 95% prediction intervals.

RESULTS: A total of 1,318,400 records for primary elective THAs performed in Italy since 2001-2023 were analyzed. The number of THAs increased by approximately 80%, rising from 68.270 in 2001 to 122.777 in 2023. Among the tested models, HPTE generally […]

PubMed · DOI

Clinical relevance of patient-reported outcome measures in the surgical management of focal chondral defects of the knee: a systematic review.

Migliorini F, Maffulli N, Memminger MK, Hofmann UK

INTRODUCTION: To evaluate clinical outcome following surgical management of focal chondral defects in the knee, patient-reported outcome measures (PROMs) are used. To give these measures meaning, parameters such as the minimal clinically important difference (MCID), patient-acceptable symptom state (PASS), minimally detectable change (MDC), clinically important difference (CID) and substantial clinical benefit (SCB) have been introduced. This systematic review investigated the MCID, SCB, CID, PASS and MDC of the most commonly used PROMs for assessing patients following surgical repair of focal chondral defects of the knee.

METHODS: This systematic review was conducted in accordance with the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and the recommendations of the Cochrane Handbook for Systematic Reviews of Interventions. All clinical studies investigating tools to assess the clinical relevance of PROMs in the surgical repair of focal chondral defects of the knee were reviewed. In April 2025, the following databases were accessed: PubMed, Web of Science and Embase. The PROMs of interest included: the International Knee Documentation Committee (IKDC) questionnaire, the Knee Injury and Osteoarthritis Outcome Score (KOOS) and its related subscales activities of daily living (ADL), pain, quality of life (QoL), sports and recreational […]

PubMed · DOI

Simultaneous versus staged bilateral total hip arthroplasty: a meta-analysis.

Soler F, Hernández J, Lamo-Espinosa JM, Benlloch M, Mariscal G

BACKGROUND: There is debate regarding the optimal timing of bilateral total hip arthroplasty (THA), with simultaneous or staged approaches considered. Cost-effectiveness is an important factor that influences resource allocation. The main aim of this study was to assess the cost implications of bilateral simultaneous (sim-THA) versus staged (st-THA) total hip arthroplasty. The secondary objective was to evaluate the efficacy and safety of sim-THA compared those with of st-THA.

METHODS: A systematic review and meta-analysis were conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines. Studies comparing simultaneous and staged bilateral THA in terms of cost, complications, length of hospital stay, and patient outcomes were eligible. Odds ratios (ORs), mean differences (MD), and standard mean differences (SMD) with 95% confidence intervals (CIs) were calculated. The risk of bias was assessed using the Methodological Index for Non-randomised Studies (MINORS). Meta-analyses were performed using Review Manager version 5.4 (Cochrane, Oxford, United Kingdom).

RESULTS: A total of 20 observational studies including 13,984 patients were included. Sim-THA was associated with significantly lower total costs (SMD -0.54, 95% CI -0.92 to -0.16; p = 0.005) and shorter hospital stay (MD -2.90, 95% CI -4.38 to -1.42; p = 0.0001) than tho […]

PubMed · DOI

Long-term outcomes of the modified Dunn procedure in moderate and severe slipped capital femoral epiphysis: a prospective case series with 7-year follow-up.

Fahmy M, Abdelazeem AH, Shawky MA

INTRODUCTION: Slipped capital femoral epiphysis (SCFE) is the most common hip disorder in adolescents and may lead to femoroacetabular impingement, early osteoarthritis, and long-term functional disability if inadequately treated. While in situ pinning remains the standard treatment for mild slips, it fails to correct the deformity in moderate and severe cases, potentially predisposing to degenerative changes. The modified Dunn procedure (MDP) was developed to restore proximal femoral anatomy through surgical hip dislocation while preserving vascular supply. The aim of the study is to evaluate the long-term radiological and functional outcomes of the MDP in patients with moderate (14 cases) and severe (10 cases) SCFE, and to assess the incidence of avascular necrosis (AVN), osteoarthritis, and other complications.

METHODS: A prospective case series was conducted between August 2015 and January 2019 at a single tertiary institution. A total of 24 hips with moderate-to-severe SCFE and open physis were treated using the MDP via surgical hip dislocation. Mild and acute-only slips were excluded. MDP was used as a primary procedure, performed early in severe slips and in selected moderate slips after clinical assessment. Patients were followed clinically and radiologically for a mean duration of 84 ± 2.6 months (range 80-88 months). Functional outcomes were assessed using the Harris […]

PubMed · DOI

The presence of a positive blood culture should be regarded a risk factor for DAIR failure in cases of haematogenous total knee arthroplasty infection.

Pedemonte-Parramón G, Reynaga E, Molinos S, de Los Ríos JD, Vivero A, García-Oltra E, López-Pérez V, Paredes R et al.

PURPOSE: Late acute haematogenous total knee arthroplasty (LAH TKA) infections represent approximately 20% of all prosthetic infections. Debridement, antibiotics, and implant retention (DAIR) is a widely accepted treatment; however, it carries a significant risk of failure. The study aims to identify risk factors for DAIR failure in LAH TKA infections, focusing on blood cultures (BC) as a potential contributor.

METHODS: A retrospective cohort study was conducted on 37 cases of LAH TKA infections from 2015 to 2023. Patients were divided into two groups: 20 in the success group (SG) and 17 in the failure group (FG). The study analyzes various factors, including demographics, comorbidities, serum and joint fluid biomarkers, blood and intraoperative cultures, prior infection or antibiotic use, and surgical and post-surgical variables.

RESULTS: Positive BC were more frequent in the FG compared with the SG (P = 0.03), and were also more common in patients with a history of prior infection (P = 0.03). Logistic regression identified positive BC as the only significant predictor of DAIR failure (odds ratio (OR) 12, 95% confidence interval (CI) 1.1-18, P = 0.04), even after adjusting for other variables. Positive intraoperative cultures were more frequent in the FG, particularly in those with a prior infection history (P = 0.08), even if they were receiving antibiotics (P = 0.05).

CON […]

PubMed · DOI

Iatrogenic femoral neck fractures or separation of the proximal femoral epiphysis during closed reduction of irreducible femoral head fracture-dislocations in children: a review of 12 cases.

Lu Y, Xue C, Canavese F, De Salvo S, Yao H, Huang D, Lin R, Zheng J et al.

OBJECTIVES: This study aimed to analyze the radiologic characteristics of irreducible pediatric femoral head fracture-dislocations to prevent potential iatrogenic femoral neck fractures (FNF) or separation of the proximal femoral epiphysis (SPFE).

METHODS: This is a retrospective review of patients who were skeletally immature and diagnosed with traumatic hip dislocations combined with femoral head fractures. The collected data included patient demographics, fracture classification, fragment ratio, combined injuries, urgent reductions, treatment strategies, complications, and final outcomes.

RESULTS: We treated 12 patients with femoral head fractures and dislocations; 11 out of 12 patients underwent urgent closed reduction (91.7%). Six of the patients failed closed reduction and experienced FNF (n = 2, 33.3%) or SPFE (n = 4, 66.7%). Five patients presented with avascular necrosis of the femoral head after open reduction with

internal fixation via a surgical hip dislocation approach, and two patients required further surgical treatment. Analysis of radiographs and computed tomography (CT) scans of irreducible femoral head fracture-dislocations revealed that the fractured femoral head was perched on the sharp angle of the posterior wall of the acetabulum, with a fragment ratio of 18-27%. After recognizing the irreducibility, one case with a fragment ratio of 26% underwent immed […]

PubMed · DOI

Antibiofilm-coated intramedullary nailing for tibial and femoral diaphyseal fractures at high-risk infection: outcomes of 103 consecutive patients.

Vicenti G, Migliorini F, Loiodice M, Scarpino A, D'attis P, Schäfer L, Colasuonno G, Solarino G

BACKGROUND: Fracture-related infection (FRI) is a devastating complication of intramedullary nailing of tibial and femoral diaphyseal fractures. The present study evaluates the clinical efficacy of an antibacterially coated intramedullary nail in patients with high-risk tibial and femoral diaphyseal fractures. Open fractures, fractures requiring fasciotomy or closed fractures with Tscherne grade II or III soft-tissue damage were considered fractures at high risk of infection.

METHODS: A single-centre observational study was conducted between January 2022 and December 2024, in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines. Patients who underwent definitive intramedullary nailing with Bactiguard (Bactiguard AB, Tullinge, Sweden)-coated Natural Nail (Zimmer Biomet, Warsaw, IN, USA) for tibial or femoral diaphyseal fractures at high risk of infection were included. The primary outcome was the incidence of FRI, defined according to the internationally validated consensus criteria. Secondary outcomes included time to union, rates of nonunion, malunion, unplanned secondary procedures and functional outcomes at a minimum follow-up of 12 months.

RESULTS: A total of 103 patients were enrolled: 55 with tibial fractures and 48 with femoral fractures. The combined cohort comprised 90 males and 13 females, with a median age of […]

PubMed · DOI

Orthopaedic publications receive fewer citations than general medicine articles: a scientometric analysis of temporal trends.

Migliorini F, Vaishya R, Maffulli N, Eschweiler J, Kobbe P, Betsch M, Schäfer L, Oliva F et al.

BACKGROUND: Orthopaedics, as a surgical speciality with a structurally narrower citing audience and lower citation density than general medicine, is systematically disadvantaged in any cross-disciplinary comparison relying on absolute citation counts. Whether this disadvantage is stable or has changed over time has not been previously quantified.

METHODS: The 100 most-cited articles indexed in the Web of Science Core Collection were identified for medicine and orthopaedics in 2014 and 2024. Citation data were extracted from a single snapshot taken on 19 January 2026. Total citations and citation velocity, defined as citations per year since publication, were compared between fields and across time points using Mann-Whitney U tests. Ratio-based analyses and distributional characteristics were also assessed.

RESULTS: Median citation counts were substantially higher in medicine than in orthopaedics in both years, reaching 5055 versus 288 in 2014 and 720.5 versus 34.0 in 2024, corresponding to 17.6- and 21.2-fold differences, respectively. Citation velocity followed the same pattern, with medicine maintaining an approximately 17- to 21-fold advantage at both time points. Within-field temporal analysis showed that citation velocity remained broadly stable in medicine between cohorts, whilst orthopaedics demonstrated reduced early citation accrual in the more recent cohort. Citatio […]

PubMed · DOI

Sleep quality and nocturnal pain in patients with elbow disorders: a prospective multicenter study.

Schneider MM, Nietschke R, Migliorini F, Schoch C, Burkhart K, Hollinger B, Kimmeyer M, Lehmann L et al.

BACKGROUND: Mental health disorders have become a growing global concern. Sleep quality is known to have a crucial impact on mental well-being. Poor sleep contributes to increasing economic costs owing to reduced

productivity as well as the development of mental disorders. Conversely, anxiety and depression often lead to sleep disturbances. Pain is a key factor affecting sleep quality. Studies have shown that musculoskeletal disorders, especially in the shoulder, may cause sleep disturbances. Research on the relationship between elbow pain and sleep quality remains limited. This study aims to evaluate the impact of common elbow pathologies on sleep quality.

METHODS: This prospective multicenter cohort study used self-reported outcome measures in patients with elbow disorders, including the Pittsburgh Sleep Quality Index (PSQI), to assess sleep quality. The PSQI was compared with other outcome measures.

RESULTS: A total of 664 patients (284 female, 380 male) were included, of whom 81% reported nocturnal pain. The mean age was 47.9 ± 14.2 years, body mass index (BMI) 26.9 ± 6.7 kg/m2 and symptom duration 15.8 ± 7.3 months. The most common diagnoses were lateral epicondylar pain (66%), medial epicondylar pain, arthritis, ulnar nerve entrapment and biceps/triceps tendinopathies. The overall PSQI was 7.3 ± 3.8, disabilities of arm, shoulder and hand questionnaire (DASH) 35.0 ± 19. […]

PubMed · DOI

A novel figure-of-eight polymeric cerclage wiring technique for greater trochanter fixation in revision arthroplasty and conversion of failed proximal femur fracture fixations to total hip arthroplasty.

Leggieri F, Martín Cocilova FN, Secci G, Stimolo D, Zanna L, Civinini R, Innocenti M

INTRODUCTION: Periprosthetic femoral fractures occur in 0.8-10% of primary total hip arthroplasties (THA) and up to 17.5% for revision THA (revTHA), with isolated greater trochanter (GT) fractures being the most predominant pattern. This study aimed to evaluate a novel figure-of-eight cerclage wiring technique for greater trochanter fixation in hip replacement cases.

METHODS: These retrospectively collected data included 36 hips (36 patients) undergoing revision THA or conversion of failed intertrochanteric fracture to THA with figure-of-eight cerclage wiring between April 2021 and June 2023. Patients under 18 years, with incomplete records, or lost to follow-up before 24 months were excluded. Primary outcomes included Trendelenburg sign, periprosthetic infections, and hip dislocation rates. Secondary outcomes evaluated THA stem subsidence and trochanteric migration using calibrated digital imaging. Statistical analysis included descriptive statistics, independent t-tests, linear regression, and correlation analyses with significance set at $p < 0.05$.

RESULTS: Mean follow-up was 404.76 ± 22.51 days. Patient demographics showed mean age 76.5 ± 12.3 years, body mass index (BMI) 25.8 ± 4.5 kg/m2, with 73.8% female patients. At final follow-up, one patient (2.8%) experienced post-surgical infection, one patient (2.8%) had dislocation, and three patients (8.3%) exhibited positive T […]

PubMed · DOI

Minimal clinically important difference (MCID), patient-acceptable state (PASS), minimally detectable change (MDC), and substantial clinical benefit (SCB) in patients who have undergone arthroscopic surgery for femoroacetabular impingement: a systematic review.

Migliorini F, Maffulli N, Schäfer L, Jeyaraman M, Betsch M, Mahmoud M

INTRODUCTION: Hip arthroscopy has been established as a successful management for femoroacetabular impingement (FAI) syndrome. Patient-reported outcome measures (PROMs) are frequently used to assess treatment results in patients. However, clinically important outcome values have recently been calculated to better understand and interpret PROMs and define successful treatment. This systematic review aimed to investigate the minimal clinically important difference (MCID), patient-acceptable state (PASS), substantial clinical benefit (SCB), and minimally detectable change (MIC) of the most commonly used PROMs for assessing patients who have undergone arthroscopic surgery for FAI.

MATERIAL AND METHODS: This study was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses: the 2020 PRISMA statement. In October 2025, the following databases were accessed: PubMed, Web of Science, and Embase. All clinical studies investigating tools to assess the clinical relevance of PROMs used to evaluate patients who have undergone arthroscopic surgery for FAI were accessed.

Only studies that assessed the minimal MCID, PASS, SCB, and/or MIC were considered.

RESULTS: The methodological quality of the 21 included studies was assessed, and the overall risk of bias was found to be low-to-moderate. The main characteristics of the included studies were analyzed and […]

PubMed · DOI

Mattress impact on lifestyle and back pain: a 25-year orthopedic literature review guideline.

Gianfilippo C, Filippo P, Stefano P, Alessandro A, Andrea F, Corrado C, Leonardo P, Giulia D et al.

BACKGROUND: Sleep quality is influenced by several modifiable environmental factors, including the sleep surface. Low back pain, morning stiffness, shoulder discomfort, and muscle tightness on awakening may indicate inadequate bedding support. This systematic review evaluated whether mattress characteristics influence spinal alignment, sleep quality, and low back pain.

METHODS: A systematic literature search and qualitative synthesis were conducted for studies published from 1999 to 2023. PubMed, Cochrane Library, Web of Science, Scopus, EMBASE, PsycINFO, ScienceDirect, and Google Scholar were searched using terms related to mattress ergonomics, pain, vertebral-column alignment, sleep, and quality of life. Eligible studies evaluated adult or young-adult populations and reported outcomes related to low back pain, spinal alignment, pressure distribution, sleep comfort, or sleep quality. Methodological quality and risk of bias were appraised using literature evaluation and grading of evidence (LEGEND) and Critical Appraisal Skills Programme (CASP) tools.

RESULTS: A total of 47 studies were included. The evidence most consistently favored medium-firm mattresses over very firm or very soft surfaces. Medium-firm support was associated with better pressure distribution, improved sleep quality, reduced morning stiffness, and low back pain intensity. Zonal support, air-adjustable syst […]

PubMed · DOI

Association between Kaplan fiber injury and anterolateral tibial subluxation in patients with anterior cruciate ligament injury.

Pan B, Chen S, Lin Y, Lin D, Zhou Z, Lin B, Zhou L, Ma J et al.

BACKGROUND: Kaplan fibers (KFs) are critical stabilizers of the knee, yet their role in anterior tibial subluxation of the lateral compartment (ATSLC) in anterior cruciate ligament (ACL) injury remains unclear. This study is to determine whether KF injury is independently associated with increased ATSLC on magnetic resonance imaging (MRI) in patients with an injured ACL, and to identify risk factors for high-grade ATSLC (≥ 4.5 mm).

PATIENTS AND METHODS: This retrospective study enrolled 132 patients with ACL injury who underwent MRI within 12 weeks of trauma. MRI was used to evaluate the integrity of the ACL, KFs, anterolateral ligament (ALL), bony morphology, and ATSLC. Two trained observers independently assessed all images using standardized criteria. Agreement between observers was calculated using standard reliability metrics. Multivariable linear regression was performed to evaluate the independent association between KF injury and ATSLC. Logistic regression and receiver operating characteristic analysis were used to identify predictors of high-grade anterior tibial subluxation (≥ 4.5 mm) in patients with combined ACL and anterolateral complex injuries.

RESULTS: Of the 132 patients, 35.6% had KF injuries. These patients exhibited higher rates of ALL injury and distinct bony morphologic features. KF injury independently explained 15.1% of anterior tibial subluxation vari […]

PubMed · DOI

Fibrous dysplasia: clinical and pathologic characteristics, surgical approaches, and outcomes in a large Italian cohort.

Mordenti M, Trisolino G, Moroni A, Parisi SC, Nanni S, Staals EL, Righi A, Gambarotti M et al.

Fibrous dysplasia (FD) is a rare benign bone disease caused by postzygotic activating mutations in the GNAS gene, leading to replacement of normal bone with fibrous tissue. It may be monostotic or polyostotic, often asymptomatic but sometimes associated with pain, deformity, or fractures. Management is mainly surgical in symptomatic or complicated cases. This retrospective study analyzes a large FD cohort to identify determinants of treatment and outcomes, with a focus on pain and its evolution. The primary aim was to assess factors influencing

surgical indication, while the secondary aim was to evaluate predictors of outcome in surgically treated patients, particularly symptom persistence at follow-up. A total of 227 patients were included. Mean age at diagnosis was 35.5 years (range 2-86 years); 79.3% had monostotic and 20.7% polyostotic FD. Pain was the most common presentation (62.6% at diagnosis; 78.4% during disease course), followed by incidental findings (21.6%), fractures (11.5%), and mass (4.4%). Surgery was performed in 51.6% of patients (127 lesions). Surgical patients were younger and had larger lesions and higher Mirels' scores (all $p < 0.001$), although Mirels' score showed only moderate predictive accuracy for surgery (area under the curve (AUC) 0.71). At a mean follow-up of 4.5 years, persistent pain was present in 24.1% of patients, with no significant differen […]

PubMed · DOI

Glycemic variability improves risk prediction and reveals a mortality threshold in orthopedic trauma patients with sepsis.

Long Z, Wang W, Wang S, Han T, Lin F, Li F, Qin Y, Lei M

BACKGROUND: Sepsis is a life-threatening complication in patients with orthopedic trauma, associated with significant mortality. Dysregulated glucose metabolism is a hallmark of critical illness, but the specific role of glycemic variability (GV), beyond hyperglycemia, in this high-risk population remains inadequately explored.

METHODS: This retrospective cohort study utilized 1946 patients from the Beth Israel Deaconess Medical Center between 2008 and 2022. We included adult intensive care unit (ICU) patients with a primary diagnosis of orthopedic trauma and concurrent sepsis. GV was quantified as the coefficient of variation of all blood glucose measurements during the ICU stay. The primary outcome was all-cause mortality at in-hospital, 30-day, 90-day, and 365-day. Restricted cubic spline (RCS) analysis was used to determine the detailed relationship between GV and mortality and the optimal GV threshold for dichotomization. Propensity score matching and multivariable logistic regression were employed to control for confounders.

RESULTS: The study cohort experienced substantial mortality, with rates of 16.0% (in-hospital), 19.8% (30-day), 26.4% (90-day), and 33.9% (365-day). RCS analysis revealed a significant nonlinear, J-shaped relationship between GV and all mortality end points ($p < 0.001$), characterized by an initial modest increase in risk followed by a sharp escalati […]

PubMed · DOI

Fibular allograft augmentation shows potential benefits in treating medial column comminuted proximal humeral fractures with osteoporosis: a secondary analysis of a randomized controlled trial.

Chai Y, Shi T, Jiang H, Chen Y, Xu Z, Zhu Y, Ruan Z, Chen Y et al.

BACKGROUND: Fibular allograft (FA) augmentation is proposed to address the lack of medial column support in plate-fixed proximal humeral fractures; however, the role of FA augmentation in these fractures remains inadequately elucidated. Our previous clinical trial did not show additional benefit in inserting an FA when treating adults with medial column comminuted proximal humeral fractures. Nonetheless, it is possible that some subgroups may still derive benefits from FA. This study aimed to investigate whether there were any subgroups benefit from FA.

MATERIALS AND METHODS: Our randomized controlled trial (RCT) recruited adults with a medial column comminuted proximal humeral fracture and randomly allocated them to receive a surgery fixed with a locking plate (LP group) or with an LP augmented with an FA (FA group). This study was a subsequent subgroup analysis of the previous randomized controlled trial, in which we included 72 participants who completed follow-up at the time of 12 months for the current subgroup analysis. The primary outcome was the Disabilities of the Arm, Shoulder and Hand (DASH) score at 12 months postoperatively. The secondary outcomes included the DASH score at other time points, changes in neck-shaft angle and humeral head height. We used generalized linear models for subgroup analysis to evaluate the effect of intervention (FA or LP) on 12-month DAS […]

PubMed · DOI

Thicker polyethylene inserts for varus deformity versus thinner inserts for flexion contracture in total knee arthroplasty: a retrospective comparative study.

Lee SH, Kwon HM, Lee WS, Park KK

BACKGROUND: The success of total knee arthroplasty (TKA) depends on achieving precise mechanical alignment and stable soft tissue balance, with the thickness of the final polyethylene (PE) insert serving as a critical component. However, the impact of preoperative factors on PE thickness remains unclear. In this study, we investigated the effects of preoperative knee alignment and clinical deformities on PE thickness.

MATERIALS AND METHODS: The medical records of patients who underwent primary TKA between August 2024 and July 2025 were retrospectively evaluated. Preoperative radiographic parameters, including the hip-knee-ankle (HKA) angle, medial proximal tibial angle, lateral distal femoral angle (LDFA), joint-line convergence angle, weight-bearing line ratio (WBLR), and posterior tibial slope, were measured, along with the flexion contracture (FC) as a clinical factor. Intra-observer reliability was assessed using intraclass correlation coefficients (ICC). Linear regression analyses were performed to evaluate the effects of these preoperative factors on PE thickness.

RESULTS: This study included 233 knees (158 patients). The intra-observer reliability of the radiographic parameters was excellent (ICC > 0.9). Multivariable linear regression revealed that preoperative HKA angle (β = 0.033, $p < 0.001$), WBLR (β = -0.008, $p < 0.001$), and LDFA (β = 0.065, $p = 0.003$) were signifi […]

PubMed · DOI

PediLoc® blade plate yields faster consolidation rates than LCP pediatric condylar plate in distal femoral osteotomies: a retrospective comparison and description of surgical technique.

Tschudi S, Delay C, Krautwurst B, Grisch D, Dreher T

BACKGROUND: The PediLoc® Locking Cannulated Blade Plate System was originally designed for proximal femoral osteotomies but has also been used for multidimensional correction of distal femoral deformities. The present study delineates the surgical technique and undertakes a retrospective analysis of the consolidation rates of distal femoral osteotomies performed using the aforementioned system, comparing them with those performed using the DePuy Synthes® 90° LCP Pediatric Condylar Plate. We hypothesized that the semi-rigid design of the blade plate results in faster osteotomy healing rates compared with other locking screw plate systems.

MATERIALS AND METHODS: This retrospective single-centre study included patients who underwent distal femoral osteotomy with the PediLoc® Blade Plate (Blade Plate) and the DePuy Synthes® 90° LCP Pediatric Condylar Plate (90° LCP-PCP) at the University Children's Hospital Zurich (2020-2024). Consolidation rates were assessed using "Regenerating Bone Defects Using New Biomedical Engineering" (REBORNE), "Radiographic Union Score for Tibial Fractures" (RUST) and modified RUST (mRUST) scores at 6 weeks and 3 months postoperatively and compared using using Wilcoxon signed-rank test. For the comparison between both intervention groups, the unpaired t-test was used for normally distributed data and the Mann-Whitney U test for non-normally distributed a […]

PubMed · DOI

Thin cortical translucent area of the iliac wing: morphometric analysis and implications for surgical safety.

Yüceda■ Gündo■du H, Pasl■ B, Erçakmak Güne■ B

BACKGROUND: The iliac wing is a frequently used fixation site in the surgical management of pelvic and acetabular fractures. Achieving safe implant placement in this region necessitates a comprehensive understanding of local morphology. However, the morphometric characteristics and surgical safety margins of the thin cortical translucent area (PLA) located in the central iliac wing have not been sufficiently defined. This may pose a significant challenge in evaluating the risk of cortical perforation, particularly during implant placement. This study aimed to evaluate the morphometric characteristics of the PLA and its relationship to fixation corridors.

METHODS: Fifty-one unpaired dry hip bones were examined. The thin cortical translucent region was identified using photoluminescence. Photograph-based measurements were conducted utilising ImageJ software on standardised photographs. Dimensions, area, distances to reference landmarks and relationships with the iliac pillar and supraacetabular corridors were assessed. The safe angular deviation range for anterior inferior iliac spine (AIIS)-origin screw fixation was calculated. Cortical thickness was measured using a digital thickness gauge.

RESULTS: The vertical and horizontal diameters and area of the PLA were 56.24 ± 16.56 mm, 59.92 ± 24.83 mm and 28.10 ± 16.47 cm2, respectively. Mean distances to the anterior superior ilia […]

PubMed · DOI

Efficacy and safety of MIPO versus ORIF in distal tibial fractures: a systematic review and meta-analysis.

Hassaan AM, Hassan AA, Abdelkhalik WS, Saad AM, Hosny AS, Elfeky BAEH, Mandour I, Ragheb MG et al.

BACKGROUND: Distal tibial fractures are challenging to manage owing to limited soft-tissue coverage and compromised blood supply. Minimally invasive plate osteosynthesis (MIPO) and open reduction and internal fixation (ORIF) are among the widely used surgical options, but the optimal approach remains unclear. MIPO preserves periosteal blood flow with smaller incisions, while ORIF offers direct visualisation but increases soft-tissue damage. With growing comparative evidence, an updated evaluation of these techniques is necessary.

AIM: To compare the efficacy and safety of MIPO versus ORIF in adult patients with distal tibial fractures.

METHODS: Following PRISMA guidelines and the Cochrane Handbook, we searched in PubMed, Embase, Cochrane, Scopus, and Web of Science till November 2025. Randomised trials and cohort studies comparing MIPO and ORIF were included. Data extraction was performed independently by two reviewers. Risk of bias was assessed using RoB 2 for RCTs and the Newcastle-Ottawa Scale for observational studies. Random-effects models were used to estimate pooled mean differences and odds ratios, and heterogeneity was assessed using I2. Sensitivity and subgroup analyses were conducted by study design.

RESULTS: Nine studies involving 530 patients met our inclusion criteria. No significant differences were found between MIPO and ORIF in AOFAS scores, union time, oper […]

PubMed · DOI

Comparative efficacy and safety of aspirin, low-molecular-weight heparin, and rivaroxaban for deep venous thrombosis prophylaxis after hip arthroplasty for femoral neck fracture: a single-center randomized controlled trial.

Miao S, Yue Q, Zhang X, Lu J, Zhang Y, Lu N, Wei L

OBJECTIVE: To evaluate the efficacy and safety of aspirin, low-molecular-weight heparin (LMWH), and rivaroxaban for deep vein thrombosis (DVT) prophylaxis after total hip arthroplasty (THA) or bipolar hemiarthroplasty (BHA) for femoral neck fracture, with subgroup analysis stratified by surgical type.

METHODS: This single-center randomized controlled trial (RCT) was conducted between June 2024 and June 2025. A total of 217 consecutive patients with femoral neck fracture undergoing hip arthroplasty were initially screened, and 161 eligible patients were finally enrolled and randomly assigned to the aspirin group, LMWH group, or rivaroxaban group at a 1:1:1 ratio. All patients received standardized intermittent pneumatic compression (IPC) and graduated compression stockings for mechanical DVT prophylaxis, as well as a unified postoperative rehabilitation protocol. The primary outcome was the incidence of lower extremity DVT within 12 weeks postoperatively, confirmed by color Doppler ultrasound. The secondary outcome was the incidence of bleeding-related complications graded by the International Society on Thrombosis and Hemostasis (ISTH) criteria. All patients were followed up at 6, 8, and 12 weeks postoperatively, with routine bilateral lower extremity ultrasound performed at 12 weeks regardless of symptoms. Subgroup analysis was performed on the basis of surgical type (THA ver […]

PubMed · DOI

Does the surgical sequence of lumbar spine fusion and total hip arthroplasty affect the dislocation and revision risk? A meta■analysis and systematic review.

Liu F, Jia C, Wu X, Zhou Y

BACKGROUND: Patients who underwent primary total hip arthroplasty (THA) after lumbar spine fusion (LSF) had a higher incidence of complications compared with those who had not previously undergone LSF. This study aimed to determine whether the timing of THA before or after LSF impacts the incidence of THA complications.

METHODS: Following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched PubMed, Web of science, Embase, and Cochrane Library from their inception until 1 October 2025. Data from all eligible published studies meeting the inclusion criteria were extracted and subsequently analyzed using a random-effects model to calculate the rates of all-cause revision, dislocation, periprosthetic joint infections, and aseptic loosening.

RESULTS: Our analysis included 10 studies involving 27,605 patients who underwent THA followed by LSF and 32,002 patients who received LSF prior to THA. There are no significant differences in the rates of dislocation (odds ratio [OR] 1.19, 95% confidence interval (CI) 0.73-1.86, p = 0.45), periprosthetic joint infections (OR 0.91, 95% CI 0.52-1.62, p = 0.76), and aseptic loosening (OR 0.89, 95% CI 0.74-1.08, p = 0.23) between the two surgical sequences, while for all-cause revision, the OR was (OR 1.01, 95% CI 0.70-1.47, p = 0.94), which reached statistical significance.

CONCLUSIONS: Compare […]

PubMed · DOI

Effects of anterior cruciate ligament resection and total knee arthroplasty on lower limb alignment and tibiofemoral rotation.

Fan Y, Chen S, Jiang L, Chen Z, Wu M, Wang C

BACKGROUND: The anterior cruciate ligament (ACL) is a key structure that restricts excessive anterior translation of the tibia and maintains rotational stability. This study aims to dynamically and quantitatively assess the direct effects of ACL resection and total knee arthroplasty (TKA) on lower limb alignment and tibiofemoral joint rotation under passive, unloaded conditions and in the pathological state of osteoarthritis.

METHODS: This retrospective study included 110 patients with varus knee osteoarthritis who underwent Mako for robotic-assisted TKA. Dynamic intraoperative measurements of the lower limb mechanical axis (flexion angle at maximum extension; varus angle) and tibiofemoral rotation at multiple flexion angles (min to max flexion) were recorded at three intervals: pre-ACL resection, post-ACL resection and post-TKA. The variation in rotation amplitude was calculated for successive flexion intervals.

RESULTS: Post-ACL resection, a significant decrease in the flexion angle at the maximum extension and varus angle (both $p < 0.001$) was observed, which was accompanied by increased tibial internal rotation at flexion angles of 60° and above ($p < 0.05$). Varus alignment was successfully corrected following TKA ($p < 0.001$). However, tibiofemoral rotational kinematics were significantly modified: the amplitude of internal rotation decreased in the initial flexion arc (min […]

PubMed · DOI

Comparison of Bankart repair with remplissage and Latarjet procedure for anterior shoulder instability: a systematic review and meta-analysis.

Nong K, Liu G, Jiang S, Li Q, Kwabena BR, Yuan D, Xiao W, Zhang K et al.

BACKGROUND: Anterior shoulder instability is the most common type of shoulder instability. For cases with subcritical glenoid bone loss, both Bankart repair with remplissage (BR) and the Latarjet procedure are viable surgical options. The aim of this study was to compare the postoperative outcomes in patients undergoing BR and Latarjet procedure.

METHODS: This review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. The methodological index for nonrandomized studies (MINORS) was used to characterize the methodological quality. A literature search was conducted using PubMed, the Cochrane Library, Embase, Web of Science, and Scopus. The analyzed outcomes included postoperative Rowe score, visual analog scale (VAS) score, the Single Assessment Numeric Evaluation (SANE) score, range of motion (ROM), return-to-sport (RTS) rate, RTS time, recurrence rate, revision rate, and complications. Subgroup analysis was performed on the basis of follow-up duration.

RESULTS: Overall, 12 studies involving 1186 patients were included. The meta-analysis showed that BR was associated with a significantly lower complication rate than the Latarjet procedure. No significant differences were observed in functional outcomes, ROM, recurrence, revision, RTS, or RTS time. Subgroup analysis showed a significant interaction between f […]

PubMed · DOI

Risk of periprosthetic joint infection within 1 year following robotic-assisted versus conventional primary total knee arthroplasty: a propensity-score-matched cohort study.

Tian S, Li Y, Mu W, Ji B, Zhang X, Cao L, Zhao J

BACKGROUND: Robotic-assisted total knee arthroplasty (RA-TKA) is increasingly being adopted for its ability to enhance bone-resection accuracy and component alignment. However, whether these technical gains influence the risk of periprosthetic joint infection (PJI) remains uncertain, especially in the context of prolonged operative duration. This study aimed to compare the 1-year rate of PJI following conventional total knee arthroplasty (cTKA) and RA-TKA in a propensity-score-matched cohort.

METHODS: We retrospectively reviewed 1284 consecutive patients who underwent primary TKAs at a single centre between 2021 and 2023. The patients were stratified according to surgical technique (cTKA versus RA-TKA) and subsequently matched 1:1 using propensity score analysis (age, sex, body mass index [BMI], American Society of Anesthesiologists [ASA] score, Charlson Comorbidity Index [CCI] score, CCI components and smoking), resulting in 522 pairs (1044 patients) for the final comparative analysis. Operative time and 1-year PJI were assessed using multivariable logistic regression. Infections were stratified according to timing: ≤ 90 days and from 90 days to 1 year after surgery.

RESULTS: The 1-year rate of PJI was 0.77% (4/522) after RA-TKA and 0.96% (5/522) after cTKA (P = 1.000). All PJIs in patients who underwent RA-TKA occurred within 90 days, whereas PJIs in patients who underwent […]

PubMed · DOI

Measured resection, gap balancing, and force-sensor-guided total knee arthroplasty result in different femoral and tibial rotational alignment but similar combined knee rotation and clinical outcomes: a randomized controlled trial.

Hernández-Hermoso JA, Nescolarde L, Yañez-Siller F, Calle-García J, Pérez-Andrés R, García-Perdomo D

BACKGROUND: Different alignment philosophies and balancing methods may alter femoral and tibial component rotation in distinct ways within the same knee. This study aimed to (1) determine which of three surgical techniques-measured resection (MR), gap balancing with computer-assisted surgery (GB-CAS), or a force-sensor soft-tissue balancing device (FS-STB)-most closely reproduces native femoral, tibial, and combined rotational alignment in mechanically aligned total knee arthroplasty (TKA), and (2) assess whether differences in rotational alignment affect outcomes at 5 years postoperatively.

MATERIAL AND METHODS: A total of 60 patients undergoing primary mechanical alignment TKA were randomly assigned to one of three surgical approaches (n = 20 per group): MR, GB-CAS, or FS-STB. Blinded observers assessed the Knee Society Score (KSS), Western Ontario MacMaster Universities Osteoarthritis Index (WOMAC) score, and hip-knee-ankle angle preoperatively and at the 5-year follow-up visit. Pre- and postoperative two-dimensional (2D)-computed tomography scans were used to measure femoral rotation (BFA), tibial rotation, and combined femur-tibia rotation (TE_PTCA and BC_PTCA). Statistical analyses included paired t-tests, one-way analysis of variance, and effect size calculations.

RESULTS: Femoral rotation remained unchanged in the MR and GB-CAS groups, but decreased slightly (1° exter […]

PubMed · DOI

Risser growth plate injury in unstable paediatric pelvic fractures: a multicentre retrospective study.

Oransky M, Aulisa AG, Roncoroni A, Zoppi AR, Mata M, Rohayem MA, Falciglia F, Toniolo RM

INTRODUCTION: Unstable pelvic fractures in children are rare but severe injuries and are associated with high-energy trauma. Unlike adults, children have open growth plates, particularly the Risser growth plate (RGP), which is vulnerable to injury. The presence of growth plates, such as the triradiate cartilage and Risser's plate, represents point of vulnerability that determines distinctive fracture patterns, similar to those seen with Salter-Harris growth plate injuries. Lateral compression fracture is the most common mechanism and results in an irreducible crushing of the sacral alae, causing a rotational deformity. Both vertical and lateral compression

displaced pelvic fractures require careful evaluation of the posterior iliac apophysis-associated injury. Failure to treat Risser growth plate injury (RGPI) can lead to severe long-term problems such as limb length discrepancy and scoliosis. This study aimed to determine how commonly RGPI occurs in unstable paediatric pelvic fractures, link it to specific fracture patterns and propose a subgroup type to improve treatment.

MATERIALS AND METHODS: This multicentre retrospective study included 40 children aged up to 12 years with unstable pelvic fractures (AO/OTA 61B and 61C). The patients were treated between 1987 and 2022 at trauma centres in Italy, Argentina and Venezuela. We reviewed patient demographics, injury mechanisms a […]

PubMed · DOI

Total femur replacement: function, quality of life, and gait: a multicentric EMSOS study.

Valentini M, Svehlik M, Leithner A, Bergovec M, EMSOS TFR Study Group, Smolle MA

BACKGROUND: Total femur replacements (TFRs) involve the replacement of the whole femoral bone, reconstruction of the hip and knee joints, and soft tissue reconstruction for function and joint stability. The aim of this multicentric European Musculo-Skeletal Oncology Society (EMSOS) study is to report on the functional outcome, quality of life, as well as gait analysis in patients with TFR.

MATERIALS AND METHODS: Retrospective data from 12 international participating centers were collected. In total, 65 patients [median (IQR) age at surgery 40 (18-56) years, 60% male] who received a TFR owing to oncologic or nononcologic indications between 1 January 1990 and 31 March 2024 were included. Patient-related outcome measures (PROMs), range of motion (ROM) of the hip and knee, and gait videos were collected.

RESULTS: At a median (IQR) follow-up of 28 (14-126) months, the median (IQR) percentage Musculoskeletal Tumor Society (MSTS) score was 63% (43-73%), 12-Item Short-Form Health Survey (SF-12) was 70% (55-79%), Harris Hip Score (HHS) was 63% (52-77%), and Oxford Knee Score (OKS) was 63% (52-79%); the median (IQR) extension-to-flexion ROM were 80° (70-97.5°, hip) and 70° (40-90°, knee). Gait videos were available for 24/65 patients (37%). The median (IQR) total Edinburgh Visual Gait Score of the operated side was 8.5 (6-13), with deviations from norm in all patients (24/24). Compens […]

PubMed · DOI

Vascularized versus nonvascularized free fibular grafts in reconstruction of post-traumatic critical long bone defects, a comparative study.

Fouaad AA, Massoud AH, Shaban M, Abdel-Naby M, Abulsoud MI

BACKGROUND: Reconstruction of critical-sized long bone defects is a complex orthopedic challenge. Free fibular grafts, vascularized (FVFG) and nonvascularized (NVFG), are established reconstructive options, but comparative clinical outcomes remain uncertain.

PURPOSE: To compare clinical and radiological outcomes of FVFG and NVFG in patients with post-traumatic critical bone defects more than 10 cm.

METHODS: A randomized controlled trial was conducted with 50 patients assigned equally to FVFG or NVFG groups. The primary outcome was time to union, while secondary outcomes included graft hypertrophy, functional scores (DASH and LEFS), complication rates, and donor site morbidity.

RESULTS: The mean time to union was 5.78 months in the FVFG group and 6.17 months in the NVFG group, showing no statistically significant difference ($p = 0.447$). Rates of graft hypertrophy, functional recovery, and complications were comparable between the groups.

CONCLUSIONS: Both FVFG and NVFG provide effective reconstruction for critical bone defects, with nearly similar healing times and functional outcomes. NVFG may represent a less complex alternative in selected cases.

PubMed · DOI

Five-year outcomes of medial open-wedge high tibial osteotomy with lateral supplemental lag screw in patients with obesity.

Tian YH, Lee KH, Huang CW, Lin SY, Yang JC

BACKGROUND: Obesity contributes to the accelerated progression of knee osteoarthritis. Medial open-wedge high tibial osteotomy (MOWHTO) is a joint-preserving surgical intervention for unicompartmental knee osteoarthritis; however, its efficacy in patients with obesity remains controversial. This study aimed to evaluate the 5-year clinical and radiographic outcomes of MOWHTO with lateral supplemental lag screw fixation in patients with obesity.

MATERIALS AND METHODS: This retrospective cohort study included patients with obesity who underwent MOWHTO between 2017 and 2020, with a minimum follow-up of 5 years. All procedures were performed by a single surgeon using three-dimensional printed, patient-specific instrumentation, locking plates, and a lateral supplemental lag screw. Clinical outcomes were assessed using the visual analog scale (VAS) and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC). Radiographic parameters, including the weight-bearing line percentage (WBL%) and medial proximal tibial angle (MPTA), were evaluated.

RESULTS: Significant improvements in VAS and WOMAC scores were observed at all postoperative time points ($p < 0.001$) and were accompanied by improved radiographic alignment, with WBL% shifting toward the Fujisawa point and increased MPTA values. At 5 years, mild regression in alignment was noted; however, the overall correction was […]

PubMed · DOI

Outcomes of surgical fixation of osteochondral fracture: a systematic review.

Pang AS, Oviya R, Tan JJ, Lim KSA, Hui JHP, Tan SHS

BACKGROUND: Dealing with knee osteochondral defects presents a substantial clinical challenge due to the variable nature of repair outcomes and intricate biomechanical complexities inherent to the knee joint. Numerous surgical alternatives exist for addressing knee osteochondral lesions, yet there is limited evidence available for comparing their respective outcomes.

METHODS: According to the Preferred Reporting Items for Systematic Review and Meta-analyses (PRISMA) guidelines, a systematic literature search was conducted to identify publications from inception to September 2023 on PubMed, Cochrane, and Medline academic search engines with the MeSH terms "osteochondral" AND ("patella" OR "patellar" OR "patellofemoral" OR "femoral condyle lesions" OR "trochlear lesions"). Inclusion criteria were clinical human studies, English language, subjects who underwent surgical management (fixation, reconstruction, or debridement) for knee osteochondral lesions with reported treatment outcomes, and a minimum sample size of ten patients. The methodological index for non-randomized studies (MINORS) was used to evaluate the non-randomized studies' methodological quality. Main outcome measurements Pediatric International Knee Documentation Committee (Pedi IKDC) scale, Tegner-Lysholm Scoring Scale, Tegner Activity Score, Visual Analog Scale, Knee Society Score (KSS) Score, Kujala Score, and F […]

PubMed · DOI

Do modular tapered long stems differ in stability and function? A prospective comparison of two stems in complex femoral revision THA.

Fahmy M, Shawky MA

BACKGROUND: Revision total hip arthroplasty (rTHA) in the setting of substantial proximal femoral bone loss is technically challenging. Modular tapered fluted stems provide predictable diaphyseal fixation while allowing independent adjustment of version, offset, and limb length. Among these, two commonly used systems-modular tapered fluted stem Type A (Revitan™) and Type B (LIMA Modulus™)-have limited direct comparative evidence. This study aimed to prospectively compare radiographic stem subsidence (primary outcome), as well as functional outcomes, complications, survivorship, and secondary outcomes, between Type A and Type B modular long stems in femoral rTHA.

METHODS: In this single-center randomized prospective study, 110 patients undergoing femoral revision rTHA were randomly assigned to receive either Type A (n = 55) or Type B (n = 55) stems. All procedures were performed by experienced revision surgeons under standardized perioperative and rehabilitation protocols. Radiographs were analyzed for stem subsidence, osseointegration, and limb-length restoration. Functional outcomes were assessed using the Harris Hip Score (HHS), Oxford Hip Score (OHS), and European Quality of Life Visual Analogue Scale (EQ-VAS) at baseline and final follow-up (mean 61.4 months). Complications and stem survivorship were recorded prospectively. Statistical analysis included paired and unpaired […]

PubMed · DOI

Clinical comparison of single posterolateral plate with medial-cannulated-screw fixation and double-plate fixation for extra-articular distal humerus fractures.

Jung HS, Chu MS, Lee JS

BACKGROUND: Double-plate fixation is the gold standard for extra-articular distal humerus fractures, but it carries a substantial risk of postoperative ulnar neuropathy. Fixation using a single posterolateral plate with a medial cannulated screw may reduce ulnar neuropathy while maintaining fracture stability. This study aimed to compare the clinical outcomes of single-plate-with-medial-screw fixation versus double-plate fixation for extra-articular distal humerus fractures.

MATERIALS AND METHODS: Fifty-six patients who underwent surgery for extra-articular distal humerus fractures (Arbeitsgemeinschaft für Osteosynthesefragen/Orthopaedic Trauma Association [AO/OTA] classification A2 or A3) between January 2018 and August 2024 were divided into a double-plate group and a single-plate-with-medial-screw group. We conducted a retrospective, nonrandomized comparative study. The double-plate fixation was used in 30 patients from January 2018 to October 2021, while the single-plate fixation with a medial screw was used in 26 patients from November 2021 to August 2024. All surgeries were performed using a posterior paratricipital approach. Bony union, radiographic healing, and loss of reduction were evaluated. Postoperative pain scores (visual analog scale at 2 days after the operation), operative time (minutes), elbow range of motion, elbow function (Mayo Elbow Performance Score [MEP […]

PubMed · DOI

A study on the association between tibial plateau fractures and intra-articular soft-tissue injuries under valgus injury mechanisms.

Duan S, Wang S, Zhu T, Li Q, Zhang M

BACKGROUND: While recent investigations have focused on injury mechanism classifications of tibial plateau fractures (TPFs), the association between valgus TPFs and concomitant soft-tissue damage involving menisci and ligaments remains insufficiently elucidated. This study aimed to characterize intra-articular soft-tissue injuries associated with various valgus TPFs and assess the predictive value of lateral plateau depression (LPD) and widening (LPW). Additionally, the analysis extended to other injury mechanisms.

MATERIALS AND METHODS: This study included adult patients with acute tibial plateau fractures who had complete imaging data, excluding patients with open fractures and multiple fractures throughout the body. Imaging examinations were used to assess the fracture mechanism and intra-articular soft-tissue damage.

RESULTS: The study retrospectively analyzed the clinical data of 138 patients with valgus injury TPFs in our hospital. The study compared the incidence of TPFs with intra-articular soft-tissue damage under different injury mechanisms. The result demonstrated that the incidence of valgus hyperextension TPFs combined with medial collateral ligament injuries was relatively high (61.5%) compared with TPFs with other injury mechanism. Multivariable logistic regression and smooth curve fitting revealed significant dose-response relationships of LPD (odds ratio [OR] […]

PubMed · DOI

The coronoid as the key fragment of trans-ulnar fracture-dislocations of the elbow: Insights from a retrospective cohort comparison using the coronoid-centric Mayo classification system.

Henssler L, Kerschbaum M, Zanklmaier M, Wieczorek LA, Alt V, Klute L

BACKGROUND: Trans-ulnar fracture-dislocations of the elbow are rare injuries with complex fracture patterns and variable outcomes. Traditional classification systems offer limited prognostic value. A recently introduced coronoid-centric Mayo classification distinguishes injury subtypes based on coronoid attachment and identifies trans-ulnar basal coronoid (TUBC) fractures as a particularly challenging entity. This study evaluated outcomes across Mayo fracture types and explored factors associated with inferior results in TUBC injuries.

MATERIALS AND METHODS: In this retrospective cohort study, surgically treated trans-ulnar elbow fracture-dislocations managed at a level I trauma center between 2010 and 2022 were identified and classified

according to the Mayo system. Demographic data, injury characteristics, surgical management, radiographic outcomes, and complications were recorded. Functional outcomes were assessed after a minimum follow-up of 12 months using the Mayo Elbow Performance Score (MEPS); Oxford Elbow Score (OES); Quick Disabilities of Arm, Shoulder and Hand Questionnaire (QuickDASH); European Quality of Life Five-Dimension, Five-Level Version (EQ-5D-5L); and range-of-motion measurements. Radiographs were analyzed for union, instability, heterotopic ossification, and post-traumatic osteoarthritis (OA).

RESULTS: A total of 52 patients were included (14 trans-olecr […]

PubMed · DOI

Feasibility study of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

Hinz N, Althoff G, Thomaschewski M, Bonke LA, Eberenz A, Münch M, Behrends L, Männel G et al.

BACKGROUND: Minimally invasive stabilization of acetabular and pelvic ring fractures using endoscopic techniques has become increasingly important. The logical advancement of conventional endoscopic techniques is a robot-assisted approach, which benefits from the advantages of robot-assistance systems (e.g., more degrees of freedom for instruments and improved visualization). The aim was therefore to investigate the feasibility of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

MATERIALS AND METHODS: In this two-part feasibility study, four different plate osteosyntheses were performed endoscopically using the Hugo™ robot-assisted surgery system, first on ten synthetic pelvic models and then on ten human cadavers. During robot-assisted dissection for the preperitoneal approach, identification of ten relevant anatomical landmarks was also assessed. In both parts, the success, number of drilling errors, and time for each plate were described as learning curves and analyzed using linear regression.

RESULTS: The infrapectineal plate could be successfully performed in 100% of synthetic models, the posterior column plate in 100%, the suprapectineal plate in 90%, and the superior pubic ramus plate in 80%. Learning curves could be observed for the number of drilling errors per plate (e.g., 0.67 to 0, p = 0.009) and the time r […]

PubMed · DOI

Favorable subjective clinical outcomes after revision hip arthroscopy for femoroacetabular impingement syndrome despite a high rate of capsular defects and increased capsular thickness.

Cao Z, Gao G, Lin W, Zhu Y, Zhou X, Wang J, Xu Y

BACKGROUND: The capsular healing status and capsular thickness changes in patients with femoroacetabular impingement syndrome (FAIS) following revision hip arthroscopy are poorly documented, and their relationship with subjective clinical outcomes remains unclear. The purpose of this work is to evaluate the incidence of capsular defects and changes in capsular thickness using magnetic resonance imaging (MRI) following hip arthroscopy in patients with FAIS, and to compare the subjective clinical outcomes between patients with and without capsular healing after revision hip arthroscopy.

PATIENTS AND METHODS: Consecutive patients with FAIS who underwent revision hip arthroscopy between 2013 and 2023 were included. Patients were categorized into two groups on the basis of capsular healing status after revision hip arthroscopy. Patient-reported outcomes (PROs) were collected preoperatively and at minimum 2-year follow-up, including the modified Harris Hip Score (mHHS), International Hip Outcome Tool-12 (iHOT-12), Hip Outcome Score-Activity of Daily Living Scale (HOS-ADL), Hip Outcome Score-Sport Specific Subscale (HOS-SSS), and visual analog scale (VAS). Achievements in minimal clinically important difference (MCID) and patient acceptable symptom state (PASS), patient satisfaction, and re-revision rates were evaluated and compared between groups. Capsular thickness was assessed by […]

PubMed · DOI

From bench to bedside: a novel suture-augmented prosthesis for greater trochanteric fixation in osteoporotic hip arthroplasty.

Yang M, Yang J, Wang X, Wu Z, Bian S, Sheng R, Liao F, Qin K et al.

BACKGROUND: Stable fixation of the greater trochanter during hip arthroplasty for unstable osteoporotic intertrochanteric fractures remains a significant challenge. Conventional techniques depend on securing the bone-implant interface, which is particularly compromised in osteoporotic bone. Here, we propose a novel conceptual approach that establishes a direct mechanical bridge from the abductor tendon to the prosthesis, thereby reducing reliance on the fragile bone-implant interface.

METHODS: In this two-stage translational study, we progressed from biomechanical validation to clinical application. First, using a decalcified caprine femoral model simulating osteoporosis, three fixation constructs were compared: locking plate (LP), suture-augmented locking plate (LPSA), and Kirschner-wire tension band (KWTB). Ultimate load and construct stability were evaluated. Biomechanical testing confirmed the principle of suture-mediated load sharing and highlighted the intrinsic weakness of screw fixation in osteoporotic bone. Guided by these results, we designed a novel femoral prosthesis that eliminates screw fixation, employs sutures as the primary load-bearing element, and incorporates integrated suture anchor tunnels. This prosthesis was then assessed in a retrospective series of 15 consecutive elderly patients with osteoporotic intertrochanteric fractures. Clinical outcomes were ev […]

PubMed · DOI

Updating radiographic parameters for the healthy growing hip: are current parameters still valid?

Hütter K, Smolle MA, Butter S, Schroedter R, Tschauner S, Kraus T

BACKGROUND: This study aimed to evaluate whether the acetabular angle (AA) by Tönnis and lateral center-edge angle (LCEA) by Wiberg-key radiographic parameters in pediatric hip assessment-have changed in a contemporary pediatric population compared with historical reference values, considering trends in earlier skeletal maturation and body composition.

MATERIALS AND METHODS: This retrospective diagnostic accuracy study assessed changes in AA and center-edge angle (CE) angles in a contemporary cohort. A total of 1774 anteroposterior pelvic radiographs (3548 hips) from patients aged 0-18 years without hip pathology were analyzed. Radiographs were obtained between 2006 and 2018 and measured using the Supervisely digital platform. Standardized anatomical landmarks were applied. Data were stratified by age and sex and compared with historical values using two-sample t-tests ($p < 0.05$). To minimize measurement bias, two independent observers conducted all assessments using standardized digital protocols.

RESULTS: A total of 1774 patients aged 0-18 years (mean age, 8.30 ± 5.20 years; 666 females, 1108 males) were evaluated. AA values did not show statistically significant differences in 11 of 16 age groups compared with historical data ($p > 0.05$). In contrast, LCEAs were significantly higher in the contemporary cohort, especially in the 5-8, 9-12, and 12-16 age groups (all $p < 0.01$) […]

PubMed · DOI

In-hospital deep vein thrombosis after tibial plateau fractures: incidence, laterality/anatomy, and risk factors in a multicenter retrospective cohort of 3366 patients.

Yang S, Li G, Li Y, Long Y, Zhang J, Liu L, Zhao Z, Sun C et al.

BACKGROUND: We aimed to determine the cumulative in-hospital incidence, anatomic distribution, and predictors of duplex ultrasonography-detected lower-extremity deep vein thrombosis (DVT) in patients hospitalized with tibial plateau fractures under routine thromboprophylaxis and protocolized surveillance.

METHODS: We retrospectively analyzed consecutive adults (≥ 18 years) admitted to two level-I trauma centers (November 2014-January 2024). Bilateral duplex ultrasonography was performed on admission/preoperatively (as soon as feasible after arrival and prior to surgery) and repeated serially during hospitalization (approximately every 5-7 days and when clinically suspected). The primary outcome was cumulative in-hospital DVT from admission to discharge; isolated intermuscular calf-muscle vein thrombosis (soleal/gastrocnemius) was recorded descriptively but excluded from the primary endpoint. Multiple injuries (polytrauma) were defined as ≥ 1 additional acute traumatic lesion beyond the index tibial plateau fracture documented on admission. Pulmonary embolism (PE) was assessed only when clinically suspected and confirmed by computed tomography pulmonary angiography (CTPA). Multivariable logistic regression identified independent predictors; receiver operating characteristic (ROC) analysis evaluated discrimination for key continuous predictors.

RESULTS: Among 3366 patients, 675 […]

PubMed · DOI

Retained articulating spacers (1.5 stage) in chronic knee periprosthetic joint infections: an analysis of associate factors and outcomes in a single center cohort of patients.

Balato G, Ascione T, Festa E, De Mauro D, Di Gennaro D, Rosa D, Mariconda M

BACKGROUND: Two-stage revision remains the gold standard for chronic periprosthetic joint infection (PJI), but it can significantly impact patients' quality of life and functional recovery. The 1.5-stage revision, involving indefinite spacer retention, has emerged as a potential alternative in selected high-risk patients. Our questions were: (1) What are the early functional outcomes, including their infection-free survival? (2) Which pre- and perioperative factors are associated with spacer retention? and (3) What is the mechanical survivorship of retained spacers?

MATERIALS AND METHODS: This was a retrospective observational study including 108 consecutive patients with chronic PJI between 2016 and 2020 who were followed prospectively for clinical outcomes. Functional status and quality of life were assessed using the EuroQol-5-Dimension-5-Level (EQ-5D-5L), Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and Knee Society Score (KSS) before spacer placement and prior to scheduled reimplantation. Infection-free status was defined as the absence of clinical, radiographic, and laboratory signs of infection recurrence over a 24-month follow-up. Univariate and multivariate logistic regression were used to identify factors associated with spacer retention.

RESULTS: All patients initially underwent articulating spacer implantation as part of a planned two-st […]

PubMed · DOI

Moderate leg length discrepancy: a long-term risk factor for hip osteoarthritis?

Aprato A, Donis A, Via RG, Scandurra A, Tuè F, Audisio A, Fusini F, Massè A

BACKGROUND: Leg length discrepancy (LLD) has been implicated as a biomechanical factor contributing to hip osteoarthritis (OA), yet the extent of its influence remains unclear. This study examines the correlation between moderate LLD (≥ 10 mm) and hip OA progression, focusing on asymmetrical OA distribution in patients over 65 years old.

MATERIALS AND METHODS: A retrospective analysis was conducted from a database of 1672 full-length standing X-rays. Patients under 65 years, with deformities, unilateral limb issues, or prosthetics, were excluded; therefore, the study group was composed of 220 patients. Tibial and femoral lengths were measured bilaterally, and hip OA severity was assessed using the Tönnis classification. Statistical analyses included Pearson's Chi-squared test and linear regression to explore correlations between LLD and OA distribution.

RESULTS: Among the sample, 18% showed an LLD ≥ 10 mm. A significant correlation was found between LLD and the asymmetrical distribution of hip OA ($p = 0.002$), with higher OA severity observed in the hypometric limb. Linear regression analysis suggested that each millimeter of LLD corresponded to a 0.74-unit change in OA severity difference between hips.

CONCLUSIONS: This study highlights a significant association between moderate LLD and contralateral hip OA in the elderly, emphasizing the biomechanical impact of asymmetrical […]

PubMed · DOI

Authorship, titles and open access as drivers of citation performance in orthopaedics: a scientometric analysis.

Migliorini F, Vaishya R, Rivera F, Eschweiler J, Kobbe P, Betsch M, Oliva F, Maffulli N

BACKGROUND: Bibliometric analyses are increasingly used to explore how scientific knowledge is created, disseminated, and perceived. In orthopaedics, research output has expanded rapidly over the past decade, yet the factors determining whether an article achieves wide visibility and scholarly impact remain poorly understood. Beyond the inherent quality of a study, elements such as authorship patterns, title construction, and open access (OA) availability may play an essential role in shaping citation performance. However, evidence in this field is still limited and sometimes contradictory, highlighting the need for large-scale, field-specific analyses.

METHODS: Orthopaedic publications from 2010 to 2020 were identified in Scopus using the keyword 'orthopaedic'. After duplicate removal, 97,806 unique articles were included with complete data on authorship, titles, citation counts, study design, and OA status. Citation rates were normalised per year since publication. Associations between bibliographic features and citation performance were assessed using multiple linear regression, while differences across title styles and study designs were evaluated with comparative statistical testing. Exploratory modelling was performed to identify combinations of authorship and title characteristics linked to the highest predicted citation rates.

RESULTS: Larger author teams were associa […]

PubMed · DOI

Interpreting nonsignificant findings: Reassessing the safety of self-locking standalone cages in octogenarian patients undergoing ACDF.

Abudayeh A, Fishchenko I

PubMed · DOI

A novel, simple, and affordable technique for arthroscopic soft-tissue tenodesis of the long head of the biceps tendon.

Fossati C, Ioppoli A, Ravaglia R, Vaja F, Menon A, Randelli PS

BACKGROUND: The purpose of this study was to describe a novel, simple, and implant-free arthroscopic soft-tissue tenodesis technique of the long head of the biceps tendon (LHBT) to the rotator cuff using an absorbable monofilament suture and to evaluate its clinical and functional outcomes at a minimum 1-year follow-up.

METHODS: A retrospective case series of 23 patients (mean age 58.0 ± 7.8 years) who underwent arthroscopic rotator cuff repair with concomitant LHBT soft-tissue tenodesis between June 2021 and June 2023 was analyzed. Functional outcomes were assessed using Constant-Murley Score (CMS), American Shoulder and Elbow Surgeons (ASES) score, Single Assessment Numeric Evaluation (SANE), visual analog scale (VAS) for pain, and the Long Head of Biceps (LHB) score. Supination strength was measured with a handheld dynamometer and compared with the contralateral side. Incidence of Popeye deformity and tenderness over the bicipital groove were recorded.

RESULTS: Objective Popeye deformity was observed in 13% of patients, with subjective concern reported by only 4.3%. Supination strength and LHB scores were similar to the contralateral side (means of 104.65 versus 104.64 N; LHB 93.7 versus 94.6 points). The mean CMS and ASES scores were 90.3 ± 12.4 and 89.6 ± 15.9 points, respectively. The SANE score averaged 87.4 ± 20.9, and the VAS for pain was low (1.65 ± 2.59 cm).

CONCL […]

PubMed · DOI

ORIF of three- and four-part proximal humeral fractures: a retrospective study of long-term clinical and complication outcomes of two-plate fixation methods.

Conteduca J, Gasbarra E, Rausa I, Casto A, Valentino L, Solarino G, Meccariello L, Rollo G

BACKGROUND: The aim of this study was to compare clinical outcomes and complication rates during long-term follow-up (≥ 6 years) after treatment for three- and four-part proximal humeral fractures using two different types of locking plates.

MATERIALS AND METHODS: A total of 113 patients with three- and four-part proximal humeral fractures who underwent surgery between September 2012 and January 2019 were enrolled retrospectively. Data for 49 patients [9 males, 40 females; mean age [standard deviation] (SD) 68.9 (5.8) years} treated with a PGR (intrauma) plate (group A) and 51 patients [10 males, 41 females; mean age (SD): 66.0 (7.5) years] treated with a PHILOS humeral plate (PHP, group B) were available at the last follow-up. The mean follow-up periods in groups A and B were 10.8 and 10.2 years, respectively, with a minimum follow-up of 6 years. At the final follow-up evaluation, functional outcomes were assessed using the Oxford Shoulder Scale (OSS), Simple Shoulder Test (SST), and Disabilities of the Arm, Shoulder and Hand (DASH) questionnaires. X-ray evaluation was also performed, and complications were recorded.

RESULTS: The mean OSS score in the PGR and PHP groups at follow-up was 42.9 and 39.1, respectively ($p > 0.05$). The SST score (PGR: 7.5 ± 2.0; PHP: 7.3 ± 4.0) and DASH score (PGR: 22.1 ± 5.5; PHP: 20.5 ± 4.0) were similar in both groups ($p > 0.05$). In our serie […]

PubMed · DOI

Incidence rates and treatment of the transcervical fracture of the neck of femur in Italy: is total hip arthroplasty an increasingly preferred approach? A population study on trends between 2001 and 2023 based on 1,120,770 hospital discharge records.

Ciminello E, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L, Cuccu A et al.

INTRODUCTION: Transcervical femoral neck fractures (TFNFs) are among the most devastating fragility fractures in the elderly. TFNF are associated with excess 1-year mortality rates ranging from 15% to 30%. Treatments include conservative methods, internal fixation, and arthroplasty (partial or total hip arthroplasty). This study aims to analyze the changes in incidences of TFNF in the Italian population between 2001 and 2023 and the evolution of the choices of treatment.

MATERIALS AND METHODS: Using hospital discharge record (HDR) data from 2001 to 2023, records with ICD9-CM codes for femoral neck fractures (820.0 and 820.1) among diagnoses were selected and categorized into four treatment groups: total arthroplasty, partial arthroplasty, fixation, and conservative. Time series were analyzed with stratification by sex and age.

RESULTS: The extracted data included 1,120,724 records of TFNFs, with 871,161 cases treated surgically (total or partial arthroplasty or internal fixation) and 249,563 treated conservatively; the average patient age was 79.1 years, with a higher proportion of women (72.8%). Partial hip arthroplasty was the preferred treatment overall. For younger patients, in the age classes < 45 and 45-54 years, fixation was the most chosen treatment. Over time, the use of the conservative treatment decreased from 27.5% in 2001 to 14.6% of cases in 2023. The use of par […]

PubMed · DOI

Does latissimus dorsi tendon transfer provide a normal scapular rhythm and external rotation strength in posterosuperior massive irreparable rotator cuff tears? A kinematic analysis in a retrospective cohort.

Calvo C, Fiumana G, Brigo A, Ruggiero ED, Bonfatti R, Donà A, Micheloni GM, Giorgini A et al.

BACKGROUND: Posterosuperior massive irreparable rotator cuff tears (PMIRT) are rare and disabling conditions. When conservative treatment fails, latissimus dorsi tendon transfer (LDTT) is a viable surgical option for symptom relief in carefully selected patients. However, its effectiveness in restoring glenohumeral function and its influence on scapulothoracic rhythm remain subjects of ongoing debate.

PURPOSE: The purpose of this study was to evaluate the clinical outcomes of LDTT and assess its impact on scapulothoracic rhythm using kinematic and electromyographic analysis.

MATERIAL AND METHODS: A total of 18 patients with PMIRT underwent LDTT. Functional scores consisted of Constant-Murley score (CMS), America Shoulder and Elbow Surgeons (ASES) score, and Quick Disabilities of the Arm, Shoulder, and Hand (QuickDASH) score. Electromyography (EMG) activity of the latissimus dorsi muscle was evaluated in conjunction with three-dimensional (3D) kinematic tracking system. Differences in external rotation strength were compared with and without active adduction.

RESULTS: The mean age was 55.9 years (range 40-69 years), with a mean follow-up of 20.4 months (range 6-39 months). At final follow-up, the mean CMS was 68.2 (95% CI 64.9-71.5), ASES was 76.9 (95% CI 66.1-87.6), and QuickDASH was 17.6 (95% CI 9.5-25.6). A significant difference in external rotation strength was observed […]

PubMed · DOI

Is there a need for exploration in pulseless supracondylar fractures of the humerus? A systematic review and individual patient data meta-analysis.

Haoyang C, Chen JG, Lai H, Lim AKS, Tan SHS, Hui JHP

BACKGROUND: Different approaches have been proposed to treat patients with pulseless supracondylar humeral fractures (SHF). We aim to analyze the current outcomes of patients who have undergone different treatment options for pulseless SHF, namely watchful waiting versus urgent surgical exploration.

METHODS: Electronic databases including PubMed, Embase, and Cochrane Library were explored. Our study included patients from all age groups but only included English-language articles. Single case reports and case studies with adequate description of patient population, injury, and outcomes were included. An individual patient data meta-analysis was done to evaluate key outcomes of surgical exploration versus no surgical exploration in pulseless patients with SHF.

RESULTS: Overall, the data for 1070 individual patients from a total of 48 studies were included. Patients with pulseless SHF with open fractures have a higher probability of requiring vascular intervention ($p < 0.001$) as they are more likely to have disrupted arteries ($p = 0.050$) and more likely to require vascular repairs ($p < 0.001$) than patients with closed fractures. Similarly, patients with pulseless SHF with pucker sign ($p = 0.003$) and ecchymosis ($p = 0.002$) were more likely to undergo surgical exploration. However, the neurological status had no relation to them undergoing surgical exploration ($p = 0.382$), and al […]

PubMed · DOI

Increasing complexity and plateauing outcomes in AO type C distal humerus fractures: a 50-year institutional case series.

Hönck K, Wenzelburger P, Eibinger N, Sadoghi P, Puchwein P

BACKGROUND: Distal humerus fractures in adults are rare but complex injuries, particularly AO type C3 fractures, often involving comminution, instability, and osteoporotic bone. Demographic trends indicate rising incidence, especially among elderly patients, with increasing functional demands challenging modern osteosynthesis. Open reduction and internal fixation (ORIF) with bicolumnar plating remains the preferred reconstructive approach, although complications remain substantial. This study aimed to evaluate trends in fracture complexity, patient demographics, and functional outcomes over five decades, focusing on the most recent cohort (2009-2018, series E) and comparing with historical cohorts (series A-D, 1969-2008).

METHODS: This retrospective analysis included five consecutive 10-year institutional cohorts (series A: n = 43; B: n = 29; C: n = 47; D: n = 58) of 233 patients with 235 supradiacondylar distal humerus fractures (AO type C1-3) treated with ORIF. Functional outcome parameters (Jupiter and Cassebaum Scores) and demographic data were available for all cohorts; additional parameters (range of motion (ROM), QuickDASH Score) were available for series D and E, Mayo Elbow Performance Score (MEPS), Short Form (SF)-36, and strength were assessed specifically in series E (n = 56; follow-up n = 21). Cross-cohort comparisons of categorical variables used chi-square tests; […]

PubMed · DOI

Magnetic resonance imaging-based fat infiltration grading improves reparability prediction in massive rotator cuff tears.

Hu Q, Zhang G, Wang D, Tung TH, Ding J, Zhou X

PURPOSE: This study aims to investigate the extent of fat infiltration (FI) in the rotator cuff (RC) muscles in more medial slices from the Y-shaped view (Y-view) on shoulder magnetic resonance imaging (MRI) and assess whether these slices provide a more accurate prediction of the repairability of massive RC tears (massive RCTs).

METHODS: The retrospective study included 57 patients with massive RCTs who successfully underwent arthroscopic surgery between 1 January 2023 and 31 December 2023. Patients were categorized into two groups: the irreparable group and the reparable group. All patients underwent shoulder MRI covering the region from the acromion to the medial border of the scapula in oblique coronal and oblique sagittal orientations. The FI stage of the RC was assessed across three different views to determine which view most effectively predicts the repairability of patients with massive RCTs.

RESULTS: The FI stage of the infraspinatus (IS) muscle across three distinct views demonstrated a significant correlation with the repairability of massive RCTs. The FI stage was more severe in the irreparable group. Receiver operating characteristic (ROC) analysis revealed that the area under the curve for the 1/2 scapula view (0.737) and the most severe view (0.745) exceeded that of the traditional Y-view (0.644). Paired-sample

ROC curve analysis revealed a significant differe […]

PubMed · DOI

Mid-term clinical and functional outcomes after reverse shoulder arthroplasty with latissimus dorsi transfer.

Colombini AG, Rab P, Macken AA, Soares MN, Kimmeyer M, Shirinskiy IJ, Popescu IA, Lafosse L et al.

BACKGROUND: Although reverse total shoulder arthroplasty (rTSA) with concomitant latissimus dorsi transfer (LDT) has been shown to effectively treat external rotation (ER) deficits, there are limited data regarding its outcomes with modern implants and its impact on activities of daily living (ADLs) requiring ER. The purpose of this study was to assess the mid-term clinical and radiographic outcomes of rTSA with concomitant isolated LDT in patients with an ER lag sign and posterior rotator cuff deficiency.

METHODS: This retrospective cohort study with prospective follow-up included consecutive patients who underwent rTSA with concomitant isolated LDT between 2010 and 2022 with a minimum follow-up of 2 years. Primary outcomes included the resolution of ER lag sign, active ER, and the Activities of Daily Living and External Rotation (ADLER) score. Secondary outcomes included the Auto-Constant Score (CS), Subjective Shoulder Value (SSV), activities of daily living requiring internal rotation (ADLIR) score, visual analog scale (VAS) for pain, and radiographic analysis of standardized radiographs.

RESULTS: In total, 32 procedures in 32 patients were identified. Of these, 22 procedures in 22 patients (68% female, 72.9 ± 8.4 years at surgery) were available for follow-up at 4.8 ± 2.2 years postoperatively (response rate 73%). The ER lag sign resolved in 95.5% of patients, the active […]

PubMed · DOI

Partial versus full revision in cementless single-stage exchange for hip periprosthetic joint infection: a multicenter comparative study with 10-year survivorship analysis.

Mu W, Jarusriwanna A, Xu B, Brown SA, Zhang X, Dawes A, Parvizi J, Cao L

BACKGROUND: Periprosthetic joint infection (PJI) is one of the most challenging complications following total hip arthroplasty (THA). Traditional management typically involves full revision to ensure comprehensive infection eradication. However, for patients with well-fixed implants, partial revision in a single-stage cementless approach may provide a viable alternative, potentially preserving bone stock and reducing operative time. The relative efficacy of this approach compared with full revision remains to be fully explored. This multicenter study aims to determine whether partial revision in cementless single-stage exchange surgery offers comparable infection control outcomes to full revision for select patients with well-fixed implants.

METHODS: We conducted a retrospective, multicenter cohort study involving 226 patients who underwent cementless single-stage exchange hip arthroplasty for PJI between 1 October 2013 and 31 July 2022. Patients were divided into partial revision (n = 61) and full revision (n = 165) groups. The primary outcome was treatment success, defined as the absence of clinical symptoms and signs of infection at a minimum follow-up of 2 years.

RESULTS: The success rates were 77.0% for the partial revision group and 80.0% for the full revision group, with no significant difference (p = 0.629). Both groups showed comparable 10-year survival rates for ove […]

PubMed · DOI



Journal of Orthopaedic Trauma

Volume 40, Issue 10 · October 2026 · 15 new articles

Methodological Concern Regarding the Multivariable Analysis of Rare Implant Breakage Events.

Yang Y, Liu J

PubMed · DOI

Author's Response Re: Methodological Concern Regarding the Multivariable Analysis of Rare Implant Breakage Events.

Van Rysselberghe NL, Gardner MJ

PubMed · DOI

Treatment Effects of Bisphosphonates Compared With Oxandrolone on Burn-Induced Bone Loss and Recovery in Mice.

Tseng MC, Wise PM, Osipov B, Song H, Lin YQ, Yeh J, Gallardo O, Wang A et al.

OBJECTIVE: Severe burn injuries can lead to systemic bone loss, increasing the risk of secondary fragility fractures. Evaluating treatments aimed at mitigating post-traumatic systemic bone loss is critically important for preventing secondary fractures in patients with burn. This study evaluated the efficacy of bisphosphonate (alendronate), oxandrolone, and vitamin D/calcium supplementation in mitigating post-traumatic bone loss and promoting recovery of bone in a murine model of burn-induced systemic bone loss, in which a full-thickness thermal skin burn involving approximately 15% total body surface area was used to model severe burn injury .

DESIGN: Male C57BL/6J mice were divided into 5 groups: control, burn with no treatment, burn + alendronate, burn + vitamin D/calcium, and burn + oxandrolone. Full thickness burn injuries were induced on the back of mice by submerging in 65°C water for 20 seconds; debridement and matrix application were performed 48 hours after injury, and treatments were administered starting on the day of injury.

OUTCOME MEASURES: Whole-body and site-specific bone mineral density (BMD) and bone mineral content (BMC) were assessed weekly using dual-energy X-ray absorptiometry. Micro-computed tomography and 3-point bending were performed at weeks 2 and 6 postinjury to assess bone microstructure and mechanical properties.

RESULTS: Untreated burned mice […]

PubMed · DOI

Preoperative Proton Pump Inhibitor Use Was Associated With Increased Risk of Revision Surgery and Impaired Bone Healing After Pelvic Fracture Fixation: A Retrospective Cohort Study.

Wang J, McGowan E, Horani K, Hagedorn JC

OBJECTIVES: The aim of this study was to evaluate the association between preoperative proton pump inhibitor (PPI) use and postoperative outcomes after pelvic fracture fixation.

DESIGN: Retrospective cohort study.

SETTING: TriNetX Research Network, a federated database of electronic medical records from approximately 103 health care organizations.

PATIENT SELECTION CRITERIA: Adult patients of 18 years of age and older who underwent surgical fixation of pelvic fractures between 2000 and 2023 were identified. All operatively managed pelvic fracture types were included, encompassing OTA/AO classifications 61 A-C; however, specific fracture classification was not available within the database. Patients were stratified into cohorts based on preoperative PPI use.

OUTCOME MEASURES AND COMPARISONS: Postoperative outcomes, including sepsis, revision surgery, hardware removal, nonunion, and malunion, were compared between patients with and without preoperative PPI use at 90 days and 2 years after surgery.

RESULTS: Before matching, 2648 patients were included in the preoperative PPI cohort and 7655 patients in the control cohort. After propensity score matching, 2403 patients remained in each cohort. The mean age was 44.0 ± 17.3 years in the PPI cohort and 43.7 ± 17.0 years in the control cohort, with 37.3% and 38.6% female patients, respectively. At 90 days postoperatively, preopera […]

PubMed · DOI

Periprosthetic Femur Fractures: The Screw-In-Washer Technique for Bicortical Small-Fragment Screw Fixation Through Large-Fragment Plates. A Technical Trick and Case Series.

Mahmoud AN, Horwitz DS

When treating periprosthetic femur fractures, bicortical screw fixation around a medullary-filling femoral stem may present a technical challenge, potentially leading surgeons to opt for unicortical locking screws or cerclage wires. These solutions may not achieve desirable stability, particularly in osteoporotic bone. In this technical note, we describe a simple approach using bicortical small-fragment screws-in-washers through large-fragment plates to facilitate secure bicortical screw passage around the stem. We also present preliminary outcomes from a case series in which this technique was used.

PubMed · DOI

Factors Affecting Outcomes of Hindfoot Fusion Nails for Acute Injury: A Multicenter Study: Erratum.

Kim E, Tornetta P, Carlson JB, Schultz A, Wireman G, Ollivere B, Zheng A, Spitler C et al.

PubMed · DOI

Modified CT Sagittal Angle Measure: Accuracy in Determining Hip Stability for Posterior Wall Acetabulum Fractures and Observer Reliability and Agreement Among Inexperienced Observers.

Jaeblon T, Harrington C, Khan U, Clark K, Kaplan S, McGlone P, Panarello N, Talwar S et al.

OBJECTIVES: The aim of this study was to describe modifications to the original CT sagittal angle measure (SAM) for predicting hip stability after posterior wall acetabular fractures (PWF), and to assess modified SAM (mSAM) accuracy and resident agreement and accuracy with the mSAM relative to examination under anesthesia (EUA).

DESIGN: Retrospective review of prospectively collected data.

SETTING: Academic Level 2 Trauma Center.

INTERVENTION: Results of a modification of the sagittal CT based PWF measurement (mSAM) and gold standard EUA were collected and evaluated.

PATIENT SELECTION CRITERIA: Patients with PWFs (AO/OTA 62A.1) requiring an EUA from June 2017 through July 2025 were included.

OUTCOME MEASURES AND COMPARISONS: The accuracy of mSAMs by the lead author and residents in predicting hip stability was quantified by sensitivity, specificity, positive predictive value, and negative predictive value using EUA, performed by fellowship trained orthopaedic surgeons, as the reference standard. A receiver operating characteristic curve analysis was used to determine whether mSAM reproduced the published SAM threshold angles most closely associated with hip stability, using EUA as the reference standard. Modified SAM reliability and agreement was assessed with intraclass correlation coefficients (ICC) after measurements in 2 rounds, separated by 1 month. Resident agreement […]

PubMed · DOI

A Dual-Implant Approach to Vancouver C Femur Fractures: Improved Outcomes With Retrograde Nail and Lateral Plate Fixation Compared With Lateral Locked Plating.

Page BE, Shaath MK, Rechter GR, Nasir BA, Baker TJ, Yawman JP, Jacobs E, Langford JR et al.

OBJECTIVES: To compare lateral locked plating (LLP) with a combined treatment approach using a retrograde intramedullary nail alongside a minimally invasive lateral locked periprosthetic plating system (Smith & Nephew; Memphis, TN) in the management of Vancouver C periprosthetic femoral fractures.

DESIGN: Retrospective chart review.

SETTING: Single, academic, Level-1 Trauma center.

PATIENT SELECTION CRITERIA: All adult patients who underwent fixation of a Vancouver C periprosthetic femur fracture (AO/OTA Type 32C [IVC/D]) with a retrograde intramedullary nail and periprosthetic plating system (NP Group) or LLP (LLP group) between 2019 and 2025 with follow-up to union, reoperation, or a minimum of 1 year were included. Patients in the NP group were recommended to fully bear weight immediately while patients in the LLP group were recommended to remain non-weight-bearing for 8 weeks.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was fracture union. Secondary outcomes included implant failure, infection, and alignment immediately postoperatively and at the final follow-up. Primary and secondary outcome measures were compared between the NP and LLP groups.

RESULTS: A total of 52 patients were included. In total, 31 in the NP group (68% female, mean age 79 years (range, 67-99)) and 21 in the LLP group (67% female, 69 years, range: 53-85, P = 0.54 for sex, P = 0.26 for ag […]

PubMed · DOI

Evaluation of an AI-Driven Risk Stratification System for Clinical Deterioration in Patients With Geriatric Hip Fracture.

Lin CQ, Ruiz M, McLellan M, Gardner MJ

OBJECTIVES: To determine whether an artificial intelligence-driven Clinical Deterioration Index (CDI) could identify geriatric patients with hip fracture at risk of early postoperative complications and to establish an orthopaedic-specific cutoff that identified patients at risk of deterioration.

DESIGN: Retrospective cohort study.

SETTING: Single Level I trauma center.

PATIENT SELECTION CRITERIA: Patients younger than 65 years of age who underwent fixation of OTA/AO 31-A/B/C hip fractures between June 2022 and December 2023 were included using ICD-10 codes S72.0-, S72.1-, and S72.2-. Exclusion criteria were pathologic fracture, revision surgery, or absence of 30-day follow-up. The CDI uses 31 clinical measures to generate a score from 0 to 100 reflecting risk of rapid deterioration. Patients were separated into 2 groups based on peak CDI in the immediate 48-hour postoperative period: CDI ≥65 and CDI <65. This cutoff of 65 was established by previous institutional validation.

OUTCOME MEASURES AND COMPARISONS: Primary outcomes were in-hospital complications (cognitive changes, cardiac complications, DVT/PE, etc). Secondary outcomes included length of stay, ambulation distance, and 1-year mortality. Outcomes were compared between CDI groups.

RESULTS: Of 197 patients (age range 65-98, 66% female), 15 (7.6%) exceeded the baseline CDI threshold of 65. This group had higher comp […]

PubMed · DOI

Blood Culture Testing in Fracture-related Infections: Low Yield and Lack of Concordance With Deep Tissue Pathogens.

Merrell LA, Solasz SJ, Ganta A, Konda SR, Egol KA

OBJECTIVES: To assess the concordance between blood culture isolates and intraoperative deep tissue cultures in patients with confirmed fracture-related infection (FRI).

DESIGN: Retrospective Cohort Study.

SETTING: Academic Medical Center.

PATIENT SELECTION CRITERIA: This Institutional Review Board-approved study included patients 18 years and older diagnosed with a confirmed FRI according to the FRI Consensus Group criteria who, at the time of irrigation and debridement (I&D), underwent deep tissue culture (TC) as well as concurrent blood culture (BC) testing (in the Emergency Department or inpatient setting). The decision to perform BC testing was left to the discretion of the initial treating providers at the time of this presentation.

OUTCOME MEASURES AND COMPARISONS: Microbiological data were reviewed from the electronic medical record. Infections were classified as monomicrobial (either gram-positive or gram-negative), polymicrobial, or culture negative. Pathogen concordance between blood and intraoperative tissue cultures was analyzed.

RESULTS: Eighty four patients were included with both intraoperative deep TC and concurrent BC. This cohort had a mean age of 56.2 ± 20.3 years and consisted of 33 women (39.3%). BC were never ordered by the orthopaedic surgeon. Microbial analysis of deep tissue specimens identified 29 gram-positive infections, 18 gram-negative infect […]

PubMed · DOI

NonTobacco Nicotine Dependence Is Associated With Increased Fracture Risk in Adolescents and Young Adults.

Zhang J, Meydan Y, Chen B, Barsi J

OBJECTIVES: The aim of this study was to evaluate whether nontobacco nicotine dependence (NTND) in adolescents and young adults is associated with an increased risk of new-onset fractures over a medium-term and long-term follow-up.

DESIGN: retrospective cohort query of TriNetX network.

SETTING: Multicenter study conducted using deidentified electronic health records from the TriNetX network of participating health care organizations.

PATIENT SELECTION CRITERIA: Adolescents and young adults (≤21 years) were included if they had a diagnosis of NTND between June 1, 2005, and June 1, 2020. Exclusions: any previous tobacco use, environmental tobacco exposure, tobacco use disorder, preexisting bone/joint pathology (eg, osteoporosis, osteoarthritis, osteogenesis imperfecta, and malignant neoplasms of bone/cartilage), or previous fracture. Patients with NTND were propensity score matched 1:1 with controls without any documented nicotine or tobacco use.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was incident fracture after the most recent eligible clinical encounter documented between June 1, 2005, and June 1, 2020. Fracture risk was assessed over a fixed 5-year period after the index date and across the longest available follow-up in TriNetX. Risk ratios (RR) and 95% confidence intervals (CI) were calculated. Kaplan-Meier survival analysis was used to estimate hazard rat […]

PubMed · DOI

Outcomes of Primary Repair of Heel Pad Degloving Injuries.

Roddy E, Dalstrom D, Sangeorzan B, Benirschke S, Vu S, Keys K, Firoozabadi R

OBJECTIVES: The aim of this study was to report the outcomes of patients with heel pad degloving injuries, and to identify patient, injury, or treatment factors associated with successful repair.

DESIGN: Retrospective cohort study.

SETTING: Single level 1 academic trauma center.

PATIENT SELECTION CRITERIA: All patients admitted with an acute heel pad degloving injury between 2005 and 2024 were included.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was successful repair of a heel pad degloving injury. Univariate analysis was performed to examine risk factors for primary repair failure.

RESULTS: A total of 58 patients with acute heel pad degloving injuries were included, of whom 50 were male with average age of 35 years (range 6-69, SD 17). Forty patients were treated with an attempt at primary repair, 3 were treated primarily with a flap, and 15 were treated with primary amputation. Primary repair was successful in 15 patients (38%). Repair method (suture alone, augmentation with k-wires, and incisional wound vac) was not associated with repair success ($P > 0.05$ for all). Of the patients with failed primary repair, 6 underwent secondary amputation, whereas 19 underwent attempted salvage with a flap (13), skin graft (5), or dermal substitute (1). Of the 19 with attempted salvage, 10 (53%) had complications including persistent ulceration, infection, and/or wound br […]

PubMed · DOI

What is the Rate of Occult Vessel Abnormalities in High-Energy Tibial Plateau Fractures (OTA/AO 41)?

Zhang D, Eurich J, Pitz-Goncalves L, Schimizzi G, Erdman MK, Christiano AV

OBJECTIVES: The aim of this study was to determine the rate of occult vascular abnormalities in high-energy tibial plateau fractures using computed tomography angiography (CTA).

DESIGN: Prospective Cohort Study.

SETTING: Single Academic Level I Trauma Center.

PATIENT SELECTION CRITERIA: Adult patients with a tibial plateau (OTA/AO 41) fracture, without clinical indication for CTA, that underwent knee-spanning external fixation were prospectively enrolled.

OUTCOME MEASURES AND COMPARISONS: After external fixation, patients underwent CTA to evaluate fracture morphology and the vascular tree. Fisher exact test was used to determine correlations between arterial abnormalities and wound complications in patients with ≥90 days of follow-up.

RESULTS: A total of 26 patients with 27 high-energy tibial plateau fractures were enrolled. Patients were predominantly male (n = 16, 61.5%) with a mean age of 45.1 ± 13.3 years. All patients had a blunt injury mechanism resulting in mostly closed (n = 25, 92.6%), bicondylar (n = 22, 81.5%) tibial plateau fractures. CTA demonstrated nonatherosclerosis vascular abnormalities in 3 patients (11.5%). Two had disrupted flow in the popliteal artery, and 1 had disrupted flow in the anterior tibial artery. No patient required vascular intervention. Atherosclerosis was appreciated in 6 (26.9%) patients. One of 3 patients (33%) with nonatherosclerosis […]

PubMed · DOI

BMP-2 Enhanced Bone Regeneration Using Clinically Available Grafting Strategies in a Chronic Caprine Tibial Defect Model.

Piuzzi NS, Luangphakdy V, Ramos MS, Zhang C, Pluhar GE, Carlson CS, Bechtold JE, Muschler GF

OBJECTIVES: The aim of this study was to assess bone grafting strategies in the chronic caprine tibial defect (CCTD) model.

METHODS: Design: Preclinical investigation.Setting: Chronic caprine tibial defect model.Patient Selection Criteria: Female Spanish-Boer goats, 4 to 6 years old.Outcome Measures and Comparisons: Primary outcome was total bone volume measured by MicroCT at 12 weeks. Secondary outcomes were radial percent bone volume and moment angle of inertia on MicroCT, and cortical bone bridging on radiograph.

RESULTS: Animals with MCA + BMP-2 (4542.1 mm 3 , P = 0.026) and MCA + BMP-2+BMA (3633.1 mm 3 , P = 0.006) grafts showed higher median tBV within the central 2.5 cm of the defect compared with animals with MCA + BMA grafts (12.3 mm 3). Median tBV within the central 2.5 cm did not differ significantly between animals grafted with ACB (486.8 mm 3) and those with MCA + BMP-2 (4542.1 mm 3) or MCA + BMA + BMP-2 (3633.1 mm 3) grafts. Adding unprocessed BMA to either MCA (MCA: 2.7 mm 3 vs. MCA + BMA: 12.3 mm 3) or MCA + BMP-2 (MCA + BMP-2: 4542.1 mm 3 vs. MCA + BMA + BMP-2: 3633.1 mm 3) did not significantly increase median tBV within the central 2.5 cm.

CONCLUSIONS: In the CCTD model, BMP-2 had a positive effect on tBV, comparable with and trending greater than ACB. This study supports using a composite of MCA + BMP-2 for large segmental bone defects, whereas the […]

PubMed · DOI

A Tibial Osteotomy of the Medial Collateral Ligament Insertion to Improve Visualization of the Medial Tibial Plateau: A Cadaveric Study.

Barth KA, Tijerina RR, Amin A, Novak AA, Mayer EN, Ismaily SK, Munz JW, Choo AM et al.

OBJECTIVES: To compare visualization of the medial tibial plateau articular surface using 3 surgical approaches: medial collateral ligament (MCL) split, femoral epicondyle osteotomy of the MCL origin, and tibial osteotomy of the MCL insertion.

METHODS: Seven matched pairs of fresh-frozen adult cadaveric lower extremity specimens (14 knee specimens total) with no evidence of previous surgery were used. Through a medial approach to the proximal tibia, the MCL split technique was performed followed by a submeniscal arthrotomy. Standardized distraction of the joint was applied using a universal distractor. A three-dimensional digitizer probe was used to map the visible articular surface. These steps were repeated for the femoral osteotomy on the same extremity as the MCL split and for the tibial osteotomy on the contralateral matched pair extremity. The tibial osteotomy preserved the meniscofemoral portion of the deep MCL, posterior oblique ligament, and proximal meniscocapsular attachments. Visible articular surface area (mm 2) for each of the 3 techniques was calculated from the mapped data points and compared among techniques.

RESULTS: The mean age of the specimens was 77.1 ± 7.2 years. MCL split technique visualized 87.2 ± 28.4 mm 2 of articular surface. The medial femoral epicondyle osteotomy provided 614.7 ± 217.6 mm 2 of visualization

(677% more than MCL split, P < 0.001) […]

PubMed · DOI

AOTS

Archives of Orthopaedic & Trauma Surgery

Volume 146, Issue 1 · August 2026 · 120 new articles

Fewer poor joint awareness results with cementless TKA compared with the same-design cemented implant.

De Berardinis L, Leggieri F, Salari P, Polizzotti G, Baldini A

BACKGROUND: Comparisons of cemented and cementless TKA have focused on mean outcomes and generally reported equivalence. We hypothesized that a 3D-printed full-titanium cementless TKA and the same-design cemented implant would not differ significantly in average joint awareness but would differ in the proportion of patients with the poorest results, and we compared the two both at the mean and across the outcome distribution.

METHODS: Prospective two-cohort study of 109 primary unilateral TKAs (54 cementless, 55 cemented) for end-stage knee osteoarthritis, with complete 24-month follow-up. Exclusion criteria were valgus deformity and bilateral surgery. The primary outcome was the FJS-12, analysed both at the mean (multivariable linear regression, with propensity score overlap weighting as sensitivity analysis) and across its distribution (quantile regression). Secondary outcomes were PASS achievement (FJS-12 ≥ 33.3), OKS, and time to functional satisfaction.

RESULTS: The cementless group was younger (P < 0.001) with more men (P = 0.008). Mean FJS-12 was not different between groups after adjustment (β = 7.4, 95% CI - 1.8 to 16.6, P = 0.115), although the confidence interval did not exclude a difference exceeding the minimal clinically important difference; this finding was reproduced by overlap weighting. On quantile regression, cementless patients scored significantly higher […]

PubMed · DOI

Elective primary total hip arthroplasty in rheumatoid arthritis versus osteoarthritis: in-hospital outcomes from the national inpatient sample, 2016-2021.

Finkel B, Gan-Or H, Ankol S, Tawatha M, Behrbalk E

BACKGROUND: Whether coded rheumatoid arthritis (RA) among patients undergoing elective total hip arthroplasty (THA) is associated with excess short-term inpatient morbidity in contemporary U.S. practice remains uncertain.

METHODS: We performed a retrospective cohort study using the National Inpatient Sample (2016-2021). Adults undergoing elective primary THA were identified using ICD-10-CM/PCS codes. The exposure cohort included admissions with coded RA in any diagnosis position. The comparator cohort included admissions with narrow primary hip osteoarthritis (OA) as the principal diagnosis and no RA. Propensity scores were estimated from demographic, socioeconomic, hospital, and comorbidity variables. Inverse probability of treatment weighting (IPTW) with survey weighting was used for the primary analysis, followed by doubly adjusted regression models that included age and all variables with persistent post-weighting imbalance greater than 0.10 (female sex, osteoporosis, chronic obstructive pulmonary disease, chronic anemia, depression, and race). Additional sensitivity analyses included stricter weight truncation, a restriction analysis, conventional multivariable modeling, and sex-stratified analyses addressing residual sex imbalance.

RESULTS: A total of 274,582 admissions met eligibility criteria; 274,539 were included in the final analytic cohort after exclusion of 43 re […]

PubMed · DOI

Research trends in total knee arthroplasty in India: a five-year bibliometric analysis.

Shetty S, Vijayan S, Mohandas Rao KG, Rao SK, Nayak Y, George BM

BACKGROUND: Total Knee Arthroplasty (TKA) is one of the fastest-growing orthopaedic procedures in India, driven by demographic transitions, increased life expectancy, and the rising burden of knee osteoarthritis. With

expanding procedural volumes, periodic evaluation of scientific output is essential to identify evolving research priorities, institutional leadership, and collaborative patterns. This study aimed to perform a bibliometric analysis of Indian TKA research published over the past five years.

METHODS: Publications were retrieved from the Scopus database using a structured search strategy for TKA-related terms. The search was restricted to studies published between 1 January 2021 and 30 December 2025 and conducted in an Indian setting. Bibliometric indicators assessed publication trends, document types, journals, authorship, institutional productivity, and state-wise contributions. Network-based bibliometric visualizations were generated using VOSviewer.

RESULTS: A total of 860 publications were identified, showing a steady increase in research output with peak productivity in 2025 (n = 221). Maharashtra and Delhi were the leading contributors, and the All-India Institute of Medical Sciences was the most productive institution. Original research articles accounted for 69.5% of publications. Emerging themes included robotic-assisted surgery, AI-based modeling, and ad […]

PubMed · DOI

Round versus tape sutures with interference fit anchors perform similarly in cadaveric glenoid and greater tuberosity biomechanical testing.

Dimitroff JD, Smith C, Hanish S, Oladeji LO, Bezold W, Nuelle CW, Smith MJ

PURPOSE: To determine whether suture shape (flat-tape vs. round-wire) influences biomechanical performance at the suture-anchor-bone interface in interference-fit anchors used for rotator cuff and glenoid labrum repair.

METHODS: Four matched pairs of human cadaveric shoulders were used for the glenoid model, with six anchor sites per glenoid. Five matched pairs of cadaveric humeri were used for the rotator cuff model, with three anchor sites per humerus. Each model tested multiple suture-anchor constructs under cyclic loading followed by load-to-failure. Biomechanical outcomes included cyclic displacement, yield characteristics, ultimate failure strength, stiffness, and force at fixed displacements. A linear mixed-effects model was used to compare suture types, with suture type and anchor position as fixed effects and specimen as a random effect.

RESULTS: In the glenoid/labrum model, there were no significant differences among suture types in cyclic displacement (cycles 1, 50, or 100; $p = 0.376$, 0.101, 0.109), displacement at yield ($p = 0.406$), yield force ($p = 0.880$), force at 1 mm ($p = 0.264$) or 5 mm ($p = 0.085$), or ultimate failure strength ($p = 0.204$). Stiffness ($p = 0.050$) and force at 2 mm displacement ($p = 0.033$) differed at the overall model level, however, no significant pairwise differences were observed. In the humerus/rotator cuff model, displacement at cycle 50 ([…]

PubMed · DOI

Association of systemic bone mineral density and tibial cortical thickness with postoperative subsidence after surgical treatment of tibial plateau fractures.

Hesmerg MK, Benner JL, Joosse P, Keijser LCM

INTRODUCTION: Tibial plateau fractures are common fractures above the age of 50 years old and are frequently considered osteoporosis-associated fractures. However, the association between bone mineral density (BMD) and postoperative articular subsidence remains unclear. The primary objective was to assess whether there was an association between systemic BMD or tibial cortical thickness and articular subsidence after surgical treatment of tibial plateau fractures. Secondary objectives were to evaluate preoperative fracture displacement and the prevalence of reduced BMD in this cohort according to WHO criteria.

MATERIALS AND METHODS: In this retrospective cohort study, patients treated surgically for low-energy tibial plateau fractures between 2017 and 2024 who underwent Dual-Energy X-ray Absorptiometry (DEXA) within 12 months before or after injury were included. DEXA-derived T-scores were used to assess systemic bone density and Tibial Cortical Thickness (TCT) was used as a measure of tibial cortical bone status. Fracture subsidence and preoperative displacement were measured in the coronal plane using articular depression. Multivariable linear regression analyses were performed adjusting for age, BMI and fracture comminution. The primary outcome was also adjusted for preoperative articular depression.

RESULTS: Fifty-nine patients were included in the analysis. There were 57 […]

PubMed · DOI

Comorbidity burden and fracture pattern as predictors of early postoperative medical complications following bipolar hemiarthroplasty for displaced femoral neck fractures: a retrospective cohort study.

Miyamoto S, Yoshii Y, Ishii T

INTRODUCTION: Postoperative medical complications substantially affect outcomes after bipolar hemiarthroplasty (BHA) for displaced femoral neck fractures, with an impact comparable to surgical factors. This study aimed to identify risk factors for postoperative medical complications, focusing on comorbidities, polypharmacy, and fracture pattern.

MATERIALS AND METHODS: This retrospective cohort study included 310 hips in 297 patients undergoing BHA. Baseline characteristics, comorbidities, and medication profiles were evaluated using the Charlson Comorbidity Index (CCI), Elixhauser Comorbidity Index (ECI), and comorbidity-polypharmacy score (CPS). The primary outcome was medical complications within 30 days. Multivariable logistic regression was used to identify independent predictors, and 1-year survival was assessed using Kaplan-Meier analysis.

RESULTS: Medical complications occurred in 53.7% of hips, most commonly delirium (17.0%). In the CCI-based model, age at surgery (odds ratio [OR] 1.05; 95% confidence interval [CI], 1.01-1.08; p = 0.006), displaced transcervical fractures (AO/OTA 31B2.1-31B2.2) (OR 2.07; 95% CI, 1.18-3.61; p = 0.011), displaced basicervical fractures (AO/OTA 31B2.3-31B3) (OR 4.67; 95% CI, 1.45-15.00; p = 0.010), a medium CCI category (OR 3.58; 95% CI, 1.10-11.60; p = 0.034), and a very high CCI category (OR 12.80; 95% CI 2.27-72.10; p = 0.004) were in […]

PubMed · DOI

Hemi-hamate arthroplasties for proximal interphalangeal joint fracture dislocations: long-term follow-up in acute, subacute and chronic cases.

Cansü E, Ilgaroglu S, Msangi WJ, Yurga E

INTRODUCTION: Hemihamate arthroplasty (HHA) method is one of the recommended treatment methods for unstable proximal interphalangeal joint (PIPJ) fracture dislocations The primary goal of this method is to achieve a stable and well-aligned joint by reconstructing the palmar lip. The aim of this study is to present mid to long term results of patients treated with HHA method in the acute, subacute and chronically.

MATERIALS AND METHODS: 22 patients was operated with this technique by single surgeon between 2013 and 2025. Patients were classified as acute (< 3 weeks), subacute (3-6 weeks), and chronic (> 6 weeks) based on the time elapsed between the date of injury and the date of surgery. Patients were assessed clinically, radiologically and subjectively.

RESULTS: The mean ROM of the PIPJ was 80.6° (20-100°), the flexion was 91.6° (65-100), and the extension lag was 11° (0-50). Grip strength on the operated side was 91.7% of the average on the unaffected side (74-110). The mean rest pain score according to the VAS (0 least-10 most) was 0.6 (0-3), and the activity pain score was 2 (0-5). The mean Q-DASH scores assessing functional level were 9.55 and the mean Mayo scores were 81.4. When patients were asked about their satisfaction after surgery, 12 patients responded very good, two patients good, and four patients moderate.

CONCLUSIONS: With HHA, reasonable PIPJ ROM and good f […]

PubMed · DOI

Partial triceps tendon transfer for dynamic stabilization in posterolateral rotatory instability of the elbow: a biomechanical study.

Pucher CJ, Lanzerath F, Kahmann SL, Hackl M, Müller LP, Leschinger T

PURPOSE: Posterolateral rotatory instability (PLRI) of the elbow results from lateral collateral ligament complex (LCLC) insufficiency. In the setting of failed conventional ligament reconstruction, no established salvage procedure is available. This study biomechanically evaluated partial triceps tendon transfer to the supinator crest as a dynamic stabilization concept for PLRI.

METHODS: Nine fresh-frozen human elbow specimens underwent cyclic external rotational loading of 0.5 Nm. Specimens were tested sequentially in the native state, an LCLC/common extensor origin-deficient unstable state,

after transosseous LCLC repair, and after partial triceps tendon transfer to the supinator crest with the LCLC detached. Testing was performed under combined biceps/triceps loading conditions with external forearm rotation relative to the humerus as the primary outcome. A two-factor repeated-measures analysis of variance (RM-ANOVA) compared unstable, repaired, and tendon-transfer conditions across loading conditions. The native state was assessed unloaded and reported separately in a one-factor RM-ANOVA.

RESULTS: Intervention significantly affected external rotation ($F(2,16) = 33.31$, $p < 0.001$, $\eta 2p = 0.81$). Across loading conditions, partial triceps transfer significantly reduced external rotation compared to the unstable state (mean difference [MD]=-3.38°, $p = 0.005$), while LCLC repair […]

PubMed · DOI

Development and validation of a model to predict full thickness rotator cuff tear.

Wang T, Tian Q, Lu B, Guo J, Yin B, Zhu Y

INTRODUCTION: Full-thickness rotator cuff tear (FTRCT) often requires surgical treatment. Magnetic resonance imaging (MRI) is the diagnostic gold standard; however, it is costly and resource-intensive. We aimed to develop and validate a clinical prediction model for FTRCT in patients with RCI, with the goal of identifying those at high risk at the initial visit.

METHODS: This prospective study consecutively enrolled patients clinically diagnosed with RCI from May 2025 to September 2025. All patients underwent shoulder MRI to confirm the presence or absence of FTRCT. Potential predictors involving demographic characteristics, clinical history, and physical examination findings, were collected. Predictor selection was performed using least absolute shrinkage and selection operator (LASSO) regression, followed by multivariable logistic regression to construct a risk prediction model, and then a nomogram was developed.

RESULTS: 357 patients were included, MRI confirmed FTRCT in 166 patients (46.5%). Patients were randomly divided into a training set (n = 249) and a test set (n = 108). LASSO regression identified 10 candidate predictors, and multivariable logistic regression identified physical labor, history of trauma, age, daily pain score, pain duration, and physical examination shoulder strength as independent predictors of FTRCT. The nomogram achieved an AUC of 0.910 (95% CI, […]

PubMed · DOI

Minimally invasive circumferential fixation using TIFI and INFIX for unstable pelvic ring injuries: a prospective evaluation of clinical and radiological outcomes.

Fahmy M, Sadek FZ, Khaled SA, Said MS, Kassem E

PURPOSE: To evaluate the clinical and radiological outcomes of minimally invasive circumferential fixation using combined trans-iliac internal fixation (TIFI) and anterior internal fixation (INFIX) in unstable pelvic ring injuries.

METHODS: A prospective study was conducted between 2022 and 2023 including 30 patients with unilateral unstable pelvic fractures (Tile B2 and C1). Patients were followed for a minimum of 12 months (mean 22 ± 4.7 months). Functional outcomes were assessed using the Majeed Score and Pelvic Outcome Score. Radiological assessment included displacement measurements, reduction quality, maintenance of reduction, and time to union.

RESULTS: Thirty patients (mean age 32.2 ± 14.9 years) were included. Mean operative time was 103.5 ± 10.5 min, blood loss 335 ± 65.8 mL, and fluoroscopy time 52 ± 14.7 s. Pelvic displacement improved from 15.8 ± 4.6 mm preoperatively to 3.6 ± 2.4 mm postoperatively, with a mean correction of 12.2 ± 4.5 mm and minimal loss of reduction (0.3 ± 0.8 mm). At one year, mean Majeed and Pelvic Outcome scores were 89.1 ± 5.6 and 34.6 ± 2.6, respectively. Mean time to union was 13.4 ± 2.1 weeks, with 90% excellent-good reductions (Matta criteria) and 100% union. Complications included superficial infection (13.3%), lateral femoral cutaneous nerve irritation (13.3%), and asymptomatic heterotopic ossification (3.3%).

CONCLUSION: Circumfere […]

PubMed · DOI

Radiographic accuracy of implant positioning in unicompartmental knee arthroplasty, does an imageless robotic system make a difference?

Montagna A, Andriollo L, Cinelli V, Marescalchi M, Matascioli L, Benazzo F, Rossi SMP

BACKGROUND: Robotic-assisted technologies and evolving implant designs have been introduced in unicompartmental knee arthroplasty (UKA) to improve the accuracy and reproducibility of component positioning. However, the relative contribution of robotic assistance compared with conventional instrumentation to radiographic outcomes in experienced, high-volume centers remains unclear.

METHODS: This retrospective comparative study included 96 consecutive medial UKAs performed at a single high-volume institution. Patients were stratified into three groups: Journey I UKA implanted with NAVIO imageless robotic assistance (n = 38), Journey II Uni UKA implanted using conventional instrumentation (n = 27), and Journey II UK UKA implanted with CORI imageless robotic assistance (n = 31). Standardized standing long-leg weight-bearing radiographs were obtained preoperatively and postoperatively. Radiographic evaluation included hip-knee-ankle (HKA) angle, femoral component coronal angle (FCCA), tibial component coronal angle (TCCA), and sagittal alignment parameters. Tibial coronal alignment was interpreted according to the Cartier metaphyseal alignment philosophy. Intergroup comparisons were performed using appropriate statistical tests.

RESULTS: Baseline demographic characteristics and preoperative radiographic parameters were comparable among the three groups (all p > 0.05). Postoperativ […]

PubMed · DOI

The perforator-based axial adipofascial turnover flap combined with juncturae tendinum grafting for complex dorsal phalangeal defects.

Ozer K

INTRODUCTION: Complex dorsal finger soft tissue defects remain a challenging reconstructive problem, particularly when associated with exposure of bone and extensor apparataus. Advances in the anatomical understanding of the dorsal cutaneous vascular system of the digits have allowed adipofascial flaps in this region to evolve from random-pattern to axial-pattern designs, improving vascular reliability. However, a structured and comprehensive approach for managing these combined defects has not been clearly defined in the literature. This study aims to present a holistic loco-regional strategy that optimizes the reconstructive ladder while minimizing donor-site morbidity and technical complexity.

MATERIALS AND METHODS: This clinical series includes seven fingers with complex soft tissue defects involving the dorsal aspect of the middle and distal phalanges. All cases were managed using an ipsilateral loco-regional protocol consisting of three components: (1) an extended axial adipofascial turnover flap for soft tissue coverage, (2) tendon reconstruction using juncturae tendinum harvested from the fourth intermetacarpal space, and (3) full-thickness skin grafting from the wrist. Patients were followed for a minimum of six months. Functional outcomes were evaluated by range of motion and complication rates.

RESULTS: Complete flap survival was achieved in all cases. All fingers […]

PubMed · DOI

Operative treatment of high-grade medial collateral ligament injury and chronic medial knee instability: a systematic review.

Xia Y, Zhao Y, Dou X, Wang X, You H, Chen L, He Y, Zhang J et al.

High-grade medial collateral ligament (MCL) injury and chronic medial knee instability can compromise both valgus and rotational control, especially in cruciate or multiligament injury. This systematic review evaluated operative treatment according to clinical phenotype rather than MCL grade alone. PubMed/MEDLINE and Web of Science Core Collection were searched from January 1, 2000, to August 10, 2026; a Cochrane CENTRAL update was completed on September 7, 2026. Eligible clinical reports included operative treatment of acute high-grade MCL/medial-complex injury or chronic symptomatic medial instability. Risk of bias was assessed with RoB 2, ROBINS-I, or the Joanna Briggs Institute case-series checklist. Thirty-five reports were included: 5 randomized or quasi-randomized trials, 9 nonrandomized comparative studies, and 21 case series. Routine acute medial surgery during ACL reconstruction was not consistently superior across heterogeneous grade III populations. In contrast, selected displaced distal or Stener-like lesions were plausible repair targets because interposition can prevent healing in apposition. Chronic symptomatic valgus and/or anteromedial rotatory insufficiency was most often treated with reconstruction, with directional improvement in objective stability and patient-reported outcomes in case series. Evidence for augmentation, the optimal reconstruction construct […]

PubMed · DOI

Correction: Recovery of gait after ankle fracture surgery following early mobilization and weight-bearing.

Bauer L, Schlegel AS, Scheuer A, Böcker W, Polzer H, Baumbach SF

PubMed · DOI

Factors associated with long-term patient-reported outcomes after periacetabular osteotomy.

Shoji T, Ueki S, Kaneta H, Morita H, Kozuma Y, Adachi N

INTRODUCTION: Long-term survivorship and patient-reported outcomes after periacetabular osteotomy (PAO) have been reported; however, the relationships of long-term PROMs with preoperative cartilage degeneration and postoperative three-dimensional hip morphology remain incompletely understood. This study evaluated long-term PROMs and examined their associations with preoperative cartilage T2 values and postoperative hip morphology after PAO.

MATERIALS AND METHODS: This retrospective study included 42 hips in 42 patients with early-stage osteoarthritis who underwent PAO for developmental dysplasia of the hip and were followed for more than 10 years. Clinical outcomes were assessed preoperatively and at final follow-up, whereas the Japanese Orthopaedic Association Hip Disease Evaluation Questionnaire (JHEQ) was assessed at final follow-up. Hip morphology was evaluated using computed tomography (CT), and cartilage degeneration was assessed using preoperative T2 mapping magnetic resonance imaging. Associations with final JHEQ scores were analyzed using correlation analyses and multivariable linear regression with JHEQ scores treated as continuous outcomes.

RESULTS: The mean final total JHEQ score was 56.4 ± 19.6 points. In the multivariable linear regression model for total JHEQ (adjusted [Formula: see text]), higher preoperative acetabular T2 values ([Formula: see text], 95% CI [[…]

PubMed · DOI

Clinical, etiological, and microbiological differences between foot and non-foot osteomyelitis.

Kundakci B, Dalgic Y, Dalkir KA, Olke HC, Mirioglu A, Tekin M

INTRODUCTION: Osteomyelitis presents with heterogeneous clinical, etiological, and microbiological characteristics depending on anatomical location. While foot osteomyelitis is commonly associated with diabetic foot infections and contiguous spread, non-foot osteomyelitis often arises from post-traumatic, hematogenous, or postoperative causes. However, direct comparative analyses between these two entities remain limited.

MATERIALS AND METHODS: A retrospective analysis was conducted on 202 adult patients who underwent surgical treatment for osteomyelitis between 2011 and 2026. Patients were divided into foot (n = 89) and non-foot (n = 113) osteomyelitis groups. Clinical, etiological, microbiological, and treatment-related variables were compared. Only intraoperatively obtained deep tissue and/or bone cultures were included.

RESULTS: Patients with foot osteomyelitis were older and had a significantly higher prevalence of diabetes mellitus (49.4% vs. 5.3%, p < 0.001), which was identified as an independent predictor of foot osteomyelitis (OR 14.15, 95% CI 4.32-46.42; p < 0.001). Infections in foot osteomyelitis predominantly developed via contiguous spread (73.0%), whereas non-foot osteomyelitis showed more heterogeneous etiologies, including post-traumatic and hematogenous causes (p < 0.001). Microbiologically, Staphylococcus aureus was the most common pathogen in non-foot ost […]

PubMed · DOI

Identification of patient subgroups by hierarchical cluster analysis in femoral neck fractures and surgical outcome analysis.

Ipek E, Altunta■ Y, Balkanl■ B, Bozca MA, Unutmaz H, Demirkale ■, Eren OT

INTRODUCTION: Femoral neck fractures are a heterogeneous entity with wide variability in patient profiles and outcomes. Single-variable classification may not capture the multidimensional interactions that influence prognosis. This study aimed to identify clinically meaningful patient subgroups using hierarchical cluster analysis and to compare mortality, postoperative mobility, and complication rates between these subgroups.

MATERIALS AND METHODS: This retrospective cohort study included 672 patients treated for intracapsular femoral neck fractures classified as AO Foundation/Orthopaedic Trauma Association (AO/OTA) 31-B between 2009 and 2023, including 586 who underwent hemiarthroplasty (HA) and 86 who underwent total hip arthroplasty (THA). Unsupervised hierarchical clustering was performed using the Gower distance matrix and Ward's D2 linkage method based on 20 preoperative and intraoperative clinical and surgical variables, excluding outcome measures. The optimal number of clusters was determined by silhouette analysis. Mortality predictors were assessed using Cox proportional hazards regression, and predictors of good postoperative mobility were evaluated by binary logistic regression.

RESULTS: Two distinct clinical phenotypes were identified: a high-risk cluster (n = 607; mean age 78.3 years, American Society of Anesthesiologists [ASA] physical status III 86.2%, HA 96.4 […]

PubMed · DOI

High variability in resistance to valgus deformation among knee braces indicated for medial collateral ligament injuries: a standardized bench-model biomechanical comparison.

Pechlivanidou E, Stacheter M, Wahl P, Jacobi M

INTRODUCTION: To quantify valgus deformation and per-load mechanical stiffness of commercially available knee braces indicated for medial collateral ligament (MCL) injuries, and to evaluate whether mechanical performance is associated with material classification and qualitative stability assessment.

MATERIALS AND METHODS: Nine knee braces identified through a national market survey were tested on a custom lower limb mechanical surrogate under progressive valgus loading at 5.7, 11.4, and 19.4 Nm. Angular deformation was measured using video-based motion tracking, and per-load brace stiffness (Nm/°) was computed at each load level.

RESULTS: At maximum applied torque (19.4 Nm), per-load stiffness ranged from 1.6 to 6.5 Nm/° across the nine devices - a fourfold difference between the stiffest and most compliant brace - with median valgus deformation of 8° (range 3-12°). In an exploratory, underpowered between-category comparison, rigid braces showed numerically higher stiffness than semi-rigid and soft devices (ε^2 = 0.34-0.45); this comparison is reported for hypothesis generation only. Qualitative ratings showed only moderate, inconsistent correlations with measured performance.

CONCLUSIONS: Commercially available knee braces indicated for MCL injuries demonstrate highly variable mechanical resistance to valgus deformation under standardized bench conditions. The present sampl […]

PubMed · DOI

The role of the plantar fascia, flexor hallucis longus, and fibro-osseous tunnel in functional hallux rigidus.

Wang M, Hunter Aitchison A, Orriols AM, Ridenour RM, Easley ME, Adams SB

BACKGROUND: Functional hallux rigidus is characterized by limited first metatarsophalangeal (MTP) dorsiflexion during weight bearing conditions, but normal motion otherwise. This phenomenon alters the biomechanics of the foot, potentially leading to first MTP pain and trauma to the MTP cartilage. The purpose of this study is to assess the role of the flexor hallucis longus (FHL), plantar fascia (PF), and fibro-osseous tunnel in functional hallux rigidus.

METHOD: Fourteen cadaveric lower limbs were analyzed. The Dananberg technique was used to dorsiflex the great toe and radiographs were taken after each of the following manipulations, measuring the dorsiflexion angle of the first MTP. The plantar fascia was transected with a Z-cut, shortened approximately 10 mm, then repaired. The fibro-osseous tunnel was released. The FHL was transected with a Z-cut then shortened approximately 7 mm.

RESULTS: Average passive dorsiflexion of the first MTP is 23.8° (95% CI 19.7-27.9) prior to manipulation. Shortening the plantar fascia and FHL did not decrease the MTP dorsiflexion significantly beyond baseline ($p>0.05$). Lengthening the plantar fascia increased the dorsiflexion by 6.7° (28.1% increase) and lengthening the FHL increased dorsiflexion by 7.0° (29.4% increase). The difference between the two methods was not significant ($p>0.05$). Releasing the fibro-osseous tunnel alone increased th […]

PubMed · DOI

Mix and match strategies in revision total hip arthroplasty: a review of techniques and outcomes.

Ashkenazi I, Khury F, Konan S

PURPOSE: This review evaluates the clinical indications, surgical rationale, and outcomes of six targeted component-retention strategies in revision total hip arthroplasty (THA) designed to reduce operative morbidity and preserve host bone stock.

METHODS: This is a narrative review. A structured but non-systematic search of PubMed/MEDLINE and Scopus was used to identify contemporary clinical cohorts, joint registry data, and biomechanical analyses, with the evidence organised sequentially across acetabular and femoral subsites.

RESULTS: Acetabular retention of stable metal-on-metal shells with dual-mobility bearings minimizes surgical trauma and yields exceptionally low dislocation rates (0% to 4%). Cementing highly cross-linked polyethylene (XLPE) or dual-mobility liners into secure or revision cups successfully decouples biological fixation from alignment, ensuring high mid-term survivorship. On the femoral side, trunnion adapter sleeves protect the head-neck junction during isolated bearing exchanges, demonstrating reliable 10-year survivorship (85%). For unavoidable stem extractions with sound cement mantles, cement-in-cement revision offers a rapid pathway; utilizing high-fatigue-resistant cobalt-chromium alloys instead of stainless steel structurally mitigates the rare risk of cantilever stem fracture. Lastly, for uncontained bone loss or profound osteopenia, extended l […]

PubMed · DOI

Complications following operative versus nonoperatively managed isolated distal third ulna shaft fractures: a retrospective single-institution cohort study.

Schultz K, Wang JE, Manley B, Peek K

PURPOSE: This study aimed to compare radiographic and clinical outcomes of operatively versus nonoperatively managed isolated distal third ulna fractures.

METHODS: This is a retrospective cohort study. Operatively and nonoperatively managed isolated distal third ulna shaft fractures on forearm radiographs in adult patients with a minimum of six-week follow-up were identified at an academic level 1 trauma center over seven years from 2017 to 2024. Outcomes included rates of radiographic evidence of union, pain scores, nonunion, tendinopathies, removal of hardware, or heterotopic ossification at final follow-up. Outcomes were compared between cases managed operatively and those managed nonoperatively.

RESULTS: A total of 56 isolated distal third ulna shaft fractures were identified with a mean follow-up duration of 30.5 weeks, including 12 (21.4%) managed operatively and 44 (78.6%) managed nonoperatively. The mean age was 49.0 years in the nonoperative group and 38.6 years in the operative group ($p = 0.115$), and the proportions of males were higher in the nonoperative group (61.4% vs. 16.7%, $p = 0.006$). At final follow-up, radiographic evidence of union was present in 77.3% of nonoperative cases and 91.7% of operative cases ($p = 0.266$). The complication rate was lower in the nonoperative group (15.9% vs. 41.7%), although the difference was not statistically significant ($p = 0.0$ […]

PubMed · DOI

Radiation exposure for planning deformity corrections with hexapods: comparison of conventional X-ray, standard CT, and ultra-low-dose CT.

Schüll D, Baumgartner H, Krüger B, Thelen J, Keller G

INTRODUCTION: Hexapod systems are among the most accurate devices for correcting complex bone deformities. Postoperative planning is carried out using web-based software solutions based on at least two X-ray images in orthogonal planes. Recent studies have shown that inaccuracies in determining mounting parameters and the spatial relationship between the reference ring and the bone may lead to measurement errors. Computed tomography images potentially offer greater accuracy in this regard, but are associated with increased radiation exposure. The aim of this study was to compare radiation exposure from conventional radiography (X-ray), standard computed tomography (CT), and ultra-low-dose CT (ULD-CT) in the context of planning deformity corrections using a hexapod system and to evaluate suitability and comparability of these imaging modalities for

treatment planning.

MATERIALS AND METHODS: Between March 1, 2022, and August 31, 2024, 28 patients were treated with a hexapod system for post-traumatic deformities at a Level I trauma center. All patients for whom standardized radiographs (anteroposterior and mediolateral views), conventional CT scans, and ULD-CT scans were available were included in this retrospective analysis. The study population thus comprised n = 18 patients. The effective radiation dose (µSv) was calculated separately for each examination. The values were comp […]

PubMed · DOI

Operative versus conservative management of distal humerus fractures in patients aged 65 years and older: a target-trial-informed retrospective cohort study of five-year mortality.

Murad H, Shehadeh M, Igbaria I, Mahamid MK, Yassin M, Mahamid A

INTRODUCTION: Older patients with distal humerus fractures die more often after conservative than after operative treatment in unadjusted comparisons; whether this reflects treatment or the selection of healthier patients for surgery is unresolved. We estimated the association between operative versus conservative management and five-year all-cause mortality in patients aged ≥ 65 years.

MATERIALS AND METHODS: Retrospective comparative cohort study at two tertiary trauma centres (2012-2025). Cases were ascertained by automated screening of codes and free text, then adjudication of all uncertain records against source notes and injury radiographs, masked to treatment and vital status. Patients were classified by management initiated within 14 days as operative (n = 75) or conservative (n = 78); vital status was registry-linked. The primary estimand was the five-year restricted mean survival time (RMST) difference; principal adjusted estimates were inverse-probability-weighted (IPW) RMST and survival differences, requiring no proportional-hazards assumption. Cox, window and landmark analyses are supportive.

RESULTS: Among 153 patients (52 deaths), conservatively managed patients were older (median 83 vs. 73 years, $p < 0.001$). Unadjusted five-year mortality was 41.9% versus 18.4%: RMST difference + 11.2 months (95% CI + 5.6 to + 16.8), log-rank $p = 0.006$. After weighting for age, […]

PubMed · DOI

Identification of biomechanical alterations associated with nonunion after tibial shaft fractures using musculoskeletal simulation based on motion capture and instrumented insole data.

Wahl T, Ganse B, Warmerdam E, Hildebrand F, Hofmann UK, Praster M

INTRODUCTION: Musculoskeletal simulations based on gait analysis data provide access to parameters such as joint reaction forces, muscle forces, and push-off moments, which cannot directly be measured in clinical practice. Applying these methods may help evaluate weight-bearing patterns and identify mechanical risk factors associated with the development of nonunion. This study aimed to investigate differences in the longitudinal development of forces in the musculoskeletal system between cases with and without union in tibial shaft fractures.

MATERIALS AND METHODS: Longitudinal 3D marker-based motion capture and instrumented insole data of thirteen patients (8 union; 5 nonunion) with tibial shaft fractures were used to generate individualized musculoskeletal models using the AnyBody™ software.

RESULTS: In a linear mixed-effects model, the fractured limb showed impaired biomechanical parameters compared with the contralateral limb, including knee joint reaction forces (KJRFs), muscle forces, and push-off moments (all $p < 0.001$). During follow-up, parameters of the fractured leg increased over time, including KJRFs ($p = 0.011$), muscle forces ($p \leq 0.030$), and ground reaction force ($p = 0.008$). The union group exhibited greater muscle forces, particularly in the soleus ($p = 0.014$) and gastrocnemius muscles ($p = 0.049$). The nonunion group exhibited significantly greater asymmetry […]

PubMed · DOI

Biomechanical comparison of innovative suture techniques for repair of Achilles tendon rupture in a porcine flexor tendon model.

Weigand L, Liebrand B, Scheible S, Meffert R, Hoelscher-Doht S

INTRODUCTION: In recent years, evidence-based progressive rehabilitation protocols following the surgical treatment of Achilles tendon rupture have been shown to be safe and appear superior to more restrictive treatment regimens. These protocols mainly rely on early mobilization and weight bearing. To enable this during postoperative rehabilitation, a biomechanically stable surgical treatment must be ensured. Different solutions for suturing ruptured Achilles tendons are currently favored by orthopedic surgeons and have been investigated in the past, including promising further developments of traditional approaches such as the combination of pretension and the Bunnell technique, as well as minimally invasive or percutaneous techniques using systems like the Arthrex PARS. However, the best solution in this regard remains unclear.

METHODS: We analyzed the biomechanical stability of four different suture techniques using porcine deep flexor tendons as a biomechanical model for human Achilles tendons. The tendons were harvested, prepared and sutured according to their respective group (A: Bunnell, B: Bunnell + pretension, C: PARS-to-PARS, D: PARS + anchor). The four groups (n = 12) underwent progressive, cyclic, uniaxial tensile loading in the biomechanical testing machine (ZwickRoell Z020).

RESULTS: Group D (PARS + anchor) showed the highest maximum force at ultimate failure, r […]

PubMed · DOI

Comparative outcomes of delta-on-delta and delta-on-highly cross-linked polyethylene bearings in total hip arthroplasty: a nationwide cohort study.

Lee YK, Kang D, Kim H, Kang E, Cho J, Park JW

INTRODUCTION: The purpose of this study was (1) to evaluate the national trends of Delta ceramic and highly crosslinked polyethylene (HXLPE) liner in total hip arthroplasty (THA), and (2) to compare survivorship of Delta ceramic versus HXLPE liner when used with a Delta ceramic head.

MATERIALS AND METHODS: Using the K-NHIS database, we identified all patients who underwent primary THA using Delta ceramic head in South Korea between 2003 and 2013 (n = 16,250). Cumulative incidences of dislocation, periprosthetic joint infection (PJI), and periprosthetic fracture were evaluated according to bearing type: Delta-on-Delta ceramic (DoD) and Delta-on-HXLPE (DoHXLPE). Survivorship for revisions was calculated to determine differences. Multivariable Cox proportional hazard regression models were used for comparisons, and propensity score matching was additionally performed as a sensitivity analysis.

RESULTS: A total of 15,299 patients underwent THA with Delta ceramic head between 2003 and 2013. Delta liner was used in 14,255 patients (93.2%) and HXLPE in 1,044 patients (6.8%). Utilization of Delta liner has noticeably increased since 2007. The DoHXLPE group had a higher risk of PJI (aHR = 2.10, 95% CI = 1.29-3.41) compared to the DoD group within 1 year. There were no significant differences in risk of dislocation (aHR = 1.41, 95% CI = 0.97-2.05) and periprosthetic fracture (aHR = 0.9 […]

PubMed · DOI

A comparative study on the clinical efficacy and complications of proximal femoral nail antirotation and femoral neck system in the treatment of basicervical femoral neck fracture (AO/OTA 31-B3).

Jiang X, Lin D, Huang Y, Liu J, Lin F

OBJECTIVES: Basicervical femoral neck fractures (FNFs) are prone to internal fixation failure owing to their inherent biomechanical instability. This study compares the efficacy of femoral neck system (FNS) and proximal femoral nail anti-rotation (PFNA) in treating basicervical FNF.

MATERIALS AND METHODS: A retrospective analysis was conducted on 99 patients with basicervical FNF admitted to fuzhou second general hospital from January 2020 to June 2024. Patients were divided into two groups based on the internal fixation method used: the FNS group (53 cases, average age 63.83 ± 18.80 years) and the PFNA group (46 cases, average age 69.78 ± 16.65 years). Main observation indicators included reduction quality and internal fixation-related complications, while secondary indicators included tip-apex distance, surgical time, occult blood loss, wound length, postoperative hospital stay, mortality, and Harris hip score.

RESULTS: The average follow-up time was 24.03 ± 9.4 months. Preoperative data were comparable between groups (P > 0.05). The PFNA group had a significantly lower overall incidence of implant-related complications than the FNS group (2.17% vs. 16.98%, P = 0.035), and the distribution of implant-related complication subtypes

also differed significantly between groups (P = 0.042). No significant differences were observed in fracture reduction quality, tip-apex distance, […]

PubMed · DOI

Management and outcome of displaced acetabular fractures in the oldest-old: a retrospective analysis of orif and acute total hip arthroplasty.

Mamoutakis I, Horst K, Alabdulrahman H, Groven RVM, Bläsius FM, Hildebrand F, Berk T, Bolierakis E

PURPOSE: The optimal surgical management of geriatric acetabular fractures remains controversial. This study compared short-term outcomes following open reduction and internal fixation (ORIF) or acute total hip arthroplasty (THA) with or without additional osteosynthesis (THA ± ORIF) and evaluated the impact of a new treatment algorithm (standard operative procedure, SOP) that favors the use of THA in elderly and medically comorbid patients.

METHODS: In this retrospective cohort study, patients aged ≥ 70 years with isolated trauma-associated unilateral displaced acetabular fractures who underwent surgical treatment at a single Level I trauma center between January 2016-March 2019 (old SOP) and January 2022-December 2025 (new SOP) were included. Patients underwent either ORIF or THA ± ORIF. The transitional implementation phase of the new SOP was excluded. The primary outcome was postoperative time to gait mobilization. Secondary outcomes included pain at discharge, functional mobility, complications, mortality, quality of fracture reduction, and conversion to THA following initial ORIF. Because treatment allocation followed different institutional algorithms across the study periods, the study primarily evaluates an evolving treatment strategy rather than directly comparing ORIF and THA.

RESULTS: Sixty patients (median age 82 years, IQR 78-87; median Charlson Comorbidity Inde […]

PubMed · DOI

Race, ethnicity, and language are underreported in hip fracture CPGs.

Mulakaluri A, Snyder EM, Young B, Kamal RN, Shapiro LM

INTRODUCTION: Clinical Practice Guidelines (CPGs) are systematically developed recommendations based on peer-reviewed literature that guide patient care. Considering the impact of sociodemographic factors on patient outcomes, this study aims (1) to review the racial, ethnic, and linguistic representation of patients in literature utilized to inform the CPGs for hip fractures in older adults and (2) to assess the use of linguistically- and culturally-adapted Patient Reported Outcome Measures (PROM) in these studies.

MATERIALS AND METHODS: Publications listed under the American Academy of Orthopaedic Surgeons (AAOS) CPGs for the Management of Hip Fractures in Older Adults were identified and extracted. Variables extracted included: title, author, publication year, inclusion/exclusion criteria, country/ies of study, language of study, patient language, race and ethnicity, patient language proficiency, outcome measures utilized, translation or cultural-adaptation of outcome measures, and clinical trial registration page. The extracted variables were reported descriptively.

RESULTS: Of the 196 unique articles included, 12/196 (6.12%) reported race/ethnicity data and 9/196 (4.59%) reported patient language, of which 6/9 (66.7%) reported patient language proficiency. Of the studies that reported patient language or language proficiency, 8/9 (88.9%) specified this data in their inclu […]

PubMed · DOI

Level-dependent changes in cervical multifidus composition following anterior cervical discectomy and fusion: bridging quantitative MRI and clinical perspectives.

Folkerts T, Tripathi R, Caffard T, Verna B, Zhu J, Sama AA, Cammisa FP, Hughes AP

INTRODUCTION: Fatty infiltration (FI) of the cervical paraspinal muscles is associated with greater disease severity and less favorable outcomes in degenerative cervical spine disease. However, longitudinal data on whether muscle composition changes following anterior cervical discectomy and fusion (ACDF) are scarce. This study aimed to evaluate changes in FI and functional cross-sectional area (fCSA) of the cervical multifidus and rotatores after ACDF and to assess whether these changes differ across surgical levels.

MATERIALS AND METHODS: This single-center retrospective study analyzed patients who underwent ACDF for degenerative cervical stenosis between 2015 and 2018. Patients were included if both preoperative and postoperative cervical spine MRI (obtained 1-5 years after surgery) were available. The multifidus and rotatores colli were segmented bilaterally on T2-weighted axial images. Measurements were categorized by position within the fusion construct as upper (most cranial fused vertebra), middle (intervening vertebrae), and lower (most caudal fused vertebra) levels. FI and fCSA were quantified using Otsu thresholding. P-values were adjusted for multiple testing using the Benjamini-Hochberg procedure.

RESULTS: Thirty-five patients met inclusion criteria (median age 58 years, 40% female, 1-4 fused levels). Bilateral FI decreased significantly at the lower level (52.1 […]

PubMed · DOI

Superior reduction quality without improved outcomes: open versus closed reduction for displaced femoral neck fractures in adults under 60 years.

Chi JE, Liu CH, Lo YH, Lai CY, Chen IJ, Yu YH

INTRODUCTION: The optimal reduction strategy for displaced femoral neck fractures (DFNFs) in non-elderly adults remains controversial. Although open reduction may improve radiographic alignment, its impact on outcomes remains unclear.

MATERIALS AND METHODS: Adults aged < 60 years with DFNFs who were treated with internal fixation between 2017 and 2023 were retrospectively reviewed. Patients were categorized into closed and open reduction groups. Reduction quality was assessed using the Garden alignment index and the Haidukewych reduction criteria. Multivariate analyses were performed for the overall cohort and patients with Pauwels type III fractures. Outcomes included non-union, osteonecrosis, femoral neck shortening, and major reoperations.

RESULTS: In total, 98 patients were included, among whom 83 and 15 underwent closed and open reduction, respectively. Although open reduction achieved a higher rate of excellent reduction (53.3% vs. 16.9%, p = 0.012), it was not associated with lower rates of non-union or osteonecrosis. In the overall cohort, age was associated with non-union (odds ratio [OR], 1.09; p = 0.024), whereas poorer reduction quality was associated with major reoperation (fair: OR 7.03, p = 0.012; poor: OR 47.6, p = 0.007). In patients with Pauwels type III fractures, poorer reduction quality predicted conversion to arthroplasty (fair: OR 9.84, p = 0.014), and […]

PubMed · DOI

Longer hospital stay but similar one-year mortality and readmission after combined proximal humerus and proximal femur fragility fractures: a Propensity-Score-Matched Cohort Study.

Murphy S, Kellett C, Roberts D, McGurk C, Ahern E, O'Loughlin P

BACKGROUND: Fragility Proximal Humerus (PH) fractures are common and often associated with significant morbidity and mortality. Unlike Proximal Femur (PF) fractures, PH fractures are often omitted from established orthogeriatric multidisciplinary team (MDT) pathways. Combined Proximal Humerus and Proximal Femur fractures (PH&PF) are rare. Combined injuries would be expected to result in worse outcomes compared to isolated PH fractures due to greater injury burden. This study assessed outcomes between isolated PH and combined PH&PF fractures, exploring whether differences in care pathways may contribute to outcome.

METHODS: A retrospective propensity score-matched cohort study was performed over a 2-year period identifying patients sustaining isolated fragility PH fractures and over a 5-year period identifying patients sustaining combined PH&PF fractures secondary to low-energy trauma. Propensity score matching was performed using age, sex, comorbidity burden (as defined by Charlson Comorbidity Index (CCI)), mobility status, cognitive impairment and fracture pattern. Primary outcome was 1-year mortality. Secondary outcomes included 1-year readmission and length of stay (LoS). Mortality was analysed using Firth logistic regression, readmission using clustered logistic regression, and LoS using Gamma regression with cluster-robust standard errors.

RESULTS: Sixty-five patients we […]

PubMed · DOI

Are kneeling stress radiography and the posterior sag sign associated with subjective flexion instability after posterior-stabilized total knee arthroplasty?

A■k■n M, Ça■lar Ö, Tokgözo■lu AM, Atilla B

INTRODUCTION: Flexion instability is a common but underdiagnosed cause of patient dissatisfaction after posterior-stabilised total knee arthroplasty (TKA). Standard clinical tests have limited reliability for this condition. We investigated whether kneeling stress radiography and selected physical examination findings can help identify it.

MATERIALS AND METHODS: We retrospectively reviewed 145 posterior-stabilised TKA knees in 119 patients, divided into two groups by the presence of subjective knee instability. The posterior sag sign, 90° anterior drawer test, and kneeling stress radiographic parameters-particularly the change in tibiofemoral translation under load (Δ-TPFA)-were compared between groups. Post-hoc analyses included ROC curve analysis, diagnostic accuracy with Wilson 95% confidence intervals, and multivariable logistic regression adjusted for age, sex, BMI, follow-up duration, implant type, and lumbar pathology.

RESULTS: Thirty knees (20.7%) reported subjective instability. Mean Δ-TPFA was 1.40 ± 3.75 mm in the unstable group versus - 1.17 ± 3.28 mm in the stable group (p = 0.001; AUC 0.68, 95% CI 0.56-0.80) and remained an independent predictor on multivariable analysis (OR 1.30 per mm, 95% CI 1.11-1.52, p < 0.001). The posterior sag sign had a specificity of 82.6% (95% CI 74.6-88.4%) and was independently associated with instability (OR 3.26, 95% CI 1.54-6.91, […]

PubMed · DOI

Pelvic injuries in elderly trauma patients: epidemiology, injury characteristics, in-hospital care and factors influencing mortality.

Enocson A, Lundin A, Granqvist M

BACKGROUND: Pelvic and acetabular fractures are increasing in incidence in elderly patients. Whether the fracture contributes independently to mortality, or merely accompanies the factors that drive it, remains contested, and large national data on elderly trauma patients are limited. The primary aim of this study was to determine factors influencing 30-day mortality in elderly trauma patients, with particular focus on pelvic injuries. Secondary aims included to analyse the influence of these injuries on in-hospital care and discharge destination.

METHODS: All patients aged ≥ 60 years admitted as trauma alerts in Sweden between 1 January 2022 and 31 December 2024 were identified through the Swedish Trauma Registry (SweTrau). The primary outcome was 30-day mortality. The association between pelvic injury (pelvic or acetabular fracture) and 30-day mortality was assessed using multivariable logistic regression, adjusting for gender, ASA class, NISS, and head injury.

RESULTS: Of 11,624 patients (median age 76 years, 39.7% males), 776 (6.7%) sustained a pelvic injury. Patients with pelvic injury more frequently sustained high-energy trauma, had a higher median NISS (17 vs. 11), longer hospital length of stay (6 vs. 3 days), higher ICU admission rate (30.9% vs. 20.0%), and was less frequent discharged direct home (34.0% vs. 55.8%; all p < 0.001). Crude 30-day mortality was 16.1% vs […]

PubMed · DOI

Pain, quality of life, and functional outcomes of total hip arthroplasty in female patients with crowe type III-IV developmental dysplasia of hip: retrospective cohort study.

Güzel N, Genç AS, Gümü■ NE, Ta■k■n O, Güzel A

INTRODUCTION: The aim of this study was to investigate physical and psychosocial recovery including pain and quality of life measures, as well as lower limb functional out-comes in patients with Crowe type III-IV developmental dysplasia of the hip (DDH) who underwent total hip arthroplasty (THA) with subtrochanteric shortening osteotomy.

MATERIALS AND METHODS: This study was conducted with a total of 45 female patients aged 32-65 with Crowe type III-IV DDH. Pre-op 30-second chair-stand Test (30-s CST), 6-minute walk test (6MWT) and Visual Analogue Scale (VAS) were assessed clinically, whereas Harris Hip Score (HHS), EuroQol 5-Dimension 5-Level (EQ-5D-5 L) scale, Forgotten Joint Score-12 (FJS-12), Hip disability and Osteoarthritis Outcome Score (HOOS), and SF-36 Quality of Life Scale (SF-36) scores were measured and recorded based on patient self reports. At least two years after the surgery, the same measurements were repeated in the patients.

RESULTS: At the mid-term follow-up, a significant increase was observed in the 30-s CST and 6MWT distance compared with the pre-op period ($p < 0.01$). Furthermore, statistically significant differences were observed in the VAS, HHS, EQ-5D-5 L, FJS-12, HOOS and SF-36 scores ($p < 0.01$).

CONCLUSIONS: An evaluation after THA with subtrochanteric shortening osteotomy in female patients revealed a significant improvement in functional perform […]

PubMed · DOI

Bioabsorbable vs. titanium screws in first metatarsophalangeal joint arthrodesis: a pilot randomized controlled trial with 2-year follow-up.

Partio N, Mattila V, Waris V, Mäenpää H

INTRODUCTION: To assess the feasibility and preliminary clinical outcomes of bioabsorbable screw fixation in first metatarsophalangeal joint arthrodesis compared to conventional titanium screw fixation.

METHODS: In this prospective pilot randomized controlled trial, 30 patients were allocated 1:1 to bioabsorbable or titanium cannulated lag screw fixation. Arthrodesis was performed using a cup-and-cone technique at two Finnish hospitals. Full weight-bearing in an orthopaedic shoe was allowed after six weeks. Outcomes included AOFAS and VA-FAS FA-VAS scores, union rates, and complications assessed at 3, 6, 12, and 24 months.

RESULTS: Both groups showed significant functional improvement with no between-group differences. At 24 months, mean AOFAS scores were 89.1 (SD 2.7) in the titanium group and 90.0 (SD 0.0) in the bioabsorbable group ($p = 0.23$). Mean FA-VAS total scores (sum of pain, function, and complaints subscales) were 1948 (SE 59.4) in the titanium group and 1914 (SE 61.4) in the bioabsorbable group ($p = 0.655$). Union rate was 90% overall; nonunion occurred in one titanium and two bioabsorbable patients, all requiring revision. No implant-related complications or hardware removal surgeries were recorded. Degradation of bioabsorbable screws caused no adverse clinical or radiographic reactions.

CONCLUSIONS: Bioabsorbable screws appear safe and feasible in first metatars […]

PubMed · DOI

Intraoperative patellofemoral load varies between total knee replacement and patellofemoral joint resurfacing, and across TKR implant systems.

Phillipson L, Naybour J, Watson A, London N, Barrett D

INTRODUCTION: Patellofemoral joint loading is influenced by implant design and surgical execution during knee replacement. Altered patellofemoral mechanics have been implicated in postoperative symptoms, yet the magnitude and variability of intraoperative load changes across different replacement procedures and implant systems remain incompletely characterised. The aim of this registry study was to describe and quantify intraoperative patellofemoral joint load changes following total knee replacement (TKR) and patellofemoral joint resurfacing (PFJR), focusing on cumulative and compartment-specific load redistribution.

MATERIALS AND METHODS: This observational registry study analysed prospectively collected intraoperative patellofemoral joint load measurements from 291 knee replacement procedures. Load was recorded in medial, lateral, superior, and inferior compartments before and after trial component implantation. Absolute and percentage changes in compartmental and cumulative load were calculated. Descriptive analyses examined the distribution and variability of load change across procedure type and implant system, with implant comparisons restricted to TKR procedures. No formal hypothesis testing was pre-specified, and no adjustment for potential confounding variables was undertaken.

RESULTS: Cumulative patellofemoral load change demonstrated substantial inter-individual v […]

PubMed · DOI

Cortical hypertrophies in cementless short stem total hip arthroplasty in young patients under 60 years and elderly patients over 75 years-analysis of outcome and risk factors for cortical hypertrophies.

Schwarz PM, Stützle S, Stadler C, Holzbauer M, Pisecky L, Gotterbarm T, Luger M

BACKGROUND: Cortical hypertrophy (CH) is a common radiological finding associated with short-stem total hip arthroplasty (THA), but its clinical significance remains unclear. This study evaluates differences in outcomes between young patients under 60 years of age and elderly patients over 75 years of age who underwent cementless short stem THA. The aim was to evaluate if there are differences in these patients depending on the presence of CH.

METHODS: A retrospective analysis of 208 short-stem THAs performed between 2014 and 2017 was conducted. Patients were divided into younger (< 60 years, n = 119) and older (> 75 years, n = 89) groups. Clinical outcomes, including Harris Hip Scores (HHS) and Oxford Hip Scores (OHS), were compared between patients with and without CH. Multivariate regression analysis was performed to identify risk factors for CH, focusing on delta hip offset, canal fill index, and stem alignment.

RESULTS: Younger patients with CH demonstrated significantly better HHS (96.1 ± 7 vs. 90.8 ± 14.3, p = 0.010) and OHS (45.8 ± 5.2 vs. 42.9 ± 8.8, p = 0.028) compared to those without CH. In older patients, the presence of CH did not significantly impact outcomes. Regression analysis identified younger age (p = 0.012), higher Canal Fill Index III (p = 0.047), and stem alignment (p = 0.049) as significant predictors of CH.

CONCLUSION: Cortical hypertrophy is associ […]

PubMed · DOI

Comparative effectiveness of single-stage versus two-stage revision for hip prosthetic joint infection: emulation of the INFORM target trial using electronic health record data.

Zhang C, Li Y, Su C, Shen Y, Lu S

OBJECTIVE: To estimate the comparative effectiveness of single-stage versus two-stage revision for hip prosthetic joint infection (PJI) in Chinese tertiary hospitals by emulating the INFORM randomized trial using the target trial emulation framework.

DESIGN: Retrospective cohort study emulating a target trial, reported per the TARGET guideline. The protocol and analysis plan were written before analysis but were not prospectively registered.

SETTING: Three tertiary hospitals in China. Index revision surgeries were performed between 1 January 2018 and 30 June 2022, with follow-up data locked on 31 December 2023, ensuring a minimum potential follow-up of 18 months for every patient.

PARTICIPANTS: 754 adults with confirmed hip PJI undergoing revision surgery (280 single-stage; 474 two-stage), identified from electronic health records and confirmed by manual chart review. Treatment groups reflected the intended surgical strategy documented before or at the index operation.

MAIN OUTCOME MEASURES: Primary outcome: Harris Hip Score (HHS) at 18 months.

SECONDARY OUTCOMES: surgically treated reinfection, reoperation, mortality, and complications. The primary analysis used inverse probability of treatment weighting (IPTW) on the full cohort with multiple imputation of missing covariate and outcome data; 1:1 propensity score (PS) matching (280 pairs) was confirmatory. Patients who di […]

PubMed · DOI

Cemented dual mobility component in newly implanted porous tantalum shells for severe acetabular defects: medium-term outcomes from a high-volume single-surgeon series.

Loppini M, Di Matteo V, Bulgarelli A, La Camera F, Chiappetta K, Morenghi E, Grappiolo G

BACKGROUND: The management of severe acetabular bone defects remains challengin due to extensive bone loss and poor soft tissue quality during revision total hip arthroplasty (rTHA). The aim of this study was to assess clinical and radiological outcomes of the use of dual mobility components (DMC) cemented into a newly implanted porous tantalum (PT) acetabular shell during acetabular rTHA.

METHODS: Patients who underwent cementation of a DMC into a newly implanted PT cup between 2014 and 2020 were included. Bone defects were classified according to the Paprosky classification. The primary outcome was to determine the reintervention-free survival rate. The secondary outcome was clinical and radiographic assessment at last follow-up.

RESULTS: Thirty-six patients (36 hips), with an average age of 67.9 years (range, 46-90) were included. Paprosky type III defects (61.1%) were more common than type II defects (38.9%), with type IIIa (36.1%) and IIIb (25.0%) predominating. Median follow-up was 67.3 months (range; 17-109). The dislocation rate was 8.3% (n = 3). Four (11.1%) of 36 hips underwent acetabular component revision: 2 (5.5%) for infections, 1 (2.7%) for aseptic loosening, and 1 (2.7%) for dislocation. Cup survival rate was 89.08% (CI 69.54% to 96.38%) at 60 months. The Harris hip score (HHS) improved from 31.8 ± 12.0 to 87.0 ± 14.2 points ($p < 0.001$). The vertical and horiz […]

PubMed · DOI

From load-bearing to load-sharing: contact-driven transitions in locked plate constructs.

Hollensteiner M, Schindler M, Greinwald M, Mühling M, Baumeister D, Augat P

Locked plate osteosynthesis functions as an internal fixator, yet load-induced plate-bone and fragment-fragment contacts - key transitions from load-bearing to load-sharing - remain poorly characterized. This study quantified their effects on construct stiffness, plate strain, and interfragmentary motion under axial loading. Twenty-seven configurations were tested using epoxy-glass surrogate bone fixed with titanium LCP plates, varying fracture gap (3-9 mm), working length (49-121 mm), and plate-bone distance (0-3 mm). Digital image correlation measured strain and motion. Contact events were identified manually from characteristic transitions in the force-displacement response, supported by corresponding changes in plate-surface strain progression. Geometric parameters dominated initial stiffness (125-442 N/mm), decreasing with plate elevation and working length in elevated setups. Within the investigated load range up to 600 N, 16/27 configurations showed non-linear transitions associated with contact events: fragment-fragment contact caused pronounced stiffness increases and strain stagnation at the fracture gap, whereas plate-bone contact was associated with local plate-surface strain stagnation at the location overlying the contact while strain at the fracture gap continued to increase. Interfragmentary motion stagnated after contact, confirming contact-dependent shifts in […]

PubMed · DOI

Factors associated with periprosthetic joint infection following total knee arthroplasty: an updated systematic review and meta-analysis.

Gao H, Deng Y, Ma Y, Li X, Zhang F, Mei Y

BACKGROUND: This study aimed to systematically evaluate factors associated with periprosthetic joint infection (PJI) following total knee arthroplasty (TKA), and thereby to provide evidence-based references for clinical prevention and perioperative risk stratification.

METHODS: Computerized searches were conducted in the following databases from their inception until May 26, 2025: PubMed, Web of Science, Embase, the Cochrane Library, CINAHL, China National Knowledge Infrastructure, Wanfang Database, Chinese Scientific Journal Database, and Chinese Biomedical Literature Database. Two researchers independently screened the literature, extracted data, and assessed study quality. The methodological quality was assessed using the Newcastle-Ottawa Scale. Quantitative synthesis was performed when at least two studies reported comparable exposure definitions and sufficient comparator information; otherwise, narrative synthesis was used. Review Manager 5.4 software was used for the primary analysis. This study is registered on PROSPERO (CRD420251079339).

RESULTS: A total of 25 observational studies were included in the qualitative synthesis, of which 24 contributed to the primary quantitative synthesis. Quantitatively pooled factors associated with PJI included male sex (OR = 1.39, 95% CI 1.27-1.51), BMI ≥30 kg/m2 (OR = 5.72, 95% CI 2.65-12.36), prolonged operative time, type 2 diabet […]

PubMed · DOI

Early surgery of trochanteric femoral fractures in patients on direct oral anticoagulants (DOACs) reduces the length of stay (LOS) and the 1-year mortality rates without increasing the risk for revision surgery: a matched-pair analysis.

Weihs V, Laggner R, Humenberger M, Duma A, Frossard M, Hajdu S

BACKGROUND: It remains unclear whether early surgical fixation of trochanteric femoral fractures in patients with direct oral anticoagulation (DOAC) therapy has an impact on the long-term outcome.

METHODS: A retrospective cohort study on patients with trochanteric femoral fractures and DOAC therapy between 2016 and 2024 was conducted. Patients were divided into two groups according to their time to surgery and matched with age and ASA score as covariates. The primary outcome of interest was the impact of early surgery on the long-term outcome.

RESULTS: Of 233 included patients, 102 patients were assigned in each group after propensity score matching. Patients with early surgery had lower in-hospital complication rates (16.0% vs. 26.3%; p = 0.077) and a significantly shorter median length of stay (LOS) (11.5 days (IQR 6) vs. 15 days (IQR 6); p < 0.001). No difference in the rates of revision surgery (9.8% vs. 5.9%; p = 0.298) was observed. Kaplan-Meier curves revealed a significant difference in the mortality rates during the follow-up period (p = 0.0057) with lower in-hospital (2.0% vs. 6.9%; p = 0.088), 30-day (3.9% vs. 9.8%; p = 0.097) and 1-year mortality rates (20.6% vs. 31.4%; p = 0.079).

CONCLUSION: Early surgical fixation of trochanteric femoral fractures in patients on DOAC therapy leads to a significant reduction in the LOS without increasing the risk for peri- and […]

PubMed · DOI

Platform-specific learning curve and early tibial radiographic agreement of ROSA®-assisted medial unicompartmental knee arthroplasty in an experienced high-volume surgeon: a consecutive case series.

Petrillo S, Ardiri D, Marullo M, Romagnoli S

PURPOSE: To evaluate the platform-specific learning curve and early tibial radiographic agreement of imageless ROSA®-assisted medial unicompartmental knee arthroplasty (UKA), with short-term clinical findings assessed exploratorily.

METHODS: This retrospective consecutive case series included 30 patients who underwent ROSA®-assisted medial UKA with the Persona® Partial Knee implant between January and April 2026 at a high-volume arthroplasty referral centre. A continuous learning-curve cumulative summation (LC-CUSUM) analysis of total surgical duration was performed against an independently defined surgeon-specific historical benchmark derived from institutional operative-register skin-to-skin times for manual UKA performed by the same surgeon (20 ± 4 min). Adequate performance was defined as a mean duration of 25 min, inadequate performance as 30 min, and the reference standard deviation as 4 min. Type I and type II error targets were 0.05 and 0.10, respectively; the decision boundary was calibrated by Monte Carlo simulation over a 30-case monitoring horizon. Robotic time and a supportive comparison of cases 1-5 versus 6-30 were secondary efficiency analyses. Postoperative radiographic measurements were performed by an observer blinded to the planned and robot-validated ROSA® values. Agreement was quantified using Bland-Altman bias and 95% limits of agreement.

RESULTS: Mean […]

PubMed · DOI

Cartilage disruption patterns associated with percutaneous intramedullary screw fixation for metatarsal fractures: a cadaveric study.

Keith B, Burke J, Hynes K, Baker J, Huh J

PURPOSE: Intramedullary screw fixation is an emerging technique for operative metatarsal fractures, yet the extent and location of articular cartilage disruption from retrograde screw insertion remain undefined. This study quantifies cartilage surface disruption and maps injury locations in the metatarsal head following screw placement.

METHODS: Fourteen fresh-frozen cadaveric feet (56 lesser metatarsals) underwent standardized retrograde intramedullary screw fixation using 3.6 mm screws. Following dissection, the total articular surface area and cartilage defect were measured to calculate the percentage of disruption. The cartilage surface was divided into quadrants (dorsal-to-plantar) and thirds (medial-to-lateral) to determine screw trajectory.

RESULTS: Mean cartilage disruption increased progressively from the 2nd to 5th metatarsals: 10.17%, 11.2%, 14.93%, and 17.43%, respectively. All screws were located within the middle third in the medial-to-lateral plane. In the dorsal-to-plantar plane, the majority of screws were positioned in the 2nd and 3rd quadrants. No screws were placed in the medial or lateral thirds.

CONCLUSION: Retrograde intramedullary screw fixation results in measurable cartilage disruption, with a tendency for screw placement in the middle third in the medial-to-lateral plane and 2nd and 3rd dorsal-to-plantar quadrants. While the relative cartilage surf […]

PubMed · DOI

Locking plate fixation for proximal humerus fractures: are four proximal screws adequate? A clinical and radiological analysis of 105 consecutive cases.

Michelitsch C, Stillhard PF, Zindel C, Sommer C, Acklin YP

INTRODUCTION: Proximal humerus fractures are common injuries, and achieving stable fixation is essential to prevent complications such as varus collapse and screw perforation. While optimal screw positioning has been emphasized biomechanically, clinical evidence regarding the required number of screws in the humeral head, particularly when minimally invasive aiming devices limit placement to four screws, remains limited. This study evaluated whether fixation with four cranial screws provides adequate clinical and radiological stability.

MATERIALS AND METHODS: A retrospective cohort study was conducted including 105 patients with displaced proximal humerus fractures treated with plate osteosynthesis. Patients were grouped according to the number of proximal screws used: four screws (n = 58) or more than four screws (n = 47). The primary outcome was the Constant-Murley Score. Secondary outcomes included secondary screw perforation, head-shaft angle maintenance, restoration of medial cortical support, and avascular necrosis. Mean follow-up was 18 ± 6 months.

RESULTS: Functional outcomes were comparable between groups, with Constant-Murley Scores of 78 ± 7 in the four-screw group and 73 ± 13 in the multi-screw group (p = 0.23). Rates of screw perforation (8.6% vs. 10.6%; p = 0.72) and postoperative head-shaft angles were likewise similar, and no significant alignment change occur […]

PubMed · DOI

Nighttime intramedullary nailing of femoral fractures is not associated with worse outcomes: a 13-year Level I trauma center cohort.

Abialevich A, Benkovich V, Osinsky V, Adiniaev Y, Tzaytlin I, Heinemann Y, Acker A

BACKGROUND: Whether nighttime orthopedic trauma surgery adversely affects outcomes remains controversial. Evidence specifically addressing intramedullary fixation across femoral fracture locations remains limited. This study evaluated whether nighttime surgery was associated with a higher rate of recorded postoperative complications than daytime surgery.

METHODS: This retrospective observational cohort included adult patients with nonperiprosthetic femoral fractures treated with intramedullary fixation between January 2010 and January 2023 at a single Level I trauma center. Of 311 screened records, 23 were excluded because the patient was younger than 18 years (n = 17) or the fracture was periprosthetic (n = 6), leaving 288 procedures (242 daytime and 46 nighttime). The primary outcome was any recorded postoperative complication during available follow-up. Secondary outcomes included radiographic abnormality, delayed union (> 24 weeks), nonunion, revision surgery, and mortality. Surgeon experience was categorized as senior surgeon or resident. Exploratory fracture-location subgroup analyses and multivariable logistic regression were performed.

RESULTS: Nighttime patients were younger (35.1 ± 19.1 vs. 46.6 ± 24.6 years, p < 0.001), more frequently male (82.6% vs. 64.0%, p = 0.014), more often injured by a high-energy mechanism (80.4% vs. 56.6%, p = 0.002), and underwent surger […]

PubMed · DOI

Patient-reported outcomes after lateral unicompartmental and total knee arthroplasty: a propensity score-matched cohort study.

Hariri M, Sorbi R, Moradi B, Weishorn J, Trefzer R, Reiner T, Walker T, Nees TA

INTRODUCTION: Comparative evidence on patient-reported outcomes (PROs) after lateral unicompartmental knee arthroplasty (UKA) and total knee arthroplasty (TKA) remains limited. We compared PROs between cohorts matched for demographic characteristics and baseline knee function.

MATERIALS AND METHODS: Prospectively collected single-center data were retrospectively analyzed. Lateral UKA was performed for isolated lateral-compartment osteoarthritis and TKA predominantly for multicompartmental disease. One-to-one propensity score matching incorporated age and body mass index, exact matching on sex, and restrictions on age and preoperative Oxford Knee Score (OKS), yielding 31 pairs. Postoperative OKS was the primary outcome. Secondary outcomes included OKS improvement, Forgotten Joint Score (FJS), University of California, Los Angeles (UCLA) activity score, OKS minimal clinically important difference (MCID) achievement, and FJS patient acceptable symptom state (PASS) attainment.

RESULTS: All matching covariates achieved acceptable balance (absolute standardized mean differences < 0.20). Postoperative OKS did not differ significantly (UKA, 38.9 ± 7.0; TKA, 39.8 ± 7.5; mean paired difference [TKA minus UKA], 0.94; 95% confidence interval [CI], - 2.57 to 4.44; p = 0.590). No significant differences were detected for OKS improvement (difference, 0.77; 95% CI, - 2.62 to 4.17; p = 0.644) […]

PubMed · DOI

Pre- and postoperative 4D CT analysis of thumb CMC joint kinematics after first metacarpal extension osteotomy.

Guebeli A, Honigmann P, Dobbe JGG, Keller M

INTRODUCTION: First metacarpal extension (Wilson) osteotomy combined with modified Brunelli ligamentoplasty is performed for early thumb carpometacarpal (CMC1) osteoarthritis to improve pain and function. The purpose of this study was to quantify in vivo changes in CMC1 joint contact distribution and kinematics after this combined procedure and to determine whether postoperative patterns resemble those of healthy volunteers. We hypothesized that the procedure would reduce cumulative palmar contact during opposition-retroposition.

MATERIALS AND METHODS: In a prospective single-center observational cohort study, six adults with early CMC1 osteoarthritis (Eaton-Littler I-II) underwent standardized preoperative and 6-month postoperative four-dimensional computed tomography (4D CT) of the CMC1 joint during active opposition-retroposition. Five healthy volunteers served as reference with a single 4D CT scan. First metacarpal-trapezium motion was registered across frames. Joint contact was estimated using a 2.0 mm proximity surrogate on the trapezial surface. The primary outcome was change in cumulative palmar contact area. Secondary outcomes included change in dorsoradial contact, temporal variability of contact curves, and contact-centroid path length and mean position. Within-subject pre- versus postoperative comparisons were performed; healthy scans provided reference patterns. […]

PubMed · DOI

Minimal kyphotic progression following pediatric vertebral fractures: implications for reduced radiographic surveillance.

Wegmann S, Lenz M, Babasiz T, Eysel P, Mueller LP, Weber M

INTRODUCTION: Pediatric spinal fractures are rare injuries (14.24 per 100,000 per year), representing only 0.6-3% of all spinal injuries. While most stable compression fractures are managed conservatively, concerns exist regarding progressive vertebral collapse and kyphotic deformity. Unlike extensive adult literature, pediatric-specific data on radiographic progression following conservative management remain limited, creating uncertainty about the clinical utility of routine follow-up imaging in this radiosensitive population.

MATERIALS AND METHODS: Retrospective single-centre cohort study of 54 pediatric patients (mean age 12.3 years) with traumatic vertebral body fractures diagnosed between 2012 and 2024. Kyphosis angles were measured using Cobb methodology at baseline and follow-up intervals (1, 3, and 6 weeks). The primary outcome was the change in kyphotic angle; secondary outcomes included the identification of predictors of progression. Paired t-tests and multivariable linear regression analysis were performed.

RESULTS: Kyphosis angles remained essentially unchanged across all follow-up intervals (ΔFU1 + 0.19°, ΔFU2 + 1.27°, ΔFU3 - 0.14°), with no statistically significant differences detected (p > 0.05 at all timepoints). No clinical or radiographic factors predicted progression of kyphosis. Only three patients (5.6%) required treatment escalation. Subgroup analyses […]

PubMed · DOI

Correction: Different sagittal cutting plane orientations do not affect posterior tibial slope in open wedge high tibial osteotomy: a Sawbone model study.

Gurbanov E, Cochard B, Bauer DE, Miozzari HH, Tscholl PM

PubMed · DOI

Resident autonomy in orthopedic trauma: independent femoral IMN by senior residents yields comparable outcomes to fellowship-trained surgeons.

Balziano S, Nesher G, Ben Nun N, Segev T, Freidlander A, Prat D

INTRODUCTION: Intramedullary nailing (IMN) of femoral shaft fractures is a fundamental cornerstone procedure in orthopedic trauma and an important benchmark of resident surgical competency. While independent resident operating is essential for skill development, its safety in complex trauma cases remains debated. This study evaluated whether senior residents can safely perform femoral IMN without direct supervision compared with fellowship-trained trauma surgeons.

MATERIALS AND METHODS: A retrospective cohort analysis was performed of adult patients undergoing IMN for isolated femoral shaft fractures between 2015 and 2023 at a level I trauma center. Cases were grouped by primary surgeon: senior residents (postgraduate year ≥ 4) versus attending trauma surgeons. Demographics, fracture characteristics, operative parameters, complications, and revision rates were compared. To minimize selection bias, 1:1 propensity score matching (PSM) was performed based on preoperative covariates including age, comorbidity, fracture type, injury mechanism, and severity.

RESULTS: A total of 207 patients met inclusion criteria; 150 (75 per group) were matched. Operative duration was longer in resident-led cases (median 120 vs. 91 min, p = 0.003). Postoperative hospital stay (5.0 vs. 6.0 days, p = 0.674) and weight-bearing protocols (p = 0.218) were similar. Overall complication rates did not dif […]

PubMed · DOI

Racial and ethnic disparities in perceived health and barriers to care among adults with hip osteoarthritis: a national analysis of 16,000 + patients from the all of us program.

Khan ST, Devadiga SA, Emara AK, Brahmbhatt S, Elmenawi KA, Delaney PG, Piuzzi NS

OBJECTIVE: To explore racial and ethnic disparities in perceived health, healthcare access, and perceived barriers among adults with hip osteoarthritis (OA) in a large, nationally representative U.S.

METHODS: Cross-sectional analysis of 16,743 adults with hip OA from the All of Us Research Program. Participants self-identified as Black, Hispanic, or White. Outcomes included self-reported general, mental, physical, and social health, provider communication, and structural barriers to care. Between-group comparisons used Chi-square tests.

RESULTS: Black and Hispanic participants were younger, had lower income and education, and relied more on Medicaid than White participants (all p < 0.001). They reported higher rates of unaffordable deductibles, difficulty accessing follow-up care, insurance coverage issues, transportation challenges, and medication cost-related nonadherence (all p < 0.001) Table 3. Minority groups also more often experienced poor provider respect, trust, and communication. Across all health domains, Black and Hispanic participants reported poorer outcomes and greater emotional distress (all p < 0.001).

CONCLUSION: Significant racial and ethnic disparities exist in structural and perceptual barriers to care among adults with hip OA. These findings identify potentially modifiable barriers that warrant further study and may inform future efforts to improve equi […]

PubMed · DOI

Postoperative periprosthetic fractures associated with compressive osseointegration: a systematic review of the Zimmer Biomet Compress® device.

Hashimoto K, Nishimura S, Goto K

The Zimmer Biomet Compress® Compliant Pre-Stress (CPS) device uses a compressive osseointegration technique for endoprosthetic fixation and provides an alternative to stemed implants. Despite having benefits, such as bone stock retention, periprosthetic fractures continue to remain a concern. Therefore, this systematic review

presents the prevalence, risk factors, and outcomes of the CPS device and periprosthetic fractures after treatment. Medical databases (PubMed, Embase, and Google Scholar) were searched to identify studies reporting complications related to the CPS device. Peer-reviewed studies with at least 2 years of follow-up for fracture rates, revision results, and implant survivorship were included. The primary data collected were fracture incidence and timing of complications, and the CPS device was compared with the traditional cemented stem. This review highlights important studies, including a cohort of 221 patients and pediatric data from 36 patients. Adult populations showed a periprosthetic fracture rate of 2.7-3.9%. A distinct temporal pattern emerged: the frequency of fractures was most pronounced within the first 2 years after surgery. In the pediatric context, spindle survivorship was 86.3% at 5-10 years, despite higher rates of mechanical complications. Notably, for most fracture cases, the bone-implant interface was radiographically stable, providing easi […]

PubMed · DOI

Hip resurfacing arthroplasty at a minimum of 14 years: revision is concentrated among female patients with dysplasia and small components.

Lin YC, Chang CH, Hsieh PH, Huang YY, Chen DW, Lau NC

INTRODUCTION: Hip resurfacing arthroplasty (HRA) is a bone-preserving option for young, active patients. The influence of the underlying diagnosis on long-term outcome remains unclear, particularly the distinction between primary osteoarthritis (OA), developmental dysplasia of the hip (DDH) and non-OA conditions.

MATERIALS AND METHODS: We retrospectively reviewed 104 consecutive metal-on-metal HRAs in 101 patients (three bilateral) performed by one surgeon between 2006 and 2011, stratified into Primary OA (n = 50), DDH (n = 29) and Non-OA (n = 25: avascular necrosis [AVN] n = 15, inflammatory arthritis n = 5, post-traumatic arthritis n = 3, Legg-Calvé-Perthes disease n = 2). AVN was analysed additionally as an independent group, and exploratory analyses restricted to female patients and to cups ≤ 50 mm addressed the collinearity of dysplasia, sex and size. Survivorship used Kaplan-Meier estimation with revision as the endpoint.

RESULTS: Implant status was known for all 104 hips; the 91 unrevised assessed hips had a mean follow-up of 16.2 ± 1.5 years. Overall 15-year survivorship was 95.2%. Primary OA, Non-OA and AVN each showed 100%; DDH 82.8% (versus Primary OA p = 0.002, versus Non-OA p = 0.031). All five revisions occurred in female DDH patients with a cup ≤ 50 mm; DDH survivorship remained lower within female patients (p = 0.025) and within small cups (p = 0.006), with no […]

PubMed · DOI

Elective arthroplasty on "high-risk" calendar dates: no evidence for worse outcomes in 43,000 procedures.

Walter N, Lowenberg DW, Heiss C, Lau EC, Rupp M

BACKGROUND: Concerns regarding the timing of elective surgery remain common among patients and may influence surgical scheduling. Superstition-associated calendar dates such as Friday the 13th are frequently perceived as "high-risk" time points, yet evidence regarding their impact on arthroplasty outcomes is lacking. This study investigated whether primary total hip arthroplasty (THA) and total knee arthroplasty (TKA) performed on these dates are associated with differences in revision risk, mortality, or procedure frequency.

METHODS: A nationwide retrospective cohort study was conducted using Medicare physician service records (2009-2019). Primary THA and TKA procedures were identified in a 5% sample of Medicare beneficiaries aged ≥ 65 years. Outcomes included aseptic revision, septic revision, and mortality. Competing-risk survival analyses (Fine-Gray) and multivariable Cox regression models adjusted for demographic, clinical, and socioeconomic covariates were applied. Procedure volumes were compared between superstition-associated and control dates.

RESULTS: A total of 945 arthroplasties were performed on Fridays falling on the 13th. No significant differences were observed in aseptic revision, septic revision, or mortality compared with control dates. For Friday the 13th, hazard ratios were 0.80 (95% CI 0.44-1.47) for aseptic revision, 0.83 (95% CI 0.31-2.25) for septic r […]

PubMed · DOI

Evaluation of a novel strap fixation method for biomechanical testing of sheep Achilles tendons.

Basci O, Aydemir S, Husemoglu RB, Çeltik M

BACKGROUND: Reliable fixation of tendon specimens is a major challenge in biomechanical testing, as slippage and tissue damage can reduce accuracy and reproducibility. Although various clamp- and suture-based fixation methods have been reported, there is still no widely accepted, practical, low-cost technique that preserves tendon integrity during testing. This study aimed to compare commonly used fixation techniques and to evaluate whether a novel strap-based fixation method provides a more stable and tissue-preserving alternative for biomechanical testing of sheep Achilles tendons.

METHODS: Forty fresh-frozen cadaveric sheep Achilles tendons obtained from a local butcher were used, and the specimens were stored at - 20 °C until testing. After controlled thawing at room temperature, all tendons were repaired using the Bunnell suture technique with No. 0 Vicryl. Four different proximal fixation methods were evaluated: (1) sutured fixation with strap, (2) direct clamp fixation, (3) Kocher clamp fixation, and (4) Krackow suture fixation. All specimens were mounted on a universal testing machine and subjected to a load-to-failure protocol at a constant loading rate of 10 mm/s. Maximum force (N), displacement at failure (mm), and mechanical failure mechanisms were recorded for each sample. Data were analyzed using the Kruskal-Wallis test with Bonferroni-corrected Mann-Whitney U te […]

PubMed · DOI

Implementation and early results of same-day hip and knee replacement in a public hospital without prior same-day surgical experience, a feasibility study.

Jørgensen CC, Sveen TM, Johansen B, Sander M, Steen-Hansen C, Sundstrup MS, Rothe C, Lange KH

BACKGROUND: Hip and knee replacements are increasingly performed as same-day surgery (SDS). However, the feasibility of SDS pathways in departments without prior experience is unknown. We report on the implementation process, feasibility and early results of implementing a SDS protocol in a department without prior experience with same-day hip and knee replacement.

METHODS: The study design was based on Bowens feasibility study framework on implementation processes. Data were collected prospectively through patient reported questionnaires and medical records during, and from the local Department of Digitalisation and Analytics after implementation. Outcome data included discharge on day of surgery, 30-days primary-care contacts, 90-days readmissions and patient satisfaction.

RESULTS: Implementation was initiated as a dedicated fellowship from September 2023 to October 2024. Patients were admitted and discharged through a dedicated day-surgical department with transfer to the regular ward in case of admission. The first SDS patient underwent surgery the 4th of September 2023. By October 2024, 150 of 713 operated patients (21.0% CI:18.2-24.2) had SDS. Of these, same-day discharge was achieved in 125 (83.3% CI:76.6-88.5) patients of which 6 (4.0% CI:0.9-7.1) were readmitted within 90-days. From October 2024 to October 2025, 178 of 800 patients were planned for SDS (22.3% CI:19.5 […]

PubMed · DOI

Relationship of foot anthropometry with Modified Pirani score during Ponseti correction of idiopathic congenital talipes equinovarus: a prospective observational study.

Suresh M, Pinto DJV, Rajasekharan P

BACKGROUND: The Ponseti method is the gold standard treatment for idiopathic congenital talipes equinovarus (CTEV). Clinical monitoring during correction primarily relies on the Modified Pirani score, which evaluates deformity severity based on clinical findings. However, objective quantitative assessment of morphological foot changes during treatment remains limited. This study aimed to evaluate the relationship between serial foot anthropometric measurements and Modified Pirani scores during Ponseti correction of idiopathic clubfoot.

METHODS: A prospective observational cohort study was conducted in 30 children with idiopathic unilateral CTEV undergoing Ponseti correction at a tertiary care center between June 2024 and February 2026. Five anthropometric parameters were measured weekly for eight weeks: great toe to midheel distance (GT-MH), little toe to midheel distance (LT-MH), second toe to midheel distance (ST-MH), medial malleolus to midheel distance (MM-MH), and lateral malleolus to midheel distance (LM-MH). Modified Pirani scores were assessed simultaneously by two independent observers. Pearson correlation analysis and inter-observer reliability testing were performed.

RESULTS: Significant improvement was observed in all anthropometric parameters and Pirani scores over the study period ($p < 0.001$). Mean GT-MH increased from 80.73 ± 3.36 mm to 87.56 ± 3.41 mm, LT-MH f […]

PubMed · DOI

Union outcomes and associated factors in adolescent scaphoid fractures.

Torun Ö, Girgin AB, Ka■■kç■ M, Acar A, Çevik HB

PURPOSE: Scaphoid fractures (SF) in adolescents differ from adult fractures in fracture pattern, healing capacity, and skeletal maturity. Limited data exist on factors affecting union and healing time in this population. This study evaluated clinical, radiological, and treatment-related factors associated with union and healing time in adolescent SF.

METHODS: This study included 77 adolescents (aged 10-18 years) treated for SF. Chronological age, bone age, fracture characteristics, treatment modality, union status, and healing time were recorded. Fractures were classified by anatomical location, displacement, and time to diagnosis (< 6 weeks or ≥ 6 weeks). CT-based morphometric measurements, including scaphoid height, lateral intrascaphoid angle, and dorsal cortical angle, were obtained from initial computed tomography images. Functional outcomes were assessed using the QuickDASH and patient-rated wrist evaluation (PRWE) scores.

RESULTS: Union was achieved in 63 patients (81.8%), with a median union time of 8 weeks (range, 6-10 weeks). Union rates were highest in distal pole fractures (100%) and lowest in proximal pole fractures (33.3%). Non-displaced fractures had higher union rates than displaced fractures (92.5% vs. 56.5%), and fractures diagnosed within 6 weeks demonstrated better union outcomes. Surgical treatment, performed in 16.9% of patients, was associated with lowe […]

PubMed · DOI

The effect of surgeon volume on outcomes after total hip arthroplasty: a systematic review and meta-analysis.

Meissner N, Ortwig K, Ramos-Pascual S, Agu C, Stoeve J, Schrednitzki D, Rolvien T, Halder AM

PURPOSE: To synthesize, critically appraise and systematically review studies comparing outcomes after primary total hip arthroplasty (THA) performed by low- vs. high-volume surgeons, and to propose evidence-based threshold definitions for these categories.

METHODS: This review followed the PRISMA guidelines and was registered in PROSPERO. Medline and Embase were searched on March 24, 2025. Prospective and retrospective clinical studies were included if they reported THA thresholds for low- and high-volume surgeons. Articles not written in English or German were excluded. Outcomes were pooled and presented in forest plots. Methodological quality was assessed.

RESULTS: Of 872 identified articles, 18 were included, reporting on 796,061 patients undergoing THA. Methodological quality was high in all studies (>6 of 7 points). Thresholds used to define high- vs. low-volume were determined using arbitrary values (n = 11), tertiles (n = 1), quartiles (n = 3), percentiles (n = 2), or receiver operating characteristic curves (n = 1). High-volume surgeons had significantly lower 1-month readmission rates [odds ratio (OR) = 1.6; p = 0.033] and 3-month mortality rates (OR = 1.8; p = 0.008). In contrast, there were no significant differences between high- and low-volume surgeons for 3-month revision rates (OR = 2.0; p = 0.228), 12-month revision rates (OR = 1.8; p = 0.228), and 3-month re […]

PubMed · DOI

Hexapod-assisted limb salvage after talectomy: outcomes of tibiocalcaneal arthrodesis using the TL-hex system.

Guerrero Serrano JA, Sánchez Morata E, García López JM, Mencía González S, Vilá Y Rico J

BACKGROUND: Severe talar destruction due to infection, trauma, avascular necrosis, or neuroarthropathy represents a major reconstructive challenge. Talectomy followed by tibiocalcaneal arthrodesis (TCA) is a limb-salvage option in selected high-risk patients, although optimal fixation remains controversial. Modern hexapod external fixators offer computer-assisted multiplanar control that may improve alignment and fusion after talectomy. The aim of this study was to evaluate the clinical and radiographic outcomes of talectomy followed by TCA using the TL-HEX system in a series of high-risk patients.

METHODS: A retrospective observational study was conducted between January 2020 and January 2024 including all patients who underwent total talectomy followed by TCA using a TL-HEX circular external fixator. Indications included talar destruction secondary to infection, avascular necrosis, post-traumatic collapse, Charcot neuroarthropathy, and aseptic loosening of total ankle arthroplasty. Primary outcomes were radiographic fusion and hindfoot alignment. Secondary outcomes included time to union, functional scores (AOFAS, EFAS), pain (VAS), complications, and patient satisfaction. Preoperative and final follow-up outcomes were compared using paired statistical tests, with significance set at $p < 0.05$.

RESULTS: Thirteen patients (mean age 57 years) were included with a mean follow-u […]

PubMed · DOI

Radiographic changes in midfoot alignment after hallux valgus correction with and without concomitant second tarsometatarsal arthrodesis: a comparative study.

Kim J, Kyung MG, An Y, Kim NY, Lee KM, Lee DY

INTRODUCTION: Second tarsometatarsal (TMT) joint osteoarthritis occurs in a subset of patients with hallux valgus (HV). While second TMT arthrodesis is performed to address symptomatic arthritis, its association with midfoot alignment changes during HV correction remains unclear. This study aimed to compare changes in radiographic arch parameters between patients undergoing HV correction with and without concomitant second TMT arthrodesis.

MATERIALS AND METHODS: This retrospective comparative study included 73 feet with HV and medial column malalignment (lateral Meary's angle > 10°). The TMT group (n = 37) underwent HV correction combined with second TMT arthrodesis for symptomatic osteoarthritis, while the control group (n = 36) underwent isolated HV correction. These patients were matched to achieve comparable baseline characteristics. Radiographic parameters of the transverse arch (M2-M5 angle, C1-M2 distance) and medial column alignment (lateral Meary's angle) were evaluated preoperatively and at a minimum 1-year follow-up. Functional outcomes were assessed as a secondary measure.

RESULTS: The TMT group demonstrated greater changes in transverse arch parameters compared with the control group. The M2-M5 angle decreased by 2.1° (95% CI, 1.4° to 2.7°) in the TMT group, whereas no significant change was observed in controls (-0.2°; 95% CI, -0.7° to 0.4°). The C1-M2 distance […]

PubMed · DOI

Coronal shear fractures of the distal humerus in children and adolescents: proposal for a modified classification.

Lieber J, Esser M, Fernandez FF, Schmittenbecher PP, Schneidmueller D, Strohm P, Kaiser MM, Kraus R

PURPOSE: Coronal shear fractures (CSF) of the capitulum and trochlea humeri are rare injuries during growth age. No systematic, prospective analysis of the classification, procedure, and outcome has yet been found in the literature.

MATERIALS AND METHODS: Based on the results of a multicentre study of CSF in growing patients, we propose a redesign of the AO Paediatric Comprehensive Classification of Long-Bone Fractures (PCCF) for children and adolescents. We also assessed inter-rater agreement consistency with the modified classification and calculated the intra-class correlation coefficient (ICC) and its 95% confident interval (CI).

RESULTS: A detailed classification has been proposed that refers to the location of the fragment, the extent to which the trochlea is involved, and the distinction between single and multiple fracture injuries. This modified AO classification distinguishes between five types of injury (13-E/3.1I - 13-E/3.1IV and 13-E/3.2), replacing the current code 13r-E/8.1. Inter-rater reliability using this modified classification was moderate (ICC 0.452 (95% CI 0,284; 0,661). This is more accurate than all previous classifications tested for CSF.

CONCLUSION: A revised classification of coronal shear fractures of the distal humerus in children and adolescents is required. Our proposed classification considers the various types of this injury and their implic […]

PubMed · DOI

Correction: Prevalence and risk factors for stem subsidence following reverse shoulder arthroplasty using on-lay press-fit short stems.

Nishiura R, Nakazawa K, Manaka T, Hirakawa Y, Ito Y, Terai H

PubMed · DOI

Reverse shoulder arthroplasty vs. hemiarthroplasty for proximal humerus fracture: in-hospital outcomes in a national trauma registry.

Raizah A

BACKGROUND: Reverse shoulder arthroplasty (RSA) has rapidly supplanted hemiarthroplasty for acute proximal humerus fracture, driven by superior functional outcomes and lower revision rates. Whether this shift has altered perioperative morbidity at the population level remains poorly characterized. This study compared in-hospital outcomes between RSA and hemiarthroplasty in a national trauma cohort.

METHODS: This retrospective cohort study used the National Trauma Data Bank (2017-2022). Adults (> = 18 years) admitted with proximal humerus fracture who underwent RSA (n = 2657) or hemiarthroplasty (n = 428) during the index hospitalization were included. The primary outcome was non-home discharge; secondary outcomes were any in-hospital complication, extended length of stay exceeding 7 days, and intensive care unit (ICU) admission. Multivariable logistic regression and inverse probability of treatment weighting (IPTW) were used to address confounding by indication.

RESULTS: Among 3085 patients (median age, 73 years; 2304 [74.7%] female), RSA patients were older (median, 74 vs. 64 years; P<.001) and more frequently female (77.3% vs. 58.6%; P<.001). After multivariable adjustment, RSA was not significantly associated with non-home discharge (adjusted odds ratio [aOR], 1.20; 95% confidence interval [CI], 0.93-1.55), any complication (aOR, 1.01; 95% CI 0.58-1.77), extended length o […]

PubMed · DOI

Impact of cognitive impairment on postoperative delirium and discharge disposition following knee arthroplasty: a Japanese nationwide propensity score-matched study.

Mori Y, Tarasawa K, Tanaka H, Kanabuchi R, Mori N, Fushimi K, Aizawa T, Fujimori K

PURPOSE: As populations age, the number of older patients undergoing knee arthroplasty continues to increase. Cognitive impairment is prevalent among older adults and has been associated with adverse perioperative outcomes; however, its impact on postoperative delirium and discharge disposition following knee arthroplasty remains insufficiently defined, particularly in large Asian populations.

METHODS: We conducted a nationwide retrospective cohort study using Japan's Diagnosis Procedure Combination database from April 2016 to March 2023. Older patients who underwent primary total knee arthroplasty (TKA) or unicompartmental knee arthroplasty (UKA) were identified. Cognitive impairment was defined using ICD-10 diagnostic codes at admission. The primary outcome was postoperative delirium, and secondary outcomes included discharge to home, length of hospital stay, and perioperative blood transfusion. Propensity score matching (1:1) was performed to adjust for demographic factors, comorbidities, and surgical characteristics. Multivariable logistic regression and sensitivity analyses were conducted.

RESULTS: Among 259,319 eligible patients, 3,934 matched pairs were identified after propensity score matching. Patients with cognitive impairment had a significantly higher risk of postoperative delirium compared with those without cognitive impairment (absolute risk difference, 3.2%; […]

PubMed · DOI

Association of primary tumor surgery with survival outcomes in patients with bone metastases across multiple cancer types.

Xu PH, Shi B, Li T, Tian M, Fang M, Cheng M, Huang W, Ren F et al.

BACKGROUND: Synchronous bone metastases (BM) have a notable impact on the survival rates of patients across various cancer types. While primary tumor surgery (PTS) has shown potential benefits, its impact across different cancers and patient subgroups needs further clarification. This study aimed to investigate the survival benefits of PTS across various cancer types.

METHODS: This retrospective cohort study analyzed data from the Surveillance, Epidemiology, and End Results (SEER) database, covering 53,015 patients diagnosed with BM from 2010 to 2019. Propensity score matching

(PSM) was performed to reduce baseline imbalance. Overall survival (OS) and cancer-specific survival (CSS) were evaluated using Kaplan-Meier analysis and Cox proportional hazards regression.

RESULTS: PTS substantially improved overall and cancer-specific survival in patients with lung, kidney, breast and bladder cancers. All surgical modalities benefited breast and renal cancer survival. In prostate cancer, PTS produced no overall survival gain; survival varied by surgery type, being better after prostatectomy and worse with local resection. For liver cancer, PTS improved cancer-specific but not overall survival, and the small matched cohort necessitates circumspect interpretation.

CONCLUSION: The findings indicate that PTS can significantly enhance survival in selected patients with BM, particularly i […]

PubMed · DOI

Liquid nitrogen-treated autograft combined with megaprosthesis versus total femoral replacement for large-volume malignant femoral bone tumors: a comparative study.

Sang NTQ, Trung ND, Huy LD, Phuc LH, Thang VD, Anh LN, Hung LT, Hien TTT et al.

BACKGROUND: Limb-salvage surgery for extensive femoral osteosarcoma presents major reconstructive challenges. Conventional options such as total femoral replacement (TFR) and long-segment biological reconstructions are associated with significant complications. We evaluated the clinical outcomes of a novel hybrid approach combining a liquid nitrogen-treated autograft with a megaprosthesis (LN-MP) and compared them with those of TFR.

MATERIALS AND METHODS: This retrospective study included 28 patients (range 8-24 years old) with malignant femoral bone tumors treated between 2020 and 2024. Patients underwent either LN-MP (n = 10) or TFR (n = 18). Perioperative outcomes, functional results (Musculoskeletal Tumor Society score-MSTS), bone union, oncological control, and survival were analyzed.

RESULTS: LN-MP demonstrated better functional outcomes (MSTS 27.0 ± 2.1 vs. 24.4 ± 2.2, p = 0.014), shorter operative time, and lower blood loss compared with TFR. Length of stay and follow-up duration were comparable. Continuous disease-free survival was higher in the LN-MP group (90% vs. 44%, p = 0.058), with lower recurrence (10% vs. 28%). Bone union was achieved in 70% of LN-MP cases at a mean of 9 months, with longer graft length associated with delayed union.

CONCLUSIONS: LN-MP reconstruction may serve as a joint-preserving alternative to total femoral replacement for patients with e […]

PubMed · DOI

Tibial bone transport using external fixation in adults: a systematic review.

Giuli C, Di Pietro G, Farine F, Bocchi MB, Vitiello R, Palmacci O

INTRODUCTION: Tibial bone transport using external fixation is widely employed for the reconstruction of large and complex tibial bone defects; however, indications and surgical strategies remain highly heterogeneous. Tibial bone transport using external fixation is a demanding but reliable reconstructive technique, characterized by low failure rates and generally favorable outcomes despite frequent complications. Adequate soft-tissue management, selective docking-site grafting, and early functional rehabilitation were commonly reported. Further prospective studies are needed to standardize treatment strategies and improve clinical decision-making. MATERIALS AND METHODS: A systematic review was conducted in accordance with PRISMA guidelines in April 2025, searching PubMed/MEDLINE and the Cochrane Library. Extracted data included patient demographics, comorbidities, indications, defect length, surgical technique, frame type, external fixation time, healing index, complications, failures, and ASAMI bone and functional outcomes. Methodological quality was assessed using the MINORS score and results were synthesized descriptively.

RESULTS: A total of 3543 patients from 89 studies were included. Post-traumatic infection or osteomyelitis accounted for 58.5% of indications. The mean defect length was 7.04 cm. Bifocal bone transport was performed in 82.4% of cases, and circular exter […]

PubMed · DOI

Accuracy and risk factors for poor cup orientation in total hip arthroplasty: a comparison between supine and lateral registration using a portable navigation system.

Matsumoto S, Tetsunaga T, Yamada K, Okuda R, Masada Y, Yamamoto T, Okazaki Y, Ozaki T

INTRODUCTION: Accurate acetabular cup orientation is essential in total hip arthroplasty (THA). Portable inertial navigation may improve reproducibility, but accuracy in lateral decubitus THA may depend on the pelvic registration workflow. We compared cup placement accuracy between supine and lateral registration workflows using NAVBIT (Smith & Nephew PLC) and explored factors associated with large navigation errors.

MATERIALS AND METHODS: We retrospectively reviewed 110 consecutive primary cementless THAs performed in the lateral decubitus position using NAVBIT. The supine-registration workflow included 52 hips (supine pelvic registration followed by repositioning), and the lateral-registration workflow included 58 hips (registration and cup implantation in the lateral decubitus position). NAVBIT recorded radiographic inclination (RI) and radiographic anteversion (RA) intraoperatively; postoperative RI and RA were measured via computed tomography. Accuracy was defined as the absolute difference between navigation records and postoperative measurements. We compared navigation accuracy outcomes between propensity score-matched groups. Outliers were defined as an absolute error > 5° in RI or RA; factors associated with outliers were evaluated in the overall cohort.

RESULTS: After propensity score matching, 46 matched pairs (n = 92) were analyzed. Postoperative cup orientation w […]

PubMed · DOI

Differences in the coronal plane alignment of the knee (CPAK) measurements using long-leg radiographs and computed tomography.

Kawaguchi K, Tay ML, Young S

INTRODUCTION: Coronal plane alignment of the knee (CPAK) parameters are increasingly used to characterize native coronal alignment and guide alignment strategies in total knee arthroplasty (TKA) based on long-leg radiographs (LLR). Computed tomography (CT) based robotic TKA software enables calculation of CPAK parameters based on anatomical reference points. However, potential differences in CPAK measurements between LLR and CT remain incompletely defined. This study compared CPAK coronal alignment parameters and phenotype distributions between LLR and CT, and evaluated factors associated with measurement discrepancies.

METHODS: This was a radiographic subanalysis of a prospective randomised controlled trial including 241 patients undergoing robotic-assisted TKA. Medial proximal tibial angle (MPTA) and lateral distal femoral angle (LDFA) were measured and arithmetic hip-knee-ankle-angle (aHKA), joint line obliquity (JLO) and CPAK phenotype distribution were calculated and classified from preoperative LLR and CT. Discrepancy groups were defined as LLR = CT (≤ 2°), LLR < CT (> 2°) and LLR > CT (> 2°). Associations between discrepancy groups and preoperative deformity and flexion contracture were assessed.

RESULTS: Compared with LLR, CT demonstrated a more varus MPTA (mean 86.7° vs. 87.1°, p = 0.04), a more valgus LDFA (87.1° vs. 87.8°, p < 0.01) and a lower JLO (173.9° vs. 174. […]

PubMed · DOI

Conservative and surgical treatment of wrist osteoarthritis: a systematic review.

Boever J, Unglaub F, Wulf J, Spies CK, Langer MF, Kußmaul AC

INTRODUCTION: Wrist osteoarthritis leads to pain and functional impairment. Despite conservative treatment options, surgery remains a central therapeutic component. In Germany, the number of patients rose from over 472,000 cases in 2018 to more than 580,000 in 2024. This review examines the most common surgical treatment approaches, focusing on Proximal Row Carpectomy (PRC) and Four-Corner Fusion (4CF), discussing indications, contraindications and potential complications.

METHODS: A systematic and narrative literature review was conducted on treatment options for wrist osteoarthritis. The systematic review focusing on PRC and 4CF followed PRISMA guidelines and included searches in PubMed, Cochrane and Embase. Included were RCTs, systematic reviews and meta-analyses from January 2010 to May 2025 that compared both procedures in terms of pain, range of motion, grip strength, function, complication rates and reoperation rates.

RESULTS: Eight studies were included. Both procedures significantly reduced pain and improved function. PRC showed advantages in range of motion and lower complication rates, while 4CF was associated with higher grip strength. Partial arthrodesis (e.g., scaphotrapeziotrapezoid (STT) arthrodesis or radioscapholunate (RSL)) arthrodesis may be selectively used but are linked to higher nonunion rates. Total wrist arthrodesis or wrist arthroplasty may be consi […]

PubMed · DOI

Factors associated with selection of total hip arthroplasty versus hemiarthroplasty for femoral neck fracture in older adults: a nationwide analysis of 99,084 cases.

Maman D, Steinfield Y, Berkovich Y

BACKGROUND: Surgical management of displaced femoral neck fractures in older adults typically involves hemiarthroplasty or total hip arthroplasty (THA). Although clinical guidelines suggest that THA may be considered in selected healthier and cognitively intact patients, real-world procedure selection varies widely. This study evaluated patient factors associated with selection of THA versus hemiarthroplasty in a contemporary U.S.

METHODS: A retrospective cohort study was conducted using the 2022 Nationwide Readmissions Database (NRD). Patients ≥ 65 years hospitalized with femoral neck fracture were identified using ICD-10-CM codes. Those treated with internal fixation or non-arthroplasty procedures were excluded. Weighted analyses characterized demographics, comorbidities, and hospital utilization between THA and hemiarthroplasty groups. A multivariable logistic regression model identified factors independently associated with receiving THA. All analyses accounted for NRD survey design.

RESULTS: Among 142,013 operative cases, 99,084 met inclusion criteria (81.8% hemiarthroplasty; 18.2% THA). Patients receiving THA were younger (76.6 vs. 81.9 years), had shorter length of stay (6.06 vs. 7.24 days), and were more frequently discharged home (17.5% vs. 6.2%). Metabolic conditions were associated with increased odds of THA, including obesity (OR 1.17) and sleep apnea (OR 1.12). F […]

PubMed · DOI

Effect of distal screw number on axial deformation and fragment-specific stability in AO/OTA 2R3 C2.1 distal radius fractures.

Dietz L, Spiegel C, Kohler FC, Biedermann U, Zderic I, Feist A, Gueorguiev-Rüegg B, Lenz M et al.

PURPOSE: To investigate the biomechanical influence of distal screw number in palmar plate fixation of unstable intra-articular distal radius fractures with radiodorsal metaphyseal comminution.

METHODS: Nine matched pairs of cryopreserved human radii were prepared using a standardized AO/OTA 2R3 C2.1 fracture model with a defined radiodorsal metaphyseal defect zone. For each pair, specimens were randomized to palmar fixation using a 2.4-mm variable-angle locking plate with either four or six distal screws. After embedding, constructs were subjected to cyclic axial loading (150 N, 5,000 cycles) simulating early postoperative rehabilitation. Global axial deformation and fragment-specific three-dimensional interfragmentary motion were recorded using an optical motion-tracking system. Statistical analysis was performed using non-parametric tests ($p < 0.05$).

RESULTS: Two specimens from one donor pair were excluded, leaving 16 radii (8 pairs) for analysis. Bone mineral density and initial construct stiffness did not differ between groups. The six-screw configuration showed significantly lower axial deformation after cyclic loading ($p = 0.047$) and lower radial fragment-shaft translation at both time points ($p = 0.028$; $p = 0.045$). Radial fragment-shaft rotation was lower in the six-screw group at baseline ($p = 0.018$), but no rotational difference remained after cyclic loading. In the […]

PubMed · DOI

Crowe severity and femoral stem geometry are associated with lower-limb alignment after total hip arthroplasty for unilateral developmental dysplasia of the hip-related secondary osteoarthritis: valgus alignment trajectories.

Afacan MY, Gurcinar MG, Davulcu CD, Unlu MC, Kaynak G

BACKGROUND: Total hip arthroplasty (THA) for secondary osteoarthritis related to developmental dysplasia of the hip (DDH) restores hip mechanics. We aimed to quantify pre-to-postoperative changes in ipsilateral knee and

global lower-limb alignment after unilateral THA to determine whether alignment changes differ by Crowe classification and femoral stem geometry.

METHODS: This retrospective cohort study included 74 patients who underwent unilateral primary THA for DDH-related secondary osteoarthritis, with a mean follow-up of 45.4 months. The predefined primary radiographic outcome set comprised preoperative-to-postoperative changes and final follow-up values in coronal lower-limb alignment parameters, including mechanical and anatomic lateral distal femoral angles (mLDFA and aLDFA), mechanical medial proximal tibial angle (mMPTA), joint line convergence angle (JLCA), global offset (GO), hip-knee-ankle angle (HKA), knee mechanical axis deviation ratio (KMADR), and femoral mechanical-anatomic axis angle (FMMA). The contralateral reference limb was used for side-to-side radiographic comparison. Patients were stratified by Crowe classification (I-II vs. III-IV), and femoral stem geometry analyses were performed according to conical, wedge, and rectangular stem designs. Clinical outcomes included hip/knee visual analog scale, Harris Hip Score, Knee Society Score, Oxford Knee Score […]

PubMed · DOI

Can ChatGPT pass the polish national medical specialization examination in orthopedics and traumatology?

Maci■g B, Bujak K, Jegierski D, Stolarczyk A

INTRODUCTION: Artificial intelligence (AI) has evolved rapidly in recent years and is becoming increasingly integrated into many areas of medicine. In the medical field, these systems have attracted considerable attention because of their potential applications in clinical decision support, medical education, scientific communication, and postgraduate training. The aim of the present study was to evaluate whether ChatGPT could achieve a passing score on the Polish National Medical Specialization Examination (Panstwowy Egzamin Specjalizacyjny, PES) in orthopedics and traumatology and to determine how different prompting strategies influenced its performance.

MATERIALS AND METHODS: Authors systematically assessed the performance of ChatGPT-4 across five consecutive official PES exams (from Autumn 2023 to Spring 2025) in orthopedics and traumatology. Each exam was administered using three distinct prompting strategies: Professor prompt, Specialist prompt, Resident prompt. For each exam session (e.g., Spring 2024, Autumn 2023), the prompts were submitted to ChatGPT sequentially. The model's responses were evaluated against the official answer key published by the Polish Center of Medical Exams (Centrum Egzaminów Medycznych, CEM).

RESULTS: The results were averaged across all prompts. The overall accuracy was 81% (range: 68.33%-85.83%). The highest score (85.83%) was recorded mult […]

PubMed · DOI

Two-year outcomes following concomitant ACL and extra-articular ligament reconstruction: a nationwide analysis of over 5,000 patients.

Abid R, J Burkhart R, J Moyal A, M Adelstein J, Kodsy M, G Calcei J, E Voos J, M Apostolakos J

PURPOSE: The anterior cruciate ligament (ACL) is the most commonly injured and reconstruction ligament in the United States. Associated injuries are common, oftentimes including components of the extra-articular knee. However, complications following concomitant ACL and extra-articular ligaments (EAL) has not been defined in a large-scale, population-based sample. We aimed to analyze outcomes following concomitant ACL and EAL reconstruction using the TriNetX nationwide database.ep METHODS: Deidentified data from the TriNetX Collaborative Network was used to identify all patients who received ACL and EAL reconstruction on the same day. Demographic information including age, sex, race, and geographic data were collected. Rates of medical and orthopaedic complications were analyzed at 30, 90, and 180 days and at 2 years. A propensity matched sub-analysis comparing outcomes patients with concomitant and staged repair was performed. Patients were identified using Current Procedural Terminology (CPT) coding.

RESULTS: We identified 5944 patients who received concomitant ACL and EAL reconstruction with postoperative outcomes available at 30, 90, and 180 days. 5040 patients had data available at 2 years. Serious medical complications like surgical site infection, deep vein thrombosis, acute kidney failure, and sepsis were rare at each point of follow-up. Rates of revision surgery were […]

PubMed · DOI

Recovery of gait after ankle fracture surgery following early mobilization and weight-bearing.

Bauer L, Schlegel AS, Scheuer A, Böcker W, Polzer H, Baumbach SF

BACKGROUND: Ankle fractures are among the most common fractures in adults. Despite significant advancements in surgical techniques, the postoperative treatment regimens remain conservative. This study aims to objectively evaluate gait recovery following early mobilization and weight-bearing after surgical ankle fracture treatment using instrumented gait analysis.

METHODS: This prospective, single-armed, longitudinal study enrolled adult patients following surgical treatment for isolated ankle fractures with early pain dependent weightbearing. Gait analysis, utilizing a treadmill integrated with a pressure plate and IMU-based motion capture systems, was conducted at 6 weeks, 3-, 6-, and 12-months post-surgery, assessing spatio-temporal and kinematic parameters. Results were compared longitudinally and against a healthy control group.

RESULTS: Longitudinal data were available for 44 patients. Significant ($p<.001$) improvements of self-paced walking velocity (SPV) were observed from 6 weeks (2.2 ± 1.1 km/h), to 3 (2.9 ± 0.9 km/h) and 6 months (3.4 ± 0.9 km/h). The spatio-temporal parameters showed significant changes between 6 weeks and all further time points only at SPV. Kinematic parameters showed significant longitudinal improvements at SPV for hip, knee, and ankle motion (including all three ankle planes). Patient gait parameters at 6 and 12 months were comparable to healthy […]

PubMed · DOI

A comparative analysis of three knee arthrodesis techniques to manage failed total knee arthroplasty due to infection: intramedullary nailing, LRS external fixator, and compression plating.

Rajasekaran RB, Ponniah HS, Palanisami DR, Sundaram VP, Palanivelayutham SS, Jayaramaraju D, Dhanasekaran S, Rajasekaran S

INTRODUCTION: This study compared external fixation (EF), intramedullary nailing (IMN), and compression plating (CP) for knee arthrodesis (KA) following failed total knee arthroplasty (TKA) due to periprosthetic joint infection, focusing on complications, radiological, and functional outcomes.

METHODS: Patients aged ≥ 18 who underwent knee arthrodesis following failed TKA due to infection from 2008 to 2023 were included. Four surgeons used three techniques: EF, IMN, and CP. Retrospective data on demographics, surgical techniques, complications, and outcomes were collected. Postoperative patient-reported outcome measures (PROMs), pain, and functional status were assessed. Radiographic data was reviewed for limb shortening and union status.

RESULTS: 46 patients (mean age 61.9 years, BMI 28.2), predominantly male (84.8%) were involved in this study. Twenty patients underwent EF, 16 underwent CP, and 10 underwent IMN, all for infected primary TKA. Pseudomonas was the most common infecting organism (15.2%). Average follow-up was 29.1 months. Bony fusion was achieved in 41 knees (89.1%), with five requiring additional surgery; four eventually required amputation due to non-fusion. EF group had the shortest time to union and no major complications or amputations. Time to union was significantly shorter in the CP group compared to the IMN group ($p < 0.001$), and EF group also had a si […]

PubMed · DOI

Accuracy of iliac crest reference pin placement in total hip arthroplasty: a CT-based analysis of risk factors for cortical perforation.

Masada Y, Tetsunaga T, Yamada K, Inoue T, Okuda R, Tetsunaga T, Okazaki Y, Ozaki T

INTRODUCTION: Accurate iliac crest reference pin placement is essential for safe navigation-assisted total hip arthroplasty (THA). Although complications are uncommon, inaccurate placement may cause neurovascular injury. This study aimed to evaluate pin insertion accuracy using postoperative computed tomography (CT), identify risk factors for cortical perforation, and determine whether CT-detected cortical perforation was associated with postoperative acetabular cup positioning accuracy.

MATERIALS AND METHODS: We retrospectively analyzed 200 primary THAs performed using portable or CT-based navigation between December 2022 and July 2025. Pin position was assessed on CT 1 week postoperatively and classified as complete intraosseous insertion or cortical perforation. Risk factors were examined using univariable and multivariable logistic regression analyses. Perforation direction was evaluated using Firth's penalized logistic regression. Postoperative cup positioning accuracy was assessed using CT-based radiographic inclination (RI), radiographic anteversion (RA), and absolute errors from target angles of 40° for RI and 15° for RA.

RESULTS: Cortical perforation occurred in 70 of 200 hips (35.0%): 32 medial and 38 lateral. On a pin-by-pin basis, 113 of 400 pins (28.3%) demonstrated perforation. In multivariable analysis, smaller ASIS width, lateral decubitus positioning, and 4-m […]

PubMed · DOI

Long-term outcomes of arthroscopic single-anchor fixation versus microfracture for osteochondral fractures of the knee weight-bearing area in adolescents and young adults.

Yang W, Pan Q, Wang H, Peng Y, Meng C, Huang W

PURPOSE: Osteochondral fractures involving the weight-bearing area of the knee in young patients require stable restoration of the native osteochondral fragment whenever possible. This study aimed to evaluate the long-term clinical and radiological outcomes of arthroscopic single-anchor fixation and to compare them with those of microfracture.

METHODS: This retrospective comparative study included 24 adolescents and young adults with osteochondral fractures of the femoral condyle or trochlear region treated between 2016 and 2017. Twelve patients underwent arthroscopic single-anchor in situ fixation, and 12 underwent microfracture. The fixation technique used one double-loaded suture anchor to obtain multi-point compression of the osteochondral fragment while preserving the subchondral plate. Clinical outcomes were assessed using the Lysholm, International Knee Documentation Committee, and Tegner scores. Magnetic resonance imaging was evaluated using the MOCART 2.0 score. The mean follow-up was 93.7 months.

RESULTS: The single-anchor group showed a significantly higher intraoperative contact area ratio than the microfracture group (96.5% ± 3.2% vs. 85.8% ± 5.6%, $P < 0.001$). At final follow-up, the single-anchor group had superior Lysholm scores (91 ± 5 vs. 83 ± 8, $P = 0.009$), IKDC scores (86 ± 6 vs. 78 ± 7, $P = 0.010$), and Tegner activity levels (5.0 ± 1.1 vs. 3.9 ± 0.9, $P = 0$ […]

PubMed · DOI

Isometric points in cerclage for patellar tendon repair: a 3-dimensional, weightbearing computed tomography kinematic simulation.

Zindel C, Hodel S, Luo Y, Meisterhans M, Fucentese SF, Vlachopoulos L

PURPOSE: The gold standard for the management of ruptured patellar tendon is the surgical repair with nonabsorbable sutures and additional reinforcement e.g. cerclage wires as described by McLaughlin in 1947. Despite the early description, the exact placement of such cerclage wires remains unclear. The aim of the study was (1) to analyze how the cerclage elongates during knee flexion and (2) to define the insertion points with the most isometric behavior at the patella and tibial tuberosity (TT).

METHODS: 3D simulations of 10 healthy knees in the range from 0° to 120° of flexion using weightbearing computed tomography scans were generated. 12 insertion points were defined at the patella in equal distances along the lateral and medial border from the proximal to the distal pole. 42 insertion points were defined at the tibial tuberosity (TT) in a 3.0 × 3.0 cm large grid around its center. An automatic string simulation algorithm was used to calculate the length for all possible combinations of the cerclage from 0° to 120° flexion. The most isometric combination of insertion points was found using the isometric score.

RESULTS: The most isometric insertion at the TT is distally ($p < 0.01$) and towards the center of the TT ($p < 0.01$). The most isometric points at the TT is located 61 mm distal to the joint line and 5 mm from the anterior border of the TT in a true lateral view fluo […]

PubMed · DOI

Patient mortality exceeds implant revision risk with equivalent survival following unicompartmental versus total knee arthroplasty in octogenarians.

Innocenti M, Leggieri F, Massenzi M, Stimolo D, Akkaya M, Matassi F, Civinini R

BACKGROUND: While traditional surgical decision-making emphasizes long-term implant survivorship for octogenarians knee reconstruction, this paradigm may be inappropriate when patient mortality risk substantially exceeds revision risk. This study compared mid-term mortality between unicompartmental knee arthroplasty (UKA) and total knee arthroplasty (TKA) in octogenarians, hypothesizing that implant type would not significantly influence patient survival.

METHODS: A retrospective cohort study was conducted on 238 consecutive octogenarian patients (age ≥ 80 years) who underwent primary knee arthroplasty at a single tertiary orthopaedic center between January 2018 and December 2020. Patients with revision arthroplasty, fracture, tumor, inflammatory arthropathy, or incomplete data were excluded. The primary outcome was all-cause mortality at minimum 5-year follow-up. Secondary outcomes included age-stratified mortality (80-84, 85-89, ≥ 90 years) and identification of independent mortality predictors. Statistical analyses included independent t-tests, chi-square/Fisher's exact tests, standardized mean differences, and multivariable logistic regression adjusting for age, follow-up duration, and registry year.

RESULTS: Of 238 patients, 185 (77.7%) underwent TKA and 53 (22.3%) underwent UKA. Overall mortality was 35.3% (84/238): 33.5% in TKA versus 41.5% in UKA (OR = 1.41, 95% CI: 0 […]

PubMed · DOI

Short- to mid-term clinical outcomes and return to sport and work after arthroscopic debridement of the extensor carpi radialis brevis in patients with chronic lateral epicondylopathy.

Siebenlist S, Doucas A, Hatt FL, Lappen S, Vieider RP, Lacheta L, Kadantsev P

BACKGROUND: This study aimed to evaluate the short- to mid-term clinical outcomes following arthroscopic extensor carpi radialis brevis (ECRB) debridement in patients with chronic epicondylopathy humeri radialis (cEHR). We hypothesized that arthroscopic ECRB debridement would result in favorable short- to mid-term patient-reported outcomes, improving pain levels, strength, range of motion (ROM) and patient satisfaction.

METHODS: A retrospective case series was conducted on patients who underwent arthroscopic ECRB debridement for cEHR with symptoms > 6 months. Patients were assessed at a minimum of 12 months postoperatively using the Mayo Elbow Performance Score (MEPS), Patient-Rated Tennis Elbow Evaluation (PRTEE), Disabilities of the Arm, Shoulder, and Hand (DASH) score, and Visual Analog Scale (VAS) for pain. Pre- and postoperative scores were compared using DASH and VAS. Maximum isometric wrist extension strength and postoperative ROM were measured bilaterally. Return to sport and work, patient satisfaction, and complications were documented.

RESULTS: Of the 35 eligible patients, 30 completed follow-up (86% follow-up rate; mean age 48.4 ± 6.6 years; range: 36-65 years; 47% male). The mean duration of symptoms before surgery was 12 months (IQR 8-20, range 6-62). The mean follow-up duration was 28.13 ± 10.9 months (range 12-45). At final follow-up, median outcome scores were […]

PubMed · DOI

Complications and outcomes after operative treatment of low-energy pelvic ring and acetabular fractures.

Salonpää KB, Nurkkala J, Lantto I, Kakko S, Kaakinen T, Aitavaara-Anttila M, Liisanantti J

BACKGROUND: Fragility fracture of the pelvis (FFP) with a low-energy mechanism of injury is relatively common among the growing geriatric population. The incidence of postoperative complications after surgically treated FFP is not well documented. The aim of this study was to report complications after operatively treated FFP and to determine whether these complications would have a detrimental effect on outcomes.

METHODS: Retrospective study of 200 consecutive patients who underwent operative treatment at Oulu University Hospital, Oulu, Finland, between 1.1.2010 and 15.4.2021 for FFP with a low-energy injury mechanism. Postoperative complications were recorded, and associations with outcomes were depicted.

RESULTS: Postoperative complications were recorded in 31 (15.5%) patients. Patients with complications had higher intraoperative bleeding (OR 2.8 (1.0-7.9), p = 0.048) and higher ASA classification 3.4 (1.3-8.4), P = 0.009). Patients with complications had longer hospital stays, but no clear association between complications and mortality or discharge location was observed.

CONCLUSIONS: The incidence of postoperative complications was comparable with that reported in previous literature. Patients who experienced complications had longer hospital stays; however, no definitive association between complications and outcomes was identified.

PubMed · DOI

Minced cartilage implantation for cartilage regeneration: a survey of current clinical practices.

Yilmaz T, Leutheuser S, Niemeyer P, Behrendt P, Roessler P, Gille J, Mumme M, Vogt S

INTRODUCTION: The objective of this study is to evaluate how minced cartilage implantation (MCI) is currently applied in clinical settings, focusing on the selection criteria, procedural execution, handling of concomitant joint pathologies, and postoperative management.

MATERIAL AND METHOD: An online survey was conducted among all 4,915 members of the German Society for Arthroscopy and Joint Surgery (AGA). Invitations with a survey link were emailed, and the questionnaire was accessible anonymously from January 10 to February 20, 2024. A total of 927 responses were received and analyzed, resulting in a 19% response rate. Demographic data were described descriptively; survey answers were reported as percentages. The 25-item questionnaire, developed by AGA and the Quality Circle for Cartilage Repair & Joint Preservation (QKG e.V.), included single- and multiple-choice questions on experience with cartilage therapy, minced cartilage procedure application and postoperative care, and the overall assessment of this technique within cartilage repair.

RESULTS: MCI has emerged as one of the three most commonly used cartilage repair techniques in the knee joint. Defect sizes up to 4 cm^2 represent the largest treated group. The application of the procedure is quite heterogeneous, with some practitioners using a shaver and others manually mincing with a scalpel. Fixation methods also var […]

PubMed · DOI

Variability in United States online rehabilitation protocols after open and endoscopic hip abductor repair.

Anwunah C, Rosenberg S, Hartwell M

PURPOSE: The purpose of this study was to evaluate readily accessible U.S. based online physical therapy protocols for open and endoscopic hip abductor repairs and to assess the variability in post-operative rehabilitation protocols.

METHODS: Online physical therapy protocols from U.S. based orthopaedic institutions for gluteus medius/minimus repair were collected using a web-based query utilizing search terms "hip abductor repair rehabilitation protocol" and "gluteus repair rehabilitation protocol." Published protocols from both Electronic Residency Application Service (ERAS) academic orthopaedic surgery programs and private groups were collected. All publicly available protocols were included. Protocols were analyzed using a standardized spreadsheet to assess rehabilitation components and timelines. Descriptive statistics (median values, percentages, ranges) were calculated.

RESULTS: 54 rehabilitation protocols were assessed, including 22 (41%) from ERAS-affiliated academic programs or associated physicians. Postoperative weight-bearing was addressed in 53/54 (98%) protocols, with 41 (77%) recommending toe-touch/touchdown/flat-foot/20-lb weightbearing. Median duration of restricted weight-bearing was 6 weeks (range, 2-8 weeks). Regarding range of motion restrictions, 37 (69%) protocols limited hip flexion to 90°, 42 (78%) advised against passive adduction past neutral, and […]

PubMed · DOI

The incidence and epidemiology of patellar fractures following medial patellofemoral ligament reconstruction: a systematic review and meta-analysis.

Abelleyra Lastoria DA, Hunt A, Ross CK, Yemane M, Keynes S, Ghosh A, Hing CB

INTRODUCTION: Patellar instability can arise from a traumatic event with anatomical predisposing factors increasing the risk of dislocation. Medial patellofemoral ligament reconstruction (MPFLR) is commonly performed to treat patellar instability. The aim of this systematic review was to determine the incidence of and risk factors for patellar fractures after MPFLR.

MATERIALS AND METHODS: The protocol for this review was registered on PROSPERO with registration number CRD420251066107. The PRISMA 2020 checklist was followed. Electronic databases and conference proceedings were searched to the 15th of May 2025. Studies were eligible if they reported incidence of patellar fractures after MPFLR. A random-effects meta-analysis was performed. Included studies were appraised using tools respective of study design.

RESULTS: A total of 4,972 records were screened in the initial search. A total of 110 studies satisfied the inclusion criteria, evaluating 115,496 patients. The evidence was moderate to low in quality. The incidence of patellar fracture ranged from 0 to 9.50%, with a pooled incidence of 1.17% (95% confidence interval [CI] 0.76- 1.68%). There was no difference in incidence of patellar fractures between patients who did and did not undergo bone tunnel fixation (0.94%, 95% CI 0.69-1.24 vs. 1.00%, 95% CI 0.15-2.44%, respectively). There was no difference in fracture incidence […]

PubMed · DOI

Predictive accuracy of a general-purpose artificial intelligence model for cut-out following proximal femoral nailing.

Ben Arie G, Warschawski Y, Amzallag N, Gan-Or H, Ben-Tov T, Khoury A, Horev I, Graif N et al.

INTRODUCTION: Cut-out is the most consequential mechanical complication after proximal femoral nailing and requires prompt recognition. General-purpose artificial intelligence (AI) models with image-interpretation capability are increasingly accessible, yet their diagnostic performance for this task is unknown. We evaluated ChatGPT's accuracy for predicting cut-out following proximal femoral nailing and compared TFNA and Gamma nail subgroups.

MATERIALS AND METHODS: In this retrospective predictive-accuracy study, 989 patients (683 TFNA, 306 Gamma nail; mean age 83.8 years; cut-out prevalence 2.5%) were analysed. For each case, three baseline radiographs - the injury anteroposterior and lateral views and the immediate postoperative control radiograph, all obtained before any complication could be radiographically present - together with the full clinical data set (excluding complication-related information) were submitted to ChatGPT, which returned a binary prediction (yes/no) and probability estimate (0-100%). The follow-up radiographs on which cut-out becomes apparent were not provided to the model. Ground truth was established on subsequent follow-up radiographs by a departmental review board requiring agreement of two senior surgeons. Accuracy metrics were calculated with Wilson 95% confidence intervals (CI); exact McNemar's and Mann-Whitney U tests assessed directional bia […]

PubMed · DOI

Staged management of a compound combined pelvi-acetabular injury with perineal degloving in a young female: a case report and literature review.

Dhankhar K, Kekatpure AL, Kekatpure AL

We present the case of a 30-year-old woman, the patient who sustained a compound combined pelvi-acetabular fracture with extensive perineal internal degloving after high-energy trauma. Computed tomography revealed an unstable pelvic ring injury (Tile type C1; anteroposterior compression type III) with right transverse acetabular fracture. The occurrence of extensive internal degloving was confirmed intraoperatively. The injury was treated using a graduated, principle-based approach incorporating early soft-tissue debridement, dead-space management and temporary pelvic stabilization, infection control, and delayed definitive internal fixation. An intercurrent pin-site infection was treated successfully without progression to deep infection. The patient demonstrated progressive radiological union at 4 months with a Harris Hip Score (HHS) of 80 and a Majeed pelvic score of 55. At one-year follow-up, complete union occurred and functional recovery was achieved, characterized by HHS of 90 and a Majeed score of 70. In this case, infection-guided staging as well as extremely aggressive soft tissue treatment may allow for safe delayed fixation even where pelvi-acetabular trauma has been compromised.

PubMed · DOI

Concomitant upper extremity fractures do not worsen outcomes in conservatively managed pelvic fractures: a cohort study of 1840 geriatric patients.

Graif N, Elbaz E, Kazum E, Rachevsky G, Gurel R, Factor S

INTRODUCTION: Pelvic fractures in elderly patients are associated with significant pain, functional decline, and prolonged rehabilitation, with the majority managed non-operatively. In the conservative management paradigm, intact upper extremity function is critical for mobilization with assistive devices; concurrent upper extremity (UE) fractures may therefore complicate recovery. The aim of this study was to evaluate the clinical outcomes associated with this injury combination in patients with conservatively managed pelvic fractures.

METHODS: A retrospective cohort study was conducted at a Level 1 trauma center, including all patients aged ≥ 65 years admitted with pelvic ring or acetabular fractures managed non-operatively between January 2010 and January 2024. Patients were stratified by the presence of a concomitant UE fracture sustained during the same traumatic event. Primary outcomes were hospital length of stay (LOS) and all-cause mortality at multiple time points. Secondary outcomes included readmission rates and systemic infection rates.

RESULTS: Of 1840 patients, 1693 (92.0%) sustained isolated pelvic fractures and 147 (8.0%) had concomitant UE fractures. Groups were comparable at baseline across all measured covariates. LOS did not differ significantly between groups (11.0 ± 14.2 vs. 11.8 ± 16.9 days; p = 0.796). Mortality was similar at all evaluated time points […]

PubMed · DOI

Single double-loaded anchor repair of isolated Lafosse III subscapularis tears in patients not involved in physically demanding activities provides satisfactory functional and radiologic outcomes at minimum 5-year follow-up.

Aral F, Oklaz EB, Ahmadov A, Asiret E, Delice C, Guler Oklaz EB, Akdulum I, Kanatli U

PURPOSE: To evaluate the outcomes of single-anchor repair for isolated Lafosse III subscapularis tears in patients not engaged in physically demanding activities.

METHODS: Patients with isolated subscapularis tears who underwent arthroscopic surgery between 2014 and 2020 were retrospectively evaluated. The inclusion criteria were patients who underwent single double-loaded anchor repair for Lafosse III tears and were not involved in physically demanding activities. Assessments included the American Shoulder and Elbow Surgeons (ASES) score, Constant-Murley Score (CMS), Visual Analog Scale (VAS), range of motion, specific clinical tests, and magnetic resonance imaging encompassing tendon integrity, cross-sectional area (CSA), vertical diameter, and transverse diameters.

RESULTS: 27 patients (mean age 52.4 ± 10.3 years; mean follow-up 80.8 ± 18.6 months) were included in the study. The final follow-up demonstrated significant improvements in the ASES (44.0 ± 12.4 to 82.4 ± 14.1), CMS (49.1 ± 15.3 to 85.3 ± 11.1), VAS (5.7 ± 2.1 to 1.3 ± 2.2), and internal rotation (L4 to T8) (p < 0.001 for all). The positivity of clinical tests decreased substantially: Bear-Hug, from 93% to 18%; Belly-Press, from 89% to 15%; and Lift-Off, from 89% to 18% (p < 0.001 for all). Radiological analyses indicated maintained tendon integrity in 93% of patients and increases in CSA (1.292 ± 453 to 1.544 […]

PubMed · DOI

Shoulder injuries in rugby union: a systematic review and meta-analysis.

Doyle J, Bellamy M, Fowler K, Keiller H, Gunasekera R, Funk L

Shoulder injuries are common in Rugby Union, resulting in prolonged absence and considerable burden. Reported incidence, severity, and mechanisms vary widely, reflecting the multiple high-impact actions in the sport. Various management approaches exist to treat a wide range of pathologies. This systematic review and meta-analysis aimed to synthesise evidence on the epidemiology, mechanisms, and treatment of shoulder injuries in rugby. PUBMED, SCOPUS, and WEB OF SCIENCE were searched from database inception until 1st June 2025. From 1,099 abstracts screened, 37 studies were included. The pooled match incidence was 11.02 injuries/1,000 player-hours (95% CI 7.23-16.82), though heterogeneity was very high (I^2 = 97%) and the 95% prediction interval was wide (2.11-57.61 injuries/1,000 player-hours), indicating the pooled rate should be read as an average across diverse populations rather than a single representative value. Higher rates were seen in male elite (13.74/1,000 h) and high school/university players (12.28/1,000 h). Training incidence was considerably lower (0.20/1,000 h). Substantial between-study heterogeneity was observed across all incidence analyses, reflecting genuine variation

across playing levels, sex, and surveillance methods. Tackling was the predominant injury mechanism, most often affecting the tackler. Bankart, Latarjet, and Bristow procedures all produced fav […]

PubMed · DOI

The pre-operative physical component subscale of the 12-item Short Form Health Survey was the only factor associated with 2-year clinical outcome and satisfaction after hip abductor tendon repair.

Ebert J, Edwards P, Klinken S, Janes G

INTRODUCTION: Surgical repair in the context of symptomatic hip abductor tears unresponsive to non-operative management has demonstrated encouraging outcomes. This study sought to investigate demographic, injury/management and post-operative factors associated with clinical outcome after hip abductor tendon repair.

METHODS: A total of 104 patients undergoing repair were included. Patients were assessed pre-surgery and 2-years post-operatively with the modified Harris Hip Score (mHHS), Oxford Hip Score (OHS), physical (PCS) and mental (MCS) component subscales of the 12-item Short Form Health Survey (SF-12), and satisfaction. Regression analysis assessed the contribution of pre-operative demographic (age, sex, body mass index, PCS, MCS), injury/treatment (symptom duration, prior corticosteroid injections, gluteus minimus/medius tear grade and fatty infiltration) and post-operative (hip abduction strength symmetry) variables, to the 2-year mHHS and being 'very satisfied'.

RESULTS: Mean improvement in the mHHS and OHS from baseline to 2 years was 36.0 points (95% CI, 31.9 to 40.2; $p < 0.001$) and 17.5 points (95% CI, 15.8 to 19.3; $p < 0.001$), respectively. Seven (6.7%) patients presented with symptomatic re-tearing. Of the 97 patients with 2-year follow-up, 78 (80.4%) were 'very satisfied'. The pre-operative SF-12 PCS was the only factor associated with the 2-year mHHS (B = 0.46; […]

PubMed · DOI

Temporising external fixation reduces loss of reduction compared with plaster splinting in ankle fracture-dislocations: a systematic review and meta-analysis of cohort studies.

Suravaram V, Solomon L, Bhat S, Wittauer M

PURPOSE: The optimal temporising strategy for ankle fracture-dislocations is a highly debated topic. These unstable injuries are commonly managed with staged fixation to allow for soft tissue recovery, using either plaster splinting or ankle-spanning external fixation. This study aimed to compare the efficacy and safety of these two approaches.

METHODS: A systematic search of MEDLINE, Embase, Scopus, Cochrane CENTRAL, and Google Scholar was conducted. Comparative studies evaluating temporary external fixation versus plaster splinting in skeletally mature patients with ankle fracture-dislocations were included. Meta-analyses were performed to assess differences in loss of reduction prior to definitive fixation, soft tissue complications, and time to definitive fixation. Risk of bias was assessed using ROBINS-I tool, and certainty of evidence was appraised using GRADE.

RESULTS: Nine retrospective cohort studies comprising 1638 patients were included (589 external fixation, 1049 splinting). Loss of reduction occurred in 3% of injuries in the external fixation cohort compared with 14.6% in those managed with plaster splinting (RR 0.22; 95% CI 0.09-0.52). Rates of overall soft-tissue complications were similar between cohorts. The data regarding time to definitive fixation were highly heterogeneous across studies. Overall certainty of evidence was moderate for LOR and very low for […]

PubMed · DOI

Comprehensive ultrasound assessment of thumb metacarpophalangeal collateral ligament injuries reveals frequent concomitant abnormalities.

Bain O, Farkash U, Atzmon R, Shemesh S, Ohana N

INTRODUCTION: Ulnar collateral ligament (UCL) tears of the thumb are well-recognized injuries, while radial collateral ligament (RCL) and dorsal extensor-related abnormalities are less frequently studied. The pattern of concomitant ultrasound abnormalities in patients after thumb trauma is not well documented. The aim of this study was to characterize concomitant UCL, RCL, and dorsal extensor-related abnormalities within a selected cohort referred to thumb ultrasound after trauma, and to explore associations with injury mechanism.

MATERIALS AND METHODS: A retrospective review was performed of 42 consecutive adult patients who underwent high-resolution ultrasound of the thumb metacarpophalangeal joint between April 2014 and February 2025 for suspected ligamentous injury after acute trauma. Patients were included if ultrasound demonstrated pathologic findings in at least one stabilizing structure, UCL or RCL, and the extensor mechanism was systematically assessed. Collateral ligament injuries were graded using standard sonographic criteria, while dorsal extensor-related findings were recorded descriptively. Associations between injury severity and mechanism of injury were tested using chi-square or Fisher's exact analysis, as appropriate.

RESULTS: UCL pathology was present in 35/42 (83.3%) patients, RCL pathology in 23/42 (54.8%), and dorsal extensor-related abnormalities in 17 […]

PubMed · DOI

Holding the line: predictors of coronal and sagittal loss of reduction after tibial plateau fracture fixation.

Hesmerg MK, Benner JL, Joosse P, Keijser LCM

INTRODUCTION: Tibial plateau fractures (TPFs) are challenging fractures that often suffer from secondary Loss of Reduction (LoR) after fracture fixation. In this study, we aimed to investigate the factors and characteristics associated with LoR after surgical fixation, using measurements of both coronal and sagittal displacement.

METHODS: A retrospective analysis of 213 TPF patients (79.3% female, mean age 57.5 ± 14.4 years) surgically treated between 2017 and 2024 in a level 1 trauma center was performed. Patients were included if they had a minimum of 100-days radiological follow-up. Loss of reduction was defined as late postoperative depression ≥ 3 mm, condylar widening ≥ 5 mm, varus/valgus or tibial slope collapse > 5 degrees, breakage, loosening, or conversion to total knee arthroplasty (TKA). Radiographic images were assessed for preoperative, direct postoperative, and late postoperative (up to 1 year) displacement. Univariable and multivariable logistic regression analyses were employed to study factors associated with LoR..

RESULTS: At a mean follow-up of 325 days (SD ± 161) 77 patients (36.2%) suffered from LoR. The adjusted multivariable logistic regression model revealed that patient age (OR 1.03, p = 0.028), involvement of the posterior column (OR 2.87, p = 0.014), preoperative step-off (OR 1.08, p < 0.001), postoperative step-off (OR 1.61, p = 0.007), and postope […]

PubMed · DOI

Using an App based algorithm to determine axial rotation and pelvic tilt at the time of an AP radiograph: a sawbone study.

Boettner F, Premkumar A, Sterneder M, Aboaf C, Haralambiev L

INTRODUCTION: Axial rotation of the pelvis impacts inclination and anteversion measurements of the acetabular component on plain radiographs. Currently the axial rotation is not taken into account when providing acetabular component measurements on plain radiographs. The current study evaluates a novel App based algorithm to determine pelvic axial rotation at the time of an AP pelvis radiograph.

MATERIALS AND METHODS: AP and lateral radiographs were taken of the sawbone pelvis to calibrate the App. The sawbone was then rotated 11 times to the left (average 5.3°, range 0.2°-10.5°) and 10 times to the right (average 5.2°, range 1.2°-11.3°). The sawbone was then tilted forward 14 times (average 5.6° range 0.1°-11.6°) and 13 times backward (average 5.9°, range 0.1°-11.2°). The amount of axial rotation and tilt of the sawbone pelvis was determined using a digital caliper. AP pelvis radiographs were taken for each position. Using a novel App based calculation algorithm the amount of axial rotation and pelvic tilt was determined from the AP Pelvis radiograph.

RESULTS: The algorithm determined the axial rotation of the pelvis at the time of the AP pelvis radiograph with an average difference of 0.6° (range 0°-1.8°) compared to the digital caliper measurement. For measuring pelvic tilt, the difference was 2.3° (range 0.4°-4.5°) compared to the digital caliper measurement.

CONCLUSIONS […]

PubMed · DOI

Risk factors for implant-associated infection after bone allograft reconstruction: a 20-year retrospective cohort study.

Scheurer F, Smolle M, Hauer G, Ruckenstuhl P, Valentin T, Leithner A

AIMS: Reconstruction of bone defects is a key challenge in orthopaedic surgery. While autologous bone grafts are considered the gold standard, osseous allografts are increasingly used as an alternative as they allow for treatment of large bone defects. However, they may be associated with an increased risk of postoperative infections. This study aims to analyze the frequency of and risk factors for implant-associated infections after bone allograft transplantation.

METHODS: All patients who underwent reconstruction with bone allografts at a single orthopaedic and trauma unit between 2003 and 2023 were retrospectively included. Data on demographics, disease-associated parameters, risk factors at index surgery, and development of implant-associated infections during follow-up were documented. Potential risk factors for development of implant-associated infections during follow-up were investigated with logistic regression models.

RESULTS: A total of 220 patients were included (46.8% males; median age at surgery 46.5 years [IQR: 22-67]; n = 23 with implant-associated infections during follow-up). Factors independently associated with subsequent implant-associated infections were smoking (p = 0.001), BMI (p = 0.015) and radiotherapy (p = 0.013). Gender, age, preoperative chemotherapy, allograft material, preoperative C-reactive protein, hemoglobin and reason for surgery (tumor, n […]

PubMed · DOI

Why it's better to believe in a larger definition of the diametaphyseal junction zone in pediatric distal radius fractures.

von Schrottenberg C, Beck SM, Schwerk P, Fitze G, Schultz J

INTRODUCTION: Diametaphyseal radius fractures (DMRF) in children are characterized by special biomechanical behavior. They present with a stereotypical loss of reduction (radial translation and ulnar tilt) of the distal fragment when stabilized with elastic stable intramedullary nailing (ESIN), the standard osteosynthesis for pediatric diaphyseal forearm fractures. No consensus has been found on the definition of the diametaphyseal junction zone (DMJZ). Some authors claim that diaphyseal fractures immediately proximal to the metaphysis, as defined by the AO Pediatric Comprehensive Classification of Long Bone Fractures (AO-PCCF), can safely be stabilized with ESIN. They confine the DMJZ to a small part within the metaphysis. Other definitions of the DMJZ extend slightly more proximally, including a small part of the distal diaphysis. The forearm fracture index (FFI) calculates the ratio of the fracture's distance to the radius' growth plate over its width and defines DMRF to have an FFI between 1 and 2. The aim of this case series was to demonstrate that these particular fractures, which can be considered diaphyseal according to the AO-PCCF, still exhibit biomechanical characteristics typical of DMRF.

MATERIALS AND METHODS: Radiologic and clinical outcomes of 11 DMRF, that would be considered diaphyseal by the AO-PCCF classification but fall into the DMJZ when using the FFI, ar […]

PubMed · DOI

Optimizing initial biomechanical strength ("Time Zero") of posterior repair in total hip arthroplasty: a biomechanical comparison of suture materials and knot configurations.

Chen M, Liu X, Chen D, Li X, Du H, Sun Y, Wang H

INTRODUCTION: Posterior capsule repair (PCR) is commonly performed during total hip arthroplasty (THA) via the posterolateral approach to improve postoperative stability. Because biological healing of the repaired posterior structures requires time, the repair construct must provide sufficient immediate mechanical support after surgery. This study aimed to identify the suture material and knot configuration with favorable initial mechanical performance in a standardized in vitro model.

MATERIALS AND METHODS: This in vitro biomechanical study was conducted in two sequential phases. Phase 1 evaluated the baseline tensile properties of three commonly used suture materials: #2 Ethibond, #0 Vicryl, and 3 - 0 PGLA. Phase 2 used a standardized porcine dermal surrogate model to compare eight suture configurations, including single- and double-strand techniques. The primary biomechanical outcomes included maximum failure load, load at 2 mm gap formation, and construct stiffness during uniaxial quasi-static load-to-failure testing.

RESULTS: In Phase 1, non-absorbable #2 Ethibond showed significantly greater ultimate tensile strength (92.44 ± 8.72 N) and higher stiffness than the absorbable alternatives (P < 0.001). In Phase 2, double-strand configurations outperformed all single-strand techniques. Among them, the double-strand Nice knot achieved the highest maximum failure load (104.04 […]

PubMed · DOI

Intravenous versus oral tranexamic acid in shoulder arthroplasty.

Parmar T, Adio A, Daher M, Khan A, Horneff J, Abboud J

PURPOSE: Tranexamic acid (TXA) is used in shoulder arthroplasty to reduce blood loss. While intravenous (IV) TXA is standard, oral TXA offers advantages in cost and administration. This study compares 90-day complications and postoperative laboratory values between IV and oral TXA in total shoulder arthroplasty (TSA).

METHODS: A retrospective cohort study using TriNetX identified adults undergoing primary TSA (2005-2025) receiving TXA on the day of surgery, stratified by route (IV vs. oral). One-to-one propensity score matching was performed. Ninety-day outcomes included transfusion, thromboembolic events, cardiac and renal complications, neurologic events, infection, wound dehiscence, readmissions, emergency department visits, opioid utilization, and mortality. Postoperative laboratory outcomes included hemoglobin, erythrocyte count, hematocrit, and platelet count. Outcomes were compared using odds ratios (ORs) with 95% confidence intervals (CIs) and t tests where appropriate.

RESULTS: After matching, 5722 patients were included in each cohort. No significant differences were observed between IV and oral TXA in 90-day transfusion rates (P = 0.873) or thromboembolic events, including deep vein thrombosis and pulmonary embolism. Rates of myocardial infarction, stroke, cardiac ischemia, acute renal failure, infection, readmission, emergency department visits, opioid use, and mo […]

PubMed · DOI

Robotic-arm assistance in revision total hip arthroplasty improves the accuracy of acetabular component positioning: a cadaver-based study.

Fitzgerald K, Borukhov I, Chandra V, Smitterberg C, Mont MA, LiArno S

INTRODUCTION: Acetabular component positioning is a key factor for stability, impingement, wear, and biomechanics in revision total hip arthroplasty (THA). However, anatomic distortion and previous implants may complicate positioning. Although evidence on computed-tomography (CT)-based robotic assistance in revision THA is limited, its planning features and precision may improve surgical plan execution. This study aimed to measure the precision and accuracy of acetabular component positioning of revision robotic-arm assisted total hip arthroplasty (RA-THA) compared with a published manual primary total hip arthroplasty (M-THA) benchmark for: (a) acetabular inclination; (b) acetabular version; and (c) center of rotation (COR), with comparison by surgeon case volume.

MATERIALS AND METHODS: There were seven cadavers (nine hips) that previously underwent primary THA revised with RA-THA by independent surgeons with varying primary RA-THA case volumes, surgical approaches, and acetabular component types. Due to surgical protocol deviations, two hips were excluded leaving seven hips from six cadavers. Pre- and postoperative CT scans were compared to calculate absolute error for inclination, version, and COR; normality was established with the Anderson-Darling test and two-variance tests assessed differences in standard deviations ($\alpha = 0.05$). Mean differences between cadaver-based rev […]

PubMed · DOI

Correction: Modified rotational wedge distal metatarsal osteotomy versus chevron osteotomy for hallux valgus: long-term radiographic and clinical outcomes.

Aybar A, Gokkus K, Özden E, Çetin MH, Çetin MÜ

PubMed · DOI

Isolated compartment arthrosis of the knee: contemporary concepts, persistent controversies, and future directions in unicompartmental knee arthroplasty.

Akkaya M, Innocenti M

PubMed · DOI

Preoperative lower radiographic osteoarthritis severity in primary total knee arthroplasty increases revision risk by tenfold compared to severe osteoarthritis.

Stroobant L, Vermue H, Droogsma T, Van Onsem S, Arnout N, van Hellemondt GG, Victor J, Heesterbeek PJC

BACKGROUND: Dissatisfaction following primary total knee arthroplasty (pTKA) is higher in patients with preoperative low-grade osteoarthritis (OA), but its impact on revision risk remains unclear.

QUESTIONS/PURPOSES: This study aimed to identify predictors for revision surgery within 10 years following pTKA, focusing on radiographic severity of OA using the Kellgren-Lawrence (KL) grading system, along with other demographic and radiographic factors.

PATIENTS AND METHODS: This retrospective case-control study was conducted at two European tertiary referral centers. The case group included 142 patients who underwent aseptic revision total knee arthroplasty (rTKA) between 2007 and 2023, within 10 years following pTKA. A 2:1 control group of 284 patients who had pTKA between 2011 and 2014, with no revision surgery within 10 years, was selected. Collected data included age, sex, body mass index (BMI), side of surgery, and American Society of Anesthesiologists (ASA) classification. Radiographic data were collected, requiring a preoperative radiograph to assess the KL grade. Univariate analyses identified potential predictors for rTKA, and multivariable logistic regression assessed their relationship with revision risk.

RESULTS: Patients requiring rTKA were significantly younger at the time of pTKA. For each additional year of age, the revision risk decreased by 6% (OR 0.94, 95% CI […]

PubMed · DOI

Correction: Non-tobacco nicotine dependence and postoperative complications following operative fixation of tibial shaft fractures: a retrospective cohort analysis.

Berk A, Bogart M, Eghtedari C, Good L, Florentino S, Moyal A, Burkhart R, Ochenjele G et al.

PubMed · DOI

Comparison of pediatric proximal femoral locking plates and dynamic hip screws in the fixation of pediatric intertrochanteric fractures, a biomechanical study.

Wang JH, Shih CA, Wang CH, Lee CC, Wu KW, Wang TM, Kuo KN, Hong CK

Treatment options for pediatric intertrochanteric fractures range from conservative casting in young children to surgical fixation in older patients, most commonly using dynamic hip screws (DHS) or pediatric proximal femoral locking plates (PF-LCP). As there is limited biomechanical evidence favoring one method over another, this study aimed to compare the biomechanical stability of DHS and PF-LCP for the treatment of pediatric intertrochanteric fractures using a synthetic bone model. Sixteen synthetic composite femurs were allocated into two groups stabilized with either a DHS or a PF-LCP. To simulate Delbet type IV intertrochanteric fractures, a vertical osteotomy was created in each specimen. Biomechanical evaluation was carried out using a universal testing machine, assessing axial stiffness, vertical, horizontal, and total displacement, as well as ultimate load to failure under standardized loading conditions. Comparative statistical analyses were then performed to determine differences in biomechanical performance between the two fixation methods. Group P demonstrated significantly smaller vertical and total displacements (0.15 ± 0.15 mm and 0.41 ± 0.31 mm) compared with the DHS group (0.59 ± 0.42 mm and 0.75 ± 0.33 mm) ($p = 0.018$ and $p = 0.045$), whereas horizontal displacement showed no difference ($p = 0.142$). Axial stiffness and ultimate failure load were comparable bet […]

PubMed · DOI

Impact of severe varus deformity on the outcome of mobile-bearing unicompartmental knee arthroplasty: a comparison of patients with varus deformity of less than and more than 15 degrees using matched-pair analysis.

Mullaji AB, Gupta AK, George GK, Kazi R

INTRODUCTION: The outcome of patients undergoing UKA with larger preoperative deformity (> 15º varus) and lesser deformity is sparsely compared in the literature. This matched-pair retrospective study attempts to fill this

research gap by comparing the effect of preoperative and postoperative residual varus on outcome.

METHODS: Patients undergoing UKA with preoperative varus deformity > 15º (HKA > 15º group) were selected and matched for age and gender with those with < 15º of preoperative varus deformity (HKA < 15º group). There were 72 age and gender-matched pairs knees. HKA was measured and Forgotten Joint Score, OKS, VAS and WOMAC scores were obtained at a minimum follow up of 5 years. Postoperative residual varus was classified into 3 groups based on postoperative HKA: varus < 3º, varus 3 to 7º and varus > 7º.

RESULTS: Mean correction of deformity was significantly more ($p < 0.001$) in the HKA > 15º group (10.5º) when compared to HKA < 15º group (4.5º). There was no significant difference in outcome measures OKS ($p = 0.21$), FJS ($p = 0.35$) and VAS ($p = 0.6$) between patients in the two groups. There was no significant correlation between the preoperative and postoperative HKA of the patient with postoperative OKS, FJS and VAS in the two groups. There was no statistically significant difference between FJS ($p = 0.73$) of the three postoperative varus groups. Excellent to good […]

PubMed · DOI

A multimodal imaging approach to predict carpal tunnel syndrome after distal radius fractures.

Önüt M, Alkol N, Önalo■lu Y, Özer M, Apayd■n E, Talmaç MA

OBJECTIVE: Distal radius fractures are commonly associated with the development of carpal tunnel syndrome (CTS), yet current evidence primarily focuses on radiographic alignment parameters. The role of carpal tunnel morphology, particularly total carpal tunnel cross-sectional area (CSA) measured on computed tomography (CT), remains insufficiently investigated. This study aimed to evaluate the relationship between CSA, volar tilt, and the development of CTS in conservatively managed distal radius fractures, and to assess the predictive performance of these parameters when used in combination.

METHODS: This retrospective case-control study included 150 patients, comprising 50 patients who developed CTS and 100 controls without CTS. CSA was measured on CT scans obtained at initial presentation, while volar tilt was assessed using standard radiographs. Additional variables, including the presence of ulna fracture and fracture type, were analyzed. Group comparisons were performed using nonparametric tests. Multivariable logistic regression analysis was conducted to identify independent predictors of CTS. Diagnostic performance was evaluated using receiver operating characteristic (ROC) curve analysis.

RESULTS: Patients who developed CTS had significantly lower CSA and higher volar tilt values (both $p < 0.001$). The presence of an ulna fracture was inversely associated with CTS ($p = $ […]

PubMed · DOI

Unicompartmental knee arthroplasty and total knee arthroplasty in patients with 15 or more degrees of varus deformity: age- and gender-matched comparative study of radiographic and clinical outcomes.

Mullaji A, Chopperla S, George GK, Kazi R

INTRODUCTION: To date, few studies have directly compared matched cohorts of patients with severe varus deformity undergoing UKA versus TKA. The current study seeks to address this gap. The hypothesis was that, in patients with deformity of > 15°, postoperative PROMs following UKA would be inferior to those achieved with TKA.

METHODS: Records of patients operated for UKA and TKA from January 2016 to December 2021 were evaluated, and the data for patients with deformity of ≥ 15° were selected. The UKA patients were then matched for age and gender with TKA patients. 52 matched pairs of patients were obtained. The HKA, mLDFA, MPTA were measured. OKS, SF12, and FJS were collected. The magnitude of change in the mLDFA and MPTA was assessed using the Estimated Marginal Means.

RESULTS: The mean follow-up times for UKA and TKA were 3.7 years and 5.5 years, respectively. The mean deformity correction (ΔHKA) was significantly greater in TKA (16.3° ± 4.5°) than UKA (10.6° ± 3.1°) ($p < 0.001$), with significant changes in mLDFA and MPTA ($p < 0.001$). In UKAs, 68% of patients demonstrated a postoperative HKA between 171.2° and 175.7°, whereas in TKAs, values clustered between 174.5° and 179.8°. Mann-Whitney U test demonstrated no statistically significant differences between the UKA and TKA cohorts across all assessed PROMs, including OKS, SF-12, and FJS ($p > 0.05$ in all). 98% of the UKA an […]

PubMed · DOI

Acetabular superior wall fractures: a distinct acetabular fracture entity.

Gänsslen A, Krappinger D, Liodakis E, Clausen JD, Omar-Pacha T, Lindtner RA, Sehmisch S, Osche DB

INTRODUCTION: A transitional "grey-zone" exists between typical locations of classical trapezoidal anterior wall and large-fragment posterior wall acetabular fractures. Part of this grey zone is the superior acetabular dome, which is commonly involved in acetabular fracture patterns, especially in geriatric patients. The aim of this study was to characterize isolated acetabular superior wall fractures and to assess their relevance as a distinct fracture pattern.

METHODS: A retrospective analysis of pelvic fracture databases from three tertiary trauma centers was performed. Acetabular fractures with superior dome involvement were screened, and cases with isolated superior articular involvement were included. Overall, seven patients were identified. Clinical and radiological data were analyzed regarding fracture morphology, associated fracture characteristics (including dislocation, comminution, marginal impaction, intra-articular fragments, and femoral head injury), concomitant pelvic ring injuries, treatment, and clinical outcomes.

RESULTS: Isolated superior wall acetabular fractures occurred following both high-energy and low-energy mechanisms, including simple falls. The mean age was 48 years and four of seven patients were male. Hip dislocation or subluxation was observed in four cases (anterior-superior or purely superior). Surgical treatment was performed in six cases, p […]

PubMed · DOI

A novel biomechanical model for reproducing and analysing the causal mechanisms of PFNA "cut-in" phenomenon.

Rasappan K, Joshua K, Chou SM, Shaw LKRM, Huang D, Yew A, Kwek EBK

INTRODUCTION: "Cut-in" is a rare but serious complication associated with cephalomedullary nail fixation for intertrochanteric hip fractures, particularly with Proximal Femoral Nail Antirotation (PFNA). This phenomenon involves paradoxical superomedial migration of the helical blade through the femoral head. Although increasingly recognised, the underlying mechanisms remain incompletely defined. This study aims to describe a new superolateral tensile loading set up in a bidirectional loading model, with the aim of incorporating possible mechanical deviations introduced by abductor weakness.

MATERIALS AND METHODS: This study was conducted in two stages. An initial stage with synthetic femurs (SYNBONE® 2420) was used to calibrate the set up. Within this stage, intramedullary canal diameter was compared (12 mm vs. 18 mm). In the final stage, 5 osteoporotic synthetic femurs (Sawbones® 3503) were used. In both stages, standardised AO/OTA 31-A1.1 intertrochanteric fracture were created in the synthetic femurs and fixed with PFNA-II implants. All femurs were subjected to cyclical bidirectional loading with a universal testing machine. Each cycle consisted of axial compression (720 N) and superolateral tensile loading (120 N) at 2 Hz. This configuration was designed to simulate altered hip biomechanics postulated with the trendelenburg gait. Testing was continued until mechanical fail […]

PubMed · DOI

Routine intraoperative infection testing in presumed aseptic hip and knee revision: a matched cohort retrospective study.

Loppini M, Rocchi C, Di Maio M, Chiappetta K, Bulgarelli A, Grappiolo G

INTRODUCTION: Periprosthetic joint infections (PJIs) complicate 1-2% of primary and up to 4% of revision arthroplasties, and their diagnosis remains challenging due to the lack of a diagnostic gold standard. Differentiating PJI from aseptic failure is particularly difficult in low-grade or biofilm-related infections. Routine intraoperative microbiological screening is not recommended in presumed aseptic revisions. This study aimed to evaluate the association between routine intraoperative microbiological screening and implant survival in presumed aseptic revision arthroplasty. A secondary objective was to identify preoperative predictors of unexpected PJIs.

METHODS: Retrospectively collected data of patients undergoing presumed aseptic THA and TKA revisions between 2013 and 2019 were included. Two matched cohorts were compared: a screened group (2016-2019)

undergoing routine intraoperative microbiological sampling, and an unscreened one (2013-2015) without systematic screening. Kaplan-Meier survival analysis with log-rank testing assessed overall and infection-free survival. Univariate conditional logistic regression and receiver operating characteristic (ROC) analyses evaluated potential associations between preoperative serum markers and unexpected PJI.

RESULTS: A total of 559 patients were included, of whom 295 underwent screening and 264 did not. Unexpected infections occ […]

PubMed · DOI

Artificial intelligence advancements for orthopaedic clinical reasoning: longitudinal assessment of newer models (ChatGPT-5, Grok-3, Gemini 2.5 Flash) compared to clinicians.

Agharia S, Soroush S, Ameen D, Zhou Y

INTRODUCTION: This descriptive study aimed to longitudinally evaluate the performance of contemporary large language models - ChatGPT-5, Gemini 2.5 Flash, and Grok-3 - on orthopaedic clinical multiple-choice tasks, benchmarked against pooled clinician consensus. A secondary aim was to assess whether recent advances in generative AI translated into improved alignment with clinician consensus compared with previous AI models.

MATERIALS AND METHODS: A total of 97 multiple-choice clinical cases spanning eight orthopaedic subspecialties were sourced from OrthoBullets and previously benchmarked against aggregated responses from thousands of practising clinicians. Using identical methodology to our 2023 study of ChatGPT-3.5, ChatGPT-4, and Bard, each model was prompted with standardised case stems and response options. The primary outcome was the proportion of AI responses matching the most popular clinician response; secondary analyses assessed agreement within 10% and 20% of clinician consensus, performance on 'controversial' (< 25% margin) questions, and inter-model concordance using Cohen's kappa coefficients.

RESULTS: Gemini 2.5 Flash achieved the highest alignment with clinician consensus (69.1%), followed by Grok-3 (66.0%) and ChatGPT-5 (58.8%). None of the LLMs refused to respond to any prompts, representing a reduction from 7.2% from our 2023 study. Subspecialty analysis de […]

PubMed · DOI

Trajectory of urinary incontinence symptoms following total hip arthroplasty for hip osteoarthritis.

Kadowaki M, Fukutani S, Ushio K, Uchio Y

INTRODUCTION: Urinary incontinence (UI), a prevalent quality-of-life-limiting condition, frequently coexists with hip osteoarthritis, potentially through pelvic floor dysfunction related to obturator internus muscle involvement. Several reports suggest improvement after total hip arthroplasty (THA), although postoperative time-course changes remain unclear. This study aimed to clarify the temporal trends in UI after THA and examine association of postoperative UI trajectories with functional recovery.

MATERIALS AND METHODS: This retrospective study included 118 female patients who underwent THA for hip osteoarthritis between October 2021 and March 2024. UI, assessed using the International Consultation on Incontinence Questionnaire-Short Form (ICIQ-SF), was defined as ICIQ-SF ≥ 1 preoperatively and at 3, 6, 9, and 12 months postoperatively, and classified as stress (SUI), urge (UUI), or mixed (MUI). Hip-related patient-reported outcomes were evaluated using the Japanese Orthopaedic Association Hip Disease Evaluation Questionnaire (JHEQ). Associations with BMI, age, surgical approach (direct lateral/Bauer vs. modified Watson-Jones/OCM), and UI type were tested using Mann-Whitney U, chi-squared, and repeated-measures ANOVA tests.

RESULTS: Of 118 patients (mean age at surgery, 67 years; mean BMI, 24.3 kg/m2) preoperatively, 73 (61.9%) reported UI (mean ICIQ-SF 7.88), including S […]

PubMed · DOI

The grand piano ratio: an adjunctive intraoperative screening tool for detecting excessive femoral component external rotation in total knee arthroplasty.

Avci Ö, Öztürk A, Güler BO, Bayrak HÇ, Kaya AÖ, Do■an Y

BACKGROUND: Accurate femoral component rotation is essential for optimal outcomes in total knee arthroplasty (TKA), yet reliable intraoperative assessment remains challenging. The "grand piano sign" has been described as a qualitative visual cue, but its quantitative clinical value has not been clearly established. This study aimed to evaluate the diagnostic performance of the grand piano ratio as an intraoperative tool for detecting excessive femoral component external rotation using advanced postoperative imaging as a reference standard.

METHODS: A retrospective analysis was conducted on 170 knees undergoing primary TKA. The intraoperative grand piano ratio was measured after completion of all femoral resections and before trial component implantation using the anterior femoral resection surface. Postoperative femoral component rotation was assessed using MAVRIC-sequence magnetic resonance imaging referenced to the surgical transepicondylar axis (sTEA). Excessive femoral component external rotation was defined as postoperative external rotation greater than 3° relative to the sTEA reference. Receiver operating characteristic (ROC) analysis was performed to evaluate diagnostic performance of the grand piano ratio and determine the optimal cutoff value, while multivariable logistic regression analysis was used to identify independent predictors of excessive external rotation. […]

PubMed · DOI

The shoulder stiffness scale: a validated tool for monitoring stiffness in arthroscopic rotator cuff reconstructions.

Bouaicha S, Biner M, Selman F, Wieser K, Müller AM, Audigé L

BACKGROUND: Postoperative shoulder stiffness after arthroscopic rotator cuff repair is common and clinically heterogeneous. Established shoulder scores assess pain and function, but do not specifically isolate stiffness during follow-up. The aim of this study was to validate the newly developed Shoulder Stiffness Scale (SSS) as a pragmatic monitoring tool for postoperative shoulder stiffness after arthroscopic rotator cuff repair (ARCR).

METHODS: From a cohort of 973 primary arthroscopic rotator cuff reconstructions, patients without concomitant joint pathology of the same or contralateral arm were included. The new SSS was based on three items related to shoulder pain, subjective and objective ROM ranging from 0 to 10 points. It was collected before surgery, as well as 6 and 12 months postoperatively, along with other pain level parameters, Constant Score (CS), Subjective Shoulder Value (SSV), Oxford Shoulder Score (OSS). Changes of the SSS and its subscales over time, Cronbach's alpha for internal consistency, bottom and ceiling effects were reported. Validity was investigated through SSS associations with other parameters of similar dimension, and functional scores.

RESULTS: The selected cohort with 740 patients showed postoperative shoulder stiffness in 12.7 %. The SSS averaged 5.1 points (SD 2.5) preoperatively, and decreased significantly to mean 2.9 (SD 2.4) and 1.6 (S […]

PubMed · DOI

Isolated screw fixation of posterior wall fractures.

Clausen JD, Özbek H, Omar-Pacha T, Sehmisch S, Gänsslen A

INTRODUCTION: Posterior wall (PW) fractures are common acetabular fractures and are most often treated by open reduction and internal fixation. The gold-standard of surgical stabilization is screw fixation of large wall fragments with additional buttress plating. Some reports focussed on screw fixation alone presenting adequate results.

MATERIAL AND METHODS: From a total of 208 PW-fracture treated between 1972 and 2008, 57 patients were identified with open reduction and internal fixation (ORIF) using isolated screw fixation. These patients were analyzed regarding demographical data (patient age, sex), type of accident, injury mechanism, Injury Severity Score (ISS), concomitant pelvic ring injuries, associated injuries, perioperative parameters and long-term results.

RESULTS: 44 patients were male and 13 were female with a mean age of 36.8 years. More than 90% sustained high-energy trauma with 34 sustaining multiple injuries or polytrauma. The mean Injury Severity Score was 11.5 points. All patients had fracture dislocations, which were reduced within 24 h after admission except one patient with secondary transfer. 11 patients presented with an initial sciatic nerve deficit. 29 patients had a single PW-fragment, 16 presented with two PW-fragments and in 12 patients marginal impactions were present. Surgery was performed in average after 7 days using the Kocher-Langenbeck appr […]

PubMed · DOI

I

Injury

Volume 57, Issue 11 · September 2026 · 52 new articles

Diagnostic accuracy of shock index, revised trauma score, and injury severity score for mortality prediction in trauma: A hierarchical meta-analysis.

Raza AA, Ahmad M, Choudhary N, Rehman A, Zahid M, Rehman A, Khan W

BACKGROUND: Shock Index (SI) and Revised Trauma Score (RTS) are physiological tools available at the point of trauma triage. Injury Severity Score (ISS) is an anatomical score that requires completed injury characterization and is typically available only after initial resuscitation. Their comparative value for predicting mortality and whether ISS meaningfully adds to the acute tools remains unclear.

AIMS: The aim of this systematic review and hierarchical meta-analysis is to pool the diagnostic accuracy of SI and RTS as acute mortality-prediction tools in adult trauma patients; to benchmark these against ISS; and to formally test, in studies reporting both, whether RTS and ISS differ in diagnostic accuracy.

METHODS: Electronic databases were searched from inception to May 2026. In-hospital mortality was prespecified as the primary endpoint. Pooled sensitivity, specificity and hierarchical summary receiver operating characteristic (HSROC) analyses were performed with a bivariate random-effects model for each tool separately. For studies reporting both RTS and ISS in the same cohort, a paired comparison was performed using a bivariate model with test type as a covariate and a pooled within-study difference analysis.

RESULTS: Twelve studies (234,129 patients) were included. Pooled sensitivity/specificity were 0.679/0.763 for SI (5 studies, AUC 0.747), 0.788/0.801 for RTS (6 st […]

PubMed · DOI

Epidemiological patterns and surgical outcomes of pediatric supracondylar humerus fractures: a focus on domestic safety and seasonal variations.

Altin R, Cakirturk S, Erten RA, Ozdemir S, Yesil M, Yilmaz BK

INTRODUCTION: Supracondylar humerus fractures (SHFs) are prevalent pediatric elbow injuries. This study investigates the epidemiology, clinical characteristics, and treatment outcomes of pediatric SHFs to identify preventive strategies.

METHODS: This retrospective study included 298 pediatric patients (aged ≤14 years) treated for SHFs between December 2018 and June 2024. Demographic data, injury mechanisms, seasonal variations, and surgical details were analyzed. Radiological reduction was assessed via the Baumann angle and anterior humeral line. Functional outcomes were evaluated using Flynn's criteria.

RESULTS: The study cohort (mean age 6.1 years; 58.7% male) predominantly sustained extension-type fractures (96.3%), with the severely displaced Gartland Type III being the most frequent (49.3%). The incidence of injuries peaked during the summer months and school holidays, with the majority occurring at home (64.1%) due to simple falls (93%). Surgical intervention, consisting of closed reduction and percutaneous pinning, was necessitated in 75.8% of cases, with a mean time-to-surgery of 4.6 h. Satisfactory (excellent and good) functional outcomes were achieved in 97.9% of the patients. Overall complication rates were minimal, and transient nerve injuries (2%) fully resolved spontaneously during follow-up.

CONCLUSION: Pediatric SHFs frequently occur in domestic settings duri […]

PubMed · DOI

Force measurement in circular external fixation frames reflects construct stability: A laboratory proof-of-concept study.

Naeem H, Maroy R, Stewart T, Harwood P, Howard A

BACKGROUND: Circular external fixation frames are widely used in limb reconstruction, however determining the optimal time for frame removal remains challenging. Current approaches rely on clinical and radiological assessment, both of which have recognised limitations. Mechanical assessment of construct stability offers a

potential alternative, but existing approaches have not translated widely into clinical practice, in part due to reliance on absolute measurement thresholds.

OBJECTIVE: To determine whether force measurement within a circular external fixation frame varies systematically with fracture site stability, and to assess whether this measurement may support monitoring of healing progression.

METHODS: Three similar four-ring Ilizarov frame constructs were assembled using a Perspex bone analogue. A load cell was integrated into the frame and used to measure force during axial loading to 700 N. Models representing different stages of fracture healing were simulated using varying fracture gaps and materials of differing stiffness. Measurements were obtained across multiple sensor positions and configurations. Force measurements were compared between models using non-parametric statistical methods. Receiver operating characteristic (ROC) analysis was performed to assess separation between conditions.

RESULTS: Force measured within the frame decreased progressively with […]

PubMed · DOI

Screening for posttraumatic stress symptoms after accidental injury: Implementation of the young child PTSD screen in the acute trauma setting.

Ray M, Bailey A, Lendrum E, Vore E, Poundstone A, Beiting S, Reichert M, Moody S et al.

BACKGROUND: Post-traumatic stress symptoms (PTSS) in young children are underrecognized following unintentional injury, and validated screening tools for early childhood are rarely implemented in acute trauma settings. We evaluated the Young Child PTSD Screen (YCPS) administered at hospital discharge to determine screening positivity, associated clinical factors, and persistence of symptoms at one month.

METHODS: We conducted a retrospective cohort study of children aged 2-6 years admitted to a Level I pediatric trauma center following unintentional injury (November 2020-November 2024). Caregivers completed the 6-item YCPS at discharge. Children with ≥ 1 endorsed symptom were contacted one month later for repeat assessment. Demographic, injury, and hospitalization data were abstracted from the trauma registry and electronic health record. Multivariable logistic regression evaluated predictors of positive screening at discharge, and follow-up analyses assessed persistence of symptoms.

RESULTS: Of 193 screened patients, 186 met inclusion criteria. Over half (55.9%, n = 104) screened symptomatic or positive at discharge. On univariate analysis, greater injury severity score and longer length of stay were associated with symptomatic/positive screening. In multivariable analysis, injury severity remained the only significant predictor (aOR 1.07 per point increase in injury severit […]

PubMed · DOI

Radiographic assessment of metacarpal shortening after nonoperative and operative treatment of spiral and oblique metacarpal shaft fractures.

Stalberg Ostwald C, Peyronson F, Hailer NP, Muder D

BACKGROUND AND PURPOSE: Spiral or oblique metacarpal shaft fractures (MSFs) can be treated nonoperatively with unrestricted mobilisation; however, this approach leads to bone shortening upon healing. While an association between bone shortening and impaired hand function is suspected, empirical clinical evidence is lacking, and differing measurement methods make comparisons between studies difficult.

METHODS: This study is based on a midterm follow-up (mean 4.5 years) of 33 patients with a single spiral or oblique MSF from our previous randomised controlled trial. The original study and the present study were registered at clinicaltrials.gov (NCT03067454, NCT05869331). The main outcome in the present study was the amount of shortening following nonoperative compared with operative treatment. Bilateral radiographs were obtained, and shortening was determined by comparing the fractured metacarpal to its contralateral counterpart using the Shortening Absolute (SH-Abs) measurement method. Intra- and interrater reliability for SH-Abs were presented with intraclass correlation coefficients. Additional outcomes included extension deficit, cosmesis, and overall satisfaction.

RESULTS: 17 patients had been treated nonoperatively with unrestricted mobilisation and 16 operatively. Mean shortening was 3.9 mm (95% confidence interval [CI], 4.8-3.0 mm) and 0.9 mm (95% CI, 2.0 to -0.3 mm), r […]

PubMed · DOI

Proximal humerus fractures: A review of current management evidence and guidance on contemporary surgical strategies.

Hoskins W, Haase D

Proximal humerus fractures (PHFs) most commonly affect elderly, osteoporotic women, although younger patients sustain them through high-energy trauma. Both incidence and the associated healthcare burden continue to rise with demographic ageing. Non-operative management is preferred for undisplaced fractures and for many elderly patients, whereas a range of internal fixation methods and arthroplasty options are available for more complex injuries. Patient age, comorbidities, functional demands and fracture pattern are the primary determinants of treatment selection, implant choice and expected outcome. Absolute indications for surgery include acute fracture-dislocations, open fractures and injuries with vascular or neurological compromise. Relative indications are based on the degree of displacement and the predicted failure of non-operative treatment, and include head-splitting fractures, severe varus, valgus or posterior tilt of the humeral head, anatomical neck fractures, calcar comminution, complete displacement of the humeral head from the shaft, significant displacement of the lesser or greater tuberosity, displaced three- and four-part fractures, and large impaction defects of the humeral head, particularly when associated with instability. It is important to recognise that the displacement thresholds in routine use are pragmatic and consensus-derived rather than validate […]

PubMed · DOI

Patient centered outcomes after surgical stabilization of rib fractures: A prospective cohort study.

Laredo J, Walker M, Wilson F, Hertslet G, Heidel E, Lange K, Cavalea A, Gerard J et al.

INTRODUCTION: Rib fractures are associated with pain, impaired physical function, and reduced quality of life. Surgical stabilization of rib fractures (SSRF) is established for flail chest and ventilator-dependent patients, but its impact on longer-term patient-reported outcomes in broader injury patterns remains unclear. We evaluated whether SSRF is associated with improved physical functioning and pain outcomes compared to nonoperative management.

METHODS: We conducted a prospective observational cohort study of trauma ICU patients with rib fractures admitted between June 2023 and August 2024 at a single Level I trauma center. Patients met institutional criteria for SSRF, including flail chest, ≥ 3 contiguous rib fractures, or ventilator failure. Outcomes were assessed using the SF-36 and SF-MPQ-2 at discharge, 2 weeks, 3 months, and 6 months. Multivariable linear regression adjusted for RibScore, flail chest, ISS, age, obesity, hospital length of stay, and associated injuries. Primary outcomes were SF-36 physical functioning and SF-MPQ-2 present pain intensity (PPI).

RESULTS: Of 121 patients, 39 (32%) underwent SSRF. The SSRF group had greater chest injury severity, including higher RibScores and more pneumothorax and hemothorax, while overall injury severity and demographics were similar. There were no differences in ICU length of stay, hospital length of stay, ventilator […]

PubMed · DOI

Clinical equipoise as inclusion criterion to control for confounding in observational studies.

van der Hoeven NM, Peuker F, Michelitsch C, Smeeing DPJ, Houwert RM, Beeres FJP, Groenwold RHH, van de Wall BJM et al.

PURPOSE: This study investigates whether higher levels of expert disagreement, as a proxy for clinical equipoise, are associated with improved baseline comparability between treatment groups in patients with proximal humerus fractures. Additionally, it aimed to assess how progressively stricter inclusion criteria-based on this disagreement-affect the distribution of baseline characteristics between the treatment groups.

METHODS: We used data from the LADON Humerus study, a multicentre, prospective cohort including patients treated operatively or non-operatively in five hospitals in the Netherlands and Switzerland. An international expert panel of trauma surgeons assessed anonymized cases, blinded to the actual treatment, and each expert gave a treatment recommendation. Disagreement between the treatment given and the expert recommendations was categorized into four levels (A: complete disagreement, B: partial disagreement, C: partial agreement, D: complete agreement). Standardized differences (StDiff) for age, sex, ASA score, and AO classification were used to assess

comparability between treatment groups across disagreement levels.

RESULTS: A total of 351 patients were included in the analysis. Comparability between treatment groups improved as the level of disagreement increased. At complete disagreement (Level A), the groups were well balanced with respect to age, sex, and […]

PubMed · DOI

A tourniquet is a good thing for continued arterial bleeding in a severely traumatized lower limb.

Rey C, Makkar M, Buckley R

PubMed · DOI

Predictors of prolonged emergency department length of stay in severely injured pediatric patients.

Cevik AA, Alao DO, Farooq M, Abu-Zidan F

BACKGROUND: Prolonged emergency department length of stay (ED-LOS) in severely injured pediatric patients has been associated with adverse clinical outcomes. The determinants of the prolonged length of stay in severely injured pediatric patients remain incompletely understood.

OBJECTIVES: This study aims to identify patient-, injury-, and system-level predictors of prolonged ED-LOS among severely injured pediatric trauma patients. Data for this analysis were obtained from a multicenter trauma registry.

METHODS: We retrospectively analyzed prospectively collected data from the Abu Dhabi Trauma Registry over a 10-year period (2014-2023). Patients included were under 18 years old with an Injury Severity Score (ISS) over 12. We defined prolonged ED-LOS as more than 4 h. We analyzed demographic, clinical, institutional, and temporal variables. Multivariable logistic regression identified independent predictors of prolonged ED-LOS.

RESULTS: We included 874 pediatric patients, with a median ED-LOS of 227.5 min. Prolonged ED-LOS was independently associated with treatment in teaching hospitals (OR 2.25, 95% CI 1.43-3.55), absence of trauma team activation (OR 2.08, 95% CI 1.35-3.22), and non-local nationality (OR 1.66, 95% CI 1.12-2.45). Prolonged ED-LOS was also linked to specific injury mechanisms: road traffic collisions (OR 13.89, 95% CI 1.75-110.00) and low-height falls (OR 13. […]

PubMed · DOI

Effect of a stabilized liquid ozone-containing formulation delivered through a hyaluronic acid/poly-D,L-lactide scaffold on early osteochondral repair in a rat model.

K■zkapan TB, O■uzkaya S, M■s■r A, Eken G, Sayer G, Bayrakç■o■lu ÖF, Tirink YM, O■uzkaya H et al.

BACKGROUND: Articular cartilage has limited intrinsic repair capacity, and osteochondral defects may progress to degenerative joint disease. Ozone has anti-inflammatory properties but is highly reactive and short-lived in gaseous form. This study evaluated whether a stabilized liquid ozone-containing formulation delivered through a hyaluronic acid/poly-D,L-lactide (HA/PDL) scaffold improves early histological repair of osteochondral defects.

METHODS: Forty female Sprague-Dawley rats were randomly allocated to four groups (n = 10 each): healthy reference, untreated defect, HA/PDL scaffold, and HA/PDL scaffold impregnated with the stabilized liquid ozone-containing formulation. A 1-mm-diameter, 3-mm-deep osteochondral defect was created in the right femoral intercondylar region. Animals were euthanized on postoperative day 45. Histological repair was evaluated using the Modified Mankin score (0-14), and residual cartilage damage area was quantified in mm^2 on the index haematoxylin and eosin (H&E)-stained section using calibrated ImageJ analysis.

RESULTS: All animals completed the study. Because the healthy reference group received zero scores for all histological variables, inferential analyses were restricted to the operated groups. Total Modified Mankin scores differed significantly among groups ($p = 0.001$), with the ozone-loaded scaffold group demonstrating lower scores than […]

PubMed · DOI

Evaluation of the wound healing effects of a hydrogel containing hydroalcoholic extract of Chelidonium majus in a rat model.

Farabi MM, Valiallahi J, Rezanejad-Gatabi Z, Livani Z, Mirhoseini M

This study aimed to develop an innovative wound dressing by loading Chelidonium majus L. extract into an alginate hydrogel matrix and to assess its efficacy in wound healing. The extract of C. majus, famous for its anti-inflammatory and antioxidant features, was obtained from plant material collected in Mazandaran Province, Iran, and incorporated into sodium alginate hydrogels at concentrations of 2% and 5% w/w. The hydrogel formulations were characterized in terms of their morphological, thermal, and swelling properties, demonstrating interconnected porous structures, altered thermal behavior indicating increased amorphous content, and high moisture absorption capacity. To evaluate biological activity, the antioxidant and flavonoid content of the extract were measured, showing notable free radical scavenging ability and high levels of flavonoids. An in vivo wound healing model using adult male Wistar rats compared the effectiveness of control, bare hydrogel, C. majus-loaded hydrogels (2% and 5%), and diltiazem treatments. Wounds treated with the 5% C. majus hydrogel showed accelerated closure rates and improved tissue regeneration, as confirmed by histological examination and enhanced expression of proliferative and regenerative genes (TGF-β) alongside decreased inflammatory markers (TNF-α and IL-1β), in comparison with the control group. These results indicate that C. majus-l […]

PubMed · DOI

Fixation-specific modeling of age-frailty interactions to predict early mortality after hip fracture surgery.

Joachim K, Lin A, Gettleman B, Fice M, Hamad C, Abernethy C, Jeong S, Perrotta A et al.

BACKGROUND: Early mortality after hip fracture surgery remains high. Existing risk models treat age and frailty as independent predictors and may not reflect how physiologic reserve modifies risk across fixation strategies. We evaluated whether modeling age-frailty interactions improves estimation of 30-day mortality.

METHODS: A retrospective cohort study using the American College of Surgeons National Surgical Quality Improvement Program (ACS-NSQIP) database (2019-2023) included adults undergoing intramedullary nailing (IMN), hemiarthroplasty, or plate fixation for femoral neck, pertrochanteric, or subtrochanteric fractures. Frailty was measured using a validated 5-item modified frailty index. Generalized additive models with bivariate spline surfaces generated procedure-specific mortality risk estimates. Secondary outcomes were assessed using propensity score-matched cohorts.

RESULTS: Among 115,643 patients, 30-day mortality was 5.25%. Predicted mortality varied across age-frailty combinations, ranging from 1.41% to 28.83% for IMN, 1.26%-32.06% for hemiarthroplasty, and 0.48%-20.37% for plate fixation. IMN showed higher transfusion rates, while hemiarthroplasty demonstrated higher cardiopulmonary complication rates. Length of stay did not differ between procedures.

CONCLUSIONS: Modeling age and frailty as interacting variables produces procedure-specific mortality risk est […]

PubMed · DOI

Comparative preclinical evaluation of an investigational topical hemostatic formulation against an active commercial comparator in rat arterial, venous and liver bleeding models under normal and heparinized conditions.

Kaplan M, Çoban A, Sorar M

BACKGROUND: An ideal topical hemostatic agent should provide rapid hemostasis, maintain clot stability after manipulation, and preserve a clean operative field. These requirements become particularly relevant under anticoagulated conditions and across different bleeding anatomies.

OBJECTIVE: To evaluate the preclinical performance of a topical hemostatic agent compared with a commercial comparator and saline control in standardized rat arterial, venous, and liver bleeding models under normal and heparinized conditions.

METHODS: A total of 144 adult male Wistar albino rats were allocated to 18 groups (n = 8 per group). Standardized abdominal aortic, vena cava, and liver bleeding models were established under normal and

heparinized conditions. Vascular injuries were created with a 21 G needle and liver injury with a standardized 6 mm × 5 mm incision. Treatment was followed by a standardized 3-minute compression protocol. The primary endpoint was time to hemostasis. Liver blood loss was assessed gravimetrically. Rebleeding tendency and operative field quality were evaluated qualitatively.

RESULTS: One-way ANOVA demonstrated overall between-group differences in time to hemostasis in all model-condition combinations (all $P < 0.001$). Across all anatomical models and coagulation settings, the investigational hemostatic agent consistently showed the shortest mean time to hemostasis, […]

PubMed · DOI

Prehospital whole blood transfusion in traumatic hemorrhagic shock: a systematic review and meta-analysis.

Ribeiro Junior MAF, Possiedi RD, Pacheco LS, de Cesaro Schpchacki N, Vieira LF

BACKGROUND/OBJECTIVES: Prehospital whole blood (WB) is increasingly used for traumatic hemorrhagic shock due to logistical and physiological advantages. This study synthesized evidence, including recent RCTs, to clarify its clinical efficacy.

METHODS: Six major databases were searched through May 22, 2026. Eligible studies included RCTs and comparative cohorts of adult patients receiving prehospital WB versus standard care. Primary outcomes were 24-hour and 28/30-day mortality. Data were pooled using random-effects models to calculate risk ratios (RR) and mean differences (MD) with 95% CIs.

RESULTS: Synthesis of six studies (3 RCTs, 3 cohorts) involving 4040 patients (1558 WB; 2482 standard care) showed prehospital WB did not significantly decrease 24-hour mortality (RR 1.00; $P = 0.98$) or 28/30-day mortality (RR 1.09; $P = 0.30$). No differences were found in 6-hour or in-hospital mortality. However, WB significantly reduced red blood cell requirements within 24 h of hospital arrival (MD -0.97; $P = 0.03$), though this secondary signal lost statistical significance during sensitivity analysis. Safety outcomes (ARDS, stroke, thrombotic events) showed no significant differences, although estimates for rare events were imprecise. Certainty of evidence ranged from very low to moderate.

CONCLUSION: Prehospital WB was not associated with improved short- or long-term survival compared […]

PubMed · DOI

Fluoroscopy-assisted percutaneous management of posterior pelvic ring injuries via multiple transsacral and iliosacral screws in the supine position.

Zheng L, Shen X, Huang X, Chen J, Cao W, Yu L, Yang G, Pan Y

INTRODUCTION: When patients are positioned supine for percutaneous screw insertion to achieve posterior pelvic ring fixation, the edge of the operating table can impede screw insertion trajectories, thereby increasing the risk of cortical perforation, especially during iliosacral screw (ISS) placement. Multiple screw insertion might also potentially increase the risk of cortical perforation. The aim of this study was to evaluate the feasibility, safety, and efficacy of inserting multiple (≥2) transsacral screw (TSS) and ISS in the supine position via the application of specific surgical strategies and techniques developed in our institution.

PATIENTS AND METHODS: Patients who underwent stabilization of posterior pelvic ring injuries via percutaneous insertion of multiple TSS and/or ISS in the supine position between May 2018 and December 2025 were included in this study. Distinct screw insertion combination patterns at different vertebral levels, which were primarily determined by the variable anterior-posterior width of the transverse osseous pathway at the first sacrum, were employed for posterior pelvic ring fixation. Wound infection and neurovascular complications associated with screws insertion procedure were recorded. Postoperative CT scans were performed to assess the position of the screws. At follow-up, radiographic evaluations were performed to evaluate the complica […]

PubMed · DOI

MicroRNAs regulate viability, longevity and extracellular matrix secretions of mesenchymal stem cell-derived chondrocytes in vitro.

Al-Shaibani MBH

BACKGROUND: Chondrogenic differentiation of human adipose tissue-derived mesenchymal stem cells (ADMSCs) is a promising strategy for cartilage regeneration, but its regulatory mechanisms remain incompletely understood. MicroRNAs (miRNAs) play critical roles in post-transcriptional control of lineage commitment and tissue development. This study aimed to identify miRNAs differentially expressed during ADMSCs chondrogenesis and explore their functional roles in chondrogenic gene regulation.

METHODS: ADMSCs were induced to undergo chondrogenic differentiation under defined conditions. miRNA expression profiles were analyzed, and significantly dysregulated miRNAs were identified. Bioinformatic and literature-based analyses were used to associate these miRNAs with chondrogenic transcription factors, signaling pathways, and cellular processes.

RESULTS: Chondrogenesis was associated with distinct patterns of miRNA regulation. miR-140-5p, miR-520d-5p, miR-127-5p, and miR-218-5p were upregulated and correlated with increased expression of cartilage markers (SOX9, COL2A1, ACAN) and suppression of hypertrophy. These miRNAs were linked to TGF-β signaling, epigenetic regulation, and inhibition of terminal differentiation. Conversely, miR-145-5p, miR-101-3p, miR-194-5p, miR-199a-5p, miR-182-5p, miR-29a-5p, miR-20a-5p, miR-193b-5p, and miR-574-3p were downregulated, potentially relieving re […]

PubMed · DOI

CCBE1 promotes wound healing in diabetic rats by improving the functional impairment of AD-MSCs induced by AGEs.

Liang F, Mo J, Zhang J, Peng Z, Zhou J, Lu Y, Li Y, Wang K et al.

OBJECTIVE: The etiology of chronic and non-healing diabetic wounds is manifold; the progression of the non-healing phenotype is strongly associated with poor vascular networks. Mesenchymal stem cells (MSCs) are reckoned as a promising therapy for wound healing, especially in angiogenesis. Here, we explored the role of CCBE1 in diabetic wound healing using human adipose-derived MSCs (hAD-MSCs).

METHODS: Human adipose-derived hAD-MSCs were isolated and expanded through serial in vitro passaging. Cell proliferation and migration were evaluated using Cell Counting Kit-8 (CCK8) and scratch wound assays, respectively. collagen, and calcium-binding EGF domain-containing protein 1 (CCBE1) and vascular endothelial growth factor (VEGF) expression were analyzed via western blotting. Functional studies of CCBE1 were conducted using siRNA-mediated knockdown and plasmid-driven overexpression approaches. Type 2 diabetes mellitus was induced in Sprague-Dawley rats through a high-fat/high-glucose diet combined with streptozotocin administration. Full-thickness excisional wounds were generated on the dorsal skin, and the effects of CCBE1 modulation in hAD-MSCs on wound closure were examined. Reepithelialization and angiogenesis were analyzed using histological and immunohistochemical methods. Protein levels of CCBE1 and vascular endothelial growth factor (VEGF) were quantified by western blotti […]

PubMed · DOI

N-acetyl cysteine (NAC) preconditioning enhances the efficacy of bone marrow derived mesenchymal stem cells for healing of skin burn wounds.

Ali F, Wajid N, Nadeem MS

Burn injuries are associated with excessive oxidative stress and inflammation that impair mesenchymal stem cell (MSC) survival and regenerative function. This study investigated whether N-acetylcysteine (NAC) preconditioning could enhance the therapeutic efficacy of bone marrow-derived MSCs (BMSCs) in oxidative stress conditions and burn wound healing. In vitro, BMSCs were preconditioned with NAC (5 or 10 mM) prior to H■O■ exposure, and cell viability, proliferation, oxidative stress markers, and regenerative gene and protein expression were evaluated. In vivo, BMSCs isolated from male Sprague Dawley rats were transplanted into second-degree thermal burn wounds induced in female rats, and wound healing was assessed over 21 days using macroscopic, histological, biochemical, and molecular analyses. NAC preconditioning significantly improved BMSC viability, proliferation, and morphology under oxidative stress while reducing LDH release. NAC-treated BMSCs exhibited increased antioxidant enzyme activities (SOD, CAT, and GSH), reduced malondialdehyde (MDA) levels, and enhanced expression of PCNA, SDF-1, CXCR4, VEGF-A, and IGF-1. In vivo, transplantation of NAC-preconditioned BMSCs accelerated wound contraction, improved re-epithelialization, increased hydroxyproline content, and promoted collagen deposition and tissue regeneration. Furthermore, NAC-preconditioned BMSCs enhanced antio […]

PubMed · DOI

Temporary internal stabilisation for staged reconstruction of critical long-bone defects: A biomechanical study.

Baida MV, Povkh VI, Derkach SO, Zilnyk RR

BACKGROUND: Critical long-bone defects remain a major challenge in orthopaedic trauma and reconstructive surgery, particularly following high-energy injury, infection, and extensive segmental bone loss. Their management frequently requires staged reconstruction with temporary skeletal stabilisation between initial debridement and definitive reconstruction. External fixation remains an established strategy during this interval, while temporary internal fixation represents an alternative approach. However, biomechanical evidence regarding purpose-designed modular intramedullary constructs used in conjunction with polymethylmethacrylate (PMMA) spacers remains limited.

AIM: To evaluate the axial and bending mechanical characteristics of a modular intramedullary temporary stabilisation system used with a PMMA spacer and to compare its mechanical behaviour with selected plate and external fixation constructs in segmental long-bone defect models.

MATERIALS AND METHODS: Five synthetic long-bone models were evaluated, with one specimen assigned to each experimental configuration. Three 50-mm femoral defect models were stabilised using plate osteosynthesis, external fixation, or modular intramedullary temporary stabilisation. Two additional modular intramedullary configurations comprised a 400-mm femoral defect and a 50-mm humeral defect. Each specimen underwent sequential quasi-static […]

PubMed · DOI

Locked plate fixation versus non-operative management for three and four-part proximal humeral fractures in elderly; A randomized controlled trial.

Abouelsoud M, Wahab MMA, Kamal HM, Mohamed AAA

BACKGROUND: In elderly patients, the proximal humerus fracture is the third most prevalent fracture, following fractures of the hip and distal radius, and is significantly associated with osteoporosis. This research aimed to determine the functional and radiological outcomes of internal fixation of 3- and 4-part proximal humeral fractures by locked plates compared with the outcomes obtained by non-operative management in the elderly.

METHODS: In this prospective randomized trial, 52 elderly patients (≥65 years) with 3- or 4-part proximal humerus fractures were allocated to either nonoperative treatment with arm sling immobilization or operative fixation using a proximal humerus locking plate in Qena General Hospital from December 2020 to December 2023. Clinical and radiological assessments were conducted up to 12 months post-treatment, with functional outcomes measured by DASH and Constant scores.

RESULTS: Six months after surgery, the Conservative group had a significantly higher DASH score than the Operative group. At 12 months, the Conservative group had a lower DASH score and a higher total Constant score ($p < 0.05$). There was no significant variance in the incidence of SAI, stiffness, deformity, or AVN among the conservative and operative groups ($p > 0.05$). The complication rate was 13.4%, with 5.7% of cases related to conservative treatment and 7.7% to operative treatme […]

PubMed · DOI

Medial clavicle resection for acute traumatic posterior sternoclavicular dislocation: Experience from Grenoble Alpes University Hospital.

Datry V, Gagnepain E, Guillot M, Pirvu A, Frey G

BACKGROUND: Posterior sternoclavicular dislocations (PSCDs) are rare but potentially life-threatening injuries. Beyond the acute management of mediastinal complications, restoring optimal shoulder function remains the primary goal of treatment. Although the optimal surgical strategy remains controversial, our institutional protocol consists of systematic medial clavicle resection performed during emergency surgery, with the aim of reducing the risk of residual sternoclavicular instability. The aim of this study was to evaluate the functional outcomes and complications of systematic medial clavicle resection performed as an emergency procedure for acute traumatic PSCDs.

METHODS: A retrospective single-centre study was conducted at Grenoble Alpes University Hospital (2019-2026). All consecutive patients who underwent medial clavicle resection for acute traumatic PSCDs were included. The primary outcome was the QuickDASH score at 12 months. Secondary outcomes included the QuickDASH score at 1-month, postoperative complications according to the Clavien-Dindo classification, and return-to-work and return-to-sport outcomes.

RESULTS: Twenty-five patients (23 males, 2 females; mean age 29 years) were included. Sports injuries accounted for 84% of cases. The median time from injury to surgery was 2 days [IQR, 1-6]. The median operative time was 33 min [IQR, 24-45]. No intraoperative c […]

PubMed · DOI

Operative versus nonoperative management of pelvic ring fractures in older adults: A propensity-matched analysis of mortality and complications.

Johnson M, Sabat L, Allam J, Perdue P, Zuelzer D

INTRODUCTION: Pelvic ring fractures are an increasingly common injury in the aging population, carrying mortality comparable to hip fractures and substantial functional decline. Most geriatric fractures are managed nonoperatively, with operative fixation reserved for unstable patterns, but high-quality evidence comparing both treatment strategies is lacking. This study compared mortality and complications between operatively and nonoperatively managed pelvic ring fractures in older adults across multiple time points.

METHODS: This retrospective cohort study used the TriNetX database to identify patients aged 55 years or older with a pelvic ring fracture. Patients were assigned to operative or nonoperative cohorts and matched 1:1. Co-primary outcomes were 30-day and 1-year all-cause mortality. Secondary outcomes were mortality at 90 days and 2 years, and VTE, pneumonia, and MI at each time window. A multivariable Cox model was constructed for 1-year mortality.

RESULTS: Following matching, 5314 patients were retained in each cohort. Mortality did not differ at 30 days ($p = 0.108$) or 90 days, but was significantly lower in the operative cohort at 1 year ($p = 0.034$) and 2 years ($p < .001$). This benefit did not persist in the multivariable Cox model (HR 0.939; $p = 0.296$). VTE and pneumonia were significantly more common in the operative cohort at all time points, while MI did not […]

PubMed · DOI

Fracture-related infection after lateral versus posterolateral approaches in AO/OTA type 44B fractures: A retrospective cohort study.

Achá J, Wahab-Zuriarrain S, Pilco-Inga J, Millán-Billi A, Carrera I, De Caso J, Gómez-Masdeu M

BACKGROUND: Postoperative wound complications remain a relevant concern after open reduction and internal fixation (ORIF) of ankle fractures. The posterolateral approach has gained popularity for AO/OTA type 44B ankle fractures due to improved access to the posterior malleolus; however, concerns persist regarding a potentially increased risk of fracture-related infection (FRI) compared with the traditional lateral approach. The aim of this study was to compare postoperative infection rates between lateral and posterolateral approaches using the international FRI consensus definition.

METHODS: A retrospective cohort study was conducted including patients with AO/OTA type 44B ankle fractures treated surgically with ORIF at a Level I trauma center between 2019 and 2024. Median follow-up was 14 months. Patients were grouped according to surgical approach (lateral vs. posterolateral). The primary outcome was FRI. Secondary outcomes included minor wound complications, reoperation, and identification of patient and surgery-related risk factors. Multivariate logistic regression analyses were performed to identify factors independently associated with FRI. A predefined subgroup analysis evaluated the impact of posterior malleolar fixation in patients with a posterior malleolar fracture.

RESULTS: A total of 427 patients were included: 300 treated with a lateral approach and 127 with a […]

PubMed · DOI

Determinants of road traffic injury preventive behaviors among old pedestrians: An application of the PRECEDE model.

Bashirian S, Ayubi E, Nezafati A, Razi T, Hosseinalipour SA, Karami S, Rezapur-Shahkolai F

OBJECTIVES: Nowadays, road traffic injuries (RTIs), a major public health concern in many countries, have gained even greater significance among elderly, especially among elderly pedestrians who are one of important vulnerable groups of road users. This study aimed to identify factors influencing RTI preventive behaviors in older pedestrians using the PRECEDE model.

METHODS: A cross-sectional study was conducted in 2024 with 417 adults aged 60 years or older attending comprehensive health centers in Hamadan, located in the west part of Iran. Participants were selected using a multi-stage cluster sampling method. Data were collected via a validated, self-administered questionnaire based on the PRECEDE model, covering demographic characteristics, behavioral and environmental factors, as well as predisposing, reinforcing, and enabling factors.

RESULTS: The study population consisted of 240 (57.6%) males and 177 (42.4%) females. Multi-variable analysis showed that environmental factors ($\beta = 0.16$, $p < 0.01$) and reinforcing factors ($\beta = 0.12$, $p = 0.04$) were significant positive predictors of preventive behaviors. Additionally, education level, marital status, hearing aid use, and insurance coverage demonstrated significant associations with RTI preventive practices ($p < 0.05$).

CONCLUSIONS: The PRECEDE model offers a useful framework for understanding RTI preventive behaviors among […]

PubMed · DOI

Structures to improve the implementation of mental health practices in adult trauma centers: A Canadian nationwide practice survey.

Bérubé M, Macedo T, O'Donnell M, Laberge-Sévigny M, Moore L, Stelfox HT, Rivest M, Kureshi N et al.

BACKGROUND: Up to one third of trauma patients experience mental disorders, placing them at greater risk of morbidity and healthcare utilization. However, few studies have examined knowledge-to-practice gaps when it comes to improving their mental health. This study aims to assess how trauma program leaders perceive the implementation of structures to facilitate the adoption of recommendations aimed at improving mental health outcomes in trauma patients.

METHODS: We conducted a cross-sectional practice survey across Levels I to III Canadian trauma centers (n = 69). The survey focused on empirically supported structures for optimizing mental health practices. Responses were analyzed using descriptive statistics and content analysis.

RESULTS: The overall response rate was 62%, with 75% in Level I trauma centers, 91% in Level II centers, and 47% in Level III centers. The structures most frequently perceived as having been implemented were: referral mechanisms to a rehabilitation program to improve psychological outcomes in severely injured patients (58%), referral mechanisms to mental health resources for patients with a psychological condition that contributed to the injury (56%), protocols for screening suicidal ideation (56%) and substance use disorders (51%), protocols for the pharmacological treatment of aggression in patients with traumatic brain injury (TBI) (51%), and fo […]

PubMed · DOI

Trends, disparities, and forecasts of osteosarcoma mortality in the United States, 1999-2035.

Gu Y, Wang T, Liu B

BACKGROUND: Osteosarcoma-related mortality in the United States remains a concern, particularly in the context of population aging.

METHODS: We used the CDC WONDER Underlying Cause of Death database (1999-2023) to identify osteosarcoma deaths (ICD-10 C40-C41). Crude mortality rates and age-adjusted mortality rates (AAMRs) per 100,000 (standardized to the 2000 U.S. population) were calculated for individuals aged ≥ 15 years and stratified by sex, age group, race/ethnicity, census region, and urban-rural classification. Temporal trends, annual percent change (APC), and average annual percent change (AAPC) were estimated using Joinpoint regression. Future mortality trends through 2035 were projected using autoregressive integrated moving average (ARIMA) and exponential smoothing state-space (ETS) models.

RESULTS: From 1999-2023, the overall AAMR increased from 0.52 to 0.59 per 100,000 (AAPC 0.67; 95% CI 0.34-0.99). Men consistently had higher mortality than women. Mortality was strongly age-graded, peaking among adults aged ≥ 85 years, whereas rates in those aged 15-24 remained essentially stable. Non-Hispanic White

individuals and residents of the South had the highest AAMRs, while the Northeast showed the lowest. Nonmetropolitan areas exhibited higher AAMRs than metropolitan areas, despite faster recent increases in cities. Forecasts based on ARIMA and ETS models indicate that […]

PubMed · DOI

Disparities in traumatic brain injury-related deaths - United States, 2023.

Waltzman D, Daugherty JD, Thomas KE

INTRODUCTION: Traumatic brain injury (TBI) is a significant public health problem. Conducting regular surveillance of TBI-related deaths is important in order to determine which groups are at disproportionate risk of dying from a TBI, tracking trends over time, and informing prevention efforts.

METHODS: TBI-related deaths from 2023 were analyzed using multiple-cause-of-death data from the National Vital Statistics System by age, sex, race/ethnicity, injury intent/mechanism of injury, and geography. TBI-related deaths were identified if a death had an injury-related International Classification of Diseases, Tenth Revision underlying cause of death and a TBI-related diagnosis code in one of the multiple-cause-of-death fields.

RESULTS: In 2023, there were 68,663 TBI-related deaths in the United States. Older adults aged ≥ 75 years (82.5 per 100,000 population), males (29.1), non-Hispanic American Indian/Alaska Native (AI/AN) individuals (29.7), and several states in the West US Census region displayed the highest rates of TBI-related deaths compared to their respective groups. Suicide (rate of 7.0 [36.0% of all TBI-related deaths]) was the most common cause of TBI-related deaths in the United States in 2023, followed by unintentional falls (rate of 5.2 [31.6% of all TBI-related deaths]).

CONCLUSIONS: Results show that deaths due to TBI are common but not all groups in the Unite […]

PubMed · DOI

Understanding the outliers: A qualitative analysis of leaders' perspectives on performance and safety in trauma centers.

Hamad DM, Conn LG, Haas B, Jerath A, Nathens AB

INTRODUCTION: The factors explaining variability in trauma center performance across quality measures are not well understood. We aimed to characterize the influence of social, cultural, and organizational factors on trauma center performance.

METHODS: We used an explanatory sequential design to study ACS TQIP-participating level I and II trauma centers that were ranked as high- and low-performing (first and last decile) across multiple representative quality indicators. The Chief Safety Officers, Directors of Quality and Safety, Trauma Medical Directors, and Trauma Program Managers within these centers were approached for semi-structured interviews guided by the framework of high-reliability organizations (HROs). Qualitative description methodology was used. Salient themes were inductively developed, and themes between centers in the first and last deciles were contrasted.

RESULTS: We interviewed 28 participants across 15 centers. Of these, 8 centers were in the first decile, and 7 centers were in the last decile. All participants expressed a commitment to making care safer and to improving hospital performance. Leaders in first decile centers more strongly expressed having adopted and used theory pertinent to safety and high reliability organizations (HROs). This manifested as an increased preoccupation with failure and with the ongoing and never-ending maintenance of safe […]

PubMed · DOI

Epidemiology of hand fractures in children - A population-based study with data from the Danish National Patient Registry.

Raja A, Faergemann C, Gundtoft P, Viberg B

INTRODUCTION: Hand fractures are among the most common pediatric injuries, but despite their prevalence, comprehensive data on incidence and anatomical distribution remain sparse. The aim of this study was to report the incidence and anatomical distribution of hand fractures in children aged 0-15 years.

METHODS: This population-based register study utilized data from the Danish National Patient Registry (DNPR) over a 20-year period from 1999 to 2018. All cases of hand fractures were identified using ICD-10 codes and

stratified by age, sex, anatomical site, and treatment type. Fracture incidences were calculated based on annual population data, gathered from Statistic Denmark. Data was analyzed using descriptive statistics and results presented in the form of number of cases and percentages.

RESULTS: A total of 158,070 hand fractures were recorded, with an average annual incidence of 748 per 100,000 children/year and a 17% increase in incidence over the study period. Boys accounted for 59% of cases, and the highest incidence was observed in the 12-15-year age group (51%). Finger fractures were the most common (60%) followed by metacarpal fractures (19%). There were 1% that received surgical treatment.

CONCLUSION: We found a 17% increase in incidence over the 20-year period. The highest incidence was in the 12-15-year-olds. Boys had the majority of fractures, and finger fractu […]

PubMed · DOI

Comparison of anterior-only and combined anterior-posterior approaches for complex acetabular fractures.

Kaptan AY, Dolap MA, Göçer E, Demir M, Varol R, Bal T, Durmaz Ü, Y■ld■z AE

BACKGROUND: Complex acetabular fractures involving both columns (T-type, both-column, and anterior column with posterior hemitransverse) often require combined anterior-posterior surgical exposures. Advances in anterior intrapelvic fixation raise the possibility that the posterior column can be reduced and stabilized from the intrapelvic side alone in selected cases, avoiding an additional posterior exposure. We compared radiological and clinical outcomes between an anterior-only intrapelvic strategy and a combined anterior-posterior strategy.

METHODS: This retrospective comparative study included 31 patients with T-type, both-column, or ACPHT acetabular fractures: 12 treated through the modified Stoppa approach alone (anterior-only) and 19 treated with combined Kocher-Langenbeck and anterior intrapelvic exposures (combined). Postoperative CT-based reduction quality (Matta-based thresholds), Harris Hip Score (HHS), Majeed Pelvic Score, complications, and perioperative transfusion requirements were compared between groups.

RESULTS: Anatomic reduction was achieved in 50.0% of the anterior-only group and 36.8% of the combined group (p = 0.710). Native-hip HHS (83.0 vs. 81.0, p = 0.229) and Majeed scores (76.0 vs. 79.0, p = 0.795) were comparable between groups. Total cumulative erythrocyte suspension transfusion was higher in the combined group (median 3 vs. 2 units, p = 0.003), […]

PubMed · DOI

Return to duty after suture anchor repair of thumb metacarpophalangeal ulnar collateral ligament injuries in active duty soldiers.

Lee J, Chung JI, Yun Y, Jeong C, Lee SM, Woo I, Son S, Cheon JH et al.

OBJECTIVE: Thumb metacarpophalangeal (MCP) ulnar collateral ligament (UCL) injury, commonly referred to as "Skier's thumb," is traditionally considered a sports-related injury. However, military personnel are frequently exposed to activities that may generate similar valgus stress mechanisms. This study aimed to evaluate the clinical characteristics, injury mechanisms, and surgical outcomes of thumb MCP UCL injuries in a South Korean military referal hospital.

METHODS: A retrospective review was conducted on 35 active duty military personnel who underwent surgical treatment for acute or subacute thumb MCP UCL injuries between May 2022 and May 2025. Patients with associated avulsion fractures requiring Kirschner-wire fixation or chronic injuries requiring ligament reconstruction were excluded. Demographic characteristics, injury mechanisms, presence of Stener lesions, clinical outcomes, return-to-duty status, and postoperative complications were analyzed. Surgical repair was performed using a suture anchor technique.

RESULTS: The mean follow-up period was 12.0 months. Rifle-related injuries accounted for 60.0% of all cases and represented the most common injury mechanism. Stener lesions were identified in 18 patients (51.4%). At final follow-up, the mean Visual Analog Scale score was 0.5, the mean Quick Disabilities of the Arm, Shoulder and Hand score was 4.2, and the mean Kap […]

PubMed · DOI

Occupational recovery after work-related AO Spine type A1 compression fractures: Association between treatment strategy and sick-leave duration.

Jordá-Gómez P, Saiz-Sapena N, Vanaclocha L, Barrios C, Vanaclocha V

PURPOSE: Return-to-work (RTW) is an important indicator of functional recovery after spinal trauma, yet occupational outcomes following stable thoracolumbar compression fractures remain poorly characterized. This study evaluated factors associated with sick-leave duration and occupational recovery in patients with work-related AO Spine type A1 vertebral compression fractures.

METHODS: This retrospective cohort study included adults with work-related AO Spine type A1 thoracolumbar compression fractures without neurological deficit treated at a tertiary referral center between 2007 and 2025. Patients underwent percutaneous vertebroplasty, orthopedic bracing, or posterior spinal arthrodesis. The primary outcome was sick-leave duration, defined as the interval between injury and resolution of work capacity. Multivariable linear regression using log-transformed sick-leave duration was performed to identify factors independently associated with occupational recovery.

RESULTS: Seventy patients were included: 35 underwent vertebroplasty, 18 orthopedic bracing, and 17 spinal arthrodesis. Sick-leave duration differed significantly among treatment groups ($p < 0.001$). Median sick-leave duration was 62 days (IQR 38-94) after vertebroplasty, 144.5 days (IQR 112-186) after orthopedic bracing, and 178 days (IQR 120-327) after spinal arthrodesis. In multivariable analysis, treatment strategy […]

PubMed · DOI

Disease burden of spinal fractures attributable to road injuries among working-age population in China during 1990-2023: An analysis based on the global burden of disease 2023.

He J, Liu S, Li M, Pan Z, Liang J, Deng M, Xiang H, Cai D et al.

BACKGROUND: Spinal fractures attributable to road traffic injuries impose a substantial disease burden on the working-age population. Nevertheless, the current affairs in the disease burden of spinal fractures among this demographic in China have not been systematically investigated.

OBJECTIVE: To evaluate the disease burden of spinal fractures and their subtypes attributable to road traffic injuries among the working-age population (15-69 years) in China from 1990 to 2023, and to forecast variation tendency through 2040.

METHODS: Leveraging data from the GBD 2023, data on the incidence rate, YLDs count, ASIR, and ASYR of spinal fractures and their subtypes triggered by road traffic injuries among China's working-age population during 1990-2023 were extracted. Joinpoint regression analysis and age-period-cohort models were employed to investigate trajectories in disease burden. The Bayesian age-period-cohort model was applied to project the disease burden from 2024 to 2040.

RESULTS: From 1990-2023, the incidence and YLDs counts of spinal fractures triggered by road injuries among China's working-age population elevated by 26.24% and 53.30%, respectively. ASIR and ASYR exhibited downward trajectories, decreasing to 17.34 per 100,000 (95% UI: 9.86-27.91) and 0.91 per 100,000 (95% UI: 0.56-1.38), with reductions of 3.34% and 7.14%. The disease burden among males consistently ex […]

PubMed · DOI

Anterior fix-and-replace strategy using direct anterior and pararectus approaches for acetabular fractures in older patients.

Ivanova S, Keel F, Keel MJ, Siebenrock KA, Tannast M, Bastian JD

BACKGROUND: Most reported fix-and-replace procedures use a posterior or lateral approach for total hip arthroplasty (THA) and a Kocher-Langenbeck, anterior intrapelvic, or ilioinguinal approach for acetabular fixation, which may require separate anterior and posterior exposures. Evidence on fully anterior, single-position workflows combining anterior acetabular fixation with THA remains limited.

AIM: The aim of this study was to assess the outcomes of acute THA through the direct anterior approach (DAA) combined with pararectus in patients considered suitable for a fix-and-replace strategy.

METHODS: This retrospective single-centre case series included 20 patients aged ≥ 60 years (mean age: 82 ± 10 years) selected for a fix-and-replace strategy based on predefined indications - such as acetabular dome destruction, femoral head damage, pre-existing hip degeneration - between 2009 and 2023. All underwent single-stage pararectus fixation and DAA THA in the supine position without posterior exposure or repositioning. Acetabular reconstruction constructs were used in 19 patients and dual-mobility articulation in five patients. The primary outcome was revision surgery within two years. Secondary outcomes included surgical data, 30-day complications, mortality, postoperative weight bearing, radiographic findings, mobility, pain, and patient satisfaction.

RESULTS: Revision surgery w […]

PubMed · DOI

Persistent postoperative hypercoagulability and venous thromboembolism risk following pelvic and acetabular fractures: A prospective pilot study.

Clarke A, Tera Y, Skeith L, Othman M, Dodd A, Duffy P, Korley R, Martin CR et al.

BACKGROUND: Pelvic and acetabular (PA) fractures are severe orthopaedic injuries associated with high risk of venous thromboembolism (VTE), a potentially life-threatening complication, despite current thromboprophylaxis strategies. The duration of postoperative hypercoagulability and associated VTE risk following surgical fixation remain unknown. Thromboelastography (TEG) analysis can assess the entire clotting process and can be used to quantify hypercoagulability and increased VTE risk through maximal amplitude (MA) and coagulation index (CI) parameters.

PURPOSE: To characterize the extent and duration of postoperative hypercoagulability and VTE risk following PA fractures using serial TEG analysis.

METHODS: Adults undergoing surgical management of PA fractures were prospectively enrolled. Serial TEG analysis was performed from admission to 3 months postoperatively. Standardized thromboprophylaxis consisted of low-molecular-weight heparin for 28 days postoperatively. Screening compression ultrasound was performed to detect asymptomatic proximal deep vein thrombosis (DVT) on postoperative day 3 (POD3). Mobility was measured using Timed Up and Go (TUG) test. One-sample t-tests compared mean MA and CI values at each time point with predefined hypercoagulable thresholds (MA ≥65 mm, CI >3).

RESULTS: Thirty patients were enrolled, including 22 men (median age: 52.5 years). Most […]

PubMed · DOI

Epidemiology of sports- and recreation-related shoulder and elbow injuries in older adults presenting to emergency departments in the United States between 2010 and 2024.

Adio AA, Boufadel P, Ilyas MH, Black EM, Hill BW, Kassam HF, Khan AZ, Horneff JG et al.

BACKGROUND: Participation in sports and recreational activity among adults aged 65 years and older continues to increase, yet contemporary data on shoulder and elbow injury patterns in this population remain limited. This study aimed to characterize national trends in demographics, diagnoses, seasonality, and sport-specific injury risk.

METHODS: The National Electronic Injury Surveillance System was queried from 2010 to 2024 for adults aged ≥ 65 years presenting with sports- or recreation-related shoulder or elbow injuries. Demographics, diagnoses, injury mechanisms, temporal trends, and sport-specific fracture risk were evaluated. Logistic regression assessed sex-specific odds, and narrative text was analyzed using tidytext.

RESULTS: An estimated 248,207 shoulder and 106,031 elbow injuries occurred nationally. Males accounted for 63.8% of shoulder and 60.3% of elbow injuries. Fractures (26.5%) and strains or sprains (24.0%) predominated in shoulder injuries, while fractures represented 32.1% of elbow cases. From 2010-2024, the proportion of these injuries increased, comprising approximately 8% of all emergency visits by 2024. Bicycling was the most common mechanism for both joints, followed by general exercise and weightlifting. Shoulder and elbow injuries peaked in summer months.

CONCLUSION: Sports-related shoulder and elbow injuries in older adults are rising, with fractu […]

PubMed · DOI

Reverse-engineering the diagnostic process: An indirect finding-driven deep learning model for pelvic active bleeding detection on CT that achieves specialist-level performance and surpasses residents - a multicenter study.

Okada N, Inoue S, Hirano Y, Liu C, Mitarai S, Honda S, Sato K, Hiraki S et al.

Contrast-enhanced computed tomography (CT) is useful for detecting active bleeding in patients with pelvic trauma. However, early detection of active bleeding from CT images is challenging due to the extensive image volume, which can delay intervention. Although the development of accurate and rapid detection support systems for pelvic active bleeding is highly anticipated, a significant data imbalance between bleeding and non-bleeding images in real-world hospital settings poses a significant challenge to clinically applicable models. We reverse-engineered the physician's diagnostic process and implemented a curriculum-based deep learning algorithm using indirect findings to prevent overfitting caused by the imbalance. In this multicenter study, we developed and implemented a multistage deep learning algorithm for the detection of active bleeding using enhanced CT of pelvic trauma patients. From April 18, 2008 to December 31, 2023, we collected 258,057 slices of whole-body CT images of 2178 patients for blunt trauma with pelvic injuries from five emergency centers. Two types of classification-based deep learning models were developed and evaluated: anatomical structure extraction (ASE) and active bleeding detection (ABD) models. An algorithm using these models was implemented, and its accuracy in detecting active bleeding in the pelvis and the inference speed were evaluated. F […]

PubMed · DOI

Robot-assisted fracture reduction and treatment outcomes in type III and IV fragility fractures of the pelvis: A retrospective cohort study.

Xiao H, Zhao C, Cao Q, Sun Q, Ma Y, Jian Y, Ming Y, Wang Y et al.

INTRODUCTION: This study compared robot-assisted fracture reduction (RAFR) with manual closed reduction followed by navigation-assisted fixation in surgically treated patients with fragility fractures of the pelvis (FFP) type III or IV. Outcomes after operative versus initial non-operative management were also examined.

PATIENTS AND METHODS: This retrospective cohort study included 115 patients aged ≥ 60 years treated between March 2021 and June 2025: 39 underwent RAFR, 34 manual closed reduction with navigation-assisted fixation, and 42 initial non-operative treatment. The primary outcome was postoperative reduction quality according to the Matta criteria. Secondary outcomes included 1-year functional scores, mortality, and surgery-related complications.

RESULTS: RAFR was associated with lower odds of a worse postoperative Matta grade than manual closed reduction with navigation-assisted fixation (adjusted common OR=0.139, 95% CI 0.039-0.440; P = 0.001). Among 58 operated survivors with complete 1-year assessments, RAFR was associated with a higher Parker Mobility Score (β=2.030, 95% CI 1.093-2.966; P < 0.001), whereas adjusted differences in Barthel Index, EQ-5D-5L, and NRS were not statistically significant. In the operative versus initial non-operative comparison, 6-month mortality was 4.3% versus 15.8% (P = 0.064) and 1-year mortality was 8.6% versus 21.1% (P = 0.078), w […]

PubMed · DOI

Factors associated with U-type sacral fractures in geriatric patients: A retrospective cohort study.

Wong AW, Musick AN, Wagner RK, Lehle CH, Vise HS, van Weezenbeek MA, Patel R, Zhang BB et al.

OBJECTIVES: To identify demographic and clinical factors associated with geriatric U-type sacral fractures.

METHODS: Design: Retrospective cohort study.

SETTING: Two Academic Level 1 trauma centers. Patient Selection Criteria: Patients ≥ 65 years with sacral fractures between January 2023 and January 2025 who underwent advanced imaging (CT or MRI) were included. Outcome Measures and Comparisons: The primary outcome was U-type sacral fracture morphology (AO/OTA 55C0, 54C3), defined as bilateral sacral ala fractures with a transverse fracture through the sacral body, consistent with spinopelvic dissociation. Factors associated with U-type morphology were evaluated using univariate logistic regression, followed by multivariable logistic regression to identify independent associations.

RESULTS: 263 geriatric patients with sacral fractures were included, of whom 76 (28.9%) had U-type fractures. On multivariable analysis, spondylosis/spondylolisthesis (OR 2.08, 95% CI 1.10-3.98, p = 0.025), independent ambulation prior to injury (with assistive device use as the reference group; OR 2.18, 95% CI 1.13-4.29, p = 0.022), and an atraumatic mechanism of injury (with low-energy injury mechanism as the reference group; OR 2.34, 95% CI 1.05-5.27, p = 0.038) were independently associated with U-type fractures.

CONCLUSIONS: Nearly one-third of geriatric patients with sacral fractures who un […]

PubMed · DOI

Traumatic spinal injuries in a rural tourist area: Epidemiology, seasonal trends and implications for healthcare resource allocation.

Peppucci E, Ferlito D, Meletti L, Montalbetti A, Piras G, Bucpapaj R, Giussani CG, Zoia C

PURPOSE: Several epidemiological studies have evaluated trends and epidemiology of Traumatic Spinal Injury (TSI) but often they don't take into account the geographical variability of nations. Having identified a lack of literature on the epidemiology of rural and touristic areas we conducted this retrospective study in our hospital near Como Lake to understand how best to manage TSI patients.

METHODS: This retrospective epidemiological study included all patients with TSI admitted to Emergency Department (ED) of Gravedona ed Uniti Hospital between April 2018 and March 2024. We focused on vertebral fractures. Demographics, trauma mechanisms, regions of the spine involved, patient comorbidities, polytrauma or Traumatic Brain Injury (TBI), any surgical treatment were analysed. We then focused on differences between warm and cold seasons.

RESULTS: 140 patients were included, but only 132 were considered due to missing data. Average age was 60.0 years. TSI involved cervical spine in 23.5% of the cases, thoracic in 27.3%, lumbosacral in 37.1%, and multiple sites in 12.1%. 54.5% of the patients were treated with conservative therapies while 45.5% underwent surgery (78.3% spinal arthrodesis and 21.7% vertebroplasty). Mechanisms of trauma were as follows: accidental falls 56.8%, vehicular accidents 21.2%, sports 20.4% and violence 1.5%. There was a statistically significant increase […]

PubMed · DOI

Stress displacement and surgical benefit in Beckmann 7-9 pelvic ring injuries: A retrospective multicenter analysis.

Achebe CC, Thorne T, Mau M, Higgins TF, Rothberg DL, Haller JM, Sato E, Steffenson L et al.

INTRODUCTION: We assessed whether stress-induced displacement identifies stability-indeterminate lateral compression type 1 (LC1) pelvic ring injuries in which surgical treatment is associated with fewer early adverse or ambulatory-decline outcomes.

METHODS: This multicenter retrospective cohort study was performed at eight Level I trauma centers. Skeletally mature patients with OTA/AO 61B2 LC1 pelvic ring injuries, Beckmann scores 7-9, and documented stress examination were included; follow-up duration was modeled as a covariate rather than used as an inclusion restriction. The primary outcome was a composite of all-cause mortality, non-elective hardware failure or revision, or worsening ambulatory category. Outcomes were compared between surgical and nonoperative management within displacement strata of less than 12 mm, 12-16 mm, and greater than 16 mm.

RESULTS: The primary analytic cohort included 187 patients (98 nonoperative, 89 operative); follow-up duration was modeled as a covariate. Surgical patients had greater median displacement than nonoperative patients (14.3 vs 5.8 mm; P < 0.001). The composite outcome occurred in 41 nonoperative patients (41.8%) and 21 operative patients (23.6%; P = 0.009). In the 12-16 mm stratum, events occurred in 6 of 11 nonoperative patients (54.5%) and 3 of 20 operative patients (15.0%), corresponding to a 39.5% absolute risk reduction, […]

PubMed · DOI

Longitudinal radiographic comparison of percutaneous and open pedicle screw fixation for thoracolumbar burst fractures: A two-year follow-up study.

Wung CH, Chang CY, Hsu CC, Wu RW, Yang FS, Lu ML, Chen HT

PURPOSE: While percutaneous pedicle screw fixation (PPSF) and open pedicle screw fixation (OPSF) demonstrate comparable clinical outcomes at 1-year follow-up, the longitudinal evolution of sagittal alignment beyond one year remains poorly characterized. This study compared the longitudinal radiographic outcomes of PPSF and OPSF in patients with thoracolumbar burst fractures (TLBF) over a two-year follow-up period.

METHODS: This retrospective cohort study included 54 patients with TLBF who underwent short-segment posterior fixation using either PPSF (n = 29) or OPSF (n = 25). The primary radiographic outcome was Cobb angle. Secondary outcomes included lumbar lordosis angle (LLA), operative time, intraoperative blood loss, postoperative hospital stay, and revision surgery. Serial radiographic evaluations were performed preoperatively and at 3 months, 6 months, 1 year, and 2 years postoperatively.

RESULTS: Baseline demographic characteristics and preoperative radiographic parameters were comparable between groups. PPSF was associated with significantly lower intraoperative blood loss than OPSF (61.9 ± 34.8 mL vs. 262.7 ± 205.1 mL, P < 0.001), whereas operative time and postoperative hospital stay did not differ significantly. Both groups demonstrated significant reductions in Cobb angle over the two-year follow-up. No significant between-group differences in Cobb angle were obse […]

PubMed · DOI

A structured three-step reduction strategy for locked posterior fracture-dislocation of the proximal humerus through a standard deltopectoral approach: A retrospective case series.

Jiang Z, Xiang Y, Lei C, Fu Q, Huang M, Zhi XS, Wang HS

BACKGROUND: Locked posterior fracture-dislocation of the proximal humerus is a rare and technically demanding injury in which the humeral head remains impacted posterior to the glenoid. Although several reduction techniques have been described, the practical details of anterior-only reduction remain incompletely standardized. We evaluated a structured three-step anterior reduction strategy performed through a standard deltopectoral approach.

METHODS: We retrospectively reviewed 16 consecutive patients with acute locked posterior fracture-dislocation of the proximal humerus treated between January 2016 and January 2022. All patients underwent reduction through a single deltopectoral approach followed by locking-plate fixation supplemented with suture fixation. The reduction strategy consisted of (1) establishing an anterior reduction corridor by opening the anterior joint through an L-shaped subscapularis incision or a lesser-tuberosity fracture interval, (2) guiding the humeral head with a periosteal elevator introduced between the humeral head and glenoid, and (3) unlocking and translating the head anteriorly using gentle elevator assistance, controlled arm rotation, and, when head-shaft control was limited, direct digital manipulation. Outcomes included fracture union, avascular necrosis (AVN), recurrent dislocation, implant-related complications, range of motion, Constant-M […]

PubMed · DOI

Prehospital transfusion requirements in severe trauma: A multicenter, retrospective estimate of requirements.

Thabouillot O, Narcisse J, Girard A, Raphael V, Rozenberg E, Wolff D

PURPOSE: To estimate the proportion of severely injured trauma patients who would have met criteria for prehospital blood transfusion, assuming a benefit when prehospital management exceeds 20 min, by applying the ABC score to a multicenter French prehospital cohort.

METHODS: We performed a retrospective observational study across four prehospital services in the Paris region over one year. Among all trauma missions, we included patients transported to a major trauma center or undergoing resuscitation for traumatic cardiac arrest. Exclusions were age < 18, burned patients, and interhospital transfers. We retrospectively calculated the ABC (Assessment of Blood Consumption) score at 10-minute intervals during the intervention (until 20 min before hospital arrival). Patients were considered "eligible" if ABC ≥ 2 and total prehospital intervention duration > 20 min. Secondary endpoints included: proportion receiving tranexamic acid, time intervals, and prehospital deaths or arrests. Data were double-reviewed by two physicians, with adjudication by a third in case of discrepancy. Confidence intervals were computed using Wilson's method.

RESULTS: Of 5921 missions performed, 139 met inclusion criteria. Twenty-seven patients (19.7%, 95% CI: 13.7-26.8%) met both ABC ≥ 2 and intervention time > 20 min, thus qualifying for prehospital transfusion under our

assumptions. Median time inter […]

PubMed · DOI

Using injury prevention priority scores to investigate adult traumatic injuries in Nova Scotia, Canada.

Green D, Jessula S, Opra I, Green RS

BACKGROUND: Traumatic injuries are a primary public health concern and a leading cause of mortality, placing significant strain on healthcare systems. To address this, the objective is to develop province-wide Injury Prevention Priority Scores (IPPS) for major traumatic injuries in Nova Scotia. Allowing for the identification of which mechanisms, if prevented, would yield the greatest impact.

METHODS: We included all adults (≥20 years old) major trauma patients in Nova Scotia from April 1, 2016, to March 31, 2023, from a population-based trauma registry. We calculated Injury Prevention Priority Scores (IPPS) for each mechanism of injury. Frequency plus one of the following measures: injury severity, mortality, resource utilization, and societal cost, is included in the IPPS calculation.

RESULTS: A total of 6240 patients were included in the analysis, with falls the leading IPPS scores in two domains (Severity = 66 and Resource Utilization = 65). Motor vehicle collisions scored the highest (65) in the societal cost domain, while intentional self-harm injuries and falls shared the highest score (64) for the mortality domain.

CONCLUSIONS: Based on our investigation, falls and motor vehicle collisions are the top injury priorities for injury prevention in Nova Scotia, while intentional self-harm linked to substance use also demonstrated high mortality. High hospital costs from f […]

PubMed · DOI

Therapeutic potential of dermal matrix in orthopaedic trauma.

Christenson MM, Boike SC, Puckett HD, Agoro KS, Sullivan D, Vang S, Cunningham BP, Busel GA

BACKGROUND: The primary aim was to examine the outcomes of dermal matrix (DM) in populations with non-closable traumatic, infectious, or post-surgical wounds managed by an orthopaedic trauma service line. The secondary aim was to identify the success and complication rates of DM, as well as attempting to document time to healing.

METHODS: This was a multi-center retrospective chart review at urban community and academic affiliated trauma centers. Consecutive patients with any traumatic, infectious or post-surgical wounds treated with dermal matrix from January 2020 to August 2025. Patients were excluded if lost to follow-up and/or had ongoing treatment. Data was collected for demographics and comorbidities, alongside origin, treatment, and time to complete healing of the wound.

RESULTS: Seventy-three patients had application of DM on 75 wounds. Four patients were lost to follow-up, 9 had ongoing treatment, and 3 died. The final cohort consisted of 57 patients, with 59 wounds. There were 33 males and 24 females. The average age was 58 + /-19.4 years. Wound origins were postoperative (n = 17), traumatic (n = 30), and infectious (n = 12) etiologies. The average number of washouts prior to application was 2.3 + /- 1.2. Two wounds required salvage plastic surgery involving sural flaps. One wound had a failure of skin graft after DM with subsequent treatment by secondary healing vi […]

PubMed · DOI

State-level retrospective analysis of the burden of emergency and operative conditions in India from 1980 to 2023.

Gerardo R, Zadey S, Shetty R, Arora A, Gupta U, Rao S, Afsar AP, Staton CA et al.

INTRODUCTION: Disease burden is a crucial factor in determining healthcare policy and resource allocation. We analyzed the subnational burden of emergency and operative conditions in India from 1980 to 2023.

MATERIALS AND METHODS: We extracted mortality and disability-adjusted life-year (DALY) estimates for 31 Indian states and Union Territories from 1980 to 2023 from the Global Burden of Disease 2023 Study. We used existing expert-consensus-based classifications for defining emergency (30 conditions), operative (57 conditions), and emergency-operative conditions (7 conditions). We investigated rates per 100,000 people and proportions

attributable to the above conditions, expressed as percentages of total mortality and DALYs. We conducted a post-hoc uncertainty analysis to calculate the mean and 95% uncertainty interval (UI) for each estimate.

RESULTS: In 2023, emergency (mortality rate: 340 deaths per 100,000, 95% UI [282, 416]; DALY rate: 13,732 DALYs per 100,000 people, 95% UI [11,492, 16,277]), operative (mortality rate: 158 deaths per 100,000, 95% UI [123, 196]; DALY rate: 7284 DALYs per 100,000 people, 95% UI [5765, 9018]), and emergency-operative conditions (mortality rate: 70 deaths per 100,000, 95% UI [58, 83]; DALY rate: 3830 DALYs per 100,000 people, 95% UI [3229, 4396]) accounted for 49.76% (95% UI [43.44%, 58.14%]), 23.12% (95% UI [19.05%, 27.52%]), and 10.31% (9 […]

PubMed · DOI

Age-specific TBSA thresholds for pediatric burn complications: A 13-year single-center inpatient retrospective cohort.

Hilewitz D, Har-Shai L, Keller A, Bal M, Hayun Y, Meshulam-Derazon S, Ad-El D, Kalish E et al.

INTRODUCTION: Pediatric burns are common, yet practical tools that stratify presentation features into age-specific risk remain limited.

OBJECTIVE: To characterize pediatric burn presentations and derive clinically interpretable, age-specific total body surface area (TBSA) thresholds, with absolute-risk tables, for fever (≥38.5 °C), Intensive Care Unit (ICU) transfer, surgery, and infection, to guide emergency department (ED) triage and inpatient management.

METHODS: A retrospective cohort of children 0-18 years with acute burns admitted to a tertiary pediatric center (December 2012-May 2025). Exposures included age, TBSA, depth, anatomic site, circumferential pattern, etiology, and post-burn days. Outcomes were ICU transfer, surgery, and in-hospital infection. Fever was analyzed as a clinical sign. We estimated absolute risks by age×TBSA, assessed co-occurrence of adverse events, and derived age-specific TBSA cutoffs using Youden optimizations.

RESULTS: Among 942 children (mean age 4.6 ± 4.6 y; 57.4% male; mean TBSA 5.0 ± 5.7%), scalds predominated (74.5%). Event rates were fever 29.0%, ICU transfer 4.4%, surgery 7.7%, infection 5.2%. Delayed presentation (>24 h) was associated with more fever ($p \approx 0.05$). Operative risk varied by site: highest in lower extremities and buttocks and was more frequent with circumferential burns. ICU transfer and surgery rarely occurred without […]

PubMed · DOI

Association of early hyperosmolar therapy and operative interventions in severe traumatic brain injury.

Ahmed N, Kuo YH

BACKGROUND: Hyperosmolar therapy, hypertonic saline (HTS) or mannitol (MAN), is one of the standard management algorithms in Traumatic Brain Injury (TBI). There is a gap in information regarding the impact of the timing of starting the hyperosmolar therapy and outcomes. Our hypothesis was that early hyperosmolar therapy would reduce the operative interventions (craniotomy or craniectomy).

METHODS: The Trauma Quality Improvement Program (TQIP) database of 2019-2023 was queried for the study. Patients aged 18 years and older who suffered severe TBI, defined on Glasgow Coma Scale (GCS) score 8 or less and Abbreviated (brain) Injury Scale (AIS) score 3-5, who received HTS or MAN were included in the study. The propensity score matching method was used to compare the patients' characteristics and outcomes between the early (hyperosmolar therapy time of start ≤3 h after admission) and late groups (the therapy was started >3 h after admission).

RESULTS: A total of 673 patients fulfilled the inclusion criteria. Propensity score matching resulted in 174 pairs of patients. There were no significant differences found between the early and late groups regarding the age in years (median [IQR]: 49.5 [30.5-70] vs 53.5 [35-66], P = 0.765), GCS (median [IQR]: 3 [3-6] vs. 3 [3-6], P = 0.718), Injury Severity Score (ISS) (median [IQR]: 25[17-26] vs. 25[17-26], P = 0.85), non-reactive pupils ([…]

PubMed · DOI

Epidemiology of mountain bike-related injury presentations to emergency departments: A 15-year retrospective analysis.

Course G, Sharman JE, Zhao T, Otahal P, Tran V

OBJECTIVES: To determine the epidemiology of mountain bike (MTB)-related presentations to public hospital emergency departments (ED) in Tasmania, Australia, pre- and post-development of purpose-built MTB trails.

METHODS: Retrospective analysis of presentations to Tasmanian public hospital EDs pre- and post-MTB trail development in 2015 (total period: 1st January 2010-31 st December 2024). MTB-related injuries were determined from hospital medical records. Epidemiological characteristics were prevalence, clinical urgency (Australasian Triage Scale [ATS] categories 1-5, most to least urgent), arrival mode (ambulance vs private transport), mortality, admission rate, seasonal variation, injury profile and patient demographics.

RESULTS: For the total period, there were 6643 MTB-related presentations. Prevalence increased 3-fold post-trail development, 1.0-3.6/1000 ED presentations (n = 743-5900) with seasonal burden highest in summer. Clinical urgency increased 375% for ATS category 1 from 0.8 (0.8) mean (SD) to 3.8 (2.8) presentations/year, and 1343% for category 2 from 8.2 (4.0) to 118.3 (65.5) presentations/year. Ambulance arrivals increased 372% from 21.2 (3.6) to 100 (47.6) /year. Hospital admission rate increased 663% post-trail development, from 17.2 (7.4) to 131.3 (64.4) /year. All four deaths were post-trail development. Over the total period, 41.6% of MTB injuries were o […]

PubMed · DOI

Factors associated with the occurrence and recurrence of road traffic injuries in car and motorcycle drivers: a cross-sectional study.

Guimarães RA, de Sena KG, Valadão KA, Vera I, Maia LG, da Silva Carvalho Vila V

OBJECTIVE: To analyze the factors associated with the occurrence and recurrence of road traffic injuries (RTIs) in drivers.

METHODS: A cross-sectional study was conducted using a non-probabilistic sample recruited through virtual snowball sampling. Participants completed a questionnaire on the occurrence and recurrence of RTIs and their potential associated factors. Poisson multiple regression was used to identify factors associated with the lifetime occurrence and recurrence of RTIs, as well as their occurrence in the last 12 months.

RESULTS: The lifetime prevalence of RTIs was 14.9% (95% confidence interval [95%CI]: 13.1-16.9), and the 12-month prevalence was 1.2% (95%CI: 0.8-1.6). The prevalence of recurrence (≥2 lifetime RTIs) was 5.1% (95%CI: 4.1-6.4). Higher prevalences were observed among male individuals (lifetime: adjusted prevalence ratio [APR]: 1.37, 95%CI: 1.07-1.76), individuals with low educational attainment (lifetime: APR: 1.94, 95%CI: 1.18-3.20; last 12 months: ≤8 years: APR: 3.31, 95%CI: 1.08-10.13; 9-11 years: APR: 4.47, 95%CI: 1.71-11.67), motorcyclists (lifetime: APR: 2.39, 95%CI: 1.96-2.92; last 12 months: APR: 5.44, 95%CI: 2.86-10.32), individuals aged 18-29 years (last 12 months: APR: 5.04, 95%CI: 1.92-13.27), individuals with a National Driver's License (lifetime: APR: 1.63, 95%CI: 1.02-2.60), and those who reported running red lights in the absence o […]

PubMed · DOI

EJTE

European Journal of Trauma & Emergency Surgery

Volume 52, Issue 1 · August 2026 · 120 new articles

Renal trauma in a pediatric Level 1 trauma center: morbidity, mortality, and the role of renal scintigraphy.

Jansen SAM, Haouli B, Backes M, Tromp TN, Vriens D, Edwards MJR, Uitert AV, Nelen SD

PURPOSE: Pediatric renal trauma is uncommon but potentially serious, and is often managed conservatively. Complications may occur, and the clinical value of renal scintigraphy during follow-up remains unclear. This study aimed to assess morbidity and mortality after pediatric renal trauma and to assess the clinical value of renal scintigraphy during follow-up.

METHODS: This retrospective single-center cohort study included pediatric trauma patients (< 18 years) with renal injuries admitted to a Level 1 trauma center in the Netherlands between January 1, 2001, and December 31, 2024. Mortality and in-hospital complications were assessed, and associations between renal scintigraphy abnormalities and complications during follow-up were evaluated.

RESULTS: A total of 72 patients were included, of whom all injuries resulted from blunt trauma. Conservative management was applied in 88% of cases; embolization and nephrectomy were performed in 8% and 3% of patients, respectively. In-hospital mortality was 11% and was attributed exclusively to concomitant traumatic brain injury. Complications occurred in 25% of patients, primarily infections (19%), followed by hypertension during follow up (4%). Renal scintigraphy was performed in 53% of patients, predominantly in those with high-grade injuries; abnormalities were frequently observed (77%) but did not correlate with the occurrence of h […]

PubMed · DOI

Measurement properties of pulmonary function measurement and patient-reported outcome measures in patients with multiple rib fracture without flail segment; validation study alongside the multicenter FixCon trial.

Luttjeboer ML, Van Diepen MR, Prins JTH, Van Wijck SFM, Verhofstad MHJ, Wijffels MME, Van Lieshout EMM, FixCon study group

BACKGROUND: The reliability and validity of pulmonary function, pain and health related quality of life (HR-QoL) as outcome measures for patients with rib fractures, remain uncertain. Our objective was to evaluate measurement properties in patients with multiple rib fractures by comparing them with generic HR-QoL instruments (EuroQoL-5D-5L [EQ-5D-5L], Short Form-12 [SF-12]).

METHODS: A multicenter RCT in adults (n = 180) was performed. Spirometry was performed at 30 days, 3, 6, and 12 months after trauma. Patients completed five questions on pain, EQ-5D-5L, SF-12, and two anchor questions on improvement in pulmonary function and thoracic complaints since previous measurement. Reliability, construct validity, responsiveness and floor and ceiling effects were evaluated for all follow-up time-points.

RESULTS: EQ-5D and SF-12 (sub)scales showed adequate internal consistency. Construct and longitudinal validity were adequate for all (sub)scales, except for construct validity of the SF-12 Physical Component Summary. Ceiling effects were present for all pain questions from 30 days onwards, for EQ-5D-5L utility score (US) from 3 months onwards. Responsiveness analysis indicated that spirometry values showed trivial (tidal volume), small (inspiratory capacity, FEV1), or moderate (FVC) degree of change from 30 days to 12 months. All five pain levels, EQ-5D-5L and SF-12 (sub)scales show […]

PubMed · DOI

Prehospital severity and in-hospital mortality in trauma patients managed by the Friuli-Venezia Giulia helicopter emergency medical service: a 6-year retrospective cohort study.

Tomasino S, Pegani C, Orso D, Ursella C, Tullio A, Bacchetti G, Molinari C, Marangone A et al.

BACKGROUND: Data on in-hospital mortality among trauma patients managed by Helicopter Emergency Medical Services (HEMS) in Italy remain limited. We aimed to describe the epidemiology and in-hospital mortality of trauma patients transported after management by the Friuli-Venezia Giulia HEMS and to assess prehospital factors associated with death.

METHODS: We conducted a retrospective observational cohort study including trauma patients managed by HEMS FVG and transported to hospital, either directly by HEMS or after continuation of transport by ground emergency medical services, between 2018 and 2023. The primary outcome was in-hospital mortality. Jeffreys-prior penalized logistic regression was used to reduce sparse-data bias and instability from separation. The RTS-based model was designated as the primary model and the NACA-based model as a secondary sensitivity analysis. Model performance was assessed internally, and the functional forms of both age and RTS were examined using restricted cubic splines.

RESULTS: Of 3,357 trauma records in the source cohort, 205 prehospital deaths were excluded from the in-hospital mortality analysis, leaving 3,152 transported patients. Outcome data were available for 3,143 patients, of whom 107 died in hospital (3.4%). In the primary RTS-based model, older age, lower RTS, tracheal intubation, and prehospital CPR were associated with mortali […]

PubMed · DOI

Surgeon-perceived intraoperative complexity is associated with early adverse outcomes after major emergency abdominal surgery: a prospective cohort study.

Jensen LR, Degett TH, Solstad TU, Nielsen LS, Kokotovic D, Jensen TK, Burcharth J

BACKGROUND: Intraoperative complexity in major emergency abdominal surgery lacks a standardized definition and is rarely quantified prospectively. Whether surgeon-perceived intraoperative complexity is associated with short-term postoperative outcomes remains unknown.

METHODS: In this single-center prospective cohort study, 411 patients undergoing major emergency abdominal surgery were included. Immediately after surgery, the operating surgeon rated perceived intraoperative complexity on a 0-10 scale. The association between surgeon-perceived intraoperative complexity and the primary outcome, 30-day unplanned reoperation, was investigated using Fine-Gray regression. The model was adjusted for age, sex, body mass index, performance status, and intraoperative pathology. Secondary outcomes were 30-day mortality, length of stay, and Comprehensive Complication Index.

RESULTS: Unplanned reoperation occurred in 15% and 30-day mortality in 11%. Higher surgeon-perceived intraoperative complexity was associated with reoperation (sHR 1.20 per point; 95% CI 1.08-1.34). Complexity was also associated with prolonged LOS (β 1.47 days per unit increase; 95% CI 0.93-2.01), higher CCI category (OR 1.24 per unit increase; 95% CI 1.14-1.35), and increased 30-day mortality (OR 1.25; 95% CI 1.07-1.45).

CONCLUSION: Surgeon-perceived intraoperative complexity was associated with reoperation, morbid […]

PubMed · DOI

The challenges and opportunities of major trauma care in Germany - results of multidisciplinary expert interviews of the LeAf trauma study.

Neubert A, Hempe S, Bieler D, Jaekel C, Gaeth C, Höfer C, Bartha S, Schwojer E et al.

BACKGROUND: The recovery process of major trauma survivors involves various sectors within a complex system of different medical service providers and stakeholders who accompany and support patients throughout the recovery pathway. The aim was to explore experiences and opinions of experts, who are actively involved in the care of major trauma patients in the German healthcare system.

METHODS: Semistructured in-depth group interviews were used to investigate experiences and opinions of experts with the focus on exploration of expert perspectives on opportunities and challenges of the cross sectoral care. An exploratory analysis was performed with an artificial intelligence-based coding software. In a second step, a system map was developed, and all inter- and intra-sectoral challenges and all opportunities for the optimization of processes were mapped.

RESULTS: 39 experts from 13 medical specialties and occupational groups who were interviewed highlighted that intersectional issues especially across and within sectors pose a significant challenge to patient care. Four challenges were highlighted that significantly influence patient care: (1) Communication between sectors; (2)

Complexity of transitions between sectors; (3) Inadequate aftercare and (4) Lack of psychosocial care.

DISCUSSION: In conclusion five key learnings can be derived which could help to transform the healt […]

PubMed · DOI

Costs and health-related quality of life of non-operative and surgical treatment modalities for displaced distal radius fractures: a comprehensive overview based on the CAST study.

Barvelink B, Heemskerk SCM, Verhaar JAN, Reijman M, Colaris JW, CAST study group

PURPOSE: The aim of this study was to provide a comprehensive overview of costs and health-related quality of life (HRQoL) associated with three treatment modalities for common displaced distal radius fractures (DRFs): (1) non-operative treatment, (2) direct plate fixation and (3) delayed plate fixation in case of fracture redisplacement during follow-up.

METHODS: A cost and HRQoL analysis was performed based on data collected during the CAST study, a multicenter cluster-randomized controlled trial including adult patients with a displaced DRF, randomized between immobilization with a plaster splint or circumferential cast. Included patients were prospectively followed for one year. Operatively treated patients also continued questionnaire follow-up. From a societal perspective, healthcare costs, patient and family costs, and indirect (i.e. productivity) costs were collected. Patients completed questionnaires on HRQoL (EQ-5D-5 L), healthcare consumption and work productivity.

RESULTS: In total, 693 patients were included in this study (non-operative: 402, direct plate fixation: 165, delayed plate fixation: 126). The estimated mean total costs per patient were €2,212 for non-operative treatment, €4,339 for direct plate fixation and €4,393 for delayed plate fixation HRQoL improved between 3 and 12 months but did not reach pre-injury HRQoL in all groups.

CONCLUSIONS: This study […]

PubMed · DOI

Peri-periprosthetic femoral fractures: a single-centre case series of a rare and challenging clinical scenario.

Matteotti R, Dalla Vecchia G, Masotti A, Monticelli L, Nardi M, Fiore L

PURPOSE: Periprosthetic femoral fractures (PFFs) are an increasingly common complication of hip arthroplasty. A poorly defined scenario is a new femoral fracture after a previously treated PFF has united, which we term a peri-periprosthetic femoral fracture.

METHODS: We retrospectively reviewed 154 surgically treated PFF episodes in patients aged over 65 years at our institution between 2011 and 2021. A peri-periprosthetic fracture was defined as a new acute femoral fracture after documented radiographic union of a previously treated PFF. Clinical, surgical, and radiographic data were reviewed.

RESULTS: Seven patients (4.5% of the institutional PFF cohort) met the criteria. Mean age was 83.1 years and 6 (85.7%) were female. The mean interval to the peri-periprosthetic fracture was 37.3 months (median 27; range 7-105). By the Vancouver classification, 3 fractures were type B1, 1 type B3, and 3 type C. Treatment was individualised to fracture configuration, prosthetic stability, hardware, and bone stock. Radiographic union was documented in all 4 patients with adequate follow-up, at a mean of 4.3 months. Five patients regained ambulation; one with pre-existing limitations did not, and ambulatory status was unavailable for one. Medical complications occurred in 2 patients, and no major orthopaedic complication required reoperation. Mortality was 14.3% at one year and 42.9% at tw […]

PubMed · DOI

In-hospital mortality in hip fracture patients undergoing surgery later than 24 hours post admission might rather reflect the impact of comorbidities and treatment limitations than additional risks imposed by the mere timely delay itself.

Babayi M, Müller CW

PURPOSE: Early surgery for proximal femoral fractures is widely recommended, but the association between surgical delay and short-term mortality remains uncertain because observational studies are vulnerable to

confounding by patient characteristics and treatment limitation.

METHODS: We retrospectively studied adults with operatively treated proximal femoral fractures at a DGU-certified regional trauma center in Germany (2022-2025). Time to surgery was measured from emergency department registration to operative start. Delays > 24 h were classified as medical or non-medical where sufficiently documented. The primary outcome was in-hospital mortality. Adjusted analyses included age, sex, ASA category, fracture type, concomitant orthopaedic injury, and treatment limitation (DNR/DNI; not time-anchored to surgery) and used Firth penalized logistic regression.

RESULTS: Among 927 patients, 871 (94.0%) underwent surgery within 24 h; 53 (5.7%) died in hospital. Crude mortality was 5.6% after surgery within 24 h and 7.1% after surgery > 24 h. After adjustment, no clear independent association between surgery > 24 h and mortality was detected (aOR 1.03, 95% CI 0.29-2.92). Higher age, male sex, advanced ASA class, and treatment limitation were associated with mortality. In exploratory analyses, mortality was 19.0% among medically delayed patients and 0% among patients with non-medical d […]

PubMed · DOI

Blast injury simulation in pigs using an air-driven shock wave generator: effects of shock waves of lateral shock wave exposure.

Kiriu N, Sekine Y, Sasa R, Yamamura K, Aoki M, Nagamura T, Nishii K, Saitoh D et al.

PURPOSE: We have hypothesized that respiratory arrest caused by temporary brainstem dysfunction is the primary cause of death during the hyperacute phase of blast injury and have demonstrated that this is highly likely. To date, we have examined the effects of shock waves on pigs by exposing them to shock waves from both the dorsal and ventral sides. Therefore, in this study, we investigated the effects of lateral shock wave exposure in pigs and identified priority regions for protection against blast injuries.

METHODS: Eight male hybrid pigs (age, 10-12 weeks; mean weight, 36.6 kg) were anesthetized and fitted with a femoral arterial line and subcutaneous electrocardiography leads for continuous monitoring of vital signs and respiration. Pigs were fixed prone with the right lateral thorax aligned to the aperture. A compressed air-driven shock tube generated a shock wave (driving pressure: 3.0 MPa). Arterial blood gases were obtained before and 0, 3, 5, and 10 min post-exposure. Whole-body computed tomography was performed pre- and post-exposure. Survivors were observed for 3 h, then underwent necropsy. Regarding lung injury, since "primary blast lung injury" is defined as "radiological and clinical evidence of acute lung injury occurring within 12 h of exposure and not due to secondary or tertiary injury", this study focuses on organ injury and mortality in the hyperacute and […]

PubMed · DOI

Cumulative operative burden is associated with poorer function after segmental bone transport for lower-extremity long-bone defects.

Y■ld■z M, Köro■lu M, Karaba■ ME, Çiçek ■B, Stevenson T, Harma A

OBJECTIVE: Segmental bone transport offers biological reconstruction and limb preservation for extensive lower-extremity long-bone defects, but treatment often involves prolonged external fixation, planned stages, and unplanned reoperations. We evaluated reconstructive and functional outcomes and examined the association between cumulative operative burden and final patient-reported function.

METHODS: This retrospective single-centre study screened 44 patients treated between January 2012 and March 2023. Thirty-six patients, representing 37 reconstructed long bones, had complete final outcome ascertainment. ASAMI bone outcome was reported per reconstructed bone (n = 37); limb preservation and all functional and multivariable analyses were patient-level (n = 36). Outcomes included limb preservation, ASAMI bone and functional results, the Lower Extremity Functional Scale (LEFS), the Karlström-Olerud score, healing indices, treatment-related events, and the cumulative number of formal operations. Multivariable linear regression, bootstrap resampling, sensitivity analyses, and exploratory receiver operating characteristic analysis were performed.

RESULTS: Mean defect length was 10.32 ± 4.67 cm and the mean defect-to-bone length ratio was 26.3% ± 12.4%. Patients underwent a mean of 8.08 ± 4.21 operations (median, 7; range, 3-18). Mean follow-up after external-fixator removal was 5 […]

PubMed · DOI

The utilisation of international consensus definitions of the disease polytrauma: a systematic review.

Kim SY, Xu Z, Balogh ZJ

PURPOSE: The disease of polytrauma is associated with significant healthcare burden and poor prognosis. The validated Newcastle and the Berlin consensus definitions of polytrauma have been proposed more than a decade ago for standardising clinical, scientific and advocacy related communications. This systematic review aims to evaluate the uptake of the Newcastle and the Berlin definitions of polytrauma since their introduction.

METHODS: This systematic review (PROSPERO: CRD420251119950) adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. A literature search of original research studies reported in English on polytrauma patients from 1st January 2015 to 31st of December 2025 was conducted, using Medline, Embase, Scopus, Pubmed, CINAHL and CENTRAL. Specific focus was on the utilisation of the Berlin and the Newcastle definitions, but all explicitly stated definitions of polytrauma were analysed.

RESULTS: A total of 2354 studies were identified through initial literature search. 1491 studies were excluded after screening by title and abstract. The full text review of 863 studies was performed and resulted in 88 eligible studies after the exclusion of 732 studies with no explicit definitions of polytrauma and 43 studies on non-human subjects. Ten studies (11.4%) used the Berlin definition of polytrauma, and six studies (6.8%) rep […]

PubMed · DOI

Adhesive small bowel obstruction in the virgin abdomen: comparable phenotype, divergent management.

Cioffi SPB, Martinez-Casas I, Marrano E, Cimbanassi S, Cimino MM, Kurihara H, Mohseni S, Kaplan LJ et al.

BACKGROUND: In small bowel obstruction (SBO), the absence of prior abdominal surgery is commonly interpreted as a marker of higher-risk pathology and is often associated with a lower threshold for early operative intervention. Whether this assumption reflects differences in disease phenotype or decision-making remains uncertain.

METHODS: This post hoc analysis of a prospective, multicenter cohort study included consecutive adults with CT-confirmed mechanical SBO. Analyses were restricted to patients with adhesion-related SBO to enable comparison between virgin abdomen adhesive SBO (vaASBO) and prior-surgery adhesive SBO (psASBO). Clinical presentation, imaging findings, management strategies, and 60-day outcomes, including patient-reported outcome measures (PROMs), were evaluated. Analyses were associative and unadjusted.

RESULTS: Among 1,737 patients, 417 (24.0%) had no prior abdominal surgery; adhesions accounted for 30% of cases in this subgroup. The adhesion-restricted cohort included 982 patients, of whom 127 (12.9%) had vaASBO. Clinical and radiological features were similar between groups; however, management differed substantially. Patients with vaASBO had higher rates of direct-to-surgery (+ 15.0% points), lower use of water-soluble contrast (-19.1% points), and higher rates of nonoperative management (NOM) failure (+ 25.1% points). Among patients undergoing surgery […]

PubMed · DOI

What to do with K-wires in displaced upper extremity fractures in children - where to bury or not.

Frodl A, Borbely M, Karvouniaris N, Mühlenfeld N, Siegel M, Kuminack K, Pietsch C, Wagner FC et al.

Kirschner wires (K-wires) are the standard fixation method for displaced pediatric upper extremity fractures, yet no consensus exists on whether wire ends should be buried subcutaneously or left exposed epicutaneously. While epicutaneous K-wires offer cost-effectiveness and avoid second anesthesia, concerns remain regarding infection risk and psychological distress. To systematically review and compare complication rates between buried and exposed K-wire fixation in pediatric upper extremity fractures, including infection, refracture, skin erosion, and secondary implant displacement. We conducted a systematic review following PRISMA guidelines, searching Cochrane, PubMed, EmBase, and Ovid databases. Inclusion criteria were: modified Coleman Methodology Score (mCMS) > 60, upper extremity fractures treated with K-wire fixation, and age < 16 years. Studies including adult patients and biomechanical studies were excluded. Meta-analysis compared outcomes across 12 studies encompassing 1,977 patients. Buried K-wires demonstrated significantly lower overall infection rates (OR = 1.74;

95% CI: 1.19-2.56; p = 0.004). However, no significant difference was observed in deep surgical siteinfections (OR = 1.56; 95% CI: 0.60-4.0; p = 0.36). Exposed wires showed significantly lower skin irritation rates (OR = 0.09; 95% CI: 0.04-0.21; p < 0.00001). No significant differences were found in seco […]

PubMed · DOI

Exploring risk groups for concomitant injuries in facial fracture patients: a data-driven analysis.

Mustasilta R, Kultanen E, Puolakkainen T, Haapanen A, Thorén H, Snäll J

PURPOSE: Patients with facial fractures are a diverse group with varying injury mechanisms and extent of injury. These patients often suffer from concomitant injuries (CIs) of different severity outside the facial region, some of which are life-threatening and may require urgent care. Therefore, early identification of patients at risk for CI is important. This study aimed to investigate exploratory risk groups for concomitant injuries in patients with facial fractures, focusing on the impact of injury energy, sex and age on the prevalence and type of concomitant injury.

METHODS: In this retrospective cohort study, a data-driven analysis was used to perform statistical analysis and identify risk groups for different CIs. The study included 4170 patients with facial fractures evaluated at tertiary trauma centres from January 2013 to October 2020.

RESULTS: 70.9% of patients with facial fractures were male. Patients' mean age was 45.7 years. Of the injury mechanisms, 36.0% were classified as high-energy trauma. In all, 31.7% of facial fracture patients had some type of CI. Female patients had more CIs than male patients (34.0% vs. 30.8%). The occurrence of CIs increased progressively with age, peaking in the age group 90-94 years, where over 55% of patients experienced CIs. However, considerable variation was observed in CI types across age groups.

CONCLUSION: Due to the high C […]

PubMed · DOI

Hydroxyapatite/poly-L-lactide mesh tube versus β-tricalcium phosphate granules in locking plate fixation of proximal humeral fractures: a retrospective comparative study.

Iwata H, Takada N, Usami T, Hayashi Y, Murakami H, Kuroyanagi G

PURPOSE: Proximal humeral fractures in older adults are often accompanied by metaphyseal bone defects, leading to reduction loss after plate fixation. Fibular allograft struts provide effective augmentation. However, their availability is limited. This study aimed to compare the outcomes of using hydroxyapatite/poly-L-lactide (HA/PLLA) mesh tubes as endosteal struts and β-tricalcium phosphate (β-TCP) granules for augmentation in locking plate osteosynthesis for proximal humeral fractures.

METHODS: This retrospective study included 42 adult patients who underwent locking plate fixation for acute, closed proximal humeral fractures at a single institution between 2013 and 2022. Among them, 20 patients received endosteal strut-HA/PLLA mesh tube (group E), and 22 received β-TCP granules (group G). Demographics, radiographic outcomes (humeral head height and neck-shaft angle), range of motion, complications, and surgical parameters were compared between the two groups.

RESULTS: Group E showed a significantly smaller mean change in humeral head height than group G (1.5 ± 3.1 vs. 4.0 ± 4.0 mm, p = 0.01). No significant difference in the mean change in neck-shaft angle was observed between the two groups (p = 0.31). Furthermore, no significant differences were observed in shoulder range of motion, complication rates, operative time, or blood loss between the two groups.

CONCLUSION: T […]

PubMed · DOI

The burden of blast injuries on a level i trauma center: a comparative analysis of blast versus blunt trauma.

Quint E, Barsky G, Osyntsov A, Replyanski I, Shaked G

PURPOSE: We hypothesized that blast trauma results in a greater institutional resource burden than blunt trauma.

METHODS: This retrospective study compared blast trauma patients (N = 321) to blunt trauma patients (N = 9,172). A Propensity Score Matching (PSM) on Age, Sex and Injury Severity Score (ISS) was executed, yielding 1,481 matched blunt controls. The primary composite endpoint was Institutional Resource Burden, defined as requiring one or more of the following high-intensity resource metrics: prolonged hospital length of stay (LOS >7 days), ICU admission, or need for urgent surgical intervention (<6 hours from admission). Secondary endpoints included in-hospital mortality, shock-trauma room triage, any surgical intervention, multiple operations (≥ 3), massive transfusion protocol (MTP) activation, requirement for mechanical ventilation, and ICU/hospital LOS.

RESULTS: Blast patients demonstrated a higher rate of the primary composite institutional burden (78.2 vs. 51.2, OR = 3.42, 95% CI: 2.59 - 4.51, p <0.0001). Blast mechanism independently predicted shock-trauma room triage (OR = 10.60, 95% CI: 8.18 - 13.75, p <0.0001), urgent surgery (OR = 7.49, 95% CI: 5.73 - 9.80, p <0.0001), ≥3 operations (OR = 12.40, 95% CI: 8.59 - 17.90, p <0.0001), MTP activation (OR = 11.68, 95% CI: 6.48 - 21.05, p <0.0001), and ICU admission (OR = 5.54, 95% CI: 4.37 - 7.04, p <0.0001). Thes […]

PubMed · DOI

Comparison of the shetty test and ottawa ankle rules for foot and ankle fractures in the emergency department: a prospective validation study.

Gumustas I, Altinbilek E, Topcu H, Avlac A, Oztekin A, Karaketir M, Ozturker N

PURPOSE: The Shetty test has been proposed as a simpler alternative to the Ottawa Ankle Rules (OAR) for excluding fracture in foot and ankle trauma and was reported to have perfect sensitivity in its derivation study. We compared its diagnostic accuracy with that of the OAR against imaging in an emergency department population.

METHODS: One hundred consecutive patients with acute foot and/or ankle trauma were enrolled after at least one of two attending physicians judged radiography indicated. Both index tests were performed independently by two blinded emergency physicians; where they differed, the patient was classified positive if either was positive. The reference standard was fracture on radiography, or on CT obtained on clinical suspicion. McNemar and DeLong tests were used, and inter-test agreement Cohen kappa.

RESULTS: Fractures occurred in 26 patients (26%); 8 (30.8%) were occult, detected only on CT. The Shetty test had lower sensitivity than the OAR (76.9% vs. 92.3%) but higher specificity (67.6% vs. 18.9%). It missed 6 of 26 fractures, including two talar and one base-of-fifth-metatarsal fracture, all of which were managed with plaster-splint immobilisation. Used alone it would have avoided 56 radiographs while missing 6 fractures, versus 16 avoided and 2 missed for the OAR. Agreement was poor (kappa 0.19).

CONCLUSION: The Shetty test did not reproduce the sensit […]

PubMed · DOI

Maxillofacial fractures associated with electric scooter trauma: a cohort study from a national level I trauma center.

Tousidonis M, Gonzalez-Moure B, Fernandez-Acosta D, Rey-Valcarcel C, Perez-Diaz MD, Troulis MJ, Hintze JM, Ochandiano S et al.

PURPOSE: Electric scooters (e-scooters) have become a widely used mode of urban transportation, but their increasing popularity has been accompanied by a rise in trauma-related emergency department visits. Craniofacial injuries represent a particularly relevant subset due to the exposed riding position and the limited use of protective equipment. This study analyzed the epidemiological characteristics, fracture patterns, associated injuries, and treatment requirements of patients presenting with facial fractures after e-scooter accidents.

METHODS: A retrospective observational study was conducted at a tertiary university hospital between January 2020 and December 2024. Adult patients presenting to the emergency department with facial fractures directly related to e-scooter accidents were included. Demographic variables, helmet use, alcohol and drug involvement, fracture distribution, associated systemic injuries, and treatment modalities were assessed.

RESULTS: A total of 42 patients with facial fractures were identified. Mean age was 29.4 years, and 76.2% were male. Helmet status was documented in 10/42 patients (23.8%); among documented cases, 3/10 (30.0%) reported helmet use. Alcohol status was documented in 14/42 patients (33.3%); among documented cases, 10/14 (71.4%) were positive. Drug status was documented in 15/42 patients (35.7%); among documented cases, 8/15 (53.3%) […]

PubMed · DOI

The radial nerve bridging plate technique for humeral shaft fractures: an approach facilitating plate removal without impeding fracture union - a case series.

Putzeys G, Nugteren LHT, Reynders-Frederix P, Hartog DD

BACKGROUND: Removing a plate of the humeral shaft with the radial nerve on top of the plate is challenging and carries a high risk for iatrogenic radial nerve palsy (iRNP). This study aimed to assess the incidence of iRNPs following plate removal of a radial nerve bridging plate (RNBP) and to evaluate outcomes of this RNBP in treatment of a humeral shaft fracture.

METHODS: This was a single-center retrospective case series. Patients treated between May 2013 and December 2025 with a RNBP for a humeral shaft fracture or nonunion were included. The primary outcome measure was the incidence of iRNP after removal of the RNBP. Secondary outcome measures were the incidence of iRNP, malunion and nonunion after primary surgery.

RESULTS: Of 48 included patients, the mean follow-up duration was 24 months. In 11 patients the plate was removed, no iRNP occurred in these patients. Of 41 patients who had no primary RNP after the fracture, seven patients experienced a temporary iRNP after primary surgery. All of the RNPs fully recovered with a mean recovery time of 12.3 weeks (range: 1-32). There were three nonunions, one of which was a pathological fracture, and one was a nonunion of a surgical neck fracture in a multilevel fracture.

CONCLUSION: The RNBP technique allows safe and uneventful removal of the RNBP without iRNP, while maintaining the efficacy of plate fixation to treat humeral […]

PubMed · DOI

The limited value of routine radiographs in the follow-up treatment of proximal humerus fractures.

Elwakel AA, Willinge GJA, Goslings JC, Twigt BA

PURPOSE: Proximal humerus fractures (PHFs) are common injuries in adults and occur particularly often in older patients. In our hospital, patients are routinely scheduled for follow-up radiographs at approximately two and six to eight weeks after injury, although the management yield of this practice is uncertain.

METHODS: We performed a retrospective cohort study of consecutive adults (≥ 18 years) presenting with PHFs to the OLVG emergency department (ED) between January and April 2022. Patients with multiple fractures, pathological fractures, fractures initially treated or followed up at another hospital, and fractures other than proximal humerus fractures were excluded. The primary outcome was a documented treatment-plan change during the observed outpatient treatment period, including conversion from conservative to operative treatment, cancellation of planned surgery, or additional surgery after initial operative treatment. The primary outcome proportion was reported with a two-sided exact 95% confidence interval. Descriptive variables included fracture classification, comminution, imaging use, follow-up timing, secondary displacement, and treatment duration.

RESULTS: One hundred patients were included (mean age 60.4 ± 18.0 years; 60% female). Most fractures were one-part (63%) or two-part (33%), 45% showed comminution, and 87% of patients were treated conservatively. Th […]

PubMed · DOI

Waiting time and mortality after hip fracture: findings from a large single-center cohort study.

Sundet M, Martinsen M, Sexton J, Sundin U, Lillegraven S

PURPOSE: Time to surgery after hip fracture is a potentially modifiable factor that may influence mortality, but the evidence for an association is conflicting for waiting time < 48 h. The aim of this study was to investigate the association between waiting time to surgery and mortality, with waiting time categorized into 12-hour intervals.

METHODS: We analysed all hip fractures operated at a large Norwegian hospital between 2015 and 2023. Waiting time from admission to surgery was categorized into 12-hour intervals. Mortality at one week, one month, three months (pre-specified primary endpoint) and one year was assessed using multivariable logistic regression,

adjusting for age, sex, nursing home residency, Charlson Comorbidity Index (CCI), fracture type, calendar year of surgery, and direct-acting oral anticoagulant (DOAC) use. Model fit and discrimination were assessed for all models.

RESULTS: The study included 4136 hip fractures. At the pre-specified primary endpoint of three months, mortality was numerically higher in patients with waiting times > 12 h, but this did not reach statistical significance for any waiting-time category (adjusted odds ratio [AOR] 1.16-1.25, all p > 0.2). Among the secondary, exploratory endpoints, the one-year timepoint showed increased mortality with increased waiting: patients operated between 12 and 24 h had significantly higher one-year mo […]

PubMed · DOI

Diagnostic accuracy and predictive value of hard signs for operative or endovascular management and major injury in penetrating neck trauma: a single-centre retrospective cohort study.

Barrientos ML, Zapata CAL, Toro DAM, López CAD

PURPOSE: Hard signs guide operative decision-making in penetrating neck trauma (PNT), yet their accuracy and predictive value remain incompletely defined. We quantified the diagnostic accuracy of hard signs for major injury and for operative or endovascular management, and explored multivariable risk estimation.

METHODS: Single-centre retrospective cohort of 310 consecutive patients with PNT (June 2013-July 2025). Hard and soft signs were assigned by the attending surgeon at initial evaluation and abstracted retrospectively from the clinical record. The primary outcome was major injury (confirmed vascular or aerodigestive injury). Operative or endovascular management was analysed as a secondary outcome describing institutional practice, not as an independent diagnostic property, because the same signs informed both the index test and the treatment decision. We estimated sensitivity, specificity, predictive values and likelihood ratios with exact binomial confidence intervals, and fitted exploratory multivariable models (STARD/TRIPOD).

RESULTS: Major injury occurred in 92/310 (29.7%) and operative or endovascular management in 96/310 (31.0%). Hard signs were specific but insensitive for major injury (sensitivity 55.4%, 95% CI 44.7-65.8; specificity 88.1%, 83.0-92.1) and for management (sensitivity 59.4%, 48.9-69.3; specificity 90.7%, 85.9-94.2). Outcome frequency rose across n […]

PubMed · DOI

Epidemiology of spinal fractures in polytrauma patients.

Malan SJ, Meakes S, Fischer A, Balogh ZJ

PURPOSE: The epidemiology of spinal fractures in polytrauma, population characteristics and their hospital outcomes are not well defined. This study aimed to describe the epidemiology and outcomes of spinal fractures in a consensus defined polytrauma cohort.

METHODS: A 13-year (2009-2021) retrospective study was conducted on Newcastle Definition based polytrauma patients at a Level-1 centre. Polytrauma patients were categorised based on the presence of vertebral fractures and compared in demographics, injury mechanism, injury severity and pattern, physiological parameters, treatment aspects, and outcomes.

RESULTS: From the 1425 polytrauma patients, 632 (44.4%) sustained spinal fractures. These patients were older (48 vs. 38 years, p < 0.001) with identical injury severity (ISS = 29), were more frequently admitted (62.8% vs. 57.4%, p = 0.037) and also stayed longer in ICU (6 vs. 3 days, p < 0.001) and in hospital (14.5 vs. 12 days, p = 0.002). Mortality did not differ significantly (10.4% vs. 13.5%, p = 0.08). The lumbar spine was the most frequently fractured segment (7.3%), followed by thoracic (3.3%), cervical (3.3%) and sacral (3.1%) levels. Spinal cord injury occurred in 41 (6.5%) spinal fracture patients and was associated with a high ISS and prolonged hospitalisation [ISS:34(IQR 20.5); ICU LOS:10(IQR21.75); Hospital LOS:22(IQR41)]. Spinal fracture patients that classifi […]

PubMed · DOI

Age-dependent injury mechanisms and injury patterns in pediatric window falls.

Badina A, Deygas A, Saliba I, Corbu A, Da Costa EP, Georges S, Wicart P

PURPOSE: Pediatric defenestration is a high-energy trauma mechanism in which age influences injury patterns and clinical severity. While previous studies have focused primarily on epidemiology and mortality, detailed analyses of age-related injury distribution and surgical burden remain limited. This study aimed to characterize age-dependent injury patterns, neurological severity, and hospital resource utilization following pediatric window falls.

METHODS: We conducted a retrospective single-center cohort study including all children aged 0-18 years admitted for defenestration between 2008 and 2023. Demographic data, fall height, intentionality, injury distribution, Glasgow Coma Scale (GCS), fracture patterns, and number of surgical procedures were collected. Multivariable logistic regression identified predictors of severe neurological impairment (GCS ≤ 8), and Poisson regression assessed determinants of surgical burden.

RESULTS: A total of 333 children were included. Age distribution was bimodal, with predominantly accidental falls in young children (median 4.6 years) and intentional falls in adolescents (median 13.7 years). Younger children more frequently sustained craniofacial injuries and presented with lower admission GCS values, whereas adolescents more often sustained combined lower limb and spinal injuries. Fall height correlated negatively with GCS (ρ = -0.32) and […]

PubMed · DOI

Clinical outcomes and risk factors in the surgical treatment of distal femur fractures.

Höller LF, Höller S, Wittur P, Sehmisch S, Lehmann W, Weiser L

PURPOSE: Distal femoral fractures are rare but severe injuries with high complication rates, particularly in elderly or multimorbid patients. This study aimed to analyze complication profiles and identify risk factors for surgery-related and non-surgery-related complications following operative treatment.

METHODS: A retrospective cohort study was conducted including all adult patients undergoing surgical treatment of distal femur fractures at a Level I trauma center between 2012 and 2018. Demographic, pre-operative, operative, and postoperative characteristics were analyzed. Risk factors were assessed using univariate statistics.

RESULTS: A total of 56 patients with 60 distal femur fractures were included. Surgery-related complications occurred in 11.7% of cases, while non-surgery-related complications were observed in 58.3%. In univariate analysis, gender (p = 0.004), age (p = 0.014), BMI (p = 0.006), osteoporosis (p = 0.047), open fracture (p < 0.001), high-energetic trauma (0.031) and time to surgery (p = 0.024) were significant risk factors for surgery-specific complications. For non-surgery-related complications, need of an external fixator for primary surgical care and type of operation were significant risk factors in univariate analysis. In multivariable logistic regression analysis, need of an external fixator was the only significant risk factor for developing non-s […]

PubMed · DOI

Hip fracture surgery within 24 hours: no association between time to surgery and mortality in a real-world cohort.

Pankratz C, Wolters M, Cintean R, Gebhard F, Schuetze K, Eickhoff A

PURPOSE: Early surgery is recommended for hip fractures, and in Germany, operative treatment within 24 h has been mandatory since 2021. While delays beyond 24-48 h are associated with worse outcomes, it remains unclear whether further acceleration within a 24-hour timeframe improves survival. This study evaluated mortality and perioperative risk factors within a strictly implemented early-treatment pathway.

METHODS: This retrospective single-centre cohort study included 1,051 consecutive patients with femoral neck and trochanteric fractures treated surgically within 24 h of admission (2016-2020). Time to surgery was grouped into < 6 h, 6-12 h, and > 12-24 h. The primary outcome was one-year mortality; secondary outcome was two-year mortality. Mortality data were obtained from the national citizen registry. Multivariable logistic and Cox proportional hazards regression analyses were performed for one-year mortality.

RESULTS: Mean time to surgery was 10.2 ± 6.2 h. One-year mortality was 29.5% and increased to 40.1% at two years. Mortality did not differ significantly between time-to-surgery groups (< 6 h: 30.9%, 6-12 h: 26.4%, > 12-24 h: 31.5%; p = 0.258). This finding was consistent across all adjusted analyses. In contrast, non-surgical complications were the strongest independent predictors of mortality, particularly pneumonia (OR 2.1), renal failure (OR 5.2), and cardiac ev […]

PubMed · DOI

Small bowel obstruction in older adults: similar nonoperative success but worse outcomes across management strategies - insights from the ESTES SnapSBO study.

Bunino FM, Cimino MM, Martinez Casas I, Bellio G, Porta M, Cayre L, Mohseni S, Kaplan LJ et al.

BACKGROUND: Small bowel obstruction (SBO) is a common surgical emergency. It remains unclear whether adverse outcomes in older patients reflect impaired obstruction resolution, differences in disease phenotype, or reduced physiologic tolerance.

METHODS: We performed a prespecified age-stratified analysis of the multicenter SnapSBO cohort, which consecutively enrolled adults with radiologically-confirmed mechanical SBO, stratified as younger (< 65 years) or older (≥ 65 years). Presentation, etiology, management, and outcomes were compared using risk differences and odds ratios. Exploratory analyses evaluated patients aged 65-79 years and ≥ 80 years.

RESULTS: Among 1,731 patients, 729 (42.1%) were aged ≥ 65 years. Older adults had more non-adhesive etiologies, including neoplastic and inflammatory obstruction, and more imaging features of severe disease. They were less likely to undergo a trial of nonoperative management (67.9% vs. 72.7%; absolute risk difference, - 4.8%; P = 0.037). However, among those managed nonoperatively, resolution rates were similar (68.8% vs. 69.5%). Despite this, older adults experienced higher rates of complications (16.7% vs. 6.7%), aspiration pneumonia (3.8% vs. 1.0%), and in-hospital mortality (6.9% vs. 3.6%). Those aged ≥ 80 years showed similar management but substantially higher mortality and complications.

CONCLUSIONS: Older adults achieved s […]

PubMed · DOI

Pancreatic stent and transpapillary pseudocyst drainage via endoscopic retrograde cholangiopancreatography for children with blunt pancreatic duct injury.

Kouchi K, Fumita T, Nakata C, Kagaya K

BACKGROUND: Nonoperative management (NOM) in blunt pancreatic injury (BPI) is frequently complicated by pseudocyst formation and pancreatic atrophy. To address these complications, we have utilized a pancreatic stent via endoscopic retrograde cholangiopancreatography (ERCP) for children with PDI since 2019.

AIM: We present our experience with ERCP-based NOM for PDI and provide a literature review of stent therapy in pediatric PDI.

MATERIALS AND METHODS: Ten children with BPI treated between January 2007 and December 2025 were classified according to the 2024 American Association for the Surgery of Trauma pancreatic injury grades: IB (n = 3), IIA (n = 3), IIIA (n = 2), and IIIB (n = 2). Pancreatic stents were placed in two IIIA cases and one IIIB case. The IIIB case subsequently underwent transpapillary pseudocyst drainage (TPPD) due to a pseudocyst infection.

RESULTS: NOM was successful in all low-grade injuries (< II) without complications. One IIIB patient treated without stenting developed a pseudocyst requiring cyst-gastrostomy; 1-year follow-up demonstrated severe pancreatic atrophy. In the two IIIA stent cases, the duct injury was successfully bridged: one patient had no pseudocyst formation, and the other developed a pseudocyst that resolved spontaneously; both showed preserved pancreatic parenchyma. The IIIB patient treated with TPPD experienced rapid cyst resolution […]

PubMed · DOI

Impact of pre-injury anticoagulation on outcomes in middle-aged trauma patients: a matched-pair analysis of the TraumaRegister DGU®.

Koch DA, Faul P, Jakobi T, Schnetz M, Egert A, Münzberg M, Lefering R, Schweigkofler U et al.

PURPOSE: Pre-injury anticoagulation is a well-known risk factor associated with adverse outcomes in geriatric trauma patients, but its impact in younger populations remains unclear. The study aimed to evaluate the association between pre-injury antithrombotic therapy and clinical outcomes in trauma patients aged 40-60 years.

METHODS: Data were obtained from the TraumaRegister DGU® (2015-2023). Patients aged 40-60 years from European trauma centers with ISS ≥ 9 were included. Patients were categorized according to the presence or absence of pre-injury antithrombotic therapy. A matched-pair analysis was performed based on age group, sex,

ASA classification, and injury severity by body region. The primary endpoint was in-hospital mortality; secondary outcomes included coagulopathy, transfusion requirements, emergency surgery, and organ failure. Analyses were primarily descriptive.

RESULTS: A total of 64,215 patients were included, of whom 6.7% (n = 4,326) received antithrombotic therapy. Antiplatelet agents were most common (68%), followed by direct oral anticoagulants (DOAC) (18%) and Vitamin-K-Antagonists (VKA) (11%). Coagulopathy was more frequent in anticoagulated patients, particularly in those receiving VKA and DOAC. In unmatched analyses, mortality was slightly higher in anticoagulated patients (6.7% vs. 5.0%) and showed a non-significant trend towards higher mortality in […]

PubMed · DOI

Correction: Impact of routine use of the GFAP and UCH-L1 TBI blood test for management of mild TBI on patient care in a European emergency department.

Brüning-Wolter F, Wolff N, Schrader M, Musaelyan K, Chandran R, Hoffmann M, Marino JA, McQuiston B et al.

PubMed · DOI

From rib fixation to dynamic chest wall reconstruction: historical lessons for modern surgical stabilization of rib fractures.

Sutori D, White TW, Bauman ZM, Whitbeck SS, Spering C

PURPOSE: Surgical stabilization of rib fractures (SSRF) has re-emerged as an important component of chest wall trauma care, yet its history is often presented as a simple timeline rather than as a source of operative and biomechanical lessons. This literature, problem-oriented historical review re-examines earlier rib fixation strategies in order to identify the mechanical and operative problems they were intended to solve, why many disappeared, and which principles remain relevant to contemporary SSRF practice.

METHODS: A literature review was conducted focusing on historically important fixation techniques and conceptual milestones in the management of serious rib fractures and flail chest. PubMed/MEDLINE and Google Scholar were searched from database inception to 19 March 2026 using terms related to rib fracture fixation, flail chest, SSRF, intramedullary fixation, and history. Studies were purposively selected if they described key fixation concepts, major shifts in management strategy, recurrent technical problems still seen in modern SSRF, or contemporary guideline context. Forty-seven articles were included and synthesized using a problem-based rather than purely chronological framework.

RESULTS: Across eras, rib fixation evolved around recurring challenges: (1) restoration of chest wall stability and thoracic continuity, (2) the tension between internal pneumatic stab […]

PubMed · DOI

Patient-reported outcome measures within European cohorts of severely injured patients: a systematic review and meta-analysis.

van den Berg L, van Ditshuizen JC, Leliefeld PHC, den Hartog D, Giannakópoulos GF, Verhofstad MHJ, de Jongh MAC

PURPOSE: Severe injury affects multiple health-related domains, yet comprehensive European data on patient-reported outcomes remain limited. This systematic review and meta-analysis evaluates patient-reported outcome measures (PROMs) use and outcomes in severely injured European cohorts.

METHODS: A systematic search of four databases up to October 14, 2025, identified European studies from 2000 onward reporting PROMs in severely injured patients. Severe injury was defined as an Injury Severity Score ≥ 16, Glasgow Coma Scale ≤ 8, intensive care unit admission, spinal cord injury, traumatic amputations, or pelvic fractures. Two reviewers independently screened records, with disagreements resolved by a third reviewer. Meta-analysis was performed when ≥ 3 studies reported comparable PROMs at similar follow-up timepoints.

RESULTS: Of 2,479 studies, 119 were included. Most cohorts originated from the Netherlands (26%), Norway (18%), and Germany (16%). General severely injured cohorts were most frequently studied (61%), followed by traumatic brain injury (17%), and spinal cord injury (15%). In total, 94 PROMs were used across 277 follow-up timepoints. Health-related quality of life was assessed most frequently (63%), after that anxiety/depression (14%), post-traumatic stress (9%), and social functioning (6%). At one year follow-up, the pooled EuroQol-5D-3 L index score was 0.70 (95% […]

PubMed · DOI

ABCB5■ mesenchymal stem cells enable high-ratio cellular augmentation of bone grafts without loss of osteogenic function.

Hofmann J, Bewersdorf TN, Sommer U, Borcherding K, Thiel K, Ganss C, Gerstner M, Frank MH et al.

Autologous bone grafting remains the standard for reconstruction of bone defects but is constrained by limited graft volume and loss of osteogenic capacity when expanded with acellular materials. We hypothesized that allogeneic ABCB5■ mesenchymal stem cells (MSCs) can serve as a biologic cell source to augment bone marrow-derived MSC (BM-MSC) populations without impairing osteogenesis. Primary human BM-MSCs from five donors were combined with GMP-manufactured ABCB5■ MSCs at graded replacement ratios (0-95%) and cultured under osteogenic conditions for 21 days. Osteogenic potential was evaluated using complementary assays of mineralization, biochemical activity, and matrix composition. Robust osteogenesis was maintained across all groups, with cultures containing 50% ABCB5■ MSCs demonstrating comparable functional osteogenic activity to BM-MSC controls, including similar 99mTc-HDP uptake (2.96 vs. 2.68 MBq) and calcium-phosphate ratios (1.2 vs. 1.1). Higher substitution levels showed a progressive decline in osteogenic output, consistent with dilution of BM-MSC-driven effects. These findings establish a quantitative threshold for biologic augmentation and demonstrate that substantial replacement of BM-MSCs is feasible without loss of function. This approach may enable cell-based expansion of bone grafts and supports further evaluation in translational in vivo models.

PubMed · DOI

Surgical outcomes in an improvised underground operating room during missile attacks.

Mitchnik IY, Essa A, Heinig O, Silberzweig JG, Ben Sira Y, Rabau O

PURPOSE: Armed conflict disrupts healthcare when hospital infrastructure is vulnerable to missile attacks. During a 12-day missile strike period, our hospital established an improvised operating room (OR) in an underground parking lot. This study evaluated perioperative and short-term outcomes among patients treated in the improvised OR compared with those treated in the hospital's fortified standard OR.

METHODS: This retrospective cohort study included patients undergoing general or orthopaedic surgery during the 12-day conflict. Procedures identified via surgical logs were categorized by OR location (improvised or standard). Demographic, perioperative, and post-discharge data were collected from electronic records. Primary outcomes included operative time, estimated blood loss, transfusions, complications, hospital and ICU length of stay, ED visits, readmissions, and 3-month mortality.

RESULTS: Overall, 74 surgeries took place in the improvised OR and 77 in the standard OR. Patients in each group were comparable in age, sex, comorbidities, and anesthesia. Improvised OR surgery time was longer (mean 68.1 vs. 47.7 minutes, $p < 0.001$). There were no significant differences between groups in terms of blood loss, transfusions, postoperative complications, or hospital length of stay. After 3-month follow-up, there were no significant differences in readmissions or mortality rates […]

PubMed · DOI

Repeated scoring with the adult appendicitis score improves the sensitivity and the specificity of appendicitis diagnosis in patients with early equivocal signs of appendicitis: a secondary analysis.

Lastunen KS, Leppäniemi AK, Mentula PJ

PURPOSE: The utilization of computed tomography in the early stage of acute appendicitis may result in overdiagnosis and unnecessarily expose patients to ionising radiation. The Adult Appendicitis Score (AAS) can be used to select patients for imaging. Observation and re-scoring in the DIAMOND trial reduced the need for imaging. Now, we wanted to determine if the change in AAS (ΔAAS) can serve as a diagnostic tool to select patients for imaging even more precisely.

METHODS: Eighty-eight patients with early equivocal appendicitis participated in the observation arm of the DIAMOND trial. The data for these patients were reanalysed, and ΔAAS during the observation was calculated. The baseline AAS, final AAS, and the change in C-reactive protein (ΔCRP) were selected as reference standards.

RESULTS: Eighty-three patients with complete data were included in the analysis. The AUROC (Area Under the Receiver Operating Characteristic) values are as follows: ΔAAS, 0.932 (95% CI 0.868-0.996); baseline AAS, 0.629 (95% CI 0.498-0.760); final AAS, 0.936 (95% CI 0.886-0.987); and ΔCRP, 0.796 (95% CI 0.696-0.897). Using receiver operating characteristic curves, we established the thresholds for low (AAS ≤ -2), intermediate (AAS -1 to 0), and high (AAS ≥ 1) probability of appendicitis. The negative predictive value for the low-probability group and the positive predictive value for the high-pr […]

PubMed · DOI

Long-term patient-reported outcomes and quality of life after tibiofemoral knee dislocations in patients with open growth plates: a retrospective case series from a Level 1 trauma center.

Hepp P, Youssef Y, Henkelmann R

BACKGROUND: Tibiofemoral dislocations in children and adolescents are rare, and treatment guidelines are primarily based on adult data and algorithms. This case series of four pediatric and adolescent patients aims to add to the limited literature and highlights key aspects of diagnosis, treatment, and patient-reported outcomes.

METHODS: We conducted a retrospective-prospective single-center case series of pediatric and adolescent patients (< 18 years, open growth plates) treated for tibiofemoral knee dislocations at a Level 1 trauma center between 2017 and 2019. Clinical, radiographic, and intraoperative data were reviewed. In addition, long-term patient-reported outcomes covering both functional recovery and health-related quality of life were assessed using KOOS, IKDC, and EQ-55 D-L.

RESULTS: Four patients (2 male, 2 female), aged between 12 and 16 years, were included in this case series. The mean follow-up was 7.3 years (range 6 to 8 years). All patients sustained Schenck Type III-L dislocations. Early MRI and vascular imaging identified frequent associated injuries such as meniscal and chondral lesions, avulsion fractures, in one case, peroneal nerve injury. A staged surgical approach was used in two of the four cases. No growth disturbances or deformities were observed in the patients based on the available clinical and radiological follow-up. Functional outcomes range […]

PubMed · DOI

The need for, feasibility of, and barriers to implementation of a trauma traveling fellowship: a survey of ESTES members.

Wohlgemut JM, Gupta S, Cioffi SPB, Llaquet-Bayo H, Wohlgemut HS, Hobby J, Mohseni S, Mariani D et al.

PURPOSE: Many European surgeons lack access to high-volume trauma training. A Trauma Traveling Fellowship, supported by the European Society of Trauma and Emergency Surgery (ESTES), could broaden educational and clinical exposure to trauma care, but feasibility and implementation factors are unclear. The aims of this survey are to assess the aspirations, expectations, and perceived challenges of prospective fellows and host centres for an ESTES Trauma Traveling Fellowship.

METHODS: A cross-sectional, web-based survey was distributed to all ESTES members, capturing demographics, fellowship aspirations, perceived barriers, and host centre capacities. The survey incorporated Likert scales, free-text responses, and was guided by the Consolidated Framework for Implementation Research (CFIR) and the Reach, Effectiveness, Adoption, Implementation, and Maintenance / Practical Robust Implementation and Sustainability Model (RE-AIM/PRISM).

RESULTS: Of 903 invited, 210 responses (23.3%) were analysed. The median age was 38 years; 71% were consultants, and 29% were residents. Most (71%) would consider applying for a Fellowship, and 47% stated their centre could host Fellows. Only 23% currently access funded fellowships. Key aspirations included hands-on participation in trauma care, although hosts could more often only guarantee observational roles. Major barriers included funding and in […]

PubMed · DOI

Antegrade transcystic balloon dilatation in emergency laparoscopic cholecystectomy with MRCP validation.

Lipping E, Saar S, Kirss J, Aaspõllu H, Bahhir A, Gapejeva O, Kangur T, Rauk M et al.

PURPOSE: Endoscopic retrograde cholangiopancreatography (ERCP) remains the predominant treatment of common bile duct stones but single-stage laparoscopic management via antegrade transcystic balloon dilatation (ATBD) offers a compelling alternative. All prior studies validate success solely by intraoperative fluoroscopic cholangiography, with no prospective trial employing postoperative magnetic resonance cholangiopancreatography (MRCP).

METHODS: This single-centre prospective study enrolled 100 consecutive patients with confirmed CBD stones requiring emergency laparoscopic cholecystectomy (LC). ATBD was attempted in all patients. Primary outcome was MRCP-confirmed CBD stone clearance. Secondary outcomes included operative time and postoperative complications.

RESULTS: One hundred consecutive patients were enrolled (median age 69 years; 38% male; 56% ASA III-IV). ATBD was performed in 88 patients (88%). Postoperative MRCP was performed in 86 of 88 patients (98%), with residual stones identified in 24 (28%; 95% CI 19.5-38.2%), corresponding to an MRCP-confirmed CBD clearance rate of 72% (95% CI 61.8-80.5%). The median operative time was 90 min (IQR 75-111.5). Among the 88 patients who underwent ATBD, postoperative complications occurred in 12 (14%), most commonly pancreatitis (4.5%) and deep surgical site infection (4.5%); one patient died following severe post-procedural panc […]

PubMed · DOI

Closed reduction and percutaneous fixation versus open reduction and internal fixation for displaced medial malleolar fractures: a randomized controlled trial.

Elkohail A, Elkasaby A, Shaheen AMZ, Teama M, Raza ML, Ibrahim IA

OBJECTIVE: To evaluate the radiographic and functional outcomes of closed reduction and percutaneous fixation (CRPF) compared to open reduction and internal fixation (ORIF) in the treatment of isolated displaced medial malleolar fractures.

METHODS: This randomized controlled trial included 50 adult patients with isolated displaced medial malleolar fractures, Herscovici classification type B and C, across two hospitals. Participants were randomly assigned (1:1) to either CRPF (Group A, n = 25) or ORIF (Group B, n = 25). The principal objectives were radiological union and clinical function evaluated by the American Orthopedic Foot and Ankle Society (AOFAS) score at a 6-month follow-up. Secondary outcomes included the incidence of postoperative complications and correlations with patient demographics and mechanism of injury.

RESULTS: At 6 months, there was no significant difference between groups ($p > 0.05$); in Group A, 13 patients rated as "Good," 11 as "Fair," and 1 as "Poor," while in Group B, 15 were "Good," 9 "Fair," and 1 "Poor." AOFAS outcomes showed 13 excellent, 7 good, 4 fair, and 1 poor in Group A, versus 12 excellent, 9 good, 3 fair, and 1 poor in Group B, with no significant difference ($p > 0.05$). Complication rates were similar: wound infection (CRPF 4% vs. ORIF 12%), malunion/non-union (8% vs. 4%), hardware discomfort (12% vs. 20%), and deep venous thrombosis (4% […]

PubMed · DOI

Long-term radiological and clinical outcomes after surgical treatment of non-union: a ten-year follow-up study.

Böpple JC, Berger J, Ferbert T, Bewersdorf TN, Schmidmaier G, Findeisen S

PURPOSE: Non-unions are a major challenge in trauma surgery. They are known to affect patients' lives for years after the initial trauma. Nevertheless, most studies only examine the first two years after treatment. The question what the long-term results look like ten years after surgery was posed in light of observations made in everyday clinical practice. To answer this, our study was designed to examine all patients who underwent non-union surgery in our department ten years ago.

METHODS: All patients who underwent surgery between July 2014 and October 2015 were invited to participate in the study. During a one-site appointment, an X-ray image was taken, which was then used to assess consolidation. The patients underwent a physical examination and answered a questionnaire, covering topics such as pain on a visual analogue scale (VAS) and overall quality of life with the short-form 12 (SF-12).

RESULTS: A total of 54 patients were included in the study. The median follow-up period was 122 months (IQR: 118-124). After 10 years 88.9% patients showed osseus consolidation with a median Lane-Sandhu-Score (LSS) of

4 (IQR: 3-4). The Wilcoxon signed-rank test revealed a significant difference compared to 24 months postoperatively (Z = 4.716, p < 0.001). The difference in the consolidation showed a marginally significant result in the McNemar test with p = 0.063. A total of 51 patien […]

PubMed · DOI

Artificial intelligence for predicting 30-day mortality after emergency laparotomy.

Bedrikovetski S, Murshed I, Seow W, Bunjo Z, De Fontgalland JS, Traeger L, Vather R

PURPOSE: Accurate pre-operative risk prediction in emergency laparotomy (EL) is essential for appropriate resource allocation and improve patient outcomes. This study aimed to establish the accuracy of an artificial intelligence (AI) model in predicting 30-day mortality after EL using a national dataset.

METHODS: Data was extracted from the Australian and New Zealand Emergency Laparotomy Audit-Quality Improvement (ANZELA-QI) database from 1 July 2018 to 24 July 2023. AI models were created employing multilayer perceptron (MLP) and radial basis function (RBF) architectures. Extended AI models were developed using all available pre-operative variables and simplified AI models were developed using statistically significant variables identified through univariate analysis. We assessed the performance of AI models with that of a multivariate logistic regression (LR) and the UK-based National Emergency Laparotomy Audit (NELA) models using the area under the receiver operator characteristic curve (AUROC).

RESULTS: Data from 8293 ELs were included in the model and were randomly divided into a training dataset of 6619 (80%) and a testing dataset of 1674 (20%). There were 537 (6.5%) deaths 30 days after EL. In the testing dataset, the NELA model performed the best (AUROC 0.836), followed by the multivariate LR model (0.824), MLP simplified (0.817). MLP extended (0.802), RBF extended (0 […]

PubMed · DOI

Predicting postoperative course in Fournier's gangrene surgery: evaluation of the E-PASS score in a single-center retrospective study.

Gercek O, Senkol M, Ulucak MA, Yazar VM

BACKGROUND: The E-PASS scoring system, which incorporates preoperative data and surgical stress factors, may provide a reliable risk classification for predicting mortality risk in patients with Fournier's gangrene. This study aimed to investigate the value of the E-PASS score in predicting mortality in patients with Fournier's gangrene.

METHODS: The study included 41 patients who were diagnosed with FG clinically and underwent surgery. Preoperative demographic and clinical data, laboratory values, FGSI, and E-PASS scores were recorded. The patients were divided into two groups: 8 (19.5%) patients who developed mortality and 33 (80.5%) patients who were discharged without mortality. Demographic, clinical, and laboratory data, as well as FGSI and E-PASS scores, were compared between the groups.

RESULTS: In the mortality group, CRP, serum creatinine levels, FGSI, E-PASS PRS, and E-PASS CRS scores were observed to be significantly higher, while serum albumin and hematocrit levels were lower compared to the other group (respectively, p = 0.020, p = 0.001, p < 0.001, p < 0.001, p < 0.001, p = 0.001, p < 0.001). FGSI, E-PASS CRS, CRP, and serum albumin values were found to have diagnostic value in predicting mortality. The cut-off values were determined as 7.5 for FGSI, 0.55 for E-PASS CRS, 186.25 for CRP, and 2.92 for albumin.

CONCLUSION: Our findings show that E-PASS can provide […]

PubMed · DOI

Thoracoscopic internal fixation, open surgical internal fixation, and conservative treatment of rib fractures: differences in postoperative rehabilitation and chest volume (a randomized controlled trial).

Li T, Zhao P, Li W, Liu B

BACKGROUND: Currently, the standard treatment for multiple rib fractures is still under debate. This study explores the efficacy of three treatments (thoracoscopic internal fixation, open surgical internal fixation and conservative treatment) for multiple rib fractures and their influence on thoracic cavity volume and postoperative rehabilitation.

METHODS: Patients with multiple rib fractures admitted to Weifang People's Hospital of Shandong Province from March 2019 to March 2021 were enrolled in this study. The thoracic volume changes before and after treatment were measured by three-dimensional computed tomography (CT) reconstruction. Twenty-seven patients in the open operation group underwent internal fixation with the rib cage, and 24 patients in the non-internal fixation group underwent external fixation with the chest band. The recovery of multiple rib fractures was evaluated by comparing the difference in thoracic volume before and after treatment, the incidence of related complications and the Comprehensive Assessment of Generic Quality of Life Inventory-74 (GQOLI-74) score one month after the operation.

RESULTS: After 12 months of follow-up, thoracic volume was significantly different between the thoracoscopic and open fixation groups ($P < 0.001$). One month after the thoracoscopic internal fixation of rib fractures, the scores of physical function, psychological func […]

PubMed · DOI

What is the rate and degree of deformity in femoral diaphyseal nonunion? A CT-based retrospective cohort study.

Nabben J, Wagner RK, Groepenhoff F, Verwest TA, Bont Y, Kloen P, Janssen SJ

BACKGROUND: This study investigated the rate and degree of deformity in femoral diaphyseal nonunion using preoperative bilateral CT-scans.

METHODS: A retrospective cohort study was conducted at a single Level 1 trauma center. Patients aged > 18 years who underwent operative treatment for femoral diaphyseal nonunion between 2015 and 2025 and had preoperative bilateral femur CT-scans were included. Axial, sagittal, coronal, and length differences between the femoral nonunion and the contralateral healthy femur were measured. Torsional deformity ≥ 15° and length discrepancy ≥ 20 mm were considered clinically relevant. Thresholds of ≥ 5° and ≥ 10° were used to identify outliers in coronal and sagittal plane deformity, respectively.

RESULTS: Thirty-two patients met the inclusion criteria (median age 57 years, 63% male). Torsional deformity ≥ 15° was present in 10 patients (31%), including internal rotation in 4 (13%) and external rotation in 6 (19%). Sagittal plane deformity ≥ 10° was observed in 5 patients (16%), with apex anterior angulation in 1 (3%) and apex posterior angulation in 4 (13%). Mean sagittal plane angulation was 9.7° ± 6.7° in the affected limb and 12.5° ± 2.6° in the unaffected limb ($p = 0.017$). Coronal plane deformity ≥ 5° was present in 10 patients (31%), including varus in 8 (25%) and valgus in 2 (6%). Mean coronal angulation was 5.3° ± 6.0° in the affected li […]

PubMed · DOI

The impact of pre- versus postoperative geriatric consultation in elderly patients with hip fractures.

Aydin FB, Laane DWPM, van Rossum du Chattel A, DiStefano MT, King AH, Wignakumar T, Nwagbata A, van der Velde D et al.

PURPOSE: To determine whether preoperative geriatric consultation was independently associated with lower incidence of delirium for elderly patients with hip fractures compared with postoperative geriatric consultation.

METHODS: This retrospective cohort study was conducted at two urban Level 1 trauma centers in the United States. Patients aged ≥ 70 years with isolated proximal femur fractures treated surgically between 2018 and 2024 were included. Patients with pathological or periprosthetic fractures, non-orthopedic admissions, and those without geriatric consultation were excluded. Patients were categorized based on timing of initial geriatric consultation: preoperative versus postoperative. Delirium was selected as the primary outcome because geriatric consultation most directly addresses modifiable delirium risk factors. Secondary outcomes included the incidence of one or more in-hospital complications, length of stay, 30-day all-cause readmissions, and mortality. Multivariable logistic and linear regression analyses were used to determine the independent association between timing of initial geriatric consultation and primary and secondary outcomes.

RESULTS: Of 1776 patients included, 1382 (78%) received initial geriatric consultation preoperatively and 394 (22%) postoperatively. Median age was 84 years (IQR 78-90), 70% were female, 73% ASA III, and 87% lived at home pr […]

PubMed · DOI

Volume-dependent detection of free intraperitoneal fluid by FAST: a prospective in-vivo study with sequential fluid instillation.

Güsgen C, Nada A, Kornek MT, Willms A, Schaaf S

PURPOSE: To assess the volume-dependent sensitivity of the Focused Assessment with Sonography for Trauma (FAST) and to analyse regional detection performance under controlled in-vivo conditions.

METHODS: In this prospective single-centre study, sequential intraperitoneal instillation of 0.9% NaCl (250-1500 mL) was performed during elective laparoscopic procedures in 35 adults. Standardised FAST examinations were conducted by two experienced sonographers using transverse and longitudinal views of Morison's pouch, the Douglas recess, and Koller's space. Sensitivity was calculated for each instilled volume and anatomical region. Specificity could not be assessed due to the experimental design with mandatory fluid presence. Regional differences were tested using one-way ANOVA, and a Monte Carlo simulation (1000 resamples) evaluated detection stability across body-mass-index (BMI) groups.

RESULTS: FAST sensitivity increased with volume: 14.3% (95% CI 4.8-30.3) at 250 mL, 37.1% (21.5-55.1) at 500 mL, 74.3% (56.7-87.5) at 1000 mL, and 91.4% (76.9-98.2) at 1500 mL. The highest detection rate occurred in Morison's pouch, followed by the Douglas and Koller recesses. Longitudinal scanning of the Douglas pouch improved sensitivity by > 20% compared with the transverse view. ANOVA confirmed significant regional variability ($F(5, 153) = 4.37$, $p = 0.001$, $\eta^2 = 0.125$). Detection rates decline […]

PubMed · DOI

When demography challenges policy: the rising burden of proximal femur fractures as a role model and driver for geriatric trauma care in Germany.

Klug A, Münzberg M, Sibbel R

PURPOSE: Germany's hospital reforms will substantially reshape service delivery and capacity planning. Proximal femur fractures (PFFs), the most frequent geriatric fractures, are highly sensitive to demographic aging and therefore serve as a model condition to assess future healthcare system requirements.

METHODS: Nationwide inpatient data from the German Federal Statistical Office (2000-2023) were analyzed for proximal femur fractures (ICD-10-GM S72.0-S72.2). Demographic projections up to 2050 were combined with Poisson regression, exponential smoothing, and autoregressive integrated moving average models to forecast future incidence and case numbers. Model performance was evaluated in a hold-out period (2016-2023) using mean absolute percentage error and root mean square error. Direct costs were estimated using the official German cost-of-illness framework. Staffing and hospital bed requirements were derived from case-based activity times, profession-specific full-time equivalent coefficients, and observed department-specific occupancy rates.

RESULTS: Between 2000 and 2023, annual proximal femur fracture cases in Germany increased from 118,243 to 182,447 (+ 54.3%). Forecasts project a rise to 317,877 cases by 2050 (95% CI 262,436-373,317), corresponding to an increase of approximately 75% compared with 2023. Patients aged ≥ 80 years are projected to account for 74% of all c […]

PubMed · DOI

Orbital floor fractures: clinical characteristics, mechanisms of injury, and surgical outcomes in a 25-year single-center study.

Kasap ■, Ayd■n MS, Özler G, Ni■anc■ M, Özvurmaz O

PURPOSE: Orbital floor fractures are common injuries following blunt midfacial trauma, presenting as isolated defects or as part of complex craniofacial fractures. This study aimed to evaluate clinical characteristics, fracture patterns, management strategies, and postoperative outcomes in surgically treated patients.

METHODS: This retrospective single-center study included 56 patients who underwent surgical treatment for orbital floor fractures between 2000 and 2025. Demographic data, injury mechanisms, imaging findings, associated fractures, clinical presentation, surgical techniques, and outcomes were analyzed. Minimum follow-up was 3 months. Statistical analysis was performed using IBM SPSS Statistics version 25.0, with $p < 0.05$ considered significant.

RESULTS: Among 56 patients, 80.4% were male, with a mean age of 38.6 years (10-70). Assault (44.6%) was the most common cause, followed by in-vehicle accidents (21.4%) and falls (17.9%). Inferior wall fractures were most frequent (55.4%), followed by medial (28.6%) and lateral (16.1%) involvement; no superior fractures were observed. Common findings included periorbital edema/ecchymosis (75.0%), diplopia (26.8%), and gaze limitation (19.6%); infraorbital hypoesthesia occurred in 12.5%. Associated fractures involved the maxilla (60.7%) and zygoma (46.4%).Surgical treatment included exploration/elevation (46.4%), titanium mes […]

PubMed · DOI

Epidemiology and healthcare burden of implant-associated complications in Germany: a nationwide population-based analysis.

Anees H, Heiss C, El Khassawna T

PURPOSE: Implant-associated complications remain a major challenge in orthopaedic and trauma surgery, particularly among elderly patients requiring revision procedures or infection management. This study analysed nationwide trends and healthcare burden of implant-associated complications in Germany.

METHODS: Nationwide German hospital discharge data from 2000 to 2023 were analysed and complemented by Diagnosis-Related Group (DRG) data from 2024. Cases were identified using ICD-10-GM code T84 and its subcategories. Outcomes included temporal trends, age-standardised incidence, demographic characteristics, in-hospital mortality, length of stay (LOS), and DRG-based healthcare resource utilisation.

RESULTS: Hospitalisations for implant-associated complications increased from 45,121 in 2000 to a peak of 95,154 in 2011, followed by a decline to 72,846 in 2023. Age-standardised incidence decreased by approximately 31% between 2010 and 2023. Segmented regression identified a temporal breakpoint in 2010.6, with annual percentage changes of + 6.21% before and - 3.55% after the breakpoint. Annual hospitalisations for periprosthetic joint infection increased from 5,367 to 11,215, while osteosynthesis-associated infections increased from 2,469 to 7,059. Mean LOS decreased from 20.8 to 11.8 days, whereas in-hospital mortality remained stable. Compared with low-complexity cases, infection-r […]

PubMed · DOI

Effect of periprosthetic greater trochanter fractures (Vancouver AG) on primary stability of the femoral stem in total hip arthroplasty: a biomechanical micromotion analysis.

Bruder J, Goronzi M, Cavalcanti Kußmaul A, Wulf J, Pachmann F, Holzapfel BM, Böcker W, Kistler M

PURPOSE: Primary stability is a crucial factor for the long-term success of cementless total hip arthroplasty (THA). Periprosthetic fractures represent an important cause of loss of primary implant stability. While Vancouver AG fractures rarely lead to stem loosening, their effect on the metaphyseal anchoring of the stems has not yet been investigated. The present study aimed to evaluate the influence of a Vancouver AG fracture on the primary stability of the CLS Spotorno stem.

METHODS: Six fourth-generation composite femora were implanted with cementless CLS Spotorno stems (Zimmer, Warsaw, USA). Primary stability was determined by three-dimensional digital image correlation under dynamic loading conditions (300-1700 N, 1 Hz) simulating physiological gait. Micromotion was recorded at six standardized measurement points (proximal, mid, distal, medial and ventral) before and after a simulated Vancouver AG fracture. Data was statistically analyzed using paired one-sided t-tests ($\alpha = 0.05$).

RESULTS: The results revealed no significant increase in micromotion at the primary anchorage zone of the stem. However, a non-significant increase in micromotion was observed at the prosthesis tip (DGventral and DGmedial), where the micromotion values increased from 164.4 µm (± 59.6 µm) and 140.4 µm (± 75.6 µm) to 193.1 µm (± 48.8 µm) and 186.9 µm (± 60.5 µm), respectively.

CONCLUSION: Vanco […]

PubMed · DOI

Injury patterns, management and outcomes of hemodynamically unstable pelvic fractures at an Australian level 1 trauma centre: a 10-year registry analysis.

Louw L, Raubenheimer K, Blakeney WG, Young S, Balogh ZJ, Weber DG

INTRODUCTION/BACKGROUND: Pelvic ring injuries (PRI) in traumatic shock are associated with over 30% mortality even in high income countries' mature trauma systems, but recent Australian data showed no haemorrhage related mortality in this group. We aimed to describe the epidemiology, management and outcomes of PRI with haemodynamic instability managed in a state's only Level 1 adult trauma centre.

DESIGN: We conducted a retrospective cohort study of all trauma patients with major PRI (Abbreviated Injury Scale (AIS) > 3) and hemodynamic instability (defined as systolic blood pressure (SBP) <90 mmHg or a shock index (heart rate (HR)/SBP) > 0.7) presenting to the state adult trauma centre between 2013 and 2022. Data on demographics, injury patterns, management strategies, and outcomes were collected from the State Trauma Registry and hospital records.

RESULTS: Of 1369 patients with pelvic or acetabular fractures during the 10-year study period, 152 (11%) had major PRI with hemodynamic instability at presentation. Most were male (72%) with a median age of 35 years. Motor vehicle and motorbike crashes accounted for 52% of injuries. The median Injury Severity Score was 33. Massive transfusion protocols were activated in 45% of cases, while angioembolisation was utilised in 5.9%. Overall in-hospital mortality was 7.2% and mortality due to haemorrhage was 2%.

CONCLUSIONS: In our mat […]

PubMed · DOI

Time to surgery and its impact on length of stay and mortality for neck of femur fractures at a major trauma centre.

Tang LY, Ho SYS, Zhou AK, Geetala R, Krkovic M

PURPOSE: Neck of femur (NOF) fractures are a major public health concern. Timely surgical intervention (current guidelines is within 36 h) is associated with improved survival, shorter hospital stays, and better functional outcomes. However, surgical delays remain common in hospitals, especially in major trauma centres, and their impact on length of stay (LOS) and mortality is not well defined across different NOF fracture subtypes.

METHODS: We conducted a retrospective cohort study of 4415 patients with NOF fractures, classified into intracapsular (n = 2507) and extracapsular (n = 1908). The primary outcome data gathered were LOS (from surgery to discharge) and in-hospital mortality. Regression analyses were performed to assess associations between LOS, and mortality, adjusting for age, sex, Charlson Comorbidity Index (CCI), American Society of Anesthesiologists (ASA) score, intra-operative blood transfusion, and surgical methods.

RESULTS: For intracapsular fractures, delays in time to surgery were significantly associated with increased LOS in patients expected to have short to median lengths of stay, after adjusting for other variables ($p < 0.05$). Delayed surgery was associated with increased mortality for both intracapsular and extracapsular fractures ($p < 0.0001$). Survival curves showed significant differences for the surgical delayed group. After controlling for time to […]

PubMed · DOI

Pattern of blast injuries. A Systematic Review: Part 1-Improvised, multiple and unspecified explosive devices.

Neubert A, Seidelmann M, Gaeth C, Bieler D

PURPOSE: This systematic review aims to comprehensively identify and categorize the specific injury patterns associated with IEDs and other explosive devices across military and civilian populations to improve medical preparedness, resource planning strategies, and treatment approaches.

METHODS: A search on PubMed und Scopus was conducted on 26. March 2025 for observational studies reporting blast injuries in civilian and military populations of all ages. Studies were clustered by explosion type - Part 1 IED and multiple/unspecified explosives. A narrative data synthesis was performed. Reporting was assessed with the JBI tool for case series.

PROSPERO: CRD42023391736.

RESULTS: 33 studies with 4,373 victims (mainly military personnel) from 2002 - 2022 reported injuries caused by IEDs. The average age was 25.1 years. 97.2% were male. Injuries to the upper and lower extremities were predominant in all study populations followed by head injuries among military personnel, and thorax and abdomen injuries among civilians. 37 studies with 24,600 casualties (mean age 25.7, 85.8% males) were included in

multiple/unspecified explosives. Extremities were predominant followed by facial injuries among military personnel, and thorax injuries among civilians.

CONCLUSION: The first part of this systematic review on the pattern of blast injuries shows that injuries to the extremities, includ […]

PubMed · DOI

Clinical and laboratory predictors of mortality in Fournier's gangrene: prognostic role of the HALP score.

Yavuz A, Ersöz V, Çamöz MS, Yurdakul ZN, ■■deci Tut G, Demirci BF, Koçsoy NS, Dönmez M

OBJECTIVE: Fournier's gangrene (FG) is a rapidly progressive necrotizing soft-tissue infection associated with substantial mortality. This study evaluated the prognostic performance of the Hemoglobin-Albumin-Lymphocyte-Platelet (HALP) score and routinely available clinical and laboratory parameters in patients with FG.

METHODS: This retrospective cohort study included 236 consecutive patients who underwent surgical treatment for FG between February 2019 and May 2025. The primary outcome was in-hospital mortality. Prognostic factors were evaluated using univariate and multivariable logistic regression analyses, and discriminative performance was assessed by receiver operating characteristic (ROC) curve analysis.

RESULTS: The overall in-hospital mortality rate was 12.3%. Non-survivors had significantly lower HALP scores, hemoglobin, hematocrit, albumin, lymphocyte, and monocyte levels (all $p < 0.05$). In univariate analysis, younger age, disease confined to the perianal region, and higher albumin, monocyte, hemoglobin, hematocrit, and HALP values were associated with survival. In the primary multivariable model, advanced age (OR = 0.966, 95% CI: 0.936-0.997; $p = 0.030$) and monocyte count (OR = 5.243, 95% CI: 1.055-26.049; $p = 0.043$) remained independent predictors of survival. Hemoglobin showed the highest discriminative performance (AUC = 0.781), followed by albumin (0.767), he […]

PubMed · DOI

Surgical stabilization of rib fractures is associated with lower adjusted odds of adverse pulmonary outcomes across RibScore-defined injury severity: a single-center cohort study.

Muldiiarov V, Liu JL, Campbell A, Bouldoukian S, Daro A, Kemp SE, Matos M, Cantrell E et al.

PURPOSE: Surgical stabilization of rib fractures (SSRF) is increasingly used for selected patients with severe chest wall injury, but its relationship with pulmonary outcomes after accounting for RibScore-defined injury severity remains unclear. This study evaluated adverse pulmonary outcomes after SSRF versus nonoperative management (NONOP) among adults with traumatic rib fractures.

METHODS: Single-center retrospective cohort study at an ACS-verified Level I trauma center from January 2016 to April 2023. Adults with CT-confirmed blunt traumatic rib fractures were categorized as SSRF or nonoperative management. RibScore was calculated from initial CT imaging. The primary outcome was a composite of pneumonia, invasive mechanical ventilation for more than 48 h, or tracheostomy. Multivariable logistic regression adjusted for demographics, RibScore, thoracic and extra-thoracic injury severity, calendar year, COPD, and current smoking status.

RESULTS: Among 3,066 patients, 444 underwent SSRF and 2,622 were managed nonoperatively. SSRF patients had greater radiographic chest wall injury severity, including a higher median RibScore than NONOP patients (2 vs. 0; $p < 0.001$). Composite adverse pulmonary outcome occurred in 14.6% of SSRF patients and 16.6% of NONOP patients. In adjusted analysis, SSRF was associated with lower odds of composite adverse pulmonary outcome (aOR 0.57, 95% C […]

PubMed · DOI

Physician-staffed prehospital units for severely injured patients in the Stockholm trauma system - is there a survival benefit?

Helmersson K, Lapidus O, Alm T, Bäckström D, Rubensson Wahlin R

BACKGROUND: The role of prehospital physicians for patients with major trauma has previously been examined with mixed results. No studies have assessed the survival benefit of physician-staffed prehospital units in the context of the Stockholm trauma system.

AIM: To investigate the association between exposure to prehospital physicians and mortality for severely injured trauma patients in the Stockholm trauma system.

METHODS: A total of 11,063 trauma patients were obtained from the Swedish Trauma Registry and stratified according to physician exposure in the prehospital setting. Logistic regression was used to evaluate the association between prehospital physician exposure and 30-day mortality and Glasgow Outcome Score (GOS).

RESULTS: No significant reduction in 30-day mortality was observed in the P-EMS group compared to the EMS group (OR 1.06, p = 0.71). However, transport with helicopter emergency medical services (HEMS) was independently associated with greater odds of survival. The P-EMS group had significantly lower GOS at discharge compared to EMS (OR 1.72, p < 0.001).

CONCLUSION: The potential benefit of prehospital physician exposure for major trauma patients remains uncertain. However, HEMS transport was associated with improved survival. Further research is needed to determine which patients benefit from prehospital physicians and to evaluate the optimal allocati […]

PubMed · DOI

Clinical impact and economic burden of electric scooter-related facial fractures in Barcelona.

Lugilde-Guerbek A, López-Masramon B, García Sánchez V, Andreu-Sola V, Collado-Delfa JM, Pueyo Morillo C, Arteaga A, Grabosky-Elbaile A et al.

PURPOSE: The landscape of facial trauma has evolved with the integration of micromobility devices into urban environments. This study aimed to characterize the clinical profile of electric scooter (e-scooter) accidents compared with traditional etiologies (traffic, assaults, falls) and evaluate the economic burden imposed on the public health system.

METHODS: A cohort of 119 patients who underwent surgery for facial fractures at a Spanish tertiary hospital between 2020 and 2024 was analyzed. To accurately assess economic impact, a specific subgroup of 51 "pure facial fractures"-those with isolated injuries-was utilized for cost analysis.

RESULTS: E-scooter users (10.1% of the total cohort) were significantly younger (mean age 27.9 years, p = 0.044), had shorter hospital stays (7.7 days, p = 0.007), and lower ICU requirements (16.7%) than the traditional group. Adverse outcomes accounted for 29.4% of cases overall, with a significantly higher rate in the e-scooter group compared to other etiologies (58.3% vs. 26.2%; p = 0.039). The median cost for an e-scooter injury was €8,092; though higher than other causes, this was not statistically significant (p = 0.33).

CONCLUSION: E-scooters generate complex facial injuries with clinical severity and complication rates comparable to high-energy trauma. The high economic cost and reconstructive complexity, suggest a need for stricter […]

PubMed · DOI

Significant other reported outcome measures within European cohorts of severely injured patients: A systematic review.

Van Ditshuizen JC, Berg LVD, Verhofstad MHJ, Leliefeld PHC, Jongh MAC, Hartog DD

PURPOSE: Major trauma can have a long-term impact, leading to diminished health-related quality of life (HRQoL) for patients and their significant others (SOs). Patient reported outcome measures (PROMs) make this impact visible, providing expectation- and self-management. European studies assessing Significant Other Reported Outcome Measures (SOROMs) in cohorts of severely injured patients were reviewed.

METHODS: A systematic literature search was conducted. European studies from 2000 onwards assessing SOROMs in severely injured patient cohorts were included. Two reviewers independently screened all records, consensus was reached by a third reviewer.

RESULTS: 2,909 studies were screened, resulting in 30 inclusions. Studies originated from Western and Northern Europe, overrepresented by traumatic brain injury (n = 27, 90%). Forty-three different SOROMs were identified.

Twenty (67%) studies reported subacute and long-term reduction in HRQoL, and high prevalence (28-56%) of depression, anxiety, and stress within SOs. Outcomes of SOs and patients run parallel and are associated with coping strategies. Sixteen (53%) studies reported around 50% of unmet family needs (information and emotional support), and a disproportionate burden of care associated with loneliness and life satisfaction.

CONCLUSION: SOs of severely injured patients reported reduced HRQoL, with high levels of depr […]

PubMed · DOI

Early treatment goal uncertainty as a barrier to regional analgesia and respiratory deterioration in traumatic rib fractures: a propensity-weighted analysis.

Yi SB, Wu JH, Zhang XN, Lin DC, Zheng YL

PURPOSE: Patients with multiple rib fractures (≥ 3) admitted to the intensive care unit (ICU) are at high risk of early respiratory deterioration, for which timely and effective analgesia is essential. Although multimodal analgesic strategies are widely recommended, their real-world implementation may be influenced by variability in early clinical decision-making, including treatment goal uncertainty (TGU). We aimed to investigate the association between early TGU, analgesic strategy selection, and respiratory outcomes in critically injured patients with rib fractures.

METHODS: This single-center observational cohort study included 236 adult ICU patients with ≥ 3 radiologically confirmed rib fractures who did not require invasive mechanical ventilation at admission (2020-2024). TGU within the first 48 h was assessed using blinded electronic medical record review as a process-level marker of variability in early care prioritization. Analgesic strategies were categorized as regional-based analgesia (RA), systemic-dominant analgesia (SA), or multimodal conservative management. The primary outcome was respiratory deterioration, defined as a composite of escalation of respiratory support, endotracheal intubation, or pulmonary complications requiring intervention. Associations were evaluated using propensity score overlap weighting and adjusted regression models.

RESULTS: High TGU […]

PubMed · DOI

Multifocal necrotizing fasciitis: what makes it different? A German trauma center experience.

Keßler F, Frank J, Störmann P, Wagner N, Saman AE, Marzi I, Leiblein M

BACKGROUND: Multifocal necrotizing fasciitis (MulNF) is a rare presentation of necrotizing fasciitis (NF) characterized by involvement of more than one non-contiguous anatomical site during the same disease episode. Comparative data between monofocal and multifocal NF remain limited.

METHODS: We performed a retrospective single-center cohort study of patients treated for NF between 2015 and 2026. Patients were classified as MNF (single site) or MulNF (multiple non-contiguous sites) based on chart review. Demographics, comorbidities, microbiology, laboratory values, organ failure, ICU course, surgical burden, and outcomes were compared.

RESULTS: Forty-seven patients were included, of whom five (10.6%) had multifocal NF. Four of these showed clearly synchronous non-contiguous involvement, whereas one developed a delayed second non-contiguous focus within the first four days of the same disease episode. Compared with MNF, MulNF was associated with higher ICU burden, more frequent mechanical ventilation, and a greater number of debridements and total operations. Higher rates of septic shock, hemodynamic failure, and renal replacement therapy were observed in MulNF. No deaths occurred in the small multifocal cohort, whereas 11/42 patients with monofocal NF died during hospitalization. Given the small multifocal sample size and multiple comparisons performed, all findings should be […]

PubMed · DOI

Trauma-related retrobulbar hematoma: predictive risk factors and clinical outcome in 52 cases.

Bicsák Á, ■ahin H, Brandt L, Hassfeld S, Bonitz L

PURPOSE: Retrobulbar hematoma (RBH) is a vision-threatening emergency. This large-scale study integrates clinical data and 3D analysis to identify prognostic factors, refine diagnosis, and support guideline-based emergency management.

METHODS: This study standardizes terminology, applies AO fracture classification, and uses CT-based 3D segmentation to analyze anatomical patterns, treatment protocols, providing evidence-based guidance for diagnosis, surgical management, and postoperative care.

RESULTS: The Dortmund Maxillofacial Trauma Registry (2018-2025) recorded 52 RBHs among 7,671 trauma patients (0.7%). RBH occurred predominantly in middle-aged males and elderly females, with overproportionate incidence of anticoagulant therapy. Most fractures clustered around the orbital floor, medial wall, zygomatic surface, often near the Frankfurt horizontal plane. RBHs were mainly extraconal; poor outcome was predicted with intraconal location, optic nerve contact, orbital floor fracture, high RBH/orbital volume ratio, and aspirin use. Surgical decompression was performed in 15 cases (28.9%); 65,4% (34 of 52) retained vision. Neural network analysis confirmed fracture site and hematoma position as key prognostic factors over demographic or comorbidity variables.

CONCLUSION: In 7,671 midface fracture cases, RBHs occurred in 0.7% (n = 52), mainly after trauma. Falls predominated in ae […]

PubMed · DOI

Patterns of negative trauma laparotomy worldwide: a sub-analysis of a global dataset.

Laird WLA, Bath MF, Amoako J, Wohlgemut J, Nuño-Guzmán CM, Khajanchi M, Smith BG, Hobbs L et al.

PURPOSE: Trauma remains a major contributor to global disability. While the trauma laparotomy can be a life-saving procedure in abdominal injury, a select proportion are performed with no pathology identified intra-operatively, potentially exposing patients to unnecessary morbidity. We aimed to assess the global prevalence of negative laparotomy and identify any influencing factors at both the patient- and system-level.

METHODS: This was a secondary analysis of the GOAL-Trauma study, a multicentre prospective international observational study on trauma laparotomy patients, conducted from April to December 2024. We defined a negative laparotomy as where no intra-abdominal injuries were identified during the index operation. A multivariable logistic regression was performed to identify predictive factors for a negative laparotomy.

RESULTS: Of the 1769 patients included, 128 patients (7.2%) underwent a negative laparotomy. Regression analysis demonstrated that a penetrating mechanism of injury (OR 2.37, CI:1.53-3.69, $p < 0.001$) and a lower grade of surgeon (surgical registrar relative to consultant - OR 2.91, CI:1.90-4.47, $p < 0.001$) were independent predictors for negative laparotomy. Neither human development index nor hospital resource level impacted negative laparotomy rates.

CONCLUSION: One in every fourteen trauma laparotomies performed in our global cohort was negative, […]

PubMed · DOI

Preoperative stay under 48 h after admission for proximal femur fractures is linked to fewer complications and better outcomes.

Moskal M, Feulner S, Grimm A, Müller-Heck R, Heep H, Godolias P

PURPOSE: To evaluate the association between admission-to-surgery time (with a focus on < 48 h) and postoperative complications and discharge mobility in patients with proximal femur fractures (PFFs), using a large administrative dataset from routine inpatient care in Germany.

METHODS: In this retrospective database analysis, 128,092 patient records from external inpatient quality assurance (North Rhine-Westphalia, Germany, 2016-2020) were included. All patients underwent surgical treatment for PFFs and were stratified by time to surgery as early surgery (Group A; within 48 h of admission) or delayed surgery (Group B; more than 48 h after admission). Two cohorts were analysed separately: hip arthroplasty (HA: early 57,635; delayed 7,504) and osteosynthesis (OST: early 59,466; delayed 3,487). Associations between admission-to-surgery time, postoperative complications, pre-admission anticoagulant therapy and inability to walk at discharge were assessed using descriptive statistics and binary logistic regression. Patients unable to walk before the fracture were excluded from the mobility analysis. Reporting was guided by the STROBE statement.

RESULTS: In-hospital mortality was 4.7% after early versus 9.7% after delayed surgery (OST) and 5.8% versus 9.3% (HA). Delayed surgery was associated with higher odds of postoperative complications (OST: OR 1.21, 95% CI 1.11-1.34; HA: OR 1. […]

PubMed · DOI

Impact of routine use of the GFAP and UCH-L1 TBI blood test for management of mild TBI on patient care in a European emergency department.

Brüning-Wolter F, Wolff N, Schrader M, Musaelyan K, Chandran R, Hoffmann M, Marino JA, McQuiston B et al.

OBJECTIVES: Blood biomarkers glial fibrillary acidic protein (GFAP) and ubiquitin C-terminal hydrolase L1 (UCH-L1) were recently introduced into clinical practice for mild traumatic brain injury (mTBI) assessment. However, real world data demonstrating TBI biomarkers' impact on the emergency department (ED) practice is lacking. This study aimed to describe the experience of a single tertiary hospital ED that introduced TBI biomarkers GFAP and UCH-L1 measurements using Alinity i® TBI test into routine mTBI assessment.

METHODS: Retrospective data were extracted from hospital records of patients who had TBI test ordered in the ED. Test results and subsequent clinical care pathway data (CT order, CT scan result, admission or discharge) were analysed.

RESULTS: Data was available for 460 mTBI patients (median age 43, 54% male). Of the 216 patients (47%) with a negative TBI test, 198 did not have a CT scan, and 18 had a CT scan with no abnormalities detected. Among 244 patients with a positive TBI test, 162 had CT scans, with 12 (6.6%) showing intracranial lesions. Overall, among all 180 scanned patients every patient with intracranial lesion findings had a positive TBI test result. The test could rule out CT scans in 47% of mild TBI patients. In practice, 42% of patients avoided a CT scan based on the TBI test results with additional 12% of patients ruled out based on clinical asse […]

PubMed · DOI

Electric bicycle versus conventional bicycle crashes: comparative analysis of injury patterns, severity, and the modifying effect of body mass index.

Bauer D, Maier JP, Holzfurtner L, Pietsch C, Wagner FC, Schmal H, Erdle B, Mühlenfeld N

PURPOSE: E-bike trauma is often presumed to result in greater injury severity than conventional cycling, yet evidence remains inconsistent. Furthermore, the impact of factors such as Body Mass Index (BMI) on injury patterns is poorly understood. This study therefore evaluated whether e-bike trauma is associated with increased overall severity or distinct regional injury patterns.

METHODS: This retrospective single-center cohort study analyzed 261 adult trauma-bay patients (e-bike n = 37; conventional bicycle n = 224). Primary endpoints included Injury Severity Score (ISS) and regional maximum Abbreviated Injury Scale (AIS) scores. Secondary endpoints included physiological burden (SAPS II) and resource utilization (LOS, TISS-10). Multivariate logistic regression was used to identify independent predictors of intracranial hemorrhage, e-bike riding and extremity fractures. Event counts and events-per-variable (EPV) ratios are reported for each model.

RESULTS: E-bike riders had a significantly higher BMI than conventional cyclists (26.1 vs. 24.6 kg/m²; p = 0.029). However, the absolute difference of approximately 1.5 kg/m² is modest, and both group medians fall within the normal-to-overweight range. No statistically significant differences were observed in ISS, physiological burden, or resource utilization. Conventional cyclists had a higher proportion of patients with AIS Pelvi […]

PubMed · DOI

Effect of management in intermediate care unit versus conventional hospitalization on long-term mortality and autonomy in patients with rib fractures aged 75 years and older: a retrospective study.

Vallod E, Debord-Peguet S, Mikail N, Termoz A, Tripoz S, Abrard S, Gueret-du-Manoir M

PURPOSE: Rib fractures in older adults are associated with substantial morbidity and mortality. Management in an intermediate care unit (IMCU) may improve outcomes, but its long-term benefit remains uncertain. We compared IMCU versus conventional management regarding 12-month mortality and autonomy in patients aged ≥ 75 years with rib fractures.

METHODS: We conducted a retrospective observational cohort study in two Lyon teaching hospitals (Hôpital Edouard Herriot and Hôpital Lyon Sud, Hospices Civils de Lyon, France) between January 2015 and December 2019. The primary outcome was a composite of death or institutionalization within 12 months of hospital admission. Secondary outcomes included discharge modality, opioid prescription at discharge, hospital length of stay, unplanned readmission within 6 months, and ADL/IADL scores. Logistic regression was adjusted for hospitalization unit, age, sex, pre-existing institutionalization, and associated injuries.

RESULTS: A total of 196 patients were included: 126 (64%) in the conventional group and 70 (36%) in the IMCU group. Death or institutionalization within 12 months occurred in 31 patients (26%) in the conventional group and 8 (12%) in the IMCU group (p = 0.026). However, after adjustment, conventional hospitalization was not significantly associated with the primary outcome (OR 2.25, 95% CI 0.90-6.09; p = 0.092). Institutional […]

PubMed · DOI

The road to the Pearl-String Technique: lessons learned from early IMT modifications using RIA autograft and cancellous allograft granules.

Hückstädt M, Scholl C, Langwald S, Klauke F, Mendel T, Kobbe P, Fischer C

BACKGROUND: The induced membrane technique (IMT) is widely used for reconstruction of large bone defects; however, reported outcomes vary considerably among technical modifications. One potential mechanism impairing consolidation is graft sedimentation, particularly when low-density cancellous autograft harvested using the reamer-irrigator-aspirator (RIA) is applied. This study evaluates a standardized modification of the IMT and explores mechanical factors potentially affecting graft consolidation.

METHODS: In this retrospective single-center study, 28 patients were treated following eradication of osteomyelitis using a defined modification of the IMT. Mean follow-up was 7.4 years. Defect length ranged from 3.2 to 15.7 cm. During the second stage, RIA-harvested autologous cancellous bone was combined with lyophilized cancellous bone allograft granules. Definitive fixation was achieved using intramedullary nailing or locking plate fixation; one patient was treated with a ring fixator.

RESULTS: Primary consolidation was achieved in 57.1% of cases. Aseptic nonunion occurred in 10.7%, with secondary consolidation achieved after revision in all affected patients. Recurrent infection was observed in 35.7%; infection control was achieved in 90% of these cases following additional procedures, and reconstruction was ultimately completed using alternative strategies. Median time to fu […]

PubMed · DOI

Non-tourniquet strategy with high-position low-perfusion in distal humeral fracture surgery: reduces swelling without increasing blood loss.

Wang Z, Ai X, Ren T, Wu B, Zhang K, Li Z, Xue H, Li M et al.

INTRODUCTION: To evaluate the feasibility of non-tourniquet strategy combined with "High-position Low-Perfusion" positioning in distal humeral fractures ORIF (DHF-ORIF), and to investigate its effects on perioperative blood loss, postoperative rehabilitation, and functional outcomes.

METHODS: Prospective inclusion of patients aged 18-45 with distal humeral fractures, randomly divided into an experimental group without tourniquets (Group A) and a control group with traditional tourniquets (Group B). The Group A adopted a lateral position with high and low perfusion of the affected limb, and received intravenous injection of TXA (15 mg/kg) combined with continuous flushing with 4 ■ physiological saline during the operation; The control group received routine use of tourniquets.

PRIMARY OUTCOMES: perioperative blood loss, elbow ROM and degree of limb swelling.

SECONDARY OUTCOMES: Postoperative VAS score, Mayo elbow joint score and DASH score, and incidence of complications.

RESULTS: No significant difference was observed in total perioperative blood loss between the Group A and the Group B (240.78 mL vs. 228.53 mL, p > 0.05). However, the Group A demonstrated significantly reduced postoperative drainage volume and hidden blood loss (p < 0.05). The Group A exhibited significantly superior outcomes in terms of VAS scores and limb swelling, compared with the control group. Additi […]

PubMed · DOI

Impacts of intracranial pressure monitoring on mortality and length of stay in severe traumatic brain injury: a systematic review and meta-analysis.

Merakis M, Kim N, Velli G, Balogh ZJ

PURPOSE: This systematic review and meta-analysis examined the association between intracranial pressure monitoring (ICPm) and mortality in patients with severe traumatic brain injury (sTBI), with additional focus on functional neurological outcomes and intensive care unit (ICU) and hospital length of stay (LOS).

METHODS: This review followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and was registered with PROSPERO (CRD42025643607). MEDLINE, Embase, and the Cochrane Central Register of Controlled Trials were searched through August 1, 2025, for studies comparing ICP-monitored and non-ICP-monitored patients with traumatic brain injury. Eligible study designs included randomized controlled trials, prospective observational cohort studies, retrospective observational cohort studies, registry-based cohort studies, and case-control studies. Study quality was assessed using the Strengthening the Reporting of Observational Studies in Epidemiology checklist, Risk of Bias 2.0 for randomized trials, and the Oxford Centre for Evidence-Based Medicine Levels of Evidence framework.

RESULTS: Thirty-one studies met inclusion criteria. The mortality meta-analysis included 121,799 patients. Pooled crude analysis showed lower mortality in ICP-monitored patients than in non-monitored patients (OR 0.79, 95% CI 0.68-0.91, p = 0.002), with high heteroge […]

PubMed · DOI

Prehospital point-of-care lactate for identification of severe injury in trauma team-activated patients: a prospective multicenter cohort study.

Faul P, Hagebusch P, Bertram E, Naujoks F, Störmann P, Schweigkofler U, Koch D

BACKGROUND: Accurate prehospital trauma triage remains challenging despite structured activation criteria, particularly in patients without overt physiologic instability. Prehospital point-of-care lactate has been proposed as a physiologic adjunct to support early identification of severely injured patients. This prospective multicenter cohort study evaluated the association between prehospital lactate and injury severity in trauma team-activated patients within a structured guideline-based trauma system.

METHODS: This prospective observational multicenter cohort study included adult trauma patients transported by emergency medical services to three Level I trauma centers and managed in the trauma room following trauma team activation (TTA). Prehospital lactate was measured by capillary earlobe sampling before or during transport and was not used for clinical decision-making. The primary outcome was severe injury, defined as Injury Severity Score (ISS) ≥16. Secondary outcomes were Abbreviated Injury Scale (AIS) ≥4 in at least one body region and intensive care unit (ICU) admission > 24 hours. Diagnostic performance was assessed using receiver operating characteristic (ROC) analysis and predefined lactate thresholds of ≥2 mmol/L and ≥4 mmol/L.

RESULTS: A total of 108 patients were included, of whom 26 (24.1%) had ISS ≥ 16. Prehospital lactate levels were higher in severely inj […]

PubMed · DOI

Surgical timing and mortality in acute traumatic subdural hematoma: is the four-hour threshold confounded by injury severity?

Rakip U, Y■ld■zhan S, Canbek ■, Boyac■ MG, Korkmaz S, Öztürk KH, Aslan MZ, Aslan A

PURPOSE: Early surgical evacuation within four hours has long been advocated for acute traumatic subdural hematoma (ASDH), yet the supporting evidence is largely observational and unadjusted for injury severity. We examined whether surgical timing is independently associated with in-hospital mortality after accounting for baseline severity.

METHODS: We retrospectively analyzed 91 consecutive patients who underwent surgical evacuation for ASDH between January 2014 and December 2024. The primary outcome was in-hospital mortality; the secondary outcome was functional status at discharge measured by the Glasgow Outcome Scale (GOS). Surgical timing (≤ 4 vs. >4 h) was assessed through unadjusted comparison, stratification by Glasgow Coma Scale (GCS) score, multivariable Firth penalized logistic regression with 1000-iteration bootstrap internal validation, and sensitivity

analyses including alternative timing thresholds (1-12 h), restricted cubic spline modeling, and timing-by-severity interaction testing.

RESULTS: In-hospital mortality was 33.0% (30/91; 95% Wilson CI 24.2-43.1%). Crude analysis showed higher mortality with early surgery (50.0% vs. 15.6%, p = 0.001), but these patients had lower admission GCS and higher rates of CT-defined traumatic axonal injury (TAI) and cerebral contusion. After stratification, no statistically detectable mortality difference was observed within […]

PubMed · DOI

Association between time to transfusion and outcomes in patients with traumatic haemorrhagic shock: an analysis of a nationwide trauma registry in Japan.

Shibahashi K, Inoue K, Kato T, Sugiyama K

PURPOSE: To investigate the association of in-hospital time to transfusion on outcomes of patients with traumatic haemorrhagic shock.

METHODS: Using data from the Japan Trauma Data Bank (January 2019-December 2023), we identified adult patients (aged ≥ 18 years) with haemorrhagic shock following trauma (systolic blood pressure < 90 mmHg; shock index ≥ 1.0 upon hospital arrival) and received ≥ 10 units red blood cells within 24 h of injury. We excluded patients transferred from or to other hospitals after emergency treatment, with burn injuries or cardiac arrest, without transfusion time records, and with first transfusion > 60 min after hospital arrival. Mixed-effect multivariable logistic regression estimated the association between time to transfusion for in-hospital mortality, adjusting for patient characteristics, comorbidities, and injury severity. Heterogeneity in the association between transfusion timing and mortality was assessed through nine subgroup analyses.

RESULTS: Overall, 698 patients (median age 59 years, 66.2% male) from 132 hospitals were analysed, of whom 265 (38.0%) died during hospitalisation. Mean time to transfusion varied from 4 to 58 min across hospitals. A positive association was observed between time to transfusion and in-hospital mortality. Each 1-min delay in transfusion initiation was associated with higher in-hospital mortality (adjusted odds […]

PubMed · DOI

Neutrophil phenotyping reflects injury severity when base excess and lactate remain normal.

van Eerten FJC, Duebel LV, de Fraiture EJ, Brands SR, van Wessem KJP, Vrisekoop N, Venhuizen JH, de Groot MCH et al.

INTRODUCTION: Accurate assessment of injury severity is essential for optimal trauma care, as it dictates timing and intensity of interventional strategy. Conventional markers, such as base excess and lactate levels are usually borderline deranged in patients with midrange injury severity. This study explored whether determining circulating neutrophil phenotypes in these patients (ISS 9-35) might aid in the assessment of injury severity.

METHODS: A descriptive single-center cohort study of trauma patients ≥ 16 years between 2013 and 2023 was conducted. Data was collected from the Dutch National Trauma Registry and the Utrecht Patient-Oriented Database (< 10% missing values). Base excess, lactate levels and neutrophil phenotypes were routinely measured in emergency department blood samples. The dynamics of these parameters was analyzed in the context of increasing Injury Severity Scores (ISS).

RESULTS: 807 patients were included, of whom 591 were classified as midrange severely injured. In these patients, banded neutrophils showed a greater fold increase between ISS 9 and 35 compared with base excess and lactate (6.7, 2.7, and 1.5, respectively). Furthermore, the number of banded neutrophils exceeded the upper interquartile range of controls at lower trauma severities compared to base excess or lactate (ISS 17 vs. 30 and 39, respectively).

DISCUSSION: Although base excess and […]

PubMed · DOI

Limb occlusion pressure-based individualized tourniquet pressure management in patients undergoing internal fixation for extremity fractures: a prospective comparative feasibility study.

Yang J, Hu P, Wang L

OBJECTIVE: This study aimed to evaluate the feasibility of limb occlusion pressure (LOP)-based individualized tourniquet pressure management and to quantify the reduction in applied tourniquet pressure compared with a traditional fixed-pressure protocol in patients undergoing internal fixation for extremity fractures.

METHODS: A prospective comparative study was conducted at Civil Aviation Shanghai Hospital from January to September 2025. Patients undergoing internal fixation for extremity fractures were enrolled and allocated to either the LOP-based group (n = 171) or the traditional fixed-pressure group (n = 235) according to the attending surgical team and operative scheduling, without randomization. The LOP-based group received individualized tourniquet pressure based on preoperative LOP measurement plus a safety margin, while the control group received standardized pressures (40-50 kPa; 300-375 mmHg). The primary outcome was applied tourniquet inflation pressure. Secondary outcomes included demographic characteristics, extremity location, operative duration, and subgroup analyses stratified by extremity location, age, and sex; multivariable linear regression was performed to adjust for potential confounders.

RESULTS: The LOP-based group demonstrated significantly lower mean tourniquet pressure compared with the traditional group (32.82 +/- 3.63 kPa [246 +/- 27 mmHg] vers […]

PubMed · DOI

Impairment in activities of daily living and textbook outcome after major emergency abdominal surgery.

Mima K, Nitta H, Shikiji Y, Nakashima R, Uemura S, Morinaga T, Itoyama R, Umezaki N et al.

PURPOSE: Textbook Outcome (TO) has been a useful tool to assess benchmark patient-centred surgical outcomes in elective abdominal surgery. This study aimed to evaluate the association of preoperative activities of daily living (ADL) status with TOs following major emergency abdominal surgery.

METHODS: This retrospective study included 301 consecutive patients who underwent major emergency abdominal surgery for high-risk acute abdominal conditions, including perforated peptic ulcer, intestinal ischemia, bowel obstruction, bowel perforation, and severe intraabdominal bleeding or infection, at Kumamoto Regional Medical Center between April 2019 and March 2025. Preoperative ADL was evaluated using the Barthel Index (range, 0 to 100), with higher scores indicating greater independence. Barthel Index ≤ 85 was defined as ADL impairment. According to a recent international consensus, TO was defined as discharge to "home" without major postoperative complications (Clavien-Dindo grade ≥ III). Multivariable logistic regression analysis was conducted to assess the association of preoperative ADL with TO achievement, adjusting for clinical features and surgery-related factors.

RESULTS: Among the 301 patients, 90 (30%) had preoperative ADL impairment (Barthel Index ≤ 85) and 211 (70%) achieved TO. Multivariable analysis revealed that preoperative ADL impairment was independently associated […]

PubMed · DOI

Early calcium replacement after minimal blood transfusion is associated with adverse outcomes.

Hewgley WP, Griffin RL, Baird E, Hashmi ZG, Baker SJ, Jansen JO, Kerby J, Holcomb JB et al.

PURPOSE: Calcium disturbances are common in patients with traumatic injuries, exacerbated by shock and blood product resuscitation. Recently, empiric calcium supplementation has been recommended with the first unit of blood transfusion, despite a lack of data supporting this practice. We hypothesize that there are risks associated with calcium administration in patients receiving small volume transfusion.

METHODS: This is a retrospective review of patients admitted to our Level 1 trauma center from 2018 to 2024 who received 1-3 units of red blood cells (RBCs). Patients were stratified by calcium levels and whether they received early calcium supplementation. A logistic regression was used to estimate odds ratios for the association between calcium levels, calcium supplementation, and outcomes. The primary outcome was in-hospital mortality. Secondary outcomes included acute kidney injury, myocardial infarction, acute respiratory distress syndrome, and venous thromboembolism. The model was adjusted for age, sex, ISS and shock index.

RESULTS: A total of 2,680 patients were included. When stratified by calcium levels and supplementation status, patients were similar in terms of age, sex, mechanism and injury severity score, but shock index was higher among patients receiving calcium supplementation. After adjustment, hypocalcemic patients who received calcium supplementation had […]

PubMed · DOI

Peripheral perfusion index as a predictor of severe shock in trauma patients: a prospective observational study.

Sundar H, Purushothaman V, Mathuram SR, Chandy GM, Sharma SL, Nayak S, James JD

BACKGROUND: Early recognition of haemorrhagic shock is critical in trauma care, particularly in low- and middle-income countries where delayed referral is common. Conventional vital signs may remain normal despite ongoing hypoperfusion. The Perfusion Index (PI), derived from pulse oximetry, is a non-invasive and continuously available marker of peripheral perfusion. This study evaluated the ability of PI to identify severe haemorrhagic shock in trauma patients and compared its performance with the Shock Index (SI) and Modified Shock Index (mSI).

METHODS: This prospective, single-centre study was conducted at a Level 1 trauma centre between January and May 2024. Adult Priority 1 trauma patients were enrolled and classified as having no/mild shock (ATLS grades 1-2) or severe shock (grades 3-4). PI was recorded during initial assessment. SI, mSI, lactate, injury severity, transfusion requirement, ICU admission, and 24-hour mortality were analysed. Receiver operating characteristic (ROC) analysis and multivariable logistic regression were performed.

RESULTS: A total of 120 patients were included (mean age 40.4 ± 17.2 years; 90% male), predominantly with blunt trauma (85% road traffic injuries). Severe shock was present in 35 patients (29.2%). Median PI was significantly lower in severe shock compared with no/mild shock (0.54 [IQR 0.28-1.31] vs. 1.33 [0.71-2.16], p = 0.008). Mean […]

PubMed · DOI

A prospective study on abdominal trauma: patterns of injury, clinical presentation, organ involvement, associated injuries, and outcomes at a tertiary hospital in Mogadishu, Somalia.

Mohamud AA, Hassan AI, Ali AK, Jubur AA, Ahmed MR, Mohamed SS, Mohamed MM, Bashir AM

BACKGROUND: Trauma is a major global public health problem and one of the leading causes of death and disability worldwide, particularly among individuals younger than 44 years of age. Approximately 6 million people die each year as a result of traumatic injuries, with nearly 90% of these deaths occurring in low- and middle-income countries. Previous epidemiological studies have reported postoperative complication rates ranging from 18% to 42% and mortality rates between 8.5% and 13%.

OBJECTIVE: To evaluate the patterns of injury, clinical presentation, organ involvement, associated injuries, management strategies, and outcomes among adult patients with abdominal trauma in Mogadishu, Somalia.

METHODS: A prospective observational study was conducted from August 2023 to July 2024 at Mogadishu Somali-Türkiye Recep Tayyip Erdo■an Training and Research Hospital. A total of 146 adult patients with abdominal trauma were enrolled. Data were collected using a pretested structured questionnaire and analyzed using Stata version 16. Univariable and multivariable logistic regression analyses were performed to identify factors independently associated with in-hospital mortality.

RESULTS: The study population consisted predominantly of young adult males, with a mean age of 30.18 ± 9.2 years, and males accounted for 97.3% of all patients. Penetrating trauma was the predominant mechanism of […]

PubMed · DOI

Comparison of fluoroscopic versus electromagnetic distal locking in humeral shaft osteosynthesis using intramedullary nail: a retrospective study.

Madeja R, Szeliga J, Voves J, Pometlova J, Konde A, Janosek J, Dousa P

PURPOSE: Intramedullary nailing is a standard method for osteosynthesis of diaphyseal humeral fractures. Distal locking, however, typically requires fluoroscopic guidance, which increases radiation exposure. This study aimed to

compare standard fluoroscopy-guided distal locking with the Ezy-Aim system (Austofix, Australia) based on electromagnetic pulse detection from the perspectives of surgical times, radiation exposure, and complications.

METHODS: A retrospective study included 58 patients treated for diaphyseal humeral fractures (AO/OTA 12A, 12B) between 2019 and 2023. Group A (n = 28) received osteosynthesis using the Ezy-Aim system; Group B (n = 30) underwent fluoroscopy-guided distal locking. Data on operative time, fluoroscopy time, radiation dose, and complications were analysed.

RESULTS: In all, distal locking was not successful at the first attempt in 14.3% (4/28) of screws placed with Ezy-Aim and 20.0% (6/30) when fluoroscopy was used. Time and dose analyses were performed on the remaining 48 patients. The total surgical time was significantly longer with the Ezy-Aim system (median 58 min) compared to fluoroscopy (median 52 min, p = 0.034), primarily due to the time required for the device assembly. However, distal screw placement was faster with Ezy-Aim (median 12 min vs. 14 min, p = 0.019). Radiation exposure was significantly lower in the Ezy-Aim group (median […]

PubMed · DOI

Patterns of patient worry following major emergency abdominal surgery: a 180-day follow-up study.

Parner MK, Jensen LR, Gamst-Jensen H, Burcharth J, Kokotovic D

PURPOSE: Major emergency abdominal surgery is associated with high postoperative morbidity and mortality. Beyond physical complications, patients experience psychological distress. This study aimed to describe the distribution of patient-reported worry across independent cross-sectional samples and its association with Days Alive and out of Hospital (DAOH) over 180 days postoperatively.

METHODS: Prospective cohort at Copenhagen University Hospital Herlev (March 30, 2023-March 31, 2024). Degree of Worry (DOW) was surveyed at discharge and postoperative day (POD) 30, 90, and 180, categorized as physical, mental, and physical-generic domains of worry. DAOH was computed cumulatively and incrementally for 0-30, 30-90, and 90-180 days. Associations between high DOW (hDOW) and subsequent DAOH were tested with Mann-Whitney U.

RESULTS: hDOW was reported by 36.1% at discharge, 22.4% at POD 30, 45.0% at POD 90, and 32.0% at POD 180. Across 180 days, the most frequent worries were relapse (21.3%), disease (14.1%), functional level (12.0%) and daily living (9.1%). The proportion of physical worries was significantly higher at POD 30 than at discharge and significantly lower at POD 180; mental worries varied between time points without statistically significant differences. hDOW at POD 30 was associated with fewer cumulative DAOH by POD 90 (p = 0.008).

CONCLUSION: Patient-reported worry f […]

PubMed · DOI

Outcomes and complications of metallic vs. bioabsorbable screw fixation for displaced medial epicondylar fractures in pediatric and adolescent patients.

Jhala T, Scherer S, Esser M, Lieber J, Fuchs J, Dietzel M

PURPOSE: The use of bioabsorbable screws in orthopedics and trauma surgery is increasing. The major advantage of not having to remove the implant is of particular interest in pediatric surgery. Due to drawbacks, most notably implant fracture, bioabsorbable screw osteosynthesis has not yet surpassed the gold standard of metallic screw osteosynthesis. This single-centre study compares metallic screw and bioabsorbable screw osteosynthesis of medial humeral epicondyle fractures in children and adolescents over a two-year period.

MATERIALS AND METHODS: Retrospective analysis of surgically treated (screw fixation with either bioabsorbable magnesium-based or metallic screw) patients with a medial humeral epicondyle fracture between 2020 and 2022 at the author's institution.

RESULTS: Seventeen patients with medial epicondyle fractures underwent surgical treatment using either bioabsorbable screws (n = 8) or metallic screws (n = 9). Baseline characteristics, surgical approach, and postoperative care were similar between groups. Two complications (Clavien-Dindo I) occurred in the bioabsorbable group due to implant failure (screw breakage), though without clinical sequelae. These implant failures led to a significantly increased number of postoperative radiographs in the bioabsorbable group (median 4.8 vs. 2.5; p = 0.014). Functional outcomes (LOM, MEPI) and follow-up durations were com […]

PubMed · DOI

Characteristics and temporal trends of mobility scooter accidents in the Netherlands: a multicenter cohort study.

Stensen Z, Peters A, Keereweer D, Krul I, van Osch F, de Groot B, Barten D

PURPOSE: Mobility scooters are increasingly used by older individuals. The aim of this study was to assess the characteristics and temporal trends of mobility scooter accidents (MSAs) in the Netherlands over a 10-year period using a representative dataset derived from fourteen Emergency Departments (EDs).

METHODS: In this retrospective multicenter cohort study, we analyzed MSA-related ED visits between January 2014 and December 2023, using the Dutch Injury Surveillance System (DISS). Information about age, sex, hospital admission, type and location of injury, mechanism of injury, and operational errors was analyzed. Injury severity was classified using the maximum abbreviated injury score (MAIS). Trend analyses were only performed for ED visits of patients with MAIS2+ injuries.

RESULTS: In total, 2,702 MSA-related ED visits were identified. Forty-nine percent of patients was male and the median age was 75 years (IQR 64.0-83.0). The most common mechanism of injury was a mobility scooter rollover (34.2%). From 2014 to 2023, the number of MSA-related ED visits of patients with MAIS2+ injuries increased by 38.0%. In the 70-79 age group, this rise was 91%. The total number of traumatic brain injuries (TBI) rose steeply bij 103.0%, which includes a modest (+34.0%) increase in moderate-severe TBI. The hospital admission rate was 30.1%, with a median length of stay of 7 days (IQR 3.0 […]

PubMed · DOI

"Is the effect of intraoperative 3D imaging beneficial in plate fixation of distal radius fractures: a matched-pair analysis".

Zumsteg JS, Pieringer A, Hinkelmann MA, Isenmann K, Pape HC, Allemann F, Barthel C

BACKGROUND: Distal radius fractures (DRFs) are among the most common adult fractures and are frequently treated with volar plate fixation. While intraoperative 3D imaging has been proposed to improve detection of malreduction and implant malposition, its impact on clinical and radiological outcomes remains unclear. This study aimed to evaluate whether intraoperative 3D imaging provides advantages over conventional 2D fluoroscopy. Outcome criteria were accuracy of fracture reduction, plate positioning relative to the watershed line, complication rates, operative time, and functional outcomes following volar plate fixation of distal radius fractures.

METHODS: This retrospective single-centre study included patients aged ≥ 16 years who underwent volar plate fixation for distal radius fractures between 2016 and 2024. Patients were assigned to a 3D imaging group or a conventional 2D fluoroscopy group. To ensure comparability, patients were matched 1:1 using exact AO/OTA subgroup matching and propensity score matching for age and sex. Outcomes included operative duration, radiographic reduction quality, plate positioning, complications, and wrist range of motion. Matched-pair comparisons were performed using appropriate paired statistical tests.

RESULTS: Of 1,259 patients screened, 418 patients (209 matched pairs) were included after exact AO/OTA subgroup matching and propensity sc […]

PubMed · DOI

High-velocity gunshot chest trauma: tailoring patient care through insights from a mass casualty experience.

Azulay AA, Stein M, Tabo LY, Lebenthal A, Ruderman L, Refaely Y

OBJECTIVES: This study describes the management strategies and clinical outcomes of a cohort of patients sustaining high-velocity military rifle gunshot wounds to the thorax during a mass casualty incident (MCI). The analysis focuses on damage control surgical techniques, the incidence and timeline of venous thromboembolism (VTE), regional pain management, and early psychiatric protocols.

METHODS: A descriptive retrospective analysis was conducted on patients presenting with a single penetrating high-velocity ballistic wound to the thorax on October 7, 2023. Inclusion criteria were limited to patients with a projectile trajectory traversing the lung parenchyma, managed by the cardiothoracic surgery department. Injuries to immediately adjacent structures (subclavian vessels, brachial plexus, clavicle, and scapula) were recorded as

associated injuries. Patients presenting with multi-system trauma involving entirely separate, distant anatomical regions were excluded.

RESULTS: The cohort comprised 11 male patients (age range: 19-49 years; 9 soldiers, 2 civilians) injured by 7.62 mm assault rifle rounds. Six patients (54.5%) required emergency thoracotomy, sternotomy, or VATS within the first 24 h due to ongoing hemorrhage; primary hemorrhage control consisted of deep parenchymal lung suturing (n = 5) and temporary chest packing (n = 1). Four patients (36.4%) underwent delayed, el […]

PubMed · DOI

Elevated risk of wound and systemic complications in MRSA-colonized patients undergoing lower extremity orthopedic trauma surgery.

Areti AS, Hafeez H, Sangineni PV, Argao L, Deveza L, Dawson J

PURPOSE: Preoperative colonization with methicillin-resistant Staphylococcus aureus (MRSA) is a known risk factor for adverse outcomes in arthroplasty but remains under-investigated in orthopedic trauma. This study evaluates the impact of preoperative MRSA colonization on postoperative complication rates in patients undergoing lower extremity orthopedic trauma surgery.

METHODS: We conducted a retrospective cohort study using the TriNetX Research Network, identifying adult patients who underwent lower extremity orthopedic trauma procedures over a 20-year period. Patients with a documented MRSA diagnosis within three months before surgery were matched 1:1 to MRSA-negative controls based on demographics and comorbidities. Postoperative outcomes were assessed within three months of surgery, including surgical site infections (SSIs), wound dehiscence, organ dysfunction, thromboembolic events, and healthcare utilization. Odds ratios (ORs) and p-values were calculated, with statistical significance set at $p < 0.05$.

RESULTS: A total of 1,868 MRSA+ and 629,342 control patients were identified. After matching, MRSA+ patients remained at significantly increased risk for wound dehiscence (OR = 1.80, $p = 0.005$), pneumonia (OR = 2.33, $p < 0.001$), sepsis (OR = 3.71, $p < 0.001$), acute renal failure (OR = 1.45, $p < 0.001$), blood transfusion (OR = 1.74, $p < 0.001$), deep vein thrombosis (OR = 2 […]

PubMed · DOI

A global contrast shortage: a descriptive study exploring IV contrast utilization and outcomes in trauma patients at a level I trauma center.

Zakaria H, Johansen C, Waldrop C, Wan B, Graham C, Fetzer D, Minei J, Bruns B et al.

PURPOSE: Computed Tomography (CT) with intravenous (IV) contrast is critical to the evaluation of the trauma patient. A pandemic-related lockdown occurred in Spring 2022 in the city where iohexol iodinated contrast media) is manufactured, resulting in a global shortage. We explored the potential impact of this shortage on contrast utilization and patient outcomes.

METHODS: A retrospective study was performed of all trauma patients who underwent CT between January and December 2022 at a level I trauma center. Pre-contrast conservation (PRE) period was defined as January to April 2022, and post-conservation (POST) as May to December 2022. Demographics, utilization rates (bolus/patient) and outcomes were evaluated. Chi-square tests were performed for categorical variables and Mann-Whitney U tests for continuous variables. Statistical significance was set at $p < 0.05$.

RESULTS: 1,097 patients were included; 509 in the pre-shortage period, 588 post-shortage. 857 CTs with contrast (1.68 contrast bolus/patient) were performed pre-conservation and 979 post-conservation (1.66 contrast bolus/patient) ($p = 0.04$). Maximum Abbreviated Injury Scale (AIS) head, head/neck, and extremity were higher post-shortage ($p < 0.05$); however, mortality was higher in the pre-conservation group ($p < 0.0012$). There were 9 cases of Acute Kidney Injury (AKI, 8 PRE, 1 POST), 22 delayed diagnoses (14 PRE, 8 P […]

PubMed · DOI

Trauma stabilization points in armed conflict: operational evolution and strategic reflections from mosul to Gaza.

Marta C, Luca P, Kai-Hsun H, Areej AF, Athanasios G, Wessam EG, Jonas Z, Johan VS et al.

BACKGROUND: Forward trauma care remains a critical gap in emergency responses to armed conflict, particularly where conventional medical evacuation is disrupted. Trauma Stabilization Points (TSPs) were introduced in previous conflicts to bring life-saving care closer to the point of injury. To improve standards and adaptability, the World Health Organization convened a Technical Working Group to develop operational guidance for TSPs, drawing on diverse field experiences, including the ongoing Gaza deployments.

CASE PRESENTATION: Between February and July 2024, three TSPs were established in Gaza, managing over 4,000 consultations. Initially focused on trauma stabilization and referral, these sites quickly adapted to minor injuries and non-traumatic conditions, reflecting population needs and access challenges. Referral rates varied across sites due to hospital proximity, ambulance availability, and shifting frontlines. Security threats limited forward deployment and safe patient access, requiring high mobility and rapid relocation. Experience from Gaza highlighted key operational principles: locating TSPs near the point of injury; integrating within a functioning trauma referral pathway supported by evacuation capacity and hospital readiness; maintaining clear clinical functions and staffing standards; and using standardized documentation for quality assurance and continuity. […]

PubMed · DOI

Acute-on-chronic versus isolated acute subdural hematoma in complicated mild traumatic brain injury: association with radiological mass effect markers in a multicenter cohort.

Crombé A, Benhamed A, Seux M, Frassin L, Matichard R, L'Huillier R, Millon D, Gorincour G

PURPOSE: The clinical impact of acute-on-chronic subdural hematoma (acSDH) in mild traumatic brain injury (mTBI) remains incompletely characterized. We assessed whether acSDH, compared with isolated acute SDH (aSDH), is associated with worse neurological presentation and increased radiological markers of mass effect.

METHODS: We conducted a retrospective multicenter cohort study using the IMADIS teleradiology head trauma workflow (103 emergency departments in France, January 2020 to December 2022). Adult patients with complicated mTBI (i.e., GCS 13-15 plus intracranial hemorrhage or skull fracture) and acute SDH on non-contrast head CT were eligible. Exposure was acSDH versus aSDH based on structured radiology reports. Outcomes were (1) GCS < 15 at presentation, (2) SDH maximal thickness ≥ 7 mm, (3) radiological brain herniation, and (4) intermediate or high QueBIC risk category. Univariable and multivariable logistic regression models were used to estimate associations.

RESULTS: Among 935 included patients, 107 (11.4%) had acSDH. Patients with acSDH were older (≥ 75 years: 57.0% vs. 39.7%, p < .0001) and more frequently had dementia (8.4% vs. 3.7%, p = .0465). Compared with aSDH, acSDH was associated with GCS < 15 (37.4% vs. 21.7%, P = .0010) and confusion (41.0% vs. 25.8%, p = .0016). On CT, acSDH was associated with SDH thickness ≥ 7 mm (54.2% vs. 20.4%, OR 4.50, 95%CI 2.9 […]

PubMed · DOI

Persistent neuropathic pain symptoms after FPV drone-related thoracic blast injury without rib fractures: a pilot observational study from the Ukraine War.

Battle C, Baker E, Giamello JD, Melchio R, Dmytriiev D

PURPOSE: The aim of this exploratory pilot study was to describe First Person View drone (FPV) related, blast-mechanism thoracic trauma with no rib fractures in Ukrainian military patients sustained during the Russia-Ukraine Conflict, and report the prevalence of persistent neuropathic pain symptoms at three months among patients admitted to a rehabilitation and pain medicine service.

METHODS: A retrospective, single-centre observational, exploratory study design was used. Military patients (≥ 16 years old) with blast-mechanism thoracic injuries with no rib fractures, admitted to a hospital's rehabilitation and pain medicine service in Ukraine for ≥ 24 h were included. The PainDETECT survey was completed at three months post-injury. Participant characteristics were summarised using descriptive statistics. Exploratory univariate and multivariate analyses were performed to explore possible associations between variables and the development of persistent neuropathic pain symptoms.

RESULTS: 217 patients were included in the final analysis. Median age was 39 years (IQR: 32-47.5). 131 (60.4%) patients were admitted to critical care with 91 (41.9%) patients requiring mechanical ventilation. The median hospital length of stay was 22 days (IQR: 16-25) and critical care length of stay was 4 days (IQR: 2-5). Persistent neuropathic pain symptoms were reported in 173 out of 217 patients ([…]

PubMed · DOI

The association between pancreatic enzyme serum concentrations and blunt pancreatic trauma: a systematic review.

Sykes B, Gracias D, Bendinelli C, Balogh ZJ

PURPOSE: Blunt pancreatic trauma is uncommon. When suspected, serum amylase and lipase are measured, but their diagnostic value remains uncertain. This systematic review evaluated the diagnostic performance of serum amylase and serum lipase in detecting blunt pancreatic trauma in adults.

METHODS: MEDLINE, Embase, Scopus and Cochrane Central were searched for studies evaluating serum amylase and/or lipase in adults with blunt pancreatic trauma. Observational studies reporting diagnostic accuracy or clinical outcomes were included. Two reviewers independently screened studies and assessed risk of bias using the Newcastle-Ottawa Scale.

RESULTS: Four observational studies met inclusion criteria. A total of 4937 patients with blunt abdominal trauma, including 150 with pancreatic injury were investigated. Two studies evaluated serum lipase alone and two compared amylase and lipase. Pancreatic injuries were graded according to the American or Japanese Association for the Surgery of Trauma. Early serum amylase measurement (< 90 min post-injury) demonstrated low sensitivity. In contrast, testing performed more than 3 h after injury has shown improved diagnostic accuracy with one study suggesting combined serum amylase and lipase testing can achieve a sensitivity of 85% and specificity of 100%.

CONCLUSION: Early serum pancreatic enzymes measurement does not appear to aid initial manag […]

PubMed · DOI

Extension of the fluidics NTF-Biobank of polytraumatized patients with extracellular vesicles: improved vesicle preservation through local processing.

Weber B, Leppik L, Han J, Becker N, Wang T, Groven RVM, Balmayor ER, Zhao Q et al.

PURPOSE: Extracellular vesicles (EVs) show diagnostic relevance in trauma, prompting the development of the NTF■EV■Biobank as an extension of the established Network Trauma Research (NTF) plasma/serum biobank. After optimizing the EV isolation and characterization standard operating procedure (SOP) at a single center, this multicenter study evaluates its transferability and examines how sample transport affects EV quality before large■scale implementation in a nationwide biobank setting.

METHODS: Plasma and serum from polytrauma patients and healthy controls (n=10) were processed under the standardized NTF■Biobank protocol and distributed to three centers. EVs were isolated using the unified NTF■EV■Biobank workflow, exchanged between centers, and analyzed for protein markers, particle size and concentration (NTA), total protein content, and morphology (TEM) to compare EV quality across sites.

RESULTS: Implementation of the EV biobank protocol across participating centers, as well as inter-center transport on dry ice, was successfully performed without complications. All samples arrived at the analyzing centers in a frozen state. EVs exhibited typical vesicular morphology as confirmed by transmission electron microscopy and remained positive for CD9 and CD81, indicating that transportation did not affect morphology or surface marker expression. However, transportation did redu […]

PubMed · DOI

Whole blood first resuscitation and association of blood product utilization based on mechanism of injury.

Hadley J, Conner J, Thomas C, Llorente K, Gilani A, White B, Miller Iii P, Nunn A et al.

INTRODUCTION: Studies evaluating the effect of whole blood (WB) vs. component-based resuscitation accounting for mechanism of injury are lacking. We sought to evaluate the effects of a WB first resuscitation strategy regarding mechanism of injury.

METHODS: This is a retrospective cohort study of trauma patients who received any blood product during resuscitation from January 2016 to November 2021. The primary outcomes were total blood volume utilized within 24 h and 30-day mortality. Multivariable gamma regression with a log link was used for blood volume and logistic

regression for mortality. A likelihood ratio test evaluated whether injury mechanism modified the treatment effect.

RESULTS: Between January 2016 and November 2021, 1,013 injured patients received blood products (785 WB-first). We did not detect statistically significant effect modification by injury mechanism for either outcome (interaction p = 0.44 for blood volume, p = 0.25 for mortality). In multivariable gamma regression, WB-first resuscitation was associated with a ratio of mean total blood volume of 0.71 (95% CI 0.55 to 0.90, p = 0.005) relative to components, representing approximately 29% lower utilization. WB-first was not associated with 30-day mortality (OR 0.90, 95% CI 0.48 to 1.71, p = 0.73).

CONCLUSIONS: A WB-first resuscitation strategy is associated with lower blood volume utilized within the f […]

PubMed · DOI

Reported complications after spinopelvic fixation for U-type sacral fractures in older adults: a systematic review with narrative synthesis.

Giordano V, Pinheiro HRS, Pires RE, Varga P, Freitas A, Giannoudis PV

BACKGROUND: Spinopelvic fixation (SPF), alone or combined with iliosacral or transiliac-transsacral fixation, is often used for highly unstable posterior pelvic ring injuries and some fragility fracture patterns in older adults. However, the evidence specifically describing complications after SPF in older patients with U-type sacral fractures is limited.

METHODS: This systematic review was conducted according to the Cochrane Handbook and PRISMA recommendations. PubMed, Embase, and grey-literature sources were searched from January 1, 2000 to July 31, 2025. Eligible studies included adults aged 60 years or older with U-type sacral fractures or closely related spinopelvic dissociation/FFP IVb patterns treated with SPF, and reporting at least one complication. Because of the small number of eligible studies, probable cohort overlap, heterogeneity in fracture definitions, treatment indications, fixation constructs, and outcome reporting, findings were synthesized narratively rather than pooled quantitatively.

RESULTS: Three studies met the eligibility criteria, comprising 157 patients overall; however, only a subset underwent SPF, and two studies were produced by the same group with probable cohort overlap. The available cohorts mixed bilateral sacral fragility fractures, FFP IVb patterns, and comparative SPF-versus-trans-sacral constructs. Reported adverse events included wound […]

PubMed · DOI

Influence of allograft augmentation on surgical outcomes and complications in proximal humerus fractures: a matched-pair analysis.

Barthel C, Rudolph F, Kalbas Y, Russek R, Pape HC, Allemann F

BACKGROUND: Proximal humerus fractures predominantly affect elderly patients and pose a growing clinical burden. This study evaluates the impact of allograft augmentation on operative time, fracture reduction quality, and complication rates compared with standard fixation across different fracture morphologies.

METHODS: This retrospective matched-pair study included patients surgically treated for proximal humerus fractures with locking plate fixation at a single tertiary trauma center between 01.01.2014 and 31.12.2024. Patients treated with allograft augmentation (Group A) were matched 1:1 to patients without allograft (Group NA) using exact matching for Mayo FJD fracture morphology and sex, and propensity score matching for age. Radiological outcomes were assessed using pre- and postoperative neck-shaft angles, with normalization defined as correction into the physiological range (120°-150°). Complications and operative parameters were recorded. Paired statistical analyses were applied.

RESULTS: A total of 1,012 patients were treated during the study period; after selection and matching, 102 matched patient pairs (204 patients) were analyzed. Operative time was longer with allograft augmentation (141.3 ± 40.3 vs. 126.9 ± 50.0 min; p = 0.08), reaching statistical significance only in varus posteromedial fractures (144.4 ± 37.8 vs. 122.2 ± 39.6 min; p = 0.007). Although overa […]

PubMed · DOI

Impact of a comprehensive trauma pathway on workflow and clinical processes in a level III trauma center.

Neitzert L, Druetto C, Forno D, Molino P, Vitale D, Grio M

PURPOSE: Organized trauma systems improve coordination of care and may enhance the efficiency of trauma management, potentially improving outcomes and reducing hospital length of stay and costs. The aim of the study was to evaluate key process-of-care indicators and clinical outcomes before and after the introduction of a comprehensive trauma pathway.

METHODS: We conducted a single-center retrospective-prospective cohort study at Rivoli Hospital, Italy. Trauma patients admitted to the Emergency Department (ED) between 2024 and 2025 who met the criteria for trauma team activation were prospectively included. Outcomes were compared with a retrospective cohort from the period prior to the implementation of the dedicated trauma pathway (2023-2024).

RESULTS: A total of 465 patients were included. Implementation of a comprehensive trauma pathway was associated with significant improvements in several time-dependent quality indicators, including ED boarding time (888 vs. 1021 min; $p = 0.02$), time to imaging (48 vs. 78 min; $p < 0.01$), time to surgery (183 vs. 237 min; $p = 0.03$), time to ICU admission (226 vs. 343 min; $p = 0.02$), and time to interhospital transfer (238 vs. 305 min; $p = 0.04$). We observed a trend toward a reduction in early mortality (2% vs. 3.6%; $p = 0.41$) and 30-day mortality (3.7% vs. 5.9%; $p = 0.62$).

CONCLUSIONS: The adoption of a structured multidisciplinary trau […]

PubMed · DOI

The DRoPH score: development of a tool for predicting reduction difficulty in proximal humerus fracture osteosynthesis.

Barthel C, Rudolph F, Kraus M, Pape HC, Allemann F

BACKGROUND: Proximal humerus fractures (PHFs) are among the most common and challenging fractures due to complex anatomy and variable fracture patterns. While anatomical reduction is critical for successful outcomes, no tool currently predicts the intraoperative difficulty of achieving it. Despite advances in surgical techniques, complications persist. Existing classification systems, such as Codman's, Neer's, AO/OTA, and Mayo-FJD, focus on fracture morphology and displacement but do not address intraoperative complexity. This study aims to develop a new score for the early prediction of intraoperative reduction difficulty in proximal humerus fracture surgery, the DRoPH score (Difficulty of Reduction of Proximal Humerus Fractures).

METHODS: An online survey was conducted among experienced upper extremity surgeons in Switzerland to identify factors influencing the difficulty of reducing and retaining proximal humerus fractures. Inclusion criteria for surgeon grading comprised experience with more than 100 surgically treated proximal humerus fractures. Eligible surgeons rated the relevance of factors influencing reduction difficulty using a five-point Likert scale. A rank-based weighting system derived from expert ratings was applied, assigning two points to the three highest-ranked factors. The resulting DRoPH-Score ranged from 0 to 10, with higher values indicating greater sur […]

PubMed · DOI

Interface stability of intramedullary nail locking configurations under combined axial and torsional loading.

Marianne H, Martin W, Claus G, Anja B, Peter A

The biomechanical performance of intramedullary nails in metaphyseal regions strongly depends on distal locking configuration. This study aimed to evaluate screw-bone interface stability under combined axial and torsional cyclic loading for different distal locking configurations of intramedullary nails, focusing on the effects of screw number, spatial distribution, interscrew distance, and multiplanar versus coplanar fixation using a standardized synthetic bone model. Six distal locking configurations ($n = 5$ per group) were tested under stepwise cyclic axial compression, superimposed with a constant torsional moment (± 2 Nm, 2 Hz). Primary endpoints were cycles to failure, cycles to 1 mm axial displacement, and cycles to 0.5° rotation. All constructs failed by progressive screw cut-through within the synthetic bone. The three-screw multiplanar configuration exhibited the highest endurance (≈ 36,000 cycles to failure) and lowest micromotion (≈ 34,000 cycles to 1 mm displacement), significantly outperforming the two-screw configurations overall ($p = 0.002$). Increasing interscrew distance from 10 mm to 34 mm improved migration resistance by approximately 25%, whereas the short 10 mm spacing led to premature failure. Blocked and free-hand configurations showed comparable behavior to the standard setup, indicating no added benefit from toggle elimination alone. Locking configuratio […]

PubMed · DOI

Three-dimensional finite element analysis for internal fixation of mandibular condylar fractures: a scoping review.

Kakei Y, Sader R, Ebener P, Tanaka A, Akashi M, Kuroda R, Thoenissen P

PURPOSE: Three-dimensional finite element analysis (3D-FEA) has emerged as an essential computational tool for evaluating the biomechanical behaviors of internal fixation systems for mandibular condylar fractures. This scoping review maps and synthesizes the existing literature on 3D-FEA studies investigating the internal fixation of mandibular condylar fractures and explores, in a descriptive manner, differences in methodological focus between European and non-European research centers.

METHODS: PubMed and Cochrane Library databases were systematically searched following PRISMA-ScR guidelines. Original English-language research articles using 3D-FEA to evaluate internal fixation of mandibular condylar fractures were included. Data were extracted using a standardized charting form.

RESULTS: Thirty-six studies met the inclusion criteria. European studies (n = 12) predominantly focused on development of novel plate designs and mechanical testing protocols, whereas Asian and other nationality studies (n = 24) emphasized material comparisons between titanium and bioabsorbable systems, patient-specific modeling, and loading condition analysis. Consensus findings include: (1) double-plate fixation provides superior stability; (2) among 3D plate designs, trapezoid plates demonstrate the best performance for subcondylar and condylar base fractures, whereas alpha and lambda plates may […]

PubMed · DOI

Initial surgical management of injuries to the upper extremities for patients with multiple and/or severe injuries - a systematic review and clinical practice guideline update.

Berk T, Könsgen N, Lechler P, Imach S, Schär M, Bieler D, Horst K, Hildebrand F et al.

PURPOSE: Our objective was to update the evidence-based and consensus-based recommendations for the initial surgical management of upper extremity injuries in patients with suspected multiple and/or severe injuries based on current evidence. This guideline topic is part of the 2025 update of the German Guideline on the Treatment of Patients with Multiple and/or Severe Injuries.

METHODS: MEDLINE and Embase were systematically searched to September 2024. Further literature reports were obtained from clinical experts. Randomised controlled trials (RCTs) or observational studies reporting risk-adjusted outcomes were included if they compared early versus delayed surgical treatment for fractures, vascular injuries, or nerve injuries affecting the upper extremities in patients with multiple and/or severe injuries. Studies comparing limb salvage versus amputation or comparing amputation criteria for the upper extremities were also included. We considered patient-relevant outcomes such as mortality and limb salvage, as well as the sensitivity and specificity of scores for predicting upper extremity amputation. Risk of bias was assessed at the outcome level using ROBINS-I for observational studies and AMSTAR-2 for systematic reviews. We used available meta-analyses if possible; alternatively, we synthesised the evidence narratively. We used GRADE to rate the certainty of evidence. Expe […]

PubMed · DOI

Transosseous sutures versus vertical cerclage wire for distal pole patellar fractures: a retrospective comparative cohort study.

Laso JI, Muñoz JT, Bianchi S, Martínez D, Rojas T, Franulic N, Olivieri R

PURPOSE: To compare transosseous sutures (TOS) and vertical wire cerclage (VCW) for treating distal pole patellar fractures, evaluating clinical outcomes, patient-reported outcome measures (PROMs), radiographic outcomes, and complications.

METHODS: We conducted a retrospective comparative cohort study of patients surgically treated for patellar distal pole fractures (AO/OTA 34A1b) with either TOS or VCW between 2015 and 2023. Patient characteristics, surgical technique, postoperative data, patellar height, and complications were recorded. Functional outcomes were evaluated at a minimum of 12 months postoperatively using PROMs (Kujala, Lysholm, and Bostman scores). The primary outcome of this study was functional recovery, assessed via the Bostman score at 12 months.

RESULTS: A total of 46 patellar distal pole fractures were included (30 TOS and 16 VCW). Baseline characteristics showed no statistically significant differences between groups. No significant differences were found regarding clinical or functional outcomes. Regarding radiographic outcomes, patellar height remained stable; however, both groups showed significant patellar length variation, with elongation being significantly more pronounced in the TOS group (mean 3.77 mm) compared to the VCW group (mean 0.44 mm; p = 0.001). While overall complication rates showed no statistically significant difference, all five ca […]

PubMed · DOI

Re■fracture risk following implant removal in consolidated hip fractures: a multicenter retrospective cohort study of 575 patients.

Mohammed J, Mili-Schmidt V, Wadsten M, Juto H, Wolf O, Crnalic S, Sköldenberg O, Mukka S et al.

PURPOSE: The risk of re-fracture after implant removal in healed hip fractures remains a clinical concern. This study aimed to determine the incidence of re-fracture following implant removal after osteosynthesis of a hip fracture.

METHODS: We conducted a retrospective multicenter cohort study including patients aged ≥ 50 years who underwent implant removal between 2003 and 2023 after radiologically confirmed consolidation of a hip fracture. Patients were identified using procedural codes. Baseline variables included age, sex, ASA classification, fracture type (femoral neck or trochanteric), and implant type (pins/screws, sliding hip device [SHD], or short/long cephalomedullary nail [CMN]). Patients were followed from implant removal until re-fracture, conversion to arthroplasty, death, or end of follow-up, with a minimum follow-up of 1 year. Cox proportional hazards regression was used to assess associations between implant type and re-fracture risk, adjusting for age and sex. Because the proportional hazards assumption was violated, a time-stratified Cox regression and restricted mean survival time analyses were applied.

RESULTS: A total of 575 patients (median age 73 years, IQR 65-81) were included, with a median follow-up of 53 months (IQR 18-100). Lateral hip pain was the most common indication for implant removal (72.5%). The overall re-fracture incidence was 10.4%. Ris […]

PubMed · DOI

Physical functionality two years after major trauma: predictors from a single center cohort study.

Fetz K, Koetsenruijter J, Rodrigues Recchia D, Rutetzki J, Lefering R, Kaske S

PURPOSE: Traumatic injuries are a major global public health concern, often leading to impaired physical function and disability. Long-term physical function impairments can greatly affect a patient's health related quality of life, influencing their emotional state and social interactions. This study examines patient-reported physical functionality before and after major trauma, identifying key predictors of poor functional recovery.

METHODS: This retrospective single-center cohort study investigates physical functionality 23 months after a traumatic injury using standardized questionnaires combined with clinical data. Inclusion criteria were Injury Severity Score (ISS) ≥ 9, requiring ICU treatment, and age ≥ 18 resulting in an eligible sample of 515 patients. Health outcomes were assessed using the Trauma Outcome Profile (TOP), the Abbreviated Injury Scale (AIS), and the ISS.

RESULTS: Physical functionality significantly decreased from 94.8 (SD 11.5) to 72.9 (SD 26.7) 23 months post-trauma with almost half of participants still scoring below the critical cutoff of 80 points. The strongest predictor for functional physical impairment was pre-trauma physical functionality (OR 8.21, p < 0.001). Other determinants included age (age 50-59 versus 18-29 OR 2.75, p = 0.001), as well as an impacted lower extremity including pelvic (OR 1.60, p = 0.018).

CONCLUSION: The most importan […]

PubMed · DOI

Association between fresh frozen plasma transfusion and mortality stratified by Glasgow Coma Scale scores in isolated traumatic brain injury: a nationwide cohort study in Japan.

Nagamura T, Aoki M, Yamada K, Terayama T, Seno S, Tomura S, Kiyozumi T

PURPOSE: Fresh frozen plasma (FFP) is widely used to manage trauma-induced coagulopathy. However, its effectiveness in isolated traumatic brain injury (TBI) management remains controversial, particularly when stratified by severity based on the Glasgow Coma Scale (GCS) scores. This study aimed to examine the association between FFP administration within 24 h and mortality in patients with isolated TBI, overall and stratified by GCS on arrival.

METHODS: We conducted a multicenter retrospective cohort study using the Japan Trauma Data Bank (2019-2023). Adults with isolated TBI (intracranial AIS ≥ 3 and extracranial AIS < 3) were included. Patients were classified into FFP and non-FFP groups based on transfusion within 24 h of admission. Subgroup analyses were performed by TBI severity (severe: GCS ≤ 8; mild-to-moderate: GCS > 8), craniotomy status, and FFP dose. The primary outcome was 28-day mortality. Propensity score matching was applied to adjust for confounding.

RESULTS: Among 12,480 patients (median age 72 years [IQR: 55, 82]; 66.6% male), 513 received FFP. After matching, FFP was not significantly associated with 28-day mortality (OR 0.80; 95% CI 0.59-1.08). In subgroup analyses, FFP was associated with higher mortality in mild-to-moderate TBI (OR 2.38; 95% CI 1.07-5.31) and lower mortality in severe TBI (OR 0.61; 95% CI 0.42-0.88). No significant differences were observ […]

PubMed · DOI

Percutaneous fixation of posterior pelvic ring lesions in the elderly: current evidence and recent advances.

Liodakis E, Schreiber S, Giannoudis VP, Fritz T, Giannoudis PV

INTRODUCTION: Percutaneous fixation techniques are increasingly used in the management of posterior pelvic ring lesions in the elderly. As the incidence of fragility fractures continues to rise, a growing number of implants and surgical techniques have been developed to address the specific challenges of this vulnerable population. The goal of this scoping review is to address the following questions: 1) What are the latest advances in percutaneous fixation of posterior pelvic ring fractures? 2) What are the clinical results and current level of evidence supporting these techniques?

METHODS: A systematic search using the keywords “((Percutaneous) OR (Minimal* invasive) OR (Transsacral) OR (Innovative implant)) AND ((Pelvic fragility fracture) OR (Sacral fracture))” was performed in PubMed and Cochrane Library using PRISMA-ScR guidelines. Studies published from the 11th of August 2020 up to the 11th of August 2025 were included and assessed independently by two reviewers.

RESULTS: A total of 334 studies were found, of which 107 were included in this review following the screening process. Of these, 22 studies pursued a biomechanical approach. Various new implants (e.g. ISG-Rod System, SI-Plate, TriFix, SACRONAIL®) and technologies (e.g. 3D computer-assisted navigation/O-arm/Robot) are described in the recent literature. Using navigation significantly reduces the rate of screw […]

PubMed · DOI

Infrared thermography and dual-layer spectral CT for the detection of femoral vascular compromise: an experimental large animal study.

Hübner CT, Ricklin J, Müller AL, Klingebiel FK, Stoeck CT, Weisskopf M, Carneiro Gomes A, Cinelli P et al.

BACKGROUND: Rapid identification of vascular compromise remains a clinical challenge in the acute trauma setting. Conventional diagnostic tools such as CT angiography (CTA) and Doppler-ultrasound might provide decent anatomical and functional assessment. However, they are limited by ionizing radiation, contrast exposure or time requirements. Although not validated, infrared thermography (IRT) might offer a non-invasive alternative capable of detecting surface temperature asymmetries that may reflect underlying perfusion deficits. This study aimed to evaluate the feasibility of thermographic imaging in detecting vascular lesions.

METHODS: As part of a porcine polytrauma model (n = 28), vascular lesion of the right femoral artery was defined as a ≥ 50% diameter reduction on CTA. Temperature differences between both extremities were measured using infrared thermography. SDCT based iodine concentrations distal to the stenosis were quantified to assess perfusion changes. Correlation and regression analyses were performed between stenosis degree, temperature difference, and tissue iodine uptake.

RESULTS: Six animals fulfilled the criteria for relevant vascular compromise. This resulted in a mean vessel diameter reduction behind the lesion of 68% ±18%. The vascular lesion group demonstrated a mean inter-limb

temperature difference of 6.93 °C ± 1.78 °C compared to 0.71 °C ± 1.27 °C i […]

PubMed · DOI

Fixation strategies for periprosthetic femur fractures around knee replacements: a comprehensive single-center analysis of 102 consecutive cases.

Nomdedéu J, González-Morgado D, Minguell-Monyart J, Joshi-Jubert N, Teixidor-Serra J, Tomàs-Hernández J, Andrés-Peiró JV

PURPOSE: To evaluate 1 year mortality, complication rates, and functional outcomes after internal fixation of femoral periprosthetic knee fractures following total knee arthroplasty, and to identify predictors of adverse outcomes and increased healthcare resource utilization.

METHODS: This retrospective cohort study included 102 consecutive patients with femoral periprosthetic knee fractures classified as UCPF VB3 and VC3 treated with internal fixation at a single tertiary center between 2010 and 2023. Interprosthetic fractures and combined fixation techniques were excluded. Primary outcomes were 1 year mortality and postoperative complications. Secondary outcomes included length of stay, discharge destination, and ambulatory status at follow up.

RESULTS: Patients were markedly frail, with a mean age of 82.4 ± 10.8 years and mean Charlson Comorbidity Index of 5.6. One year mortality was 16.7% and was associated with older age, higher ASA class, higher Charlson index, impaired pre fracture ambulation, and fracture related infection. Overall complication rate was 29.4%, including 15.7% infections and 13.7% mechanical failures. Open reduction was more frequent in plate fixation and resulted in longer operative times, without increasing complication rates. Median hospital stay was 18 days, 66.3% required discharge to nursing facilities, and only 7.8% regained independent ambulati […]

PubMed · DOI

Epidemiology and management of pediatric femoral fractures at a level I trauma center: a 10-year retrospective monocentric study.

Dietz SO, Traub F, Schmidt E, Schippers P, Wegner E, Schierjott L, Gercek E, Nowak TE

PURPOSE: Pediatric femoral fractures are relatively uncommon but represent severe injuries frequently requiring inpatient trauma care. Epidemiological characteristics, fracture patterns, and treatment strategies vary between institutions and regions. This study aimed to analyze the epidemiology, fracture distribution, treatment modalities, complications, and fracture-predisposing comorbidities of pediatric femoral fractures treated at a German level I pediatric trauma center over a 10-year period.

METHODS: All patients aged 0-14 years treated for femoral fractures between January 2012 and December 2021 were retrospectively reviewed. Fractures were categorized using the AO Pediatric Comprehensive Classification of Long-Bone Fractures (PCCF). Demographic variables, fracture characteristics, management approaches, time to consolidation, complications, and relevant comorbidities predisposing to fracture were analysed descriptively in accordance with STROBE recommendations.

RESULTS: A total of 183 patients met the inclusion criteria (67.2% male; mean age 5.2 years). Children aged 2-5 years constituted the largest group (42.1%). Femoral shaft fractures were the most frequent femoral fracture type (70%). Surgical management, most frequently elastic stable intramedullary nailing (ESIN), was increasingly utilized with advancing age. The overall complication rate was 14.2%, with the ma […]

PubMed · DOI

Blood-based biomarkers GFAP/UCH-L1 for the diagnosis of mild traumatic brain injury (mTBI): a single-center implementation experience.

Maegele M, Breidenbach L, Kaufmann J, Keles Y, Uhl E, Schroeer P

INTRODUCTION: Mild traumatic brain injuries (mTBI) affect millions of people worldwide every year as one of the most common clinical presentations in the emergency department. Diagnosis is mainly based on clinical criteria and computed tomography scans. The use of computed tomography causes high costs, long waiting times in daily clinical practice and radiation exposure. GFAP (glial fibrillary acidic protein) and UCH-L1 (ubiquitin carboxyl-terminal hydrolase-L1) turned out to be potential biomarkers for the diagnosis of mTBI. This study retrospectively evaluates the possible use of these biomarkers combined as negative predictors for excluding brain

injuries in patients with suspected mTBI in the emergency department.

METHODS: Adult patients (n = 320) registered in the emergency department at a level 1 trauma emergency center in Germany (Cologne Merheim Medical Center/CMMC) between 11/2023 and 04/2024, with suspected mTBI, Glasgow Coma Scale (GCS) score 13-15 and within 12 h after trauma were considered. All evaluable patients underwent cranial CT (cCT) scans and blood tests for GFAP and UCH-L1 serum concentrations.

RESULTS: Biomarkers GFAP and UCH-L1 were tested positive in 261 patients (82%) while CT detected intracranial injuries in only 29 patients (9%). Biomarkers combined had a sensitivity of 97% and a negative predictive value (NPV) of 98% in mTBI diagnosis with a nega […]

PubMed · DOI

Resuscitative endovascular occlusion of the aorta restores cerebral metabolic markers of ischaemia induced by haemorrhagic shock.

Bader SE, Magnuson A, Brorsson C, Wallin G, Löfgren N, Löfgren F, Blind PJ, Öman M et al.

BACKGROUND: Resuscitative endovascular balloon occlusion of the aorta (REBOA) is used as an adjunct in haemorrhagic shock (HS) to restore proximal perfusion. Its cerebral metabolic effects during haemorrhagic shock, particularly in the presence of elevated intracranial pressure (ICP), remain incompletely characterised.

METHODS: In a controlled porcine model, HS was induced by controlled bleeding to a mean arterial pressure of approximately 40 mmHg. Animals were allocated to a normal ICP group or an experimentally elevated ICP group. Total REBOA (zone 1) was applied for 90 min. Cerebral metabolism was assessed using intracerebral microdialysis with measurements of lactate, pyruvate, and lactate-pyruvate ratio (LPR). Cerebral haemodynamics and ICP were continuously monitored.

RESULTS: HS was associated with a statistically significant increase in LPR in both groups, indicating cerebral metabolic disturbance. Following aortic occlusion, LPR gradually decreased toward baseline in both groups. Animals with elevated ICP demonstrated a transient delay in metabolic normalisation during the early post-occlusion phase. Statistically significant differences between groups were limited to the first 10 min following occlusion. Overall metabolic trajectories were similar thereafter.

CONCLUSIONS: Total REBOA restored cerebral metabolic markers of ischaemia during haemorrhagic shock, as ref […]

PubMed · DOI

Delayed single-stage surgical management using a peroneus brevis muscle flap for fracture-related infection after lateral malleolus fixation: a retrospective case series.

Dyreborg K, Jørgensen AI, Peterlin AAN, Gottlieb H

BACKGROUND: Fracture-related infection (FRI) following lateral malleolus fracture fixation complicated by soft tissue breakdown represents a challenging clinical problem with no established standard of care. This study evaluates a treatment strategy consisting of prolonged antibiotic suppression until fracture union, followed by delayed single-stage surgical management.

METHODS: This retrospective consecutive case series included 11 patients treated between 2020 and 2023 at a multidisciplinary orthopaedic infection unit. All patients were managed according to a standardized protocol involving oral antibiotic suppression until radiologically confirmed fracture consolidation. This was followed by a single-stage surgical procedure including implant removal, radical debridement, deep tissue sampling, local antibiotic therapy, and soft tissue reconstruction using a distally based peroneus brevis muscle flap with split-thickness skin grafting.

RESULTS: Eleven consecutive patients (median age 58 years) were treated according to the protocol. Six patients underwent antibiotic suppression for a median duration of 147 days prior to surgery. Among patients available for outcome assessment, infection control was achieved in 10 of 10 surviving patients. One patient died of unrelated causes within 14 days postoperatively and was not classified as treatment failure. Soft tissue complication […]

PubMed · DOI

Mixed reality improves agreement on surgical approach selection and patient positioning in tibial plateau fracture planning compared to CT, 3DCT and 3D printing.

Dust T, Henneberg JE, Hartel M, Reiter A, Kylies J, Korthaus A, Streckenbach A, Keller J et al.

PURPOSE: Tibial plateau fractures require precise preoperative planning, particularly regarding treatment concept, patient positioning, and surgical approach. Mixed reality (MR) may enhance three-dimensional understanding of fracture morphology beyond conventional imaging and physical 3D models. This study investigated whether MR improves agreement on key preoperative planning decisions compared to computed tomography (CT), 3D computed tomography (3DCT), and 3D-printed fracture models.

METHODS: Twelve orthopaedic trauma surgeons (6 junior, 6 senior) evaluated 22 surgically treated tibial plateau fractures (AO/OTA type B or C). Each case was assessed in four steps using (1) CT, (2) 3DCT reconstructions, (3) physical 3D-printed models, and (4) MR visualization using Microsoft Hololens 2. Raters selected (i) treatment concept, (ii) patient positioning, and (iii) surgical approach. Interobserver agreement was analysed using Fleiss' kappa (κ) and percentage match (PM).

RESULTS: Mixed Reality (MR) yielded the highest interobserver agreement for both surgical approach (PM 32%, κ = 0.30) and patient positioning (PM 57%, κ = 0.35), particularly among junior surgeons (approach PM 39%, κ = 0.28; positioning PM 57%, κ = 0.39). Compared to CT, 3DCT, and 3D printing, MR demonstrated consistent improvements, while agreement on treatment concepts remained high across all modalities (PM > 82% […]

PubMed · DOI

Timing of fracture fixation for femur and pelvis fractures in patients with severe traumatic brain injury - an analysis of the TraumaRegister DGU®.

Bayer TP, Teuben MPJ, Halvachizadeh S, Shehu A, Lefering R, Pfeifer R, Pape HC, Jensen KO

PURPOSE: Fracture fixation timing and strategy in polytrauma patients with traumatic brain injury (TBI) remain controversial. This study investigates treatment patterns and outcomes for femoral and/or pelvic fractures stratified by TBI severity.

METHODS: Patients in the TraumaRegister DGU® (2016-2022) with pelvic and/or femoral fractures (AIS ≥3) and TBI (head AIS ≥3) were included. Strategies were non-operative management (NOM), early total care (ETC), and damage-control orthopedics (DCO). Outcomes included treatment allocation, fixation timing, and in-hospital mortality.

RESULTS: 985 patients were included (mean age 52.5, SD 26.3 years; ISS 27.8, SD 8.1). Allocation was NOM in 320 (32.5%), ETC in 336 (34.1%), and DCO in 329 (33.4%) patients. Head AIS was 3 in 48.5%, 4 in 31.1%, and 5 in 20.3%. NOM patients were older, had the highest ISS and estimated mortality, and showed the largest proportion of critical TBI (AIS 5: NOM 30.9%, ETC 14.3%, DCO 16.1%). Femoral ETC was mainly performed within the first day (median 0, IQR 0-1 days), whereas pelvic ETC was delayed with increasing TBI severity (median 3, IQR 0-5 days for head AIS 3; 5, IQR 0-7 days for AIS 4). Observed mortality was 37.2% after NOM, 9.2% after ETC, and 10.3% after DCO.

CONCLUSION: ETC in patients with moderate TBI (AIS 3) was associated with reduced observed mortality relative to NOM and matching DCO. Increasi […]

PubMed · DOI

From single to dual platelet inhibition FIBTEM assay: fibrinogen replacement thresholds for critical bleeding after trauma.

Latona A, Winearls J, Ho A, Mitra B

PURPOSE: The FIBTEM assay in Rotational Thromboelastometry (ROTEM®) measures fibrin-based clot formation. The reagent transitioned from single to dual-platelet inhibition with addition of tirofiban to cytochalasin D, offering more complete platelet suppression. The aim of this study was to assess the impact of this change on transfusion thresholds in trauma.

METHODS: In this retrospective cohort study, we included adult patients with major trauma across Queensland who presented between 2022 and 2025 and had results available for paired ROTEM and conventional coagulation tests on arrival. Patients were sub-grouped into single and dual-platelet inhibition assay groups. Correlations

between FIBTEM-A5 and Clauss fibrinogen were analysed, comparing correlation strength and diagnostic accuracy for fibrinogen levels of < 2.0 g/L.

RESULTS: The relationship between FIBTEM-A5 and Clauss fibrinogen was similar using single platelet inhibition assays (R^2 0.61;95%CI: 0.51-0.70) and dual-platelet inhibition assays (R^2 0.70;95%CI: 0.62-0.77; p = 0.94). Predicted FIBTEM-A5 amplitudes at 2 g/L fibrinogen were 7.83 mm (95%CI: 7.42-8.23) and 7.54 mm (95%CI: 7.22-7.87). Diagnostic performance for detecting fibrinogen < 2.0 g/L was comparable between assays, for both FIBTEM-A5 ≤ 10 mm and FIBTEM-A5 ≤ 8 mm (all p > 0.05). For the dual-platelet inhibition assay, sensitivity and specificity were 97. […]

PubMed · DOI

Pattern of blast injuries. A systematic review: Part 2 - Landmines, unexploded ordnance and terrorism.

Neubert A, Gaeth C, Seidelmann M, Bieler D

PURPOSE: This systematic review analyzes the patterns and distribution of blast injuries from landmine explosions, UXOs and terror attacks including suicide bombing, in civilians and military personnel.

METHODS: A search was conducted in two databases on 26. March 2025 for observational studies reporting blast injuries in civilian and military populations of all ages. Studies were clustered by explosion typ. A narrative data synthesis was performed. The JBI tool for case series was used.

PROSPERO: CRD42023391736.

RESULTS: 126 studies were included in the overall systematic review. 45 studies on terror & suicide bombings and 12 studies on landmines & UXOs are analyzed in the present paper. Studies on landmines from 1980 to 2006 included 13,270 individuals (age: 24.0 years (Range 8.2 years - 32.2 years); mainly male (88.9%)) in post-conflict low- and middle-income countries. Lower extremity injuries including amputations were most frequent, followed by eye injuries. Studies on terrorist attacks with a total of 12,518 mainly civilian victims (average age: 33,1 years; 61.6% males) reported on incidents in post-conflict regions and high-income cities mainly from the early 2000s. The injury pattern was multidimensional, with extremity injuries most frequent (28%-30%), including fractures, lacerations, and traumatic amputations (18%-20%).

CONCLUSION: This Systematic Review Part 2 […]

PubMed · DOI

Is nail plate fixation the way forward for distal femur fractures with long segmental metaphyseal comminution? - A study of 56 patients.

Mahesan H, Sundaram VP, Devendra A, Ramesh P, Dheenadhayalan J, Rajasekaran S

BACKGROUND: Mechanical instability due to long metaphyseal comminution and medial bone gap after lateral locking plate results in varus collapse and implant failure. We believe that fixation in the form of Nail and plate combination provides better stability and prevents complications.

MATERIALS: We performed a retrospective analysis of 56 distal femur fractures with long metaphyseal comminution treated in our institution with nail and plate construct from 2017 to 2023. The mean age was 43.1 years and the mean follow up was 24 Months (4-36 months). Of these 56 patients, 16 were closed injuries underwent primary fixation, and 40 were open injuries requiring staged reconstruction. At the final follow up, patients were assessed for shortening, knee range of movements, Sander et al. functional score, radiological union and loss in alignment.

RESULTS: The mean length of metaphyseal comminution was 10.15 cm(5-18 cm). Primary Iliac crest bone grafting was done in 25 patients. Of the 56 patients, 51 achieved primary union and the mean time to union was 6 months. Of the remaining, 3 patients united after secondary bone grafting and 2 patients had nonunion. The mean loss in aLDFA at the final follow up was 1.75° in 7 patients. Shortening of the limb was noted in 16 patients, and 4 patients had significant shortening of 2 cm. Three patients developed deep SSI, of which one patient requi […]

PubMed · DOI

Impact of modified ERAS protocol implementation on length of stay and complications in elderly hip fracture patients: a before-and-after study.

Liu Y, Ou G, Xie M, Wu F, Wang X, Yang B

AIM: Hip fractures in elderly patients pose significant healthcare challenges with high complication rates and prolonged hospital stays. Enhanced Recovery After Surgery (ERAS) protocols have shown promise in various surgical specialties, but their application in elderly hip fracture patients, particularly in Traditional Chinese Medicine (TCM) hospital settings, remains limited.

METHODS: This retrospective before-and-after study was conducted at a TCM hospital from July 2022 to July 2025. A total of 300 elderly hip fracture patients (≥ 65 years) were included: 146 in the control group (July 2022 - October 2023) receiving conventional care and 154 in the ERAS group (February 2024 - July 2025) receiving modified ERAS protocol incorporating TCM elements. Primary outcomes were length of hospital stay and perioperative complications within 30 days. Secondary outcomes included functional recovery [Harris Hip Score (HHS), Modified Barthel Index (MBI)], time to first mobilization, pain scores, and healthcare costs.

RESULTS: The ERAS group demonstrated significantly shorter length of stay compared with controls (12.38 ± 3.75 vs. 16.24 ± 5.10 days; mean difference 3.86 days, 95% CI: 2.84-4.88; P < 0.001). Total perioperative complications were reduced (7.79% vs. 15.07%; risk difference: -7.28%, 95% CI: -14.5% to - 0.1%; P = 0.047). The ERAS group showed superior functional outcomes with […]

PubMed · DOI

Prehospital endotracheal intubation and 30-day survival in severe trauma: a matched cohort study from Navarra, Spain.

Zulet Murillo D, Ferraz Torres M, Fortún Moral M, Belzunegui Otano T, Tamayo Rodríguez I

PURPOSE: Prehospital endotracheal intubation (ETI) is frequently used in major trauma, but its association with 30-day survival remains debated. We evaluated the association between prehospital ETI and 30-day survival and assessed whether sex modifies the age-related increase in mortality risk.

METHODS: We conducted a retrospective cohort study using the Navarra Major Trauma Registry (2010-2019). Patients aged ≥15 years with New Injury Severity Score (NISS) ≥15 were included; patients who died before hospital admission or had incomplete records were excluded. To mitigate confounding by indication, intubated and non-intubated patients were matched 1:1 using prehospital Glasgow Coma Scale (GCS), respiratory distress, and prehospital cardiac arrest as matching variables. Because the proportional hazards assumption for Cox regression was not met, parametric Weibull survival models were fitted, including a sex-by-age interaction term.

RESULTS: Overall, 1,909 patients were included; 212 (11.1%) underwent prehospital ETI. Matching yielded 190 pairs (n=380). In the matched cohort, prehospital ETI was not independently associated with 30-day survival (adjusted HR 0.91, 95% CI 0.65-1.27). Mortality risk was independently associated with lower GCS (per 1-point increase: HR 0.86, 95% CI 0.82-0.91) and respiratory distress (HR 7.63, 95% CI 4.34-13.4). A significant sex-by-age interaction […]

PubMed · DOI

Mechanical and manual cardiopulmonary resuscitation in traumatic cardiac arrest: a retrospective observational study.

El-Menyar A, Ahmed K, Howland IR, Mollazehi M, Mekkodathil A, Rizoli S, Cameron P, de Oliveira Manoel AL et al.

BACKGROUND: Prehospital Traumatic cardiac arrest has a high fatality rate despite advances in trauma systems and resuscitation strategies. Mechanical chest compression devices have been increasingly adopted to optimize cardiopulmonary resuscitation (CPR) during Emergency Medical Services (EMS) transportation. We aimed to evaluate the impact of CPR (mechanical during transportation versus manual CPR only at the scene) on survival among trauma patients.

METHODS: A retrospective analysis was conducted for patients who received CPR at the scene and during transportation to the hospital between 2016 and 2024.

RESULTS: A total of 610 patients sustaining blunt traumatic cardiac arrest were included. The mean age was 34 ± 12.6 years; 94.6% were male, and 5.4% were female patients. Two hundred twenty (36.1%) received mechanical

CPR during transportation to the hospital, and 390 (63.9%) received manual CPR only at the scene. Compared with the manual CPR group, the mechanical CPR group had higher rates of primary chest injury (30.5% vs. 20.5%; p = 0.006), bystander CPR (15.9% vs. 6.7%; p = 0.001), adrenaline administrations (95% vs. 26.4%; p = 0.001), and initial non-shockable rhythm (80% vs. 27.2%; p = 0.001). The two groups were comparable in median Injury Severity Score (ISS), head Abbreviated Injury Score (AIS), and total transport time. Mechanical CPR was associated with a markedly […]

PubMed · DOI

Cervical spine clearance in pediatric trauma patients - Consensus algorithm of the Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU).

Bolte J, Rüther H, Brecht P, Wiersbicki DW, Youssef Y, Jarvers JS, Disch AC

PURPOSE: Although cervical spine injuries in pediatric patients are rare, clinical and radiological examinations are clinical routine. Currently, no standardized decision algorithm does exist for detection of these injuries in the European region. The proposed algorithm is aiming to safely identify significant injuries through consistent diagnostics while minimizing immobilization and radiation exposure at the same time.

METHODS: The Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU) developed an algorithm based on own scientific investigations and considering primary and secondary literature. The algorithm was developed through a formal consensus process and thereby aligns with international recommendations.

RESULTS: The algorithm uses the initial pediatric Glasgow Coma Scale (pGCS) to group patients to three treatment pathways. Resulting recommendations for imaging, clinical re-assessment, and further management include the child's age, trauma mechanism, and clinical evaluations.

CONCLUSION: Despite their rarity, potential cervical spine injury in children requires accurate diagnosis. Clinical examinations can be challenging, especially in children with compromised consciousness. Increased radiation exposure from X-rays and computer tomography (CT) scans must be considered, particularly as magnetic resonance im […]

PubMed · DOI

Development of a low-cost external fixation system: an off-label solution for resource-limited settings.

Seiser M, Deininger C, Fierlbeck J, Hollensteiner M, Coles P, Reuter C, Tempfer H, Traweger A et al.

PURPOSE: External fixators are essential for the surgical management of long-bone fractures, particularly in resource-limited settings where access to commercial systems is often restricted by cost and logistics. This study aimed to develop a low-cost external fixation concept using widely available construction materials and evaluate its biomechanical performance under axial loading.

METHODS: An external fixator composed of M10 threaded rods, washers, and nuts was developed. Five configurations were tested: SP (Steinmann pins), KT (threaded K-wires), KS (smooth K-wires), and BKS (bilaterally applied, smooth K-wires). A commercial external fixator served as control (CG). Juvenile bovine ulnae with a standardized fracture gap were subjected to cyclic axial loading. Construct stiffness, elastic deformation, and plastic deformation were assessed.

RESULTS: CG showed the highest stiffness (107.80 ± 14.12 N/mm), followed by SP (87.73 ± 17.92 N/mm), BKS (83.81 ± 10.01 N/mm), KS (49.84 ± 3.80 N/mm), and KT (46.66 ± 2.39 N/mm). Post-hoc analyses showed no significant differences between CG vs. SP and CG vs. BKS, whereas KT and KS were significantly less stiff. Elastic deformation followed a similar pattern. Plastic deformation was greatest in CG, while SP and BKS showed significantly lower values.

CONCLUSIONS: The threaded-rod fixator provides relevant mechanical stability at extreme […]

PubMed · DOI